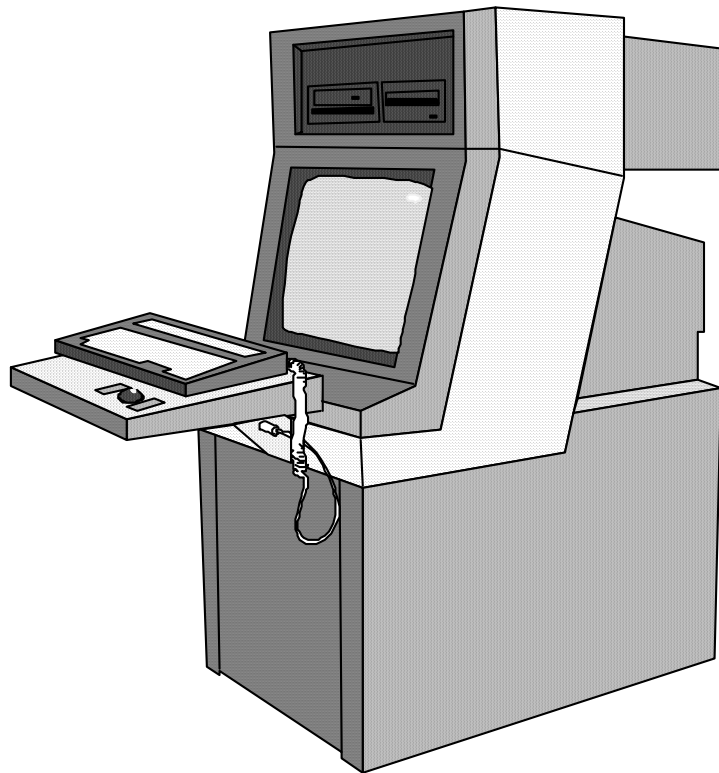


Universal Station Service



LCN Service - 1

Universal Station Service

US13-600

Release 630

CE Compliant

11/04

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Revision 630 – November 11, 2004

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About This Publication

Purpose

This publication provides instructions for use by the system service personnel, to service the Universal Station with the exception of Five/Ten-Slot Module and Dual Node Module(s). It will help you in determining how to perform service required on Universal Stations, identify spare parts, and will provide the disassembly/assembly instructions required for replacing a part.

This publication supports **TotalPlant** Solution (TPS) Network Release 521 and earlier releases. TPS is the evolution of TDC 3000^X.

This publication supports CE Compliant equipment. Any equipment designated as “CE Compliant” complies with the European Union EMC and Health and Safety Directives. All equipment shipping into European Union countries after January 1, 1996 requires this type of compliance—denoted by the “CE Mark.”

A change bar is used to indicate that a paragraph, table, or illustration has been changed or added because of updated information. A change bar is not used to indicate a correction for a minor typographical error.

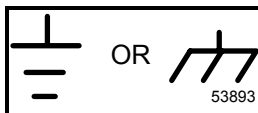
Standard Symbols

Scope

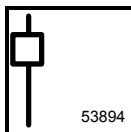
ATTENTION

CAUTION

WARNING



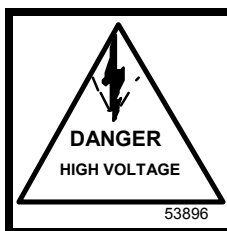
Ground connection to building safety ground.



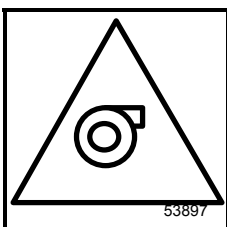
Ground stake for building safety ground.



Electrical Shock Hazard—can be lethal.



Electrical Shock Hazard—can be lethal.



Rotating Fan—can cause personal injury.

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1 INTRODUCTION

1.1 Overview

This manual provides detailed instructions for maintenance and repair of the Universal Station. Section 2 lists the various Periodic Replaceable Items (printer ribbons, etc.) that are used with the Station, and discusses optional equipment. The remaining sections address each of the ORUs regarding cleaning, adjustments, test/troubleshooting, removal/replacement, ORU part numbers, and startup. Figure 1-1 shows a typical Universal Station.

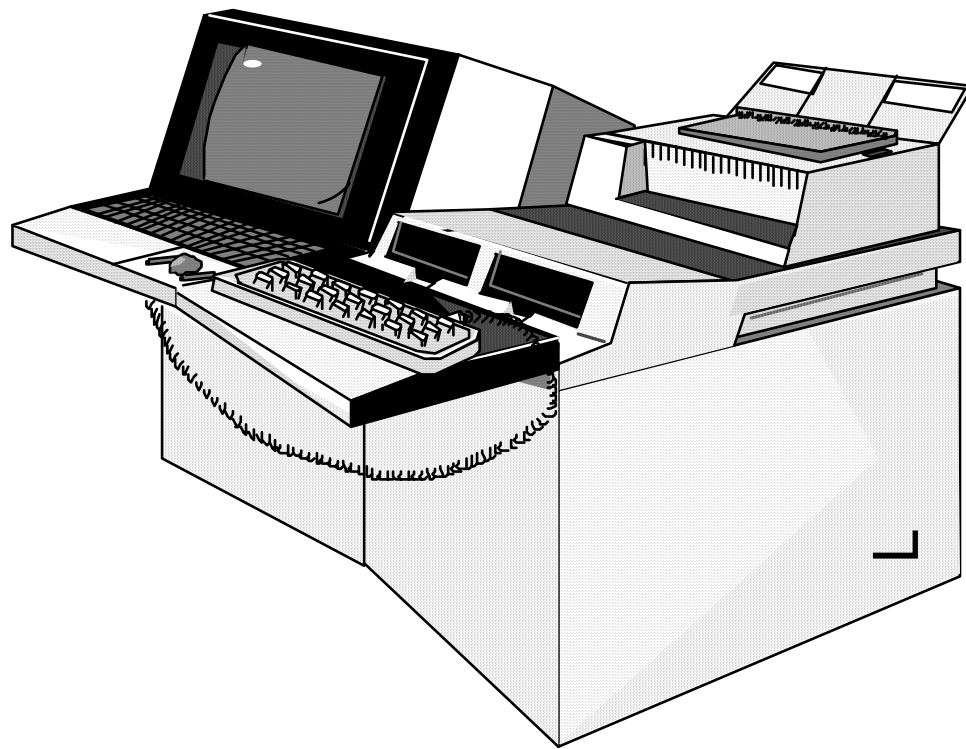


Figure 1-1 — Typical Universal Station

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1.2 Related Publications

The following publications apply to the **TotalPlant** Solution (TPS) System, or to the station ORUs and should be referred to as required and available:

Publication Title	Publication Number	Binder Title	Binder Number
<i>LCN System Installation</i>	SW20-500	LCN Installation	TPS 3025
<i>LCN System Checkout</i>	SW20-510	LCN Installation	TPS 3025
<i>System Maintenance Guide</i>	SW13-500	LCN Service-1	TPS 3060-1
<i>Maintenance Test Operations</i>	SW11-502	LCN Service-1	TPS 3060-1
<i>Five/Ten-Slot Module Service</i>	LC13-500	LCN Service-2	TPS 3060-2
<i>Dual Node Module Service</i>	LC13-510	LCN Service-2	TPS 3060-2
<i>Process Operations Manual</i>	SW11-501	Operation/Process Operations	TPS 3050

CAUTION

All assemblies used in the station are likely to contain electronically sensitive devices. Use a personal grounding strap and grounded work surfaces/equipment. Store and transport parts only in electrostatically safe containers.

2 STATION COMPONENTS

2.1 General Description

The Universal Station provides the man-machine interface to the **TotalPlant** Solution (TPS) System. As this role varies greatly from one application to the next, the station configuration can be varied to best suit each situation by the choice of optional equipment. The station(s) may differ somewhat in overall makeup from those discussed here, but the individual ORUs are the same. The various ORUs that comprise the Universal Station are shown in Figure 2-1, which represents an installation with available optional equipment.

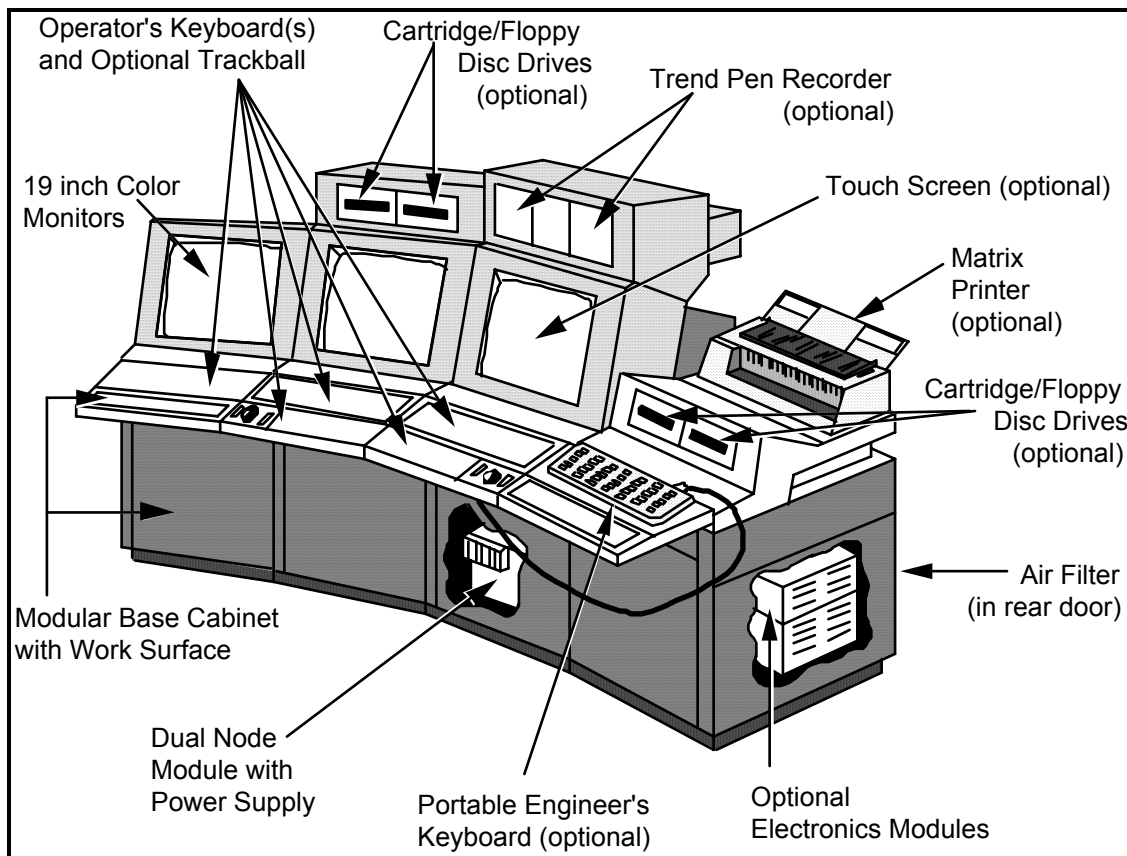


Figure 2-1 — Universal Station ORU Locations

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2.2 Periodic Replaceable Items

Table 2-1 lists the items replaced periodically. They are in alphabetical order with the applicable Honeywell part numbers. Each device within the Universal Station has its own section in this manual, along with its own cabling diagram and ORU list.

Table 2-1 — Periodic Replacement Items

Item Name	Part Number*
Air Filter, Cabinet — 74 x 17 cm (29 x 6 inches)	51201201-100
Air Filter, Console — 42 x 35 cm (16 x 13 inches)	51201201-200
Cartridge Disk, 20 MB	51195169-100
Cartridge Disk, 35 MB Multidisk 150 MB Compatible 5.25" cartridge 150 MB Multidisk 150 MB Compatible 5.25" cartridge	51195485-100 51195485-400
Cartridge Disk Device Cleaning Kit	51196485-500
Floppy Diskette	MP-MKBL10
Keyboard Blank Insert	51400759-100
Matrix Printer Ribbon — ASPI-32 (optional Long-Life)	51194335-902
Ribbon, Black — ASPI-41	51108305-700
Ribbon, Black — ASPI-46	51195091-700
Ribbon, Black—Signum 2043 (15-pack)	51196690-400
Ribbon, Black—Signum 2043 (single-pack)	51196690-401
Ribbon, Color—Signum 2043 (15-pack)	51196690-500
Ribbon, Color—Signum 2043 (single-pack)	51196690-501

* Refer to *Five/Ten-Slot Module Service* for board configuration and part numbers.

2.3 CRT Monitor Assortment

Table 2-2 lists CRT Monitors found in TPS equipment. Use this listing to help identify the type of monitor installed in your console.

Table 2-2 — CRT Monitors

	Display CRT (Electrohome)	Interlaced CRT (Aydin 8828)	Enhanced Noninterlaced CRT (Aydin 8920)	Enhanced Noninterlaced CRT (Aydin 8939)
Honeywell Part Number	51108695-100 (110 V) 51108695-200 (220 V)	51108308-100 (110 V) 51108308-200 (220 V)	51304360-100 (110 V) 51304360-200 (220 V)	51304449-100 (110 V/220 V)
US Board Complement	VDG VDG I/O ----- ----- PDG PDG I/O ----- ----- EPDG PDG I/O	VDG VDG I/O ----- ----- PDG PDG I/O ----- ----- EPDG PDG I/O	EPDG, EPDG I/O, or EPDGP I/O	EPDG, EPDG I/O, or EPDGP I/O

The EPDG2 is a replacement board for the EPDG board.

2. STATION COMPONENTS

2.4. CE Compliant Station Cabling Schematic

2.4 CE Compliant Station Cabling Schematic

The numbers in the circles of Figures 2-2 and Figure 2-3 represent item numbers in Tables 2-3 and 2-4, respectively.

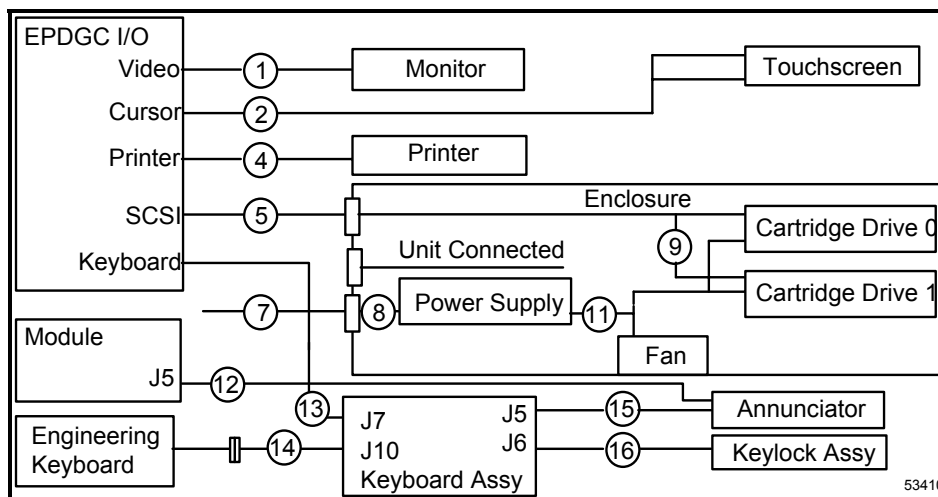


Figure 2-2 — US Cabling—Classic Furniture with Touchscreen

Table 2-3 — US Cabling—Classic Furniture with Touchscreen

Reference Number	Part Number	Application Notes
1	51308120-002	Classic Console EC Video Cable (lower)
	51308120-003	Classic Console EC Video Cable (upper)
2	51308123-300	EC Smartscreen Cable
3		Not Applicable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable — 15 meters/49.2 ft.
5	51192054-102	Cable,SCSI Shielded, 50 position
6		Not Applicable
7	51308094-600	Classic Module AC Power Cable
8	51305342-100	Power Supply Enclosure A.C. Cable
9	51305070-202	Cable, SCSI (1 or 2 Drives)
10		Not Applicable
11	51305341-100	DC Power Cable, Fan
12	51303420-586	Module Status Cable

Table 2-3 — US Cabling—Classic Furniture with Touchscreen

Reference Number	Part Number	Application Notes
13	51308126-100	Keyboard Cable
14	51304496-026	Engineer's Keyboard Cable
15	51304047-652	Annunciator Power Cable
16	51304029-500	Keylock Switch Cable
17	51305070-103	SCSI Cable (not connected in this configuration)

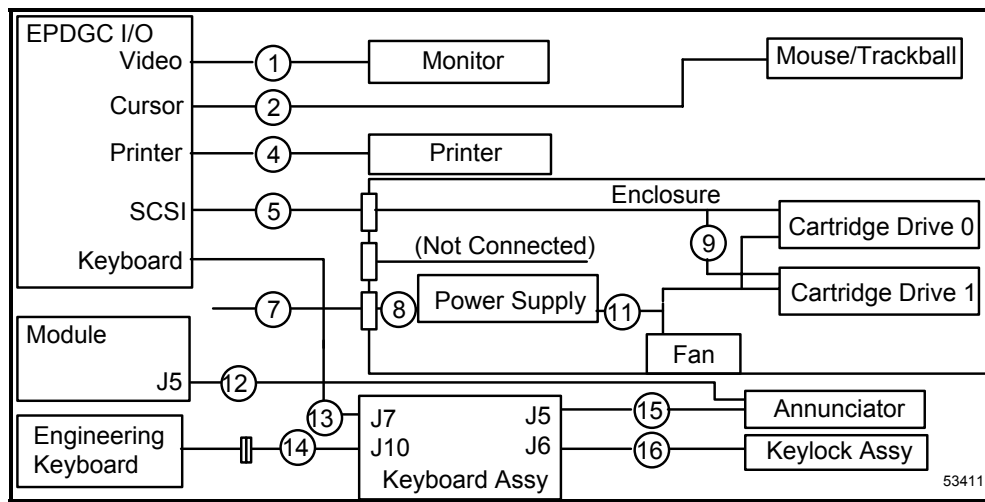


Figure 2-3 — US Cabling—Classic Furniture with Mouse/Trackball

2. STATION COMPONENTS

2.4. CE Compliant Station Cabling Schematic

Table 2-4 — US Cabling—Classic Furniture with Mouse/Trackball

Reference Number	Part Number	Application Notes
1	51308120-002	Classic Console EC Video Cable (lower)
	51308120-003	Classic Console EC Video Cable (upper)
2	51308099-100	EPDGC Mouse/Trackball Interface Cable
3		Not Applicable
4	51308103-003	Printer Cable – 3 meters/9.8 ft.
	51308103-008	Printer Cable – 8 meters/26.3 ft.
	51308103-015	Printer Cable – 15 meters/49.2 ft.
5	51192054-102	Cable,SCSI Shielded, 50 position
6		Not Applicable
7	51308094-600	Classic Module AC Power Cable (Skynet P.S.)
8	51305342-100	P.S. Enclosure AC Cable (Power Input to Skynet P.S.)
9	51305070-202	Cable, SCSI (1 or 2 Drives)
10		Not Applicable
11	51305341-100	DC Power cable, Fan
12	51303420-586	Module Status Cable
13	51308126-100	Keyboard Cable
14	51304496-026	Engineer's Keyboard Cable
15	51304047-652	Annunciator Power Cable
16	51304029-500	Keylock Switch Cable
17	51305070-103	SCSI Cable(not connected in this configuration)

3 DISPLAY CRT ASSEMBLY 51108695-100

NOTE

The 51108695-100 CRT described in this section is no longer available from the manufacturer. Spare parts are still available. Use the 51108308-100 described in Section 4 as a functional replacement for this DISPLAY CRT.

3.1 Cleaning

NOTE

Cleaning of the station color monitor CRT, touchscreen, and keyboards should be done with power off, and the station off-line; therefore, it is best to do them all at once as described below. For the touchscreen cleaning procedure, see subsection 6.1.

1. Open door below keyboards and set the main power circuit breaker located in the upper-right corner to the down (OFF) position.
2. If you have a keyboard power supply, set its power supply switch, located just to the left of the main power circuit breaker, to "0" (OFF).
3. Open station rear door (below CRT chassis) and set the Module Power Switch to OFF.
4. Switch all other peripheral equipment associated with the station, such as floppy disk drives, etc. to OFF.
5. Clean the CRT face with a mild detergent solution and soft, lint-free cloth (do not use paper wipes of any kind). It is best to spray the solution on the CRT face to dissipate any static charge before wiping with the cloth.
6. Clean the Operator/Engineer keyboard by dampening soft cloth with mild detergent and wiping keys. Do not spray detergent solution on keys as this may contaminate the switch mechanism under the keys.
7. Restore power in accordance with subsection 3.5, *Startup*.

3.2 Checks and Adjustments

3.2.1 Operator Adjustments

The checks and adjustments provided in this subsection are intended for use by the maintenance technician. The only adjustments available to the operator are listed as follows:

- Degauss—With power applied to the station, press the degauss button on the right side of the screen. Hold the button for a few seconds until colors are restored.
- Brightness and Contrast—Adjust brightness and contrast thumb wheels on the right side of the screen for the desired affect.

3.2.2 Service Checks

There are no built-in tests for the CRT monitor. Fault isolation consists of ensuring that power is applied, and determining whether a fault is caused by the CRT monitor, by the touchscreen, or some other source. The following list of symptoms and isolation procedures should be consulted before performing any maintenance technician adjustments:

SYMPTOM	FAULT ISOLATION PROCEDURE
Display black	Ensure that the power cord is plugged in. Check the power circuit breaker behind the door under the keyboard. Ensure that all LEDs on the 10-Slot Module are green. Set the power circuit breaker to OFF. Remove the rear door to the cabinet by opening and lifting off the hinge pins. Loosen the captive screws at the bottom of the CRT rear cover and pull the cover straight off the unit. Check the fuse located on the rear panel of the CRT chassis and replace, if required, with a fuse of the same type and rating (see Figure 3-3 and 3-10). Ensure that the green input cable is present and tight. Replace cover and door, and refer to subsection 3.5, <i>Startup</i> .
No display, just a raster, or "snow"	Ensure that power is applied to the 10-card module. Remove the cabinet rear door and CRT rear cover (see Display black—Fault Isolation Procedure). Check the cables on the rear of the CRT and the 10-card module to ensure that they are secure. Ensure that the cables between the PIC and VDG or PDG boards on the front of the 10-card module are present and tight. Refer to <i>Five/Ten-Slot Module Service</i> for further troubleshooting instructions. Replace cover and door.
Display just vertical or horizontal line or dot	Refer to <i>Five/Ten-Slot Module Service</i> for module troubleshooting procedures. If the fault cannot be isolated to the module, the CRT is defective.

SYMPTOM**FAULT ISOLATION PROCEDURE**

Colors not right Press the degauss button on the front of the touchscreen and hold for three to five seconds. Ensure that contrast and brightness are correctly adjusted. Remove the rear door and the CRT rear cover (see Display black—Fault Isolation Procedure). Check the red, green, and blue cables on CRT and module.

3.2.3 Maintenance Technician Adjustments

It is recommended that the following procedures be performed in the order they are listed. Refer to Figures 3-1 through 3-12 for component locations and wave shapes. Use of electrically nonconducting equipment (such as plastic screwdrivers) is recommended. If a setting cannot be adjusted within specifications, refer to subsection 3.3, *Troubleshooting*.

3.2.4 Equipment and Materials

The following list of test equipment is recommended as a minimum to fully support the CRT assembly. Additional equipment may also be used, as may equivalent substitutes for the listed items:

Digital Multimeter (DMM)—20 k ohms/Vdc, or greater, input impedance

High Voltage Probe—rated to 30 kV or greater (used with DMM)

Oscilloscope (scope)—dual channel

10:1 probe (2 required)—for use with oscilloscope

Remote Control—locally fabricated, see Figure 3-12

Resistor—10 k ohm, 2 watt

3. DISPLAY CRT ASSEMBLY 51108695-100

3.2. Checks and Adjustments

3.2.5 High Voltage

WARNING

Improper handling of the CRT assembly can cause electrical shock, fire, and x-ray hazards. Servicing of the unit should be done only by qualified personnel, using approved practices and equipment.

CAUTION

Do not check the high voltage by drawing an arc, because doing so can damage the unit. Use the DMM with the High-Voltage Probe. With the power off, discharge the CRT high voltage to chassis ground through a 10 k ohm, 2 watt resistor. Ensure that the probe is properly grounded before connecting it to the CRT.

1. With the power off, connect the High Voltage Probe to the CRT. Turn the power on and check for an indication between +24.7 kVdc and +25.1 kVdc on the DMM. If the voltage is not as specified, adjust the HV Adjust, R621, located on PWB GA2001A+MOD (see Figure 3-6).

3.2.6 Focus

1. Load the Station Diagnostics (HVTS OPIF test) so that test patterns are accessible.
2. Ensure that the input termination switches on the rear of the CRT assembly are set to 75 ohms. Select a dot or cross-hatch pattern from the Station Diagnostics.
3. Set the remote control brightness and contrast controls to the center position. Press and hold the degauss button for at least 10 seconds. If the internal degaussing coil cannot restore color purity, use an external degaussing coil.
4. With the brightness control in the center position, turn the contrast control until the display is about to beam limit (starts to become fuzzy). Adjust the focus control, located in the high voltage section on the right rear of the unit, for the best compromise focus between the center of the screen and the periphery. Repeat this step until a good compromise focus is reached.

3.2.7 Horizontal Size and Position

1. Select a dot or cross-hatch display pattern. On the CRT assembly, set switch SW01, located on GA4002A, to OVER SCAN. The display should expand beyond the edges of the CRT face. If this does not occur, adjust the Width, R559, located on GA2001A+MOD, (R558 is the fixed resistor just in front of R559) to make the display about 10% wider than the width of the CRT face.
2. Set switch SW01 to UNDER SCAN. The display should become about 95% as wide as the CRT face (this can best be seen with the signal generator set for a white field). If necessary, adjust the US Width, R556, located on GA2001A+MOD, until the display just starts to clip in one corner of the CRT.
3. Ensure that the display is horizontally centered on the CRT face. If required, use the H Center, R538, located on GA2001A+MOD (next to the high voltage power supply) to center the display.

3. DISPLAY CRT ASSEMBLY 51108695-100

3.2. Checks and Adjustments

3.2.8 Vertical Size and Position

1. With a dot or cross-hatch display pattern selected, set switch SW01 to OVER SCAN. Rotate the Sub Height, R437, located on GA2001A+MOD, fully clockwise.
2. Adjust the vertical linearity with V Line, R430, located on GA2001A+MOD, (next to R437) for the most linear display possible. Ensure that the vertical size is about 10% larger than the CRT face. If required, adjust the size with the Height, R413, located on GA 2001A+MOD.
3. Set switch SW01 to UNDER SCAN, set the Signal Generator for a white field, and ensure that the display height is 95% of the CRT face. If required, adjust the vertical size with US Height, R414, located on GA2001A+MOD, until the display just clips in the corners.
4. Ensure that the display is centered in the CRT face. If required, center the display with V. Center, R442, located on GA2001A+MOD.

3.2.9 Replacement Circuit Board Checks

When replacing the CRT assembly or printed circuit boards within it, check the following settings on the boards:

GA2001A+MOD DEFLECTION PWA (PART #51109153-300)

The Horizontal Phase Tab should be set on position 4, as indicated in Figure 3-1, for the Universal Station.

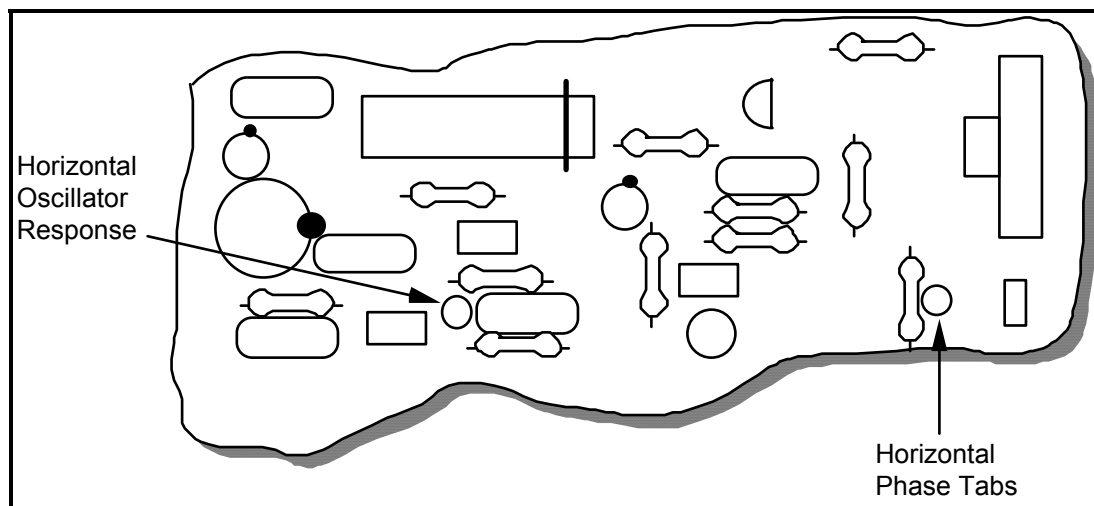


Figure 3-1 — Horizontal Phase Tabs and Oscillator Response

1640

GA6000 INPUT CIRCUIT PWA (PART #51109153-600)

The Sync Connection Mode setting for the Universal Station should be set as indicated on Diagram 3 in Figure 3-2.

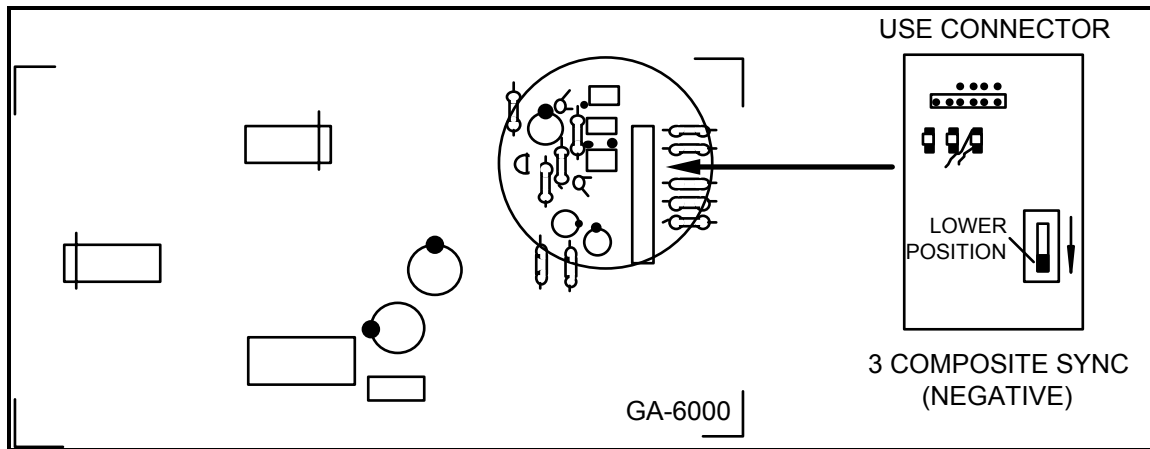


Figure 3-2 — Sync-Mode Selection

1641

3.3 Test/Troubleshooting

The troubleshooting procedures in this subsection are keyed to Checks and Adjustments in subsection 3.2, and are designed to isolate a failure down to a particular subassembly or part. Because a problem may be caused by a part that is out of adjustment, it is important to perform the checks and adjustments procedures before starting to troubleshoot. Figures 3-3 through 3-12 are included as aids to troubleshooting, and can be referred to for finding a part location, or for illustrations of the wave shapes within the CRT assembly. The procedures assume that the failure has a single cause, that the cause is not defective wiring or soldering, and that the test equipment is functioning normally.

WARNING

Certain parts used in the CRT assembly are critical, requiring replacement with exact duplicates. Failure to do so may result in electrical shock, fire, or x-ray hazards.

3.3.1 Procedure

The fault isolation procedures in Table 3-1 will aid in finding most problems and, wherever possible, should be performed before calling the Honeywell Technical Assistance Center (TAC). Refer to subsection 3.2.4 for a list of the required test equipment and materials. Unless otherwise specified, the isolation procedures in Table 3-1 assume conditions as established in the referenced *Checks and Adjustments* step.

Table 3-1 — Troubleshooting Procedures

Paragraph Reference	Symptom	Isolation Procedure
3.2.3	No High Voltage	Check wave shape at GA2001A+MOD terminal W1*. If as shown, the high voltage transformer or module is defective. If not as shown, check for 123 +5 Vdc at GA 9000 BH IN UNREG. If voltage is not as specified, GA 9000 is defective. Check wave shape at GA2001A+MOD terminal W4**. If wave shape is as shown, the HV output transistor X02 is defective. If the wave shape (WS) is not as shown, check for +12 Vdc at GA2001+MOD connector pin S3. If voltage is as specified, GA2001A+ is defective. If voltage is not as specified, GA 9000 is defective.
	No High Voltage Adjustment	GA 2001A+MOD is defective.
	No Focus Adjustment	The Screen/Focus Adjusting Module is defective.
	Just Horizontal Line	Check WS at GA 2001A+M R433 (see Figure 3-3). If as shown, check for similar WS# at yoke BRN wire (see GA 2001A+M pin U5). If the WS is as shown, the yoke is defective (check for burns). If no WS at U5, GA2001A+M is defective. If the WS at GA 2001A R433 is not as shown, check the WS at GA 2001A+M IC 501 pin 2. If the WS is not as shown, GA 2001A+M is defective. If the WS is as shown, transistors X04 and X05 are defective. Check the WS at GA 2001+M connector pin T1. If the WS is as shown, the yoke is defective. If the WS is not as shown, check the WS at connector pin T4. If the WS is as shown, transistor X01 is defective. If the WS is not as shown, GA 2001A is defective.

3. DISPLAY CRT ASSEMBLY 51108695-100

3.3. Test/Troubleshooting

Table 3-1 — Troubleshooting Procedures

Paragraph Reference	Symptom	Isolation Procedure
3.2.5.1	No Dot/ Crosshatch	Check for a negative-going brightness pulse at GA6000 transistor X11-E. If pulse is not present, GA6000 is defective. If pulse is present, GA4002A or GA1001A is defective.
	Over- Under Scan Does Not Work	With the CRT assembly turned off, check continuity across the switch. Set the switch to the OVER position, and check resistor in series with UNDER position. If switch and resistor are OK, GA2001A+MOD is defective.
	Unable To Adjust Size	GA2001A+MOD is defective.
3.2.5.2	R556/R538 Does Not Work	GA2001A+MOD is defective.
3.2.6	R437/R430 /R538 Does Not Work	GA2001A+MOD is defective.

* It may be easiest to see the wave shape at transistor X02-C.

** It may be easiest to see the wave shape at transistor X02-B.

Positive spike missing from rounded-sawtooth wave shape.

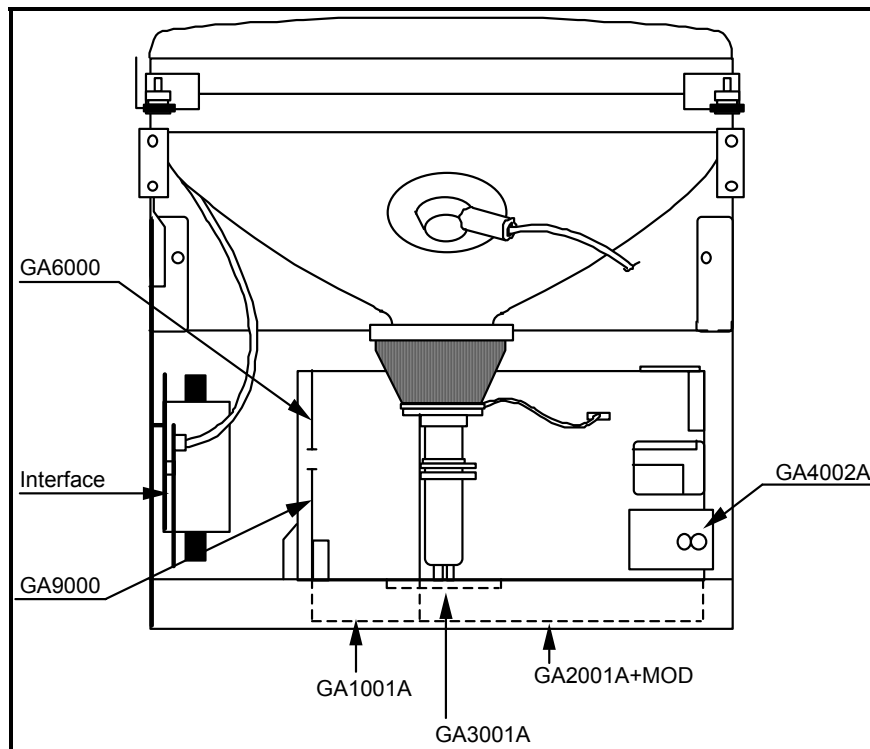


Figure 3-3 — PWA Locations 1642

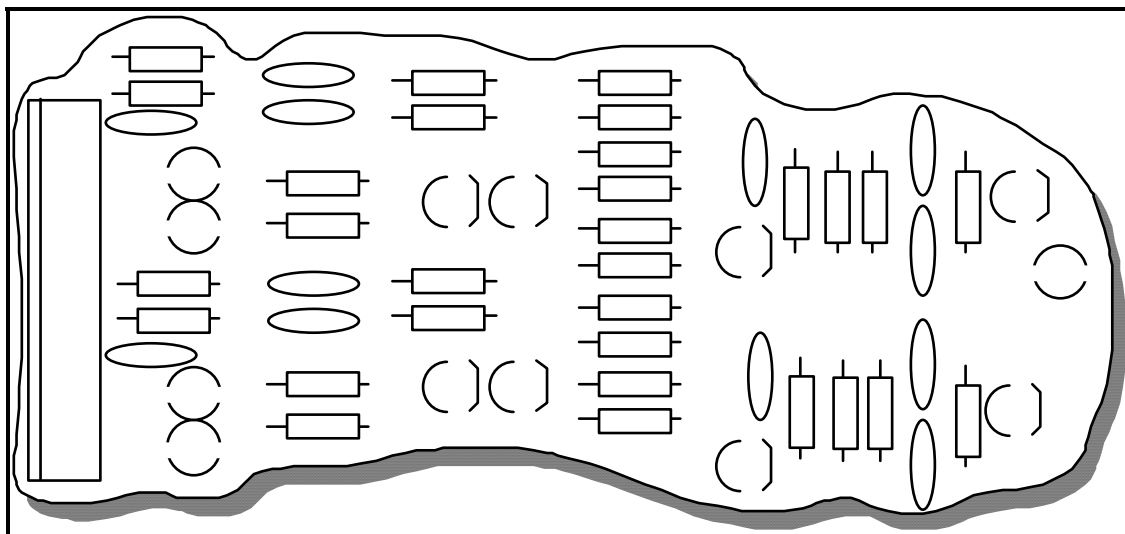


Figure 3-4 — PWA 03-170092-01, Interface (P/N 51109153-900)

1643

3. DISPLAY CRT ASSEMBLY 51108695-100

3.3. Test/Troubleshooting

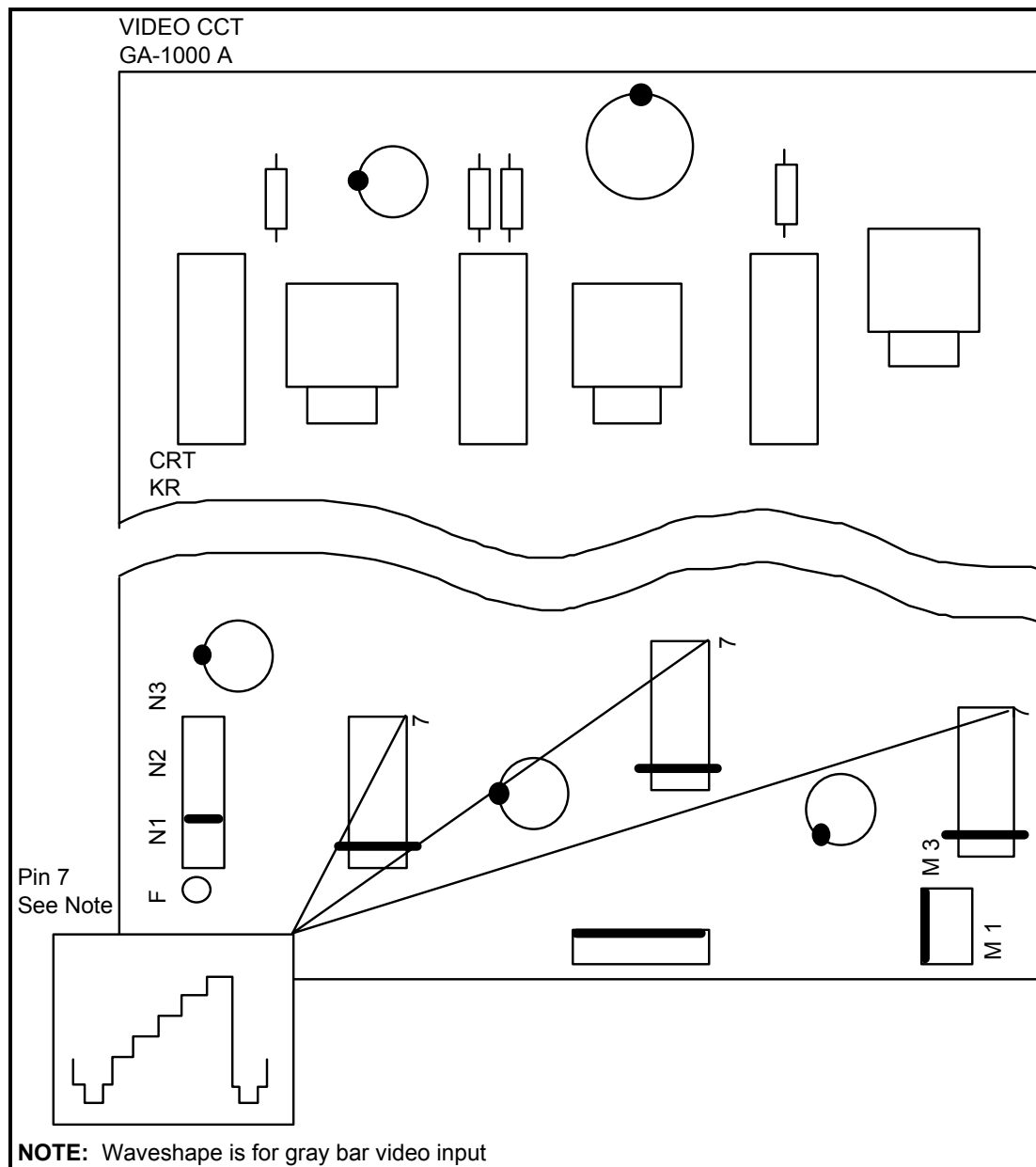


Figure 3-5 — PWA GA1001A Video (P/N 51109153-200)

1644

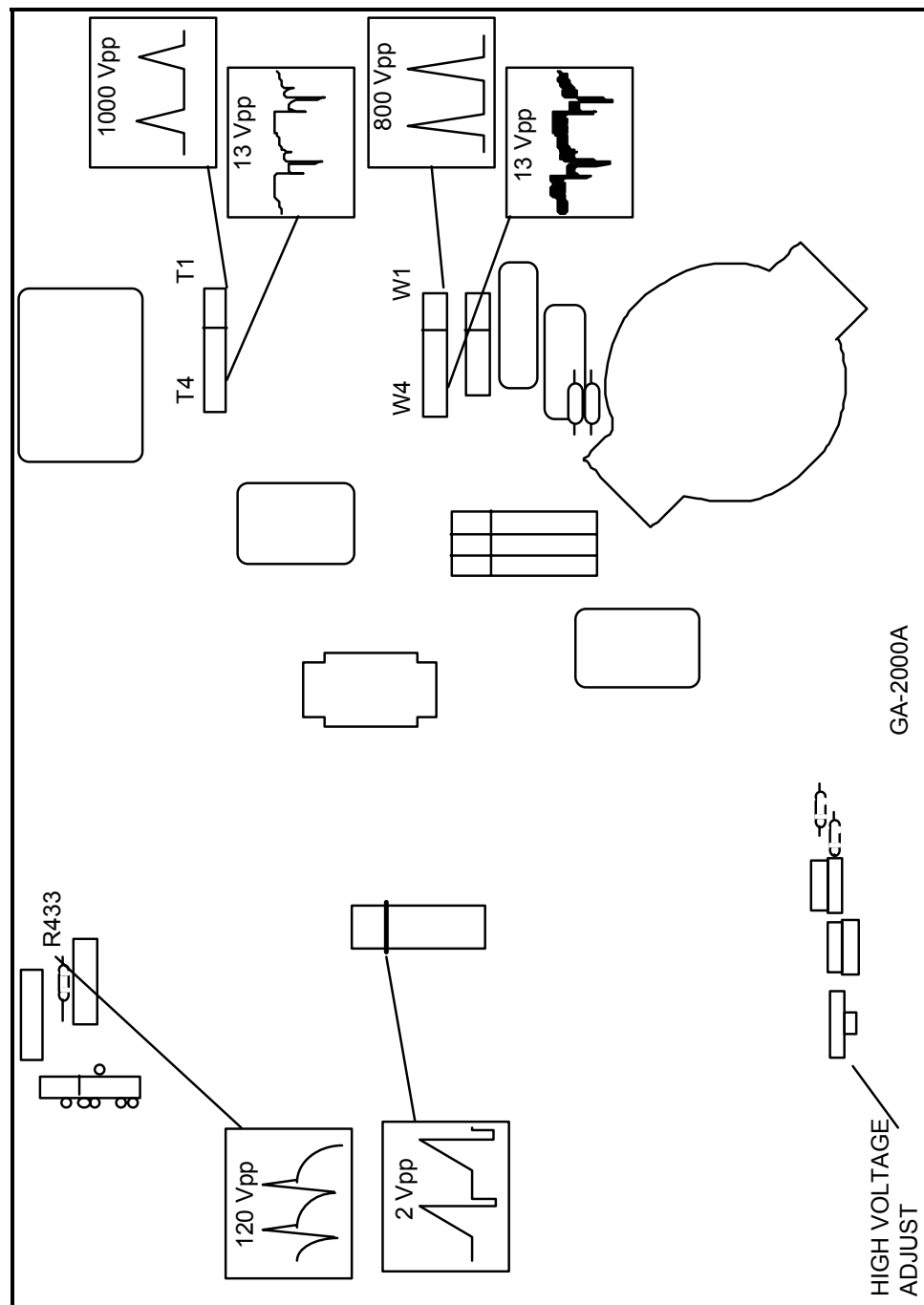


Figure 3-6 — PWA GA2001A+MOD, Deflection (P/N 51109153-300) 1645

3. DISPLAY CRT ASSEMBLY 51108695-100

3.3. Test/Troubleshooting

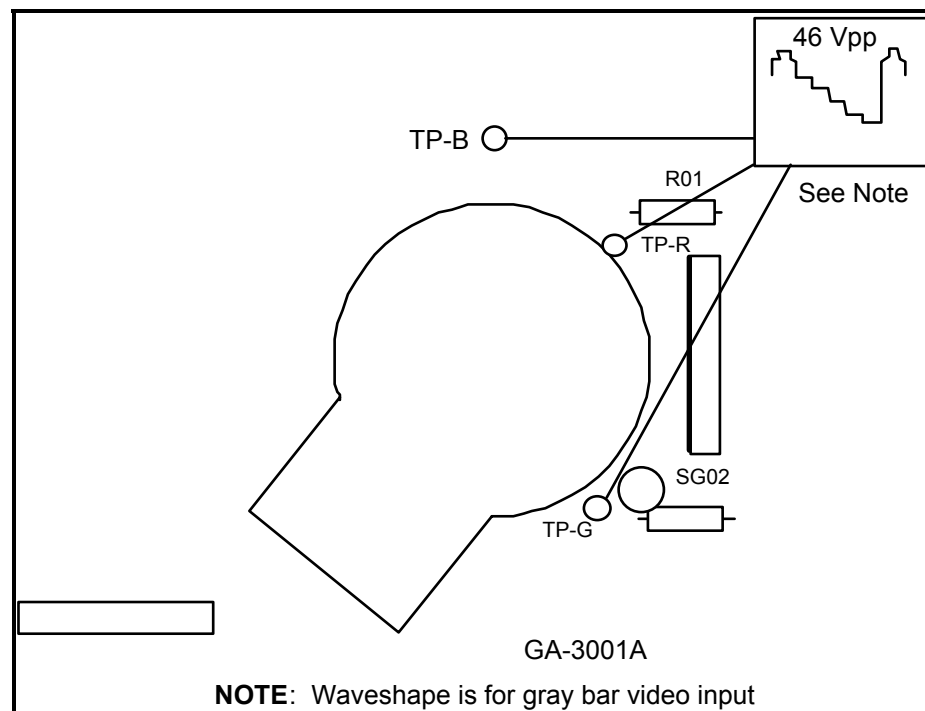


Figure 3-7 — PWA GA3001A, CRT Socket (P/N 51109153-400)

1646

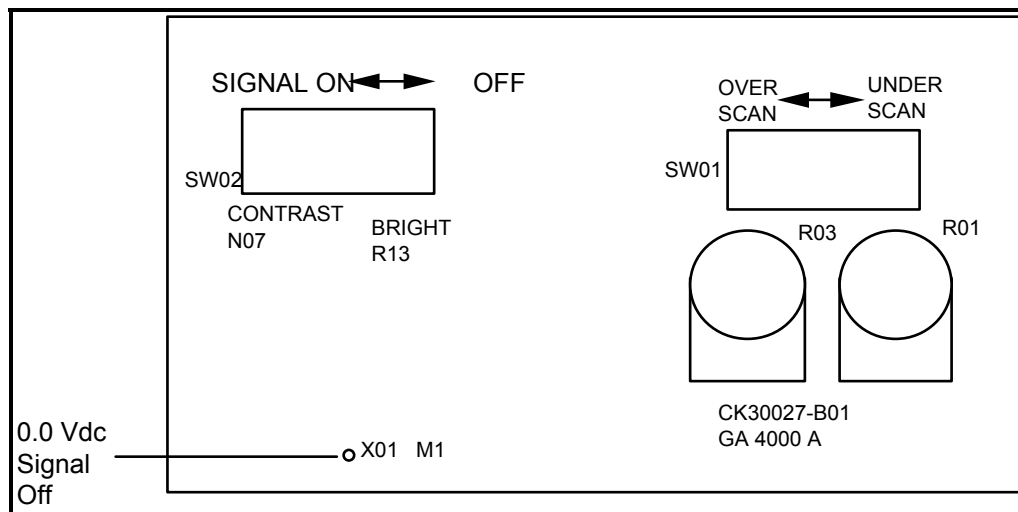


Figure 3-8 — PWA GA4002A, Control Circuit (P/N 51109153-500)

1647

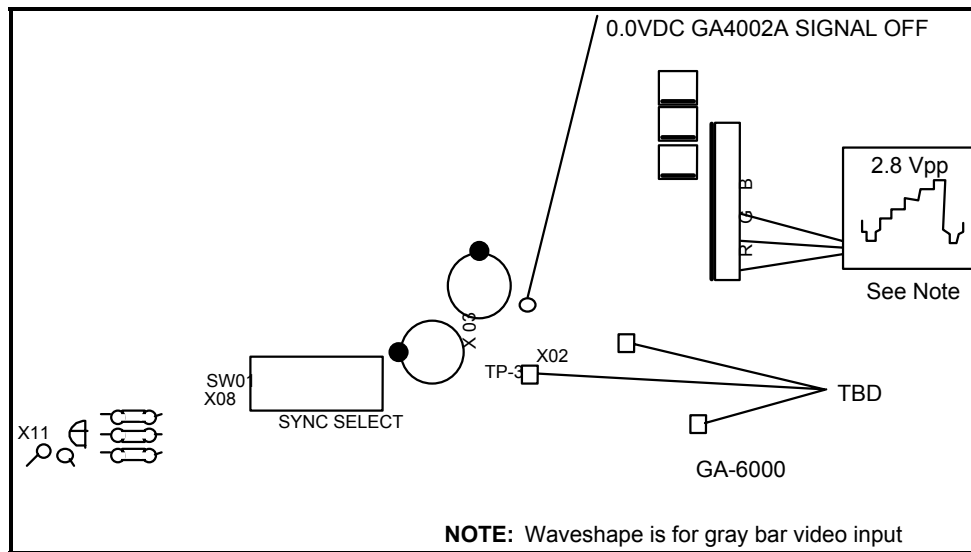


Figure 3-9 — PWA . GA6000, Input Circuit (P/N 51109153-600)

1648

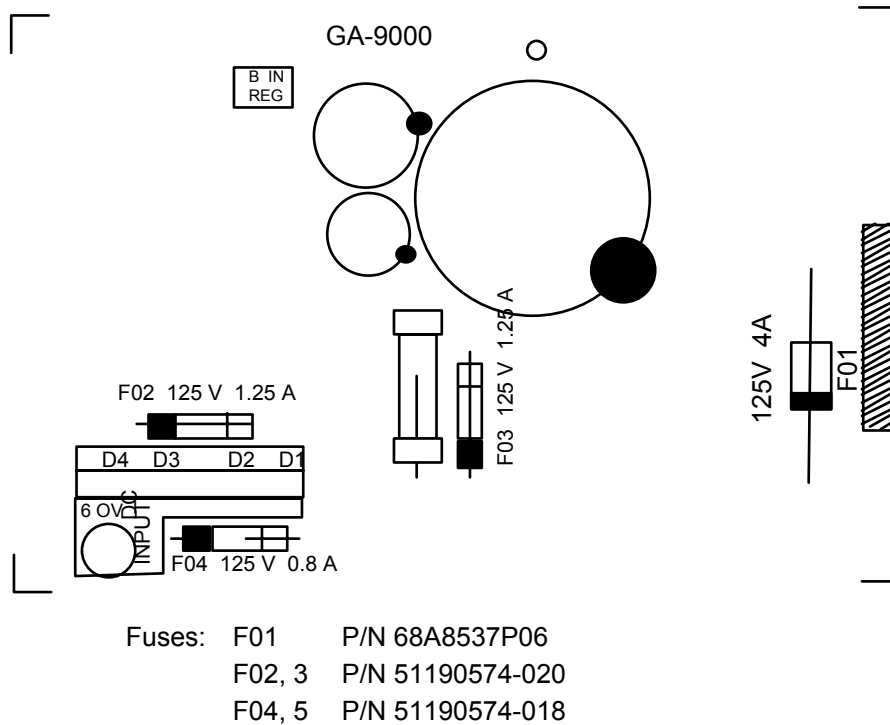


Figure 3-10 — PWA GA9000, Power Supply (P/N 51109153-700)

1649

3. DISPLAY CRT ASSEMBLY 51108695-100

3.3. Test/Troubleshooting

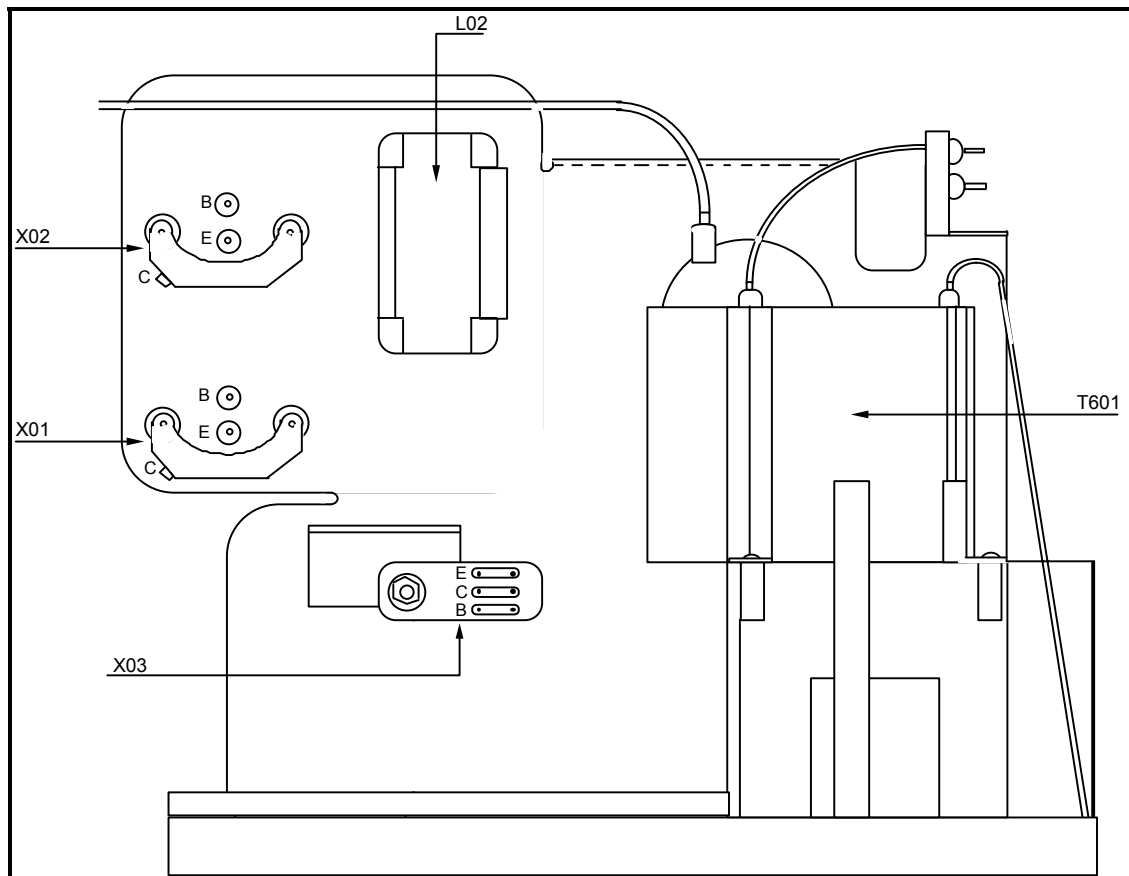


Figure 3-11 — Deflection and High Voltage Heat Sink 1650

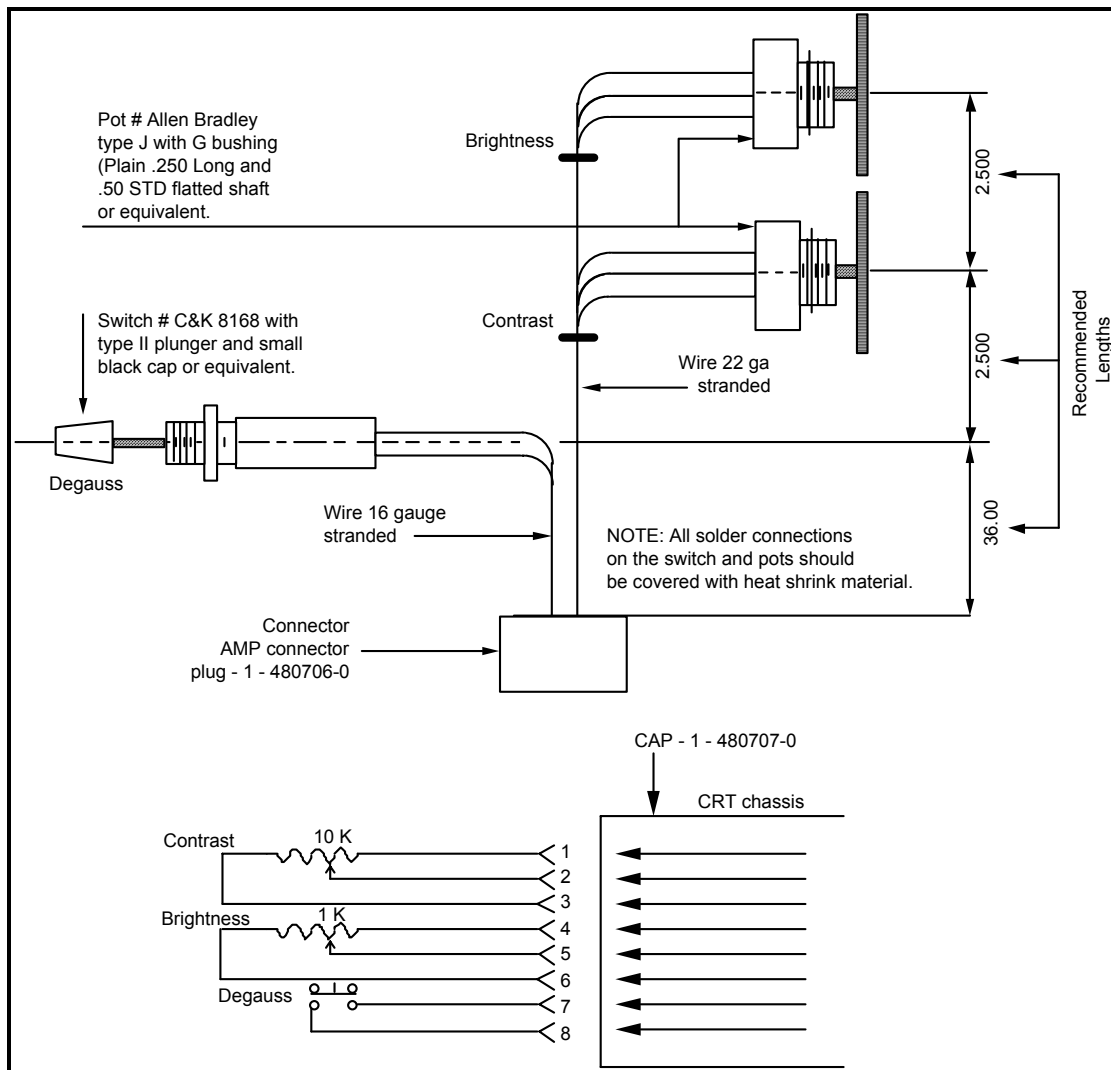


Figure 3-12 — Remote Control

1651

3.4 Removal/Replacement

Removal and replacement of the CRT assembly should be performed only by a qualified technician using the best shop practices.

WARNING

The CRT assembly contains high voltages and a vacuum in the glass CRT envelope. Ensure that the power is removed, by unplugging the power cable, before working on the unit. Use only insulated tools. Handle the CRT assembly only by the chassis; never lift by the neck or strike the glass.

3.4.1 CRT Assembly Removal

1. Lower Tier mounted— Gain access to the rear of the CRT chassis by opening the rear door of the console and lifting the door off the hinge pins. Remove the rear cover from the assembly by loosening two captive screws on the bottom of the cover and pulling the cover off.

Upper Tier mounted— Loosen the rear cover from the assembly by removing four screws. Remove the cover by pulling it straight back and away from the CRT chassis. Use care not to drop the cover on the fragile glass cathode ray tube.

WARNING

BE SURE POWER IS OFF ON THE UNIVERSAL STATION ELECTRONICS MODULE AS WELL AS ON THE CRT CHASSIS.

If power is still applied to the electronics module when the center conductor of a loose coax cable is allowed to touch ground, a cable driver in the VDG or PDG I/O card will be damaged.

2. Label all cables to the CRT chassis (if required) and remove them. These include:

- Power cable
- 3 Coax cables (red, green, blue)
- Remote cable assembly

3. Lower Tier mounted— Push the CRT chassis forward to relieve pressure on the retainers, then press on the retainer blocks to rotate the retainers down and away from the rear of the chassis (it may be necessary to unscrew the hex-head bolts slightly so that they clear the chassis rails). The CRT chassis, mounted on a slide tray, will now slide clear of the cabinet on rails and stop at a detent position.

WARNING

On Lower Tier mounted units, when the side buttons are again released in the next step, the CRT CHASSIS WILL BE FREE TO MOVE! Since the chassis is heavy with most of its weight concentrated at the front, you are strongly advised to seek assistance in removing the unit from the cabinet slide rails.

THE CHASSIS CENTER OF GRAVITY IS IN THE AREA OF THE CRT ANODE CONNECTOR. Two people can lift the unit if each of them grasps the chassis shroud on each side of the anode with one hand while pressing the side button, then steadying the chassis with the other hand while separating the slide rails.

4. Lower Tier mounted— With the assistance of a helper, again release the chassis by pressing the side buttons on the rails. Pull the unit free of the console along the slide rails, and place it on a bench for further disassembly.
5. Remove six bolts (3 on each side) to free the CRT chassis from the slide tray.

NOTE

If a touchscreen must be removed or replaced, see Section 7, *Touchscreens*, for assistance.

3.4.2 CRT Assembly Replacement

1. With the aid of an assistant, set the CRT chassis on the tray between the mounting brackets, and align it with the holes on the mounting bracket. Install six bolts (3 on each side) to secure the chassis to the tray.
2. Lower Tier— With the aid of your assistant, engage the chassis slide rails with the slide rails on the console and push the assembly beyond the detent position so that both rails lock and hold the assembly in place.
3. Lower Tier— As you move the assembly forward in this step, check the alignment of the CRT face with the touchscreen. There should be a very small gap between them. Press the detent buttons and carefully push the CRT assembly forward. Lift the retainer blocks up behind the CRT chassis to hold the chassis in position. You may need to adjust the hex-headed bolts on the retainers to maintain the proper clearance with the touchscreen. Tighten the locknuts on the retainers when you are satisfied with the alignment.
4. The CRT should also be centered in the screen. Center it by loosening the touchscreen mounting screws with a long-handled screwdriver, and moving the touchscreen slightly. You may need to readjust this setting again after power has been applied.
5. Connect the Red, Green and Blue coax cables to their respective BNC connectors and connect the remote cable assembly to its connector. Plug in the AC line cable.
6. Lower Tier mounted— Replace the rear cover and secure it with two captive screws. Reinstall the rear door of the console by sliding it back onto its hinges.

Upper Tier mounted— Be sure to engage both tracks inside the bottom of the cover, then slide the cover over the CRT chassis and secure it with four screws.

3. DISPLAY CRT ASSEMBLY 51108695-100

3.5. Startup

3.5 Startup

To prevent power surges that could damage the equipment, ensure that all US equipment is turned off. Apply power by setting the main power circuit breaker, located behind the door under the keyboard, to "1" (ON). If supplied, set the keyboard power supply switch to "1" (ON), and close the door. Open the rear door and set the module power switch to ON. After 1.75 minutes, check the module light emitting diode (LED) indicators to ensure that they are all green. The alphanumeric indicator displays the module node address when the operating system is running or not running. If any of the above conditions cannot be met, refer to the *Five/Ten-Slot Module Service* manual in the *LCN Service* binder. Close the station rear door. Apply power to any other equipment turned off at the beginning of this procedure.

Adjust CRT brightness and contrast for best resolution on the character that appears in the corner of the screen. Check the *Touchscreen to CRT Alignment* in subsection 6.4.3.

3.6 Spare Parts List

Table 3-2 — CRT Spare Parts List

Part Number	Description
* 51108695-100	CRT Purchase Specification.
* 51190574-018	Fuse 8/10 Amp. 125 V Glass Tube Time-delay
* 51190574-020	Fuse Glass 1 1/4 Amp. 125 V Tube Time-delay
* 68A8537P06	Fuse, Miniature 4.0 Amp. 125 V
51109153-100	Tube with Yoke
51109153-200	Video Drive PCB GA1001A
51109153-300	Deflective PCB GA2002A + Mod. (Fly-back Transformer not included)
51109153-400	CRT Socket PCB GA3001A
51109153-500	Control Circuit PCB GA4002A
51109153-600	Input Circuit PCB GA6000
51109153-700	Power Supply PCB GA9000
51109153-800	Harness Assembly
51109153-900	Input Module

* ORU Level Replacement Item

4 DISPLAY CRT ASSEMBLY 51108308-100

NOTE

The 51108695-100 CRT (described in Section 3) is no longer available from the manufacturer. This monitor, the INTERLACED CRT (with its remote cable assembly) is a functional replacement for the DISPLAY CRT.

Maintenance tasks in this section cover three levels:

- subsections 4.1 and 4.2 contain preventative maintenance tasks which can be performed by the station operator. These tasks include cleaning the CRT screens, keyboards, etc. and degaussing the CRT.
- subsections 4.3 and 4.4 contain the maintenance tasks required to determine if the monitor has failed and the tasks required to replace the monitor at the ORU level. These tasks may be performed by a trained technician familiar with this equipment.
- subsections 4.5 and 4.6 contain those maintenance tasks required to troubleshoot at the card and major component level. These tasks demand special skills and may require alignment and adjustment of the CRT monitor. They should **ONLY** be attempted by a skilled technician familiar with the repair of color video monitors.

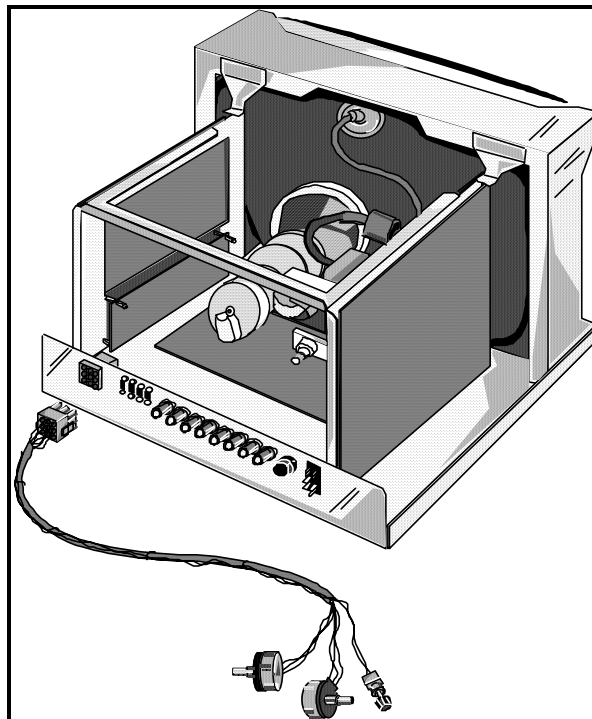


Figure 4-1 — Interlaced CRT Monitor

3321

4.1 Cleaning (Operator Qualified)

NOTE

Cleaning of the station color monitor CRT face, touch screen, and keyboards should be done with **power off**, and the station off-line; therefore, it is best to do them all at once as described below. For the touch screen cleaning procedure, see subsection 6.1.

1. Open the door below the keyboards and set the main power circuit breaker located in the upper-right corner to the down (OFF) position.
2. If you have a keyboard power supply, set its switch (located just to the left of the main power circuit-breaker) to "0" (OFF).
3. Open the station rear door (below the CRT chassis) and set the Module Power Switch to OFF.
4. Switch all other peripheral equipment associated with the station, such as floppy disk drives, etc., to OFF.
5. Clean the area around the operator station, including the floor and horizontal surfaces on or near the station that accumulate dust.
6. Clean the Operator/Engineer keyboard by dampening a soft cloth with mild detergent and wiping the keys. Do not spray detergent solution directly on the keys as this may contaminate the switch mechanism underneath.
7. Clean the CRT face with a mild detergent solution and a soft, lint-free cloth (do not use paper wipes of any kind). It is best to spray the solution on the CRT face to dissipate any static charge before wiping with the cloth.
8. Check the operating area for the addition of devices that may cause electrical interference.
9. Restore power by reversing steps 1 through 4. Then, verify the checks detailed in the start up procedures in subsection 4.7, *Startup*.

4.2 Adjustments (Operator Qualified)

The operator has access to three controls at the right of the monitor screen which adjust brightness, contrast, and control the degaussing circuit.

4.2.1 Brightness and Contrast

Adjust brightness to a higher level by moving the top thumbwheel control upwards. Increase image contrast by moving the lower thumbwheel control upwards.

4.2.2 Degaussing the CRT

The need for degaussing is indicated by poor color purity or poor convergence (colors appear to separate). Degaussing should be performed before a maintenance technician is called to perform adjustments on the monitor.

- With power applied to the station, press the degauss button on the right side of the screen below the brightness and contrast controls. Hold the button for a few seconds until colors are restored.

WARNING

ALL OTHER ADJUSTMENTS REQUIRE A SKILLED TECHNICIAN FAMILIAR WITH COLOR MONITOR SERVICING.

Any further attempts at servicing or adjusting this equipment may expose personnel to lethal high voltage shock and/or permanently damage the monitor.

4.3 Test/Troubleshooting (Technician Qualified)

There are no adjustments to this monitor at the ORU (Optimum Replaceable Unit) level. A faulty monitor is replaced in its entirety.

Use the instructions in this subsection and subsection 4.4 to diagnose, remove and replace an entire monitor which has failed.

A skilled user, however, may choose to replace circuit cards and make adjustments beyond the ORU level. See subsections 4.5 through 4.6 for assistance in identifying, removing and replacing those parts and making the necessary adjustments.

4.3.1 Preliminary Checks

Before investigating deeply into a problem, make some preliminary checks.

4.3.1.1 Check the Universal Station Electronics Module

The following checks can be made by observation without turning power off.

1. Remove the front panel of the module. Check the green LED on the power supply. If it is **NOT ON**, set the AC power switch to off and then back on (this resets the circuit breaker). If the green LED does not come on, either the power supply is defective, its load is shorted, or power is not being supplied to the module.
2. Check the red LED on the fan assembly. If it is on, the fan assembly is defective. For service on the fan, see the *Five/Ten-Slot Module Service* manual in the *LCN Service* binder.
3. Check that all lights on the circuit cards are green or yellow (no red) and that the alphanumeric display shows the US node number. If not, go to the *Five/Ten-Slot Module Service* manual in the *LCN Service* binder.
4. Check that the VDG I/O card is properly plugged into the I/O slot directly behind the VDG card or that a PDG I/O card is properly plugged into the I/O slot directly behind either a PDG or EPDG card. Note that any combination of VDG and VDG I/O or PDG and PDG I/O or EPDG and PDG I/O is acceptable (see Table 4-1). Ensure that the red, green and blue coax cables are firmly fastened to the I/O card.

Table 4-1 — Acceptable Card Combinations to Drive this CRT Monitor

Control Card	I/O Card
VDG	VDG I/O
PDG	PDG I/O
EPDG *	PDG I/O

* In this arrangement, the EPDG card takes on the characteristics of a PDG.

4.3.1.2 Exposing the CRT Chassis

1. Open the door below the keyboards and set the main power circuit breaker located in the upper-right corner to the down (OFF) position.
2. If you have a keyboard power supply, set its switch (located just to the left of the main power circuit breaker) to "0" (OFF).
3. Open the station rear door (below the CRT chassis) and set the Module Power Switch to OFF.
4. Switch all other peripheral equipment associated with the station, such as floppy disk drives, etc. to OFF.
5. Lower Tier mounted— Gain access to the rear of the CRT chassis by opening the rear door of the console and lifting the door off the hinge pins. Remove the rear cover from the assembly by loosening two captive screws on the bottom of the cover and pulling the cover off.

Upper Tier mounted— Loosen the rear cover from the assembly by removing four screws. Remove the cover by pulling it straight back and away from the CRT chassis. Use care not to drop the cover on the fragile glass cathode ray tube.

4.3.1.3 Check the CRT chassis

WARNING

Operation of this monitor outside the cabinet or with the cover removed involves a shock hazard from the monitor power supplies. Work should not be attempted by anyone who is not thoroughly familiar with necessary precautions when working around high voltage equipment.

1. Ensure that the power cord is plugged in and supplying power. Check the fuse on the back of the chassis (see Figure 4-2).
2. Check that the red, green and blue coax cables are firmly fastened to the connectors on the rear of the chassis. Check that all 75 Ω terminating resistors are switched in.

WARNING

If you **must** remove a coax cable from either the VDG/PDG I/O card or the CRT chassis while power is applied, **it is much safer to remove the connection from the VDG/PDG card FIRST!**

If the center conductor of any video cable is allowed to short to ground while under power, the video driver within the VDG/PDG I/O card will fail and force replacement of the card.

4. DISPLAY CRT ASSEMBLY 51108308-100
4.3. Test/Troubleshooting (Technician Qualified)

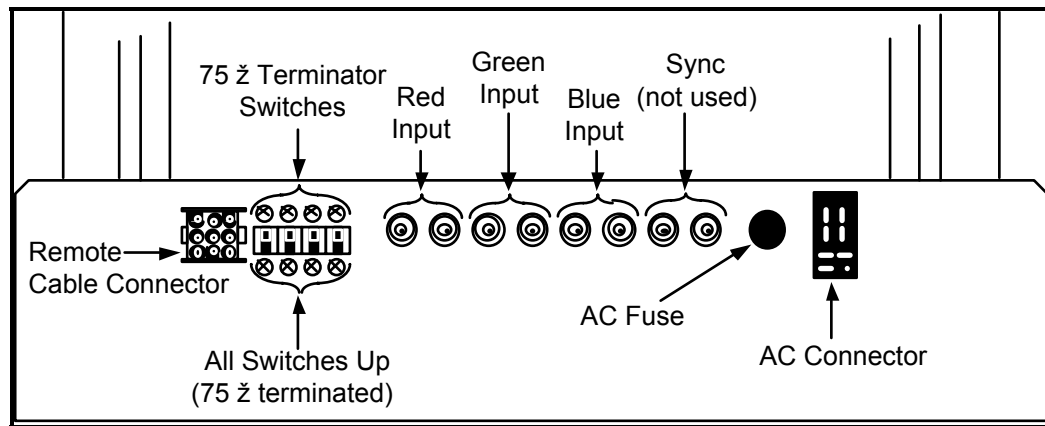


Figure 4-2 — CRT Monitor Rear Panel

3309

3. Because this application uses composite sync on the green video input, there will be no visible raster present if the green signal is missing. Be sure the **INT. SYNC/EXT. SYNC** switch (S1) in the upper left corner of the video PC card (see Figure 4-10) is set to the **INT. SYNC** position.

4.4 ORU Removal and Replacement (Technician Qualified)

If the fault with the CRT monitor cannot be found at this point, the monitor must be removed and replaced in its entirety.

4.4.1 Removing the CRT Chassis

You may already have done the first five steps below, as they are the same as those in subsection 4.3.1.2.

1. Open the door below the keyboards and set the main power circuit breaker located in the upper-right corner to the down (OFF) position.
2. If you have a keyboard power supply, set its switch (located just to the left of the main power circuit-breaker) to "0" (OFF).
3. Open the station rear door (below the CRT chassis) and set the Module Power Switch to OFF.
4. Switch all other peripheral equipment associated with the station, such as floppy disk drives, etc. to OFF.
5. Lower Tier mounted— Gain access to the rear of the CRT chassis by opening the rear door of the console and lifting the door off the hinge pins. Remove the rear cover from the assembly by loosening two captive screws on the bottom of the cover and pulling the cover off.

Upper Tier mounted— Loosen the rear cover from the assembly by removing four screws. Remove the cover by pulling it straight back and away from the CRT chassis. Use care not to drop the cover on the fragile glass cathode ray tube.

WARNING

BE SURE POWER IS OFF ON THE UNIVERSAL STATION ELECTRONICS MODULE AS WELL AS ON THE CRT CHASSIS.

If power is still applied to the electronics module when the center conductor of a loose coax cable is allowed to touch ground, a cable driver in the VDG or PDG I/O card will be damaged.

6. Label all cables to the CRT chassis (if required) and remove them. These include:
 - Power cable
 - 3 Coax cables (red, green, blue)
 - Remote cable assembly

4. DISPLAY CRT ASSEMBLY 51108308-100

4.4. ORU Removal and Replacement (Technician Qualified)

7. Lower Tier mounted— Push the CRT chassis forward to relieve pressure on the retainers, then press on the retainer blocks to rotate the retainers down and away from the rear of the chassis (it may be necessary to unscrew the hex-head bolts slightly so that they clear the chassis rails). The CRT chassis, mounted on a slide tray, will now slide clear of the cabinet on rails and stop at a detent position.

WARNING

On Lower Tier mounted units, when the side buttons are again released in the next step, the CRT CHASSIS WILL BE FREE TO MOVE! Since the chassis is heavy with most of its weight concentrated at the front, you are strongly advised to seek assistance in removing the unit from the cabinet slide rails.

THE CHASSIS CENTER OF GRAVITY IS IN THE AREA OF THE CRT ANODE CONNECTOR. Two people can lift the unit if each of them grasp the chassis shroud on each side of the anode with one hand while pressing the side button, then steadying the chassis with the other hand while separating the slide rails.

8. Lower Tier mounted— With the assistance of a helper, again release the chassis by pressing the side buttons on the rails. Pull the unit free of the console along the slide rails, and place it on a bench for further disassembly.
9. Remove six bolts (3 on each side) to free the CRT chassis from the tray.
10. If the console has a touch screen mounted in the front bezel, disconnect the touch screen cable from the back of the touch screen. If the touch screen is one of the new designs, there is also a power cable to disconnect from the touch screen.
11. Lower Tier mounted— Remove the four M5 bolts that hold the bezel in place and gently lay it face down on the keyboard (the keylock switch cable is still attached).

Upper Tier mounted— Remove the four M5 bolts holding the bezel in place and lay the bezel aside.
12. Remove the remote cable assembly from the bezel by removing the degauss switch and two potentiometers.

4.4.2 Replacing the CRT Chassis

1. First, replace the remote cable assembly you removed in subsection 4.4.1 with the remote cable assembly supplied with the replacement CRT chassis (although they appear to be identical, the cable assemblies may have different value potentiometers). Replace the bezel and touch screen by reversing the disassembly procedure. Don't forget to reconnect the touch screen connector(s).
2. With the aid of an assistant, set the CRT chassis on the tray between the mounting brackets, and align it with the holes on the mounting bracket. Install six bolts (3 on each side) to secure the chassis to the tray.
3. Lower Tier— With the aid of your assistant, engage the chassis slide rails with the slide rails on the console and push the assembly to the end of its travel. Now rotate the retainer blocks to lock the chassis in place.

4. Lower Tier— As you move the assembly forward in this step, check the alignment of the CRT face with the touch screen. There should be a very small gap between them. Press the detent buttons and carefully push the CRT assembly forward. Lift the retainer blocks up behind the CRT chassis to hold the chassis in position. You may need to adjust the hex-headed bolts on the retainers to maintain the proper clearance with the touch screen. Tighten the lock nuts on the retainers when you are satisfied with the alignment.
5. The CRT should also be centered in the screen. Center it by loosening the touch screen mounting screws with a long-handled screwdriver, and moving the touch screen slightly. You may need to readjust this setting again after power has been applied.
6. Connect the Red, Green and Blue coax cables to their respective BNC connectors and connect the remote cable assembly to its connector. Plug in the AC line cable.
7. Lower Tier mounted— Replace the rear cover and secure it with two captive screws. Reinstall the rear door of the console by sliding it back onto its hinges.
8. Upper Tier mounted— Be sure to engage both tracks inside the bottom of the cover, then slide the cover over the CRT chassis and secure it with four screws.
9. Restore power by reversing steps 1 through 4 in subsection 4.4.1. Then, verify the checks detailed in the start up procedures in subsection 4.7, *Startup*.

4.5 CRT Monitor Parts Replacement (Color Video Monitor Qualified)

A skilled user may replace other parts in the CRT monitor beyond the ORU level (see the Spare Parts in subsection 4.8). Use this subsection as a guide in removing and replacing those parts. Use subsection 4.6 for assistance in adjusting the monitor after parts have been replaced.

WARNING

THE MAINTENANCE DESCRIBED HERE MUST BE PERFORMED ONLY BY A SKILLED TECHNICIAN FAMILIAR WITH THE REPAIR OF COLOR VIDEO MONITORS.

Following the replacement of a major component such as the CRT or some printed circuit cards, the monitor alignment must be checked and it may need to be completely realigned.

Operation of this monitor outside the cabinet or with the cover removed involves a shock hazard from the monitor power supplies. Work should not be attempted by anyone who is not thoroughly familiar with precautions necessary when working around high voltage equipment.

These instructions are limited to the diagnosis of the problem and to the replacement of those items listed in the spare parts list and identified in Figure 4-3.

4. DISPLAY CRT ASSEMBLY 51108308-100

4.5. CRT Monitor Parts Replacement (Color Video Monitor Qualified)

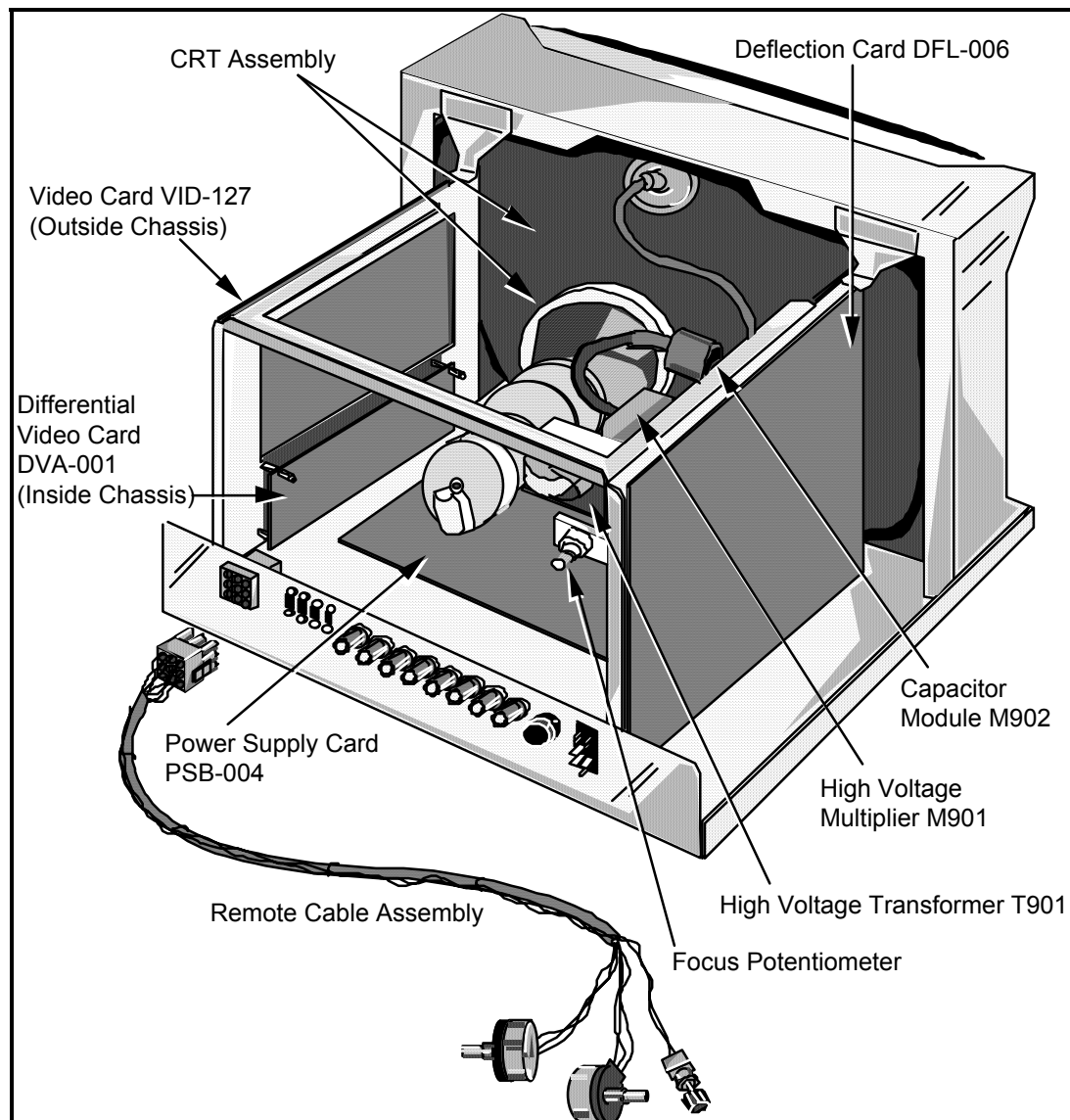


Figure 4-3 — Location of Major Components

3310

The instructions will help find the faulty part, indicate how to remove and replace it, then provide adjustment procedures, if applicable. Figure 4-4 is a line schematic (showing cabling, not wire-by-wire) detailing the interconnection of all major parts of the monitor.

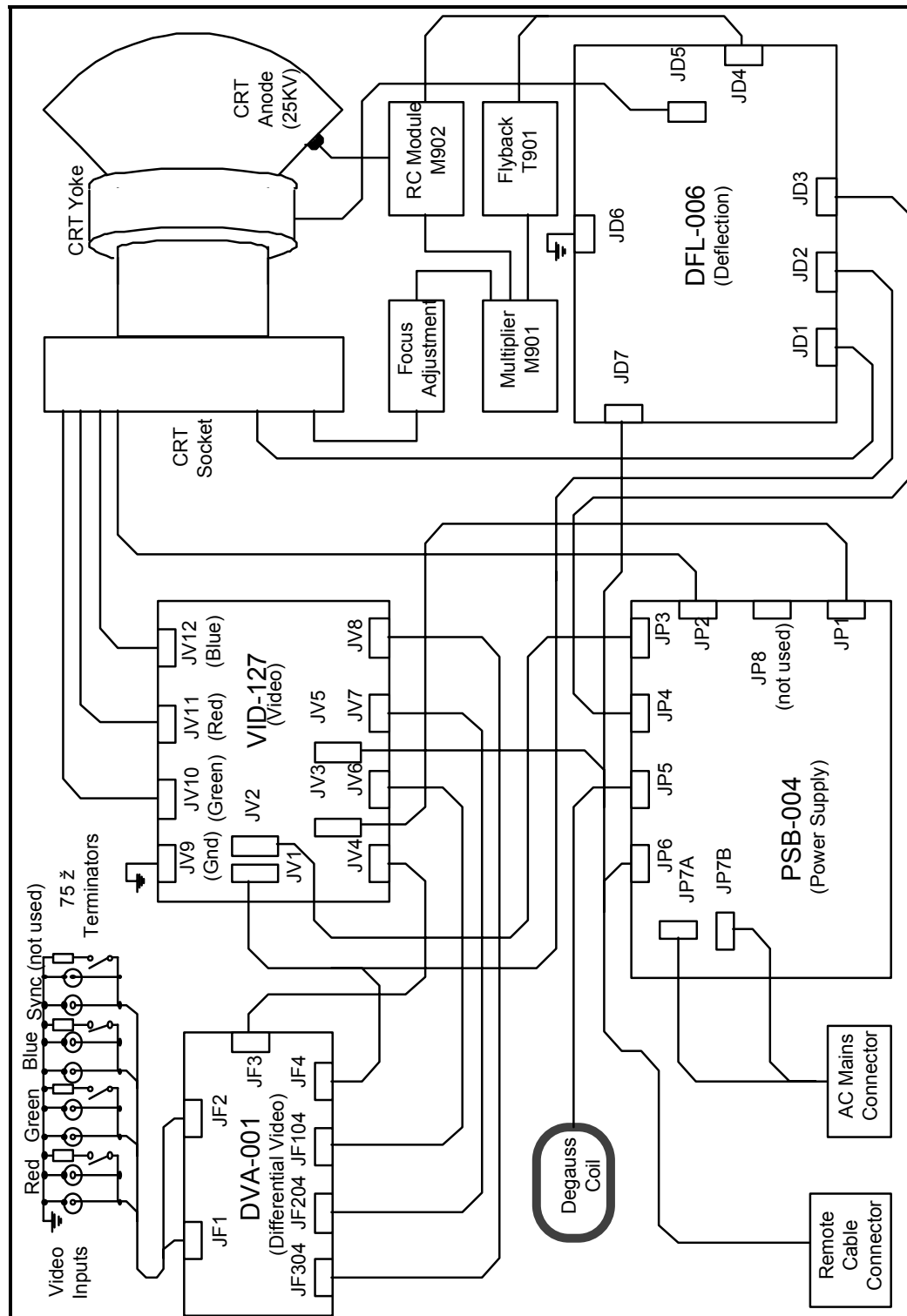


Figure 4-4 — Line Schematic of CRT Monitor 3471

4. DISPLAY CRT ASSEMBLY 51108308-100

4.5. CRT Monitor Parts Replacement (Color Video Monitor Qualified)

4.5.1 Test/Troubleshooting (Functional Partitioning)

The CRT chassis may be operated on a bench with the appropriate power cord and other test equipment.

WARNING

Operation of this monitor outside the cabinet or with the cover removed involves a shock hazard from the monitor power supplies. Work should not be attempted by anyone who is not thoroughly familiar with precautions necessary when working around high voltage equipment.

A technician's skill and his knowledge of the interaction between circuit cards and components in finding a fault cannot be confined to a few words in this manual. Instead, the manual describes each circuit card briefly, illustrates input/output signals, and lists "notes" about specific operating characteristics.

With the descriptions given, a technician familiar with video monitors should be capable of determining the cause of a fault on a printed circuit (PC) card or in other major parts, based upon his familiarity with the operation of similar circuits used in the industry.

4.5.1.1 Rear-Panel Video Signals

Red/green/blue video signals from the VDG or PDG I/O card enter BNC connectors on the rear panel. There is provision for a fourth "external sync" BNC connector, but it is not used in this application. Four switches associated with sync and the three video inputs allow selection of a 75 Ω termination for each signal (the switches should be UP or in the termination-selected position). Additional BNC connectors allow for loop-through operation to another monitor, but this feature is not used in this application.

4.5.1.2 Differential Video PC Card (DVA-001)

This card is mounted on the left side of the chassis, when viewed from the rear, and inboard to the VID-127 video PC card (see Figure 4-3). It contains three video amplifiers, each with a gain of 1 (no amplification). The purpose of this card is to terminate differential video signals from the VDG or PDG I/O cards and present a ground-based video signal compatible with the input requirements of the VID-127 card.

Input Signals to DVA-001 PC Card

The three video signals from the rear panel enter the card at JF1. All signals resemble those shown in Figure 4-5A (red and blue video signals don't have sync).

Output Signals from DVA-001 PC Card

The green/red/blue video output signals are presented on JF104/JF204/JF304 respectively. Since there is no amplification or inversion, these signals also resemble those shown in Figure 4-5A.

4.5.1.3 Video PC Card (VID-127)

This card is mounted on the left side of the chassis, when viewed from the rear (see Figure 4-3). It contains three identical wide-band dc-restored video amplifiers (one for each primary color). A sync separator circuit on the card derives internal sync from the green video input. Since internal sync is used in this application, be sure the sync selection switch (S1) is in the **INT. SYNC** position. This card also contains the horizontal oscillator and the sync generator for the power supply.

Video Amplifiers

The red/green/blue signals enter the VID-127 card at JV7-1/JV6-1/JV8-1 (Figure 4-5A) and leave the VID-127 card for the CRT at JV11-1/JV10-1/JV12-1 (Figure 4-5B).

Sync Separator Circuit

The sync separator circuit operates from the green video input via sync selection switch S1.

Sync signals are processed in the VID-127 card and are presented as:

- | | | |
|--------------------|----------------------|-------------|
| • Horizontal Drive | JV1-3 (Gnd at JV1-4) | Figure 4-6A |
| • Vertical Sync | JV1-5 | Figure 4-6B |
| • Flyback | JV1-6 | Figure 4-6C |

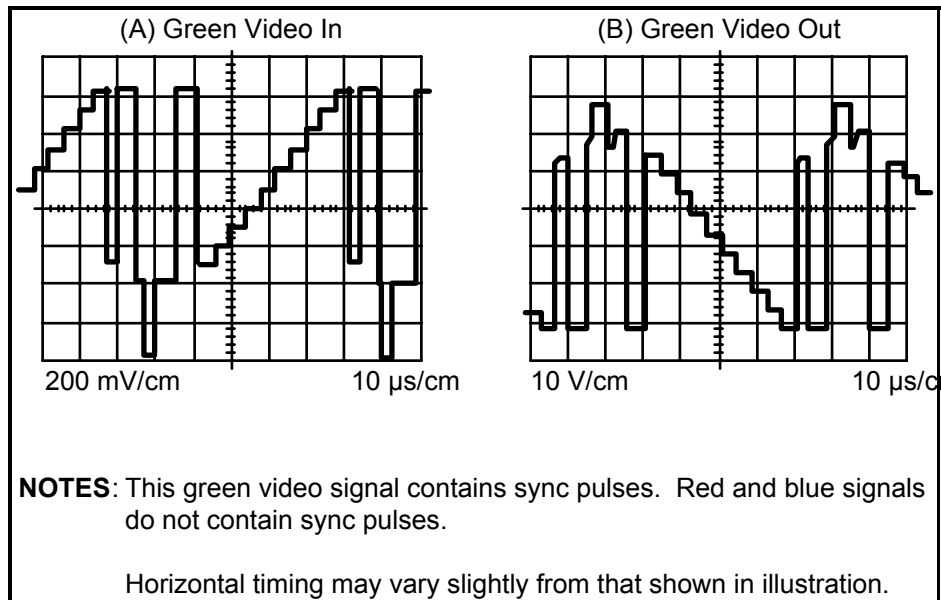


Figure 4-5 — Video Input/Output Signals

3323

4. DISPLAY CRT ASSEMBLY 51108308-100

4.5. CRT Monitor Parts Replacement (Color Video Monitor Qualified)

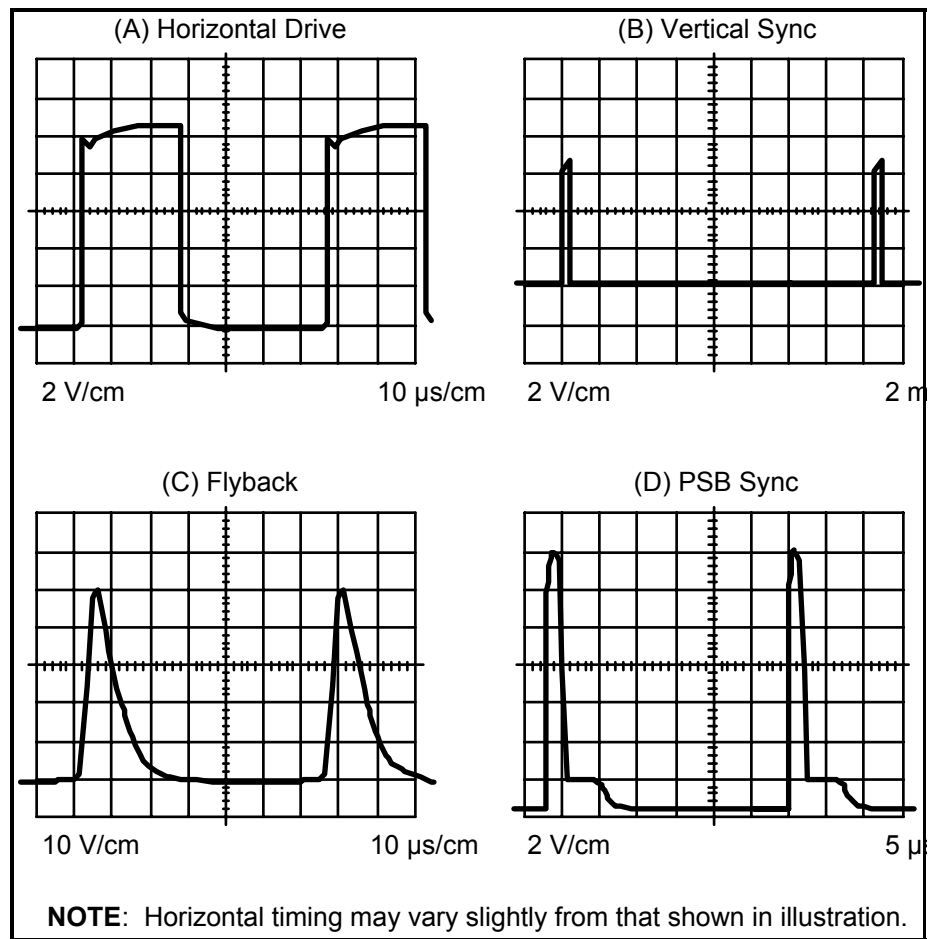


Figure 4-6 — Video PC Card Signals

3324

Horizontal Oscillator

The oscillator frequency is controlled by **H-HOLD** pot R33 (horizontal hold). An automatic phase control circuit changes the frequency of the oscillator so that the horizontal scan is in phase with the video. The **H-PHASE** (horizontal phase) control, R23, is adjusted so that the video information is centered on the raster.

Sync Generator for the Power Supply

Sync is generated for the switched-mode power supply (PSB-004). Its output consists of sync pulses at twice the horizontal frequency (SP1 in the **A** position). This signal is:

PSB Sync	JV3-3 (Return at JV3-4)	Figure 4-6D
----------	-------------------------	-------------

Input Power for VID-127 PC Card

Power enters at the following points:

+15 Vdc	JV3-1	
-15 Vdc	JV3-2	
+140 Vdc	JV2-1 (Gnd at JV2-2)	
+24 Vdc	JV2-3 (Gnd at JV2-4)	
+15 Vdc	JV2-5	

4.5.1.4 Deflection PC Card (DFL-006)

This card is mounted on the right side of the chassis, when viewed from the rear (see Figure 4-3). It contains the vertical deflection circuit, the horizontal scan and centering circuits, the pincushion and width regulator circuits, the automatic brightness limiter circuit, the regulated high voltage circuit, and various CRT biasing and blanking circuits.

Input Signals for DFL-006 PC Card

Signals enter at the following points:

• Horizontal Drive	JD2-1 (Gnd at JV1-4)	Figure 4-6A
• Vertical Sync	JD2-2	Figure 4-6B
• Flyback	JD2-3	Figure 4-6C
• ABL Sense	JD2-6	6 Vpp

4. DISPLAY CRT ASSEMBLY 51108308-100

4.5. CRT Monitor Parts Replacement (Color Video Monitor Qualified)

Output Signals From the DFL-006 PC Card

Signals are present at the following points:

• G1	JD1-1	+800 Vdc (High Z measurement)	
• G2	JD1-3	-40 Vdc (High Z measurement)	
• HV Feedback	JD4-1	DO NOT MEASURE	
• Feedback Shield	JD4-2 (Gnd at JD4-3)	0 V	
• Drive HV Xfmr	JD4-4	80 Vpp	
• Drive HV Xfmr	JD4-5	80 Vpp	
• Xfmr B+	JD4-6	+50 Vdc	
• Vert DFL Yoke	JD5-1	50 Vpp	Figure 4-7A
• Vert DFL Yoke	JD5-2	2 Vpp (+12 V level)	
• HV Interlock	JD5-3	+11 V	Figure 4-6A
• HV Interlock	JD5-4	+11 V	Figure 4-6A
• Horiz DFL Yoke	JD5-5	+1000 V	Figure 4-7B
• Horiz DFL Yoke	JD5-6	50 Vpp on 80 Vdc level	
• Gnd	JD6-1	0 V	

Input Power for DFL-006 PC Card

Power enters at the following points:

+50 Vdc	JD3-1 (Gnd at JD3-2)
-15 Vdc	JD3-3
+110 Vdc	JD3-4 (Gnd at JD3-5)
+15 Vdc	JD3-6
+24 Vdc	JD3-7

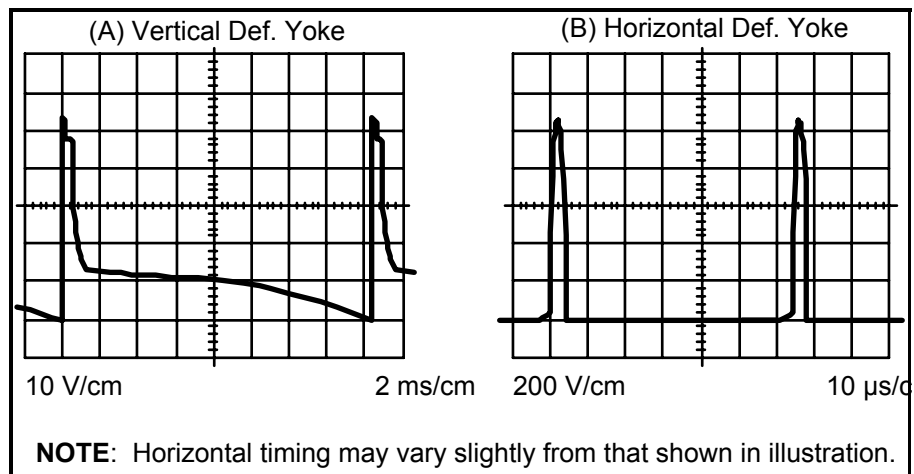


Figure 4-7 — Vertical/Horizontal Deflection Signals 3325

4.5.1.5 Power Supply PC Card (PSB-004)

This card is mounted on the bottom of the chassis, under the CRT yoke (see Figure 4-3). The power supply circuit is a switched-mode type that operates synchronously with the horizontal scanning frequency. Ac line voltage enters the card through jumper plug JP5-6 and JP7A-4 (for 115 V operation) or JP7B-4 (for 220 V operation). The power supply is designed to have a slow start up when power is applied.

Power supply board (PSB) sync is supplied to this PC card through JP1.

There are two modes of over-current protection with two corresponding trip levels. In the first mode, the duty cycle of the output pulse is reduced on a cycle-by-cycle basis until the current level is reduced. In the second mode, the trip level inhibits the output pulse for a certain time, after which the slow-start circuit tries to restart the power supply.

NOTE

Once an overload has been removed from the power supply it will operate normally. If the overload is still present, the power supply will repeatedly trip, emitting an audible chirping sound.

Input Power/Signal to PSB-004 PC Card

Power and signals enter at the following points:

PSB Sync	JP1-2 (Return at JP1-1)	Figure 4-6D
AC Line Input	JP5-6	
115 Vac Neutral	JP7A-4	Depends on JP7
220 Vac Neutral	JP7B-4	Depends on JP7

Output Power From PSB-004 PC Card

Power is supplied at the following points:

+6.3 Vdc	JP2-2 (Gnd at JP2-1)	Heater power for CRT
-15 Vdc	JP3-1	To VID-127
+15 Vdc	JP3-2	To VID-127
+24 Vdc	JP3-3	To VID-127
+140 Vdc	JP3-4, 5	To VID-127
Ground	JP3-6, 7	To VID-127
Ground	JP4-1, 2	To VID-127
B+ Scan	JP4-3, 4	+110/125/150 Vdc (depends on SP702)
+24 Vdc	JP4-5	To DFL-006
+15 Vdc	JP4-6	To DFL-006
-15 Vdc	JP4-7	To DFL-006
+50 Vdc	JP4-8	To DFL-006
115 Vac/220 Vac	JP5-1	To Degauss Switch (it must be pressed)
+105 Vdc	JP5-2	Power Lamp
AC Prim. Cmn.	JP5-3	Power Lamp return
115 Vac/220 Vac	JP6-1	Degauss Coil
115 Vac/220 Vac	JP6-2	Degauss Coil
	JP7A-1, 3 or JP7B-1, 3	AC Interlock
	JP7A-5, 6 or JP7B-5, 6	Degauss Interlock
+7 Vdc	JP8-1 (return is JP8-2)	Power Indicator (LED)

4.5.2 Parts Removal and Replacement

Recommended spare parts are listed in subsection 4.8. Use these replacement instructions to remove and replace the faulty part.

WARNING

Always disconnect ac power from the monitor before performing any replacement operation.

4. DISPLAY CRT ASSEMBLY 51108308-100

4.5. CRT Monitor Parts Replacement (Color Video Monitor Qualified)

4.5.2.1 Required Tools

- 1/4" Hex Nut Driver
- 3/8" Hex Nut Driver
- Long Nose Pliers
- Safety Goggles*
- Shop Apron*

* Required for CRT replacement

4.5.2.2 PSB-004 (Power Supply PC Card)

1. Disconnect AC power at the rear panel connector of the monitor. If necessary, locate PSB-004 using Figure 4-3.
2. Disconnect card connectors JP1 through JP6 on the power supply PC card.
3. Disconnect card connectors JP7A and JP7B taking particular note of the orientation of each connector in the wire harness (it is possible to inadvertently switch these two connectors). If necessary, label the connectors to prevent confusion.

WARNING

The inadvertent switching of the JP7A and JP7B connectors will result in damage to the monitor upon applying ac power.

4. Unsnap four nylon circuit card corner mounts and lift the card out of the chassis.
5. Install the new power supply PC card in its place, snapping in the four corners of the card.
6. Reinstall connectors JP7A and JP7B removed in step 3, making certain that they are correctly positioned on their respective pin headers.
7. Reinstall the connectors removed in step 2.
8. Reconnect the ac power connector.
9. There should be no need for adjustments after replacing this card.

4.5.2.3 DFL-006 (Deflection PC Card)

1. Disconnect ac power at the rear panel connector of the monitor. If necessary, locate DFL-006 using Figure 4-3.
2. Disconnect card connectors JD1 through JD7.
3. Unsnap four nylon circuit card corner mounts and lift the card out of the chassis.
4. Install the new deflection PC card in its place, snapping in the four corners of the card.
5. Reinstall the connectors removed in step 2.
6. Perform the following adjustments: Height, Width, Centering, CRT bias, High Voltage, Linearity and Geometry.

4.5.2.4 VID-127 (Video PC Card)

1. Disconnect ac power at the rear panel connector of the monitor. If necessary, locate VID-127 using Figure 4-3.
2. Disconnect card connectors JV1 through JV12.
3. Unsnap four nylon circuit card corner mounts and lift the card out of the chassis.
4. Install the new video PC card in its place, snapping in the four corners of the card.
5. Reinstall the connectors removed in step 2.
6. Perform the Horizontal Hold, Horizontal Phase, Cathode Bias, and Color Balance adjustments.

4.5.2.5 DVA-001 (Differential Video Card)

1. Disconnect ac power at the rear panel connector of the monitor. If necessary, locate DVA-001 using Figure 4-3.
2. Disconnect card connectors JF1 through JF4 and connectors JF104, JF204 and JF304. Take particular note of the orientation of each connector in the wire harness. If necessary, label the connectors to prevent confusion during reinstallation.
3. Remove the two Hex fasteners (use 1/4" nut driver) along the top of the card and unsnap the two nylon circuit card mounts along the lower portion of the card. Lift the card out.
4. Install the new circuit card by reinstalling the upper corners on the standoffs, then snap the lower corners onto the card mounts.
5. Reattach the two hex fasteners to secure the upper corners of the card.
6. Reinstall the connectors removed in step 2.
7. No adjustments are required after replacing this card.

4. DISPLAY CRT ASSEMBLY 51108308-100

4.5. CRT Monitor Parts Replacement (Color Video Monitor Qualified)

4.5.2.6 Cathode Ray Tube (Picture Tube)

WARNING

Do not remove, install, or otherwise handle the CRT picture tube without wearing shatterproof goggles and a shop apron. Persons not so protected should be kept at least 3 meters (10 feet) away. Also, keep the picture tube positioned in such a way that the tube screen always faces any unprotected persons in the area.

If the picture tube has been ON prior to removal, wait at least one minute after removing power before attempting to remove the high voltage connector.

1. Disconnect ac power at the rear panel connector of the monitor and wait about 1 minute. If necessary, locate the CRT using Figure 4-3.
2. Unplug the CRT tube socket connector from the CRT pin base.
3. Unplug the CRT high voltage anode connector.
4. Unplug the yoke harness connector JD5 on the DFL-006 (Deflection) circuit card and route the harness back to the CRT. Wrap the harness loosely around the neck of the CRT so it won't become entangled during CRT removal.
5. Begin removing the hex nuts that secure the mounting ears at the front of the CRT, starting with two nuts diagonally in opposite corners. Then, while holding the CRT in place, remove the remaining two nuts. Make certain the tube does not move unexpectedly.
6. Carefully lift the tube and pull it forward out of the chassis, making sure that the tube neck does not contact the chassis. Always use special care with the tube neck—if it is accidentally broken, it may cause an implosion of air rushing into the vacuum tube. An implosion, and the ensuing explosion, may cause personnel injury and/or property damage.
7. Install the new CRT reversing the above steps.
8. Perform the CRT bias, Focus, Degaussing, and Color Balance adjustments.

4.5.2.7 Focus Potentiometer

1. Disconnect ac power at the rear panel connector of the monitor and wait about 1 minute. If necessary, locate the focus potentiometer using Figure 4-3.
2. Steady the potentiometer with one hand while removing the screws and washers that secure it to the chassis. Use particular care not to drop the potentiometer or the screws and washers into the chassis.
3. While holding the potentiometer in your hand, unplug the two connections from the focus potentiometer. Take particular note of the orientation of each connector. If necessary, label the connectors to prevent confusion during reinstallation.
4. Install the new focus potentiometer, reversing the above steps.
5. Perform the Degaussing and Focus adjustments.

4.5.2.8 High Voltage Transformer (T901)

1. Disconnect ac power at the rear panel connector of the monitor and wait about 1 minute. If necessary, locate the transformer using Figure 4-3.
2. Unplug the connections from the transformer. Take particular note of the orientation of each connector. If necessary, label the connectors to prevent confusion during reinstallation.
3. Steady the transformer with one hand while removing the screws and washers that secure it to the chassis. Use particular care not to drop the transformer or the screws and washers into the chassis.
4. Install the new transformer, reversing the above steps.
5. Perform the High Voltage adjustments.

4. DISPLAY CRT ASSEMBLY 51108308-100

4.5. CRT Monitor Parts Replacement (Color Video Monitor Qualified)

4.5.2.9 High Voltage Multiplier (M901)

1. Disconnect ac power at the rear panel connector of the monitor and wait about 1 minute. If necessary, locate the multiplier using Figure 4-3.
2. Unplug the connections from the multiplier and disconnect the ground wire from the chassis ground-stud. Take particular note of the orientation of each connector. If necessary, label the connectors to prevent confusion during reinstallation.
3. Steady the multiplier with one hand while removing the screws and washers that secure it to the chassis. Use particular care not to drop the multiplier or the screws and washers into the chassis.
4. Install the new multiplier, reversing the above steps.
5. Perform the High Voltage adjustments.

4.5.2.10 Capacitor Module (M902)

1. Disconnect ac power at the rear panel connector of the monitor and wait about 1 minute. If necessary, locate the capacitor module using Figure 4-3.
2. Unplug the connections from the capacitor module. Take particular note of the orientation of each connector. If necessary, label the connectors to prevent confusion during reinstallation.
3. Steady the capacitor module with one hand while removing the screws and washers that secure it to the chassis. Use particular care not to drop the capacitor module or the screws and washers into the chassis.
4. Install the new capacitor module, reversing the above steps.
5. Perform the High Voltage adjustments.

4.6 Setup and Adjustment (Color Video Monitor Qualified)

Use Table 4-2 to determine if adjustments are required after replacing one of the printed circuit (PC) cards or the cathode ray tube (CRT).

Table 4-2 — Adjustments Required When a Major Component is Replaced

Adjustments	PSB- 004	DVA- 001	VID- 127	CRT	DFL- 006	Focus Pot.	T901	M901	M902
NO ADJUSTMENTS REQUIRED	√	√							
Degaussing				√		√			
Horiz./Vert.Hold			√						
Width					√				
Height					√				
CRT Bias			√	√	√				
Horizontal Phase			√						
Pincushion					√				
Centering Adj.					√				
Linearity					√				
Focus Adjustment				√		√			
Pincushion					√				
Color Balance			√	√					
High Voltage Adj. *					√		√	√	√

*Requires skilled knowledge, special tools and techniques.

4. DISPLAY CRT ASSEMBLY 51108308-100

4.6. Setup and Adjustment (Color Video Monitor Qualified)

4.6.1 Recommended Test Equipment and Special Tools

1. Digital Multimeter (DVM), Fluke™ 8050A or equivalent (10 meg input impedance).
2. 40 kV High Voltage Probe, either a free-standing model or compatible with a DVM.
3. Oscilloscope, TEK™ 265 or 475, 2-channel, 100/200 MHz amplifier, two 10/1 probes and a 100/1 probe.
4. Degaussing Coil, G. C. Electronics No. 9317.
5. Alignment Screwdriver, G. C. Electronics No 5009.
6. Optional Line Isolation Transformer (required when taking measurements on the primary side of T702).
7. Load the Station Diagnostics (HVTS OPIF tests for the PIC/VDG control card combination—HVTS PDGI tests for an EPDG or PDG control card) so that test patterns are accessible.

4.6.2 Safety Precautions

WARNING

Do not install, remove, or handle the picture tube in any manner unless shatter-proof goggles and a shop apron are worn. People not so equipped should be kept AT LEAST 3 meters (10 feet) away while the picture tube is being handled.

Use a line isolation transformer when taking measurements on the primary side of T702.

4.6.3 X-Radiation and Product Safety

This monitor uses components which are essential for X-radiation safety and other special characteristics. For this reason, replace all critical components indicated in this manual with the exact component found in the spare parts list.

It is recommended that only authorized factory personnel perform any repair or adjustment of the monitor's high voltage section. Should unauthorized repair become necessary, however, procedures for measurement and adjustment are given in subsection 4.6.

Note that the CRT operating voltage is 25 kV at minimum brightness.

4.6.4 Component Degaussing

The need for degaussing is indicated by poor color purity or poor convergence. Always degauss the monitor before any color purity or convergence adjustments are attempted. The CRT alone can often be degaussed by using the degauss switch and the built-in degaussing coil. If large, metallic components near the CRT (i.e., focus potentiometer or T901) have been replaced, you may need to use a commercial degaussing coil.

Other parts of the monitor may also require degaussing. This is indicated by poor color purity or static convergence which cannot be corrected by adjustments. Use a commercial degaussing coil and the following procedure:

1. Position the degaussing coil 2 to 3 meters (6 to 8 feet) from, and perpendicular to, the CRT screen and turn the degaussing coil on.
2. Turn the coil parallel to the CRT screen and, with a slow circular motion, bring the coil to the monitor.
3. Continuing the circular motion, pass the coil over the front, top, and sides of the monitor for approximately two minutes.
4. Now, with a slow circular motion, turn the coil perpendicular to the CRT screen and slowly back away 2 to 3 meters (6 to 8 feet) before switching the degaussing coil off.

4.6.5 Adjustment Procedures

NOTE

You may be required to turn this CRT monitor off, then on, while making these adjustments. The monitor's power supply has a slow start-up circuit and an electronic overload circuit that requires a 10 second wait between the time the power is turned OFF and the time it is turned back ON . If you do not wait 10 seconds, you may find the overload circuit trips, prohibiting restart.

If the power supply will not restart after the 10 second waiting period, suspect other problems. One possible problem is a shorted load on another card that prevents a successful startup.

Adjustment procedures should be followed in sequence and should only be made when required. All adjustments should be made with an insulated screwdriver. In the following procedures, it is always assumed that you have the proper line voltage and frequency available.

4. DISPLAY CRT ASSEMBLY 51108308-100

4.6. Setup and Adjustment (Color Video Monitor Qualified)

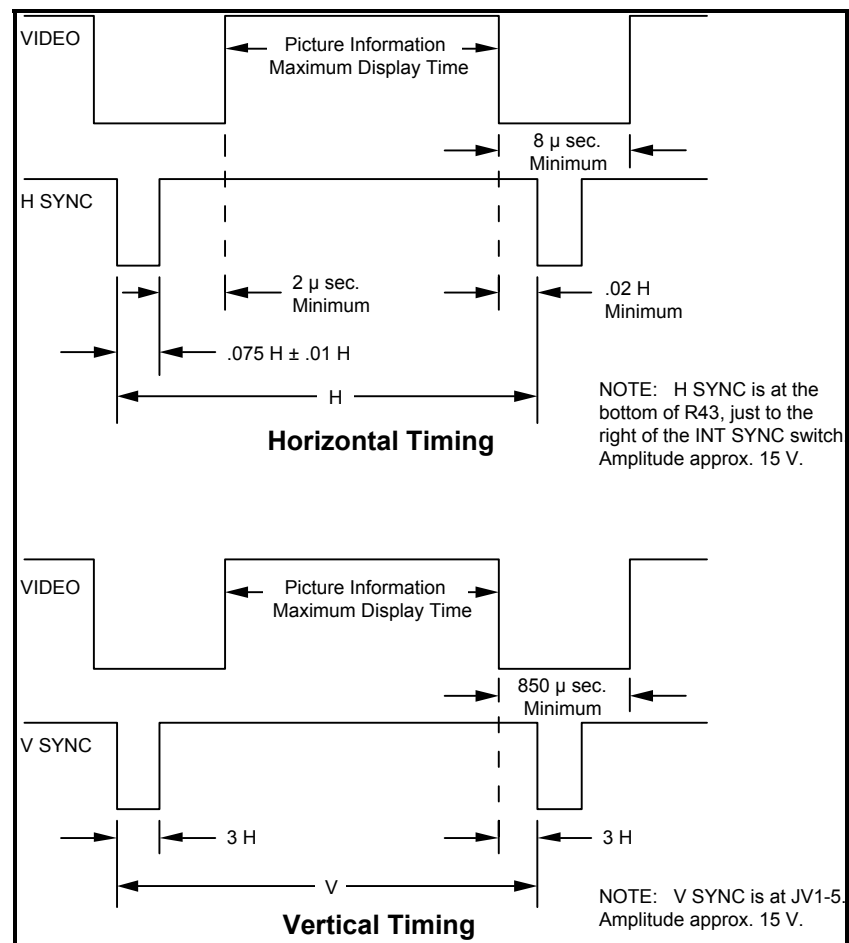


Figure 4-8 — Sync Timing Requirements

3326

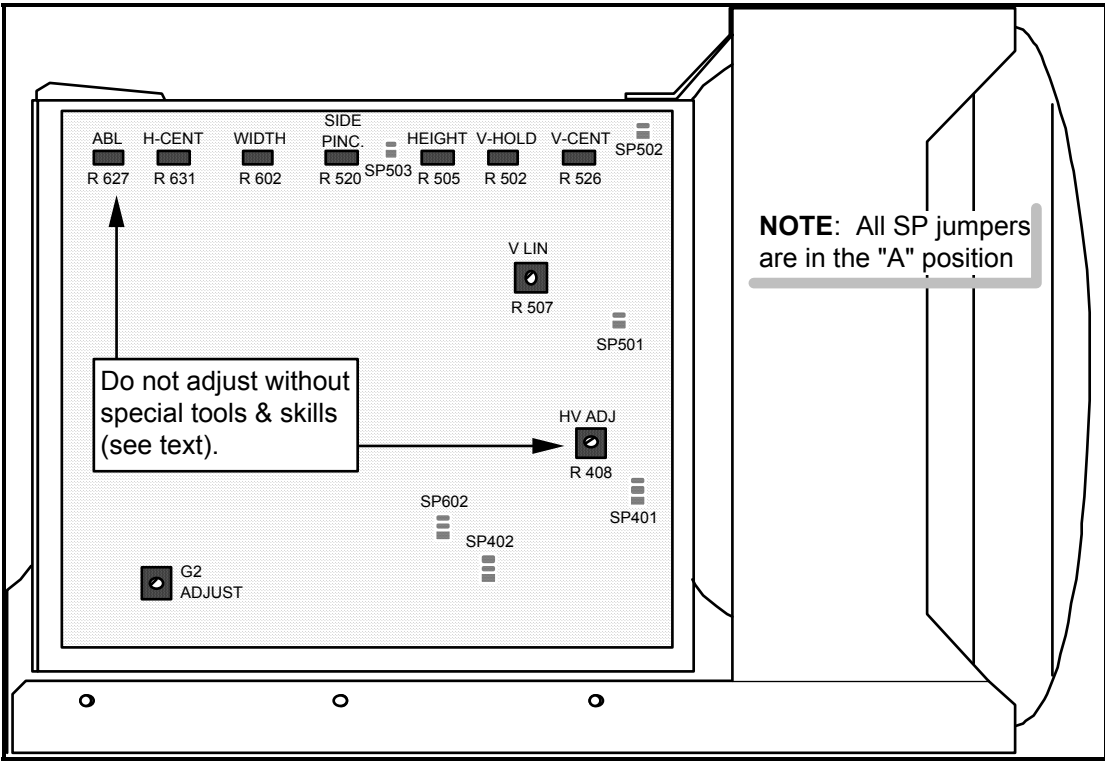


Figure 4-9 — DFL-006 Control Locations

3311

4. DISPLAY CRT ASSEMBLY 51108308-100

4.6. Setup and Adjustment (Color Video Monitor Qualified)

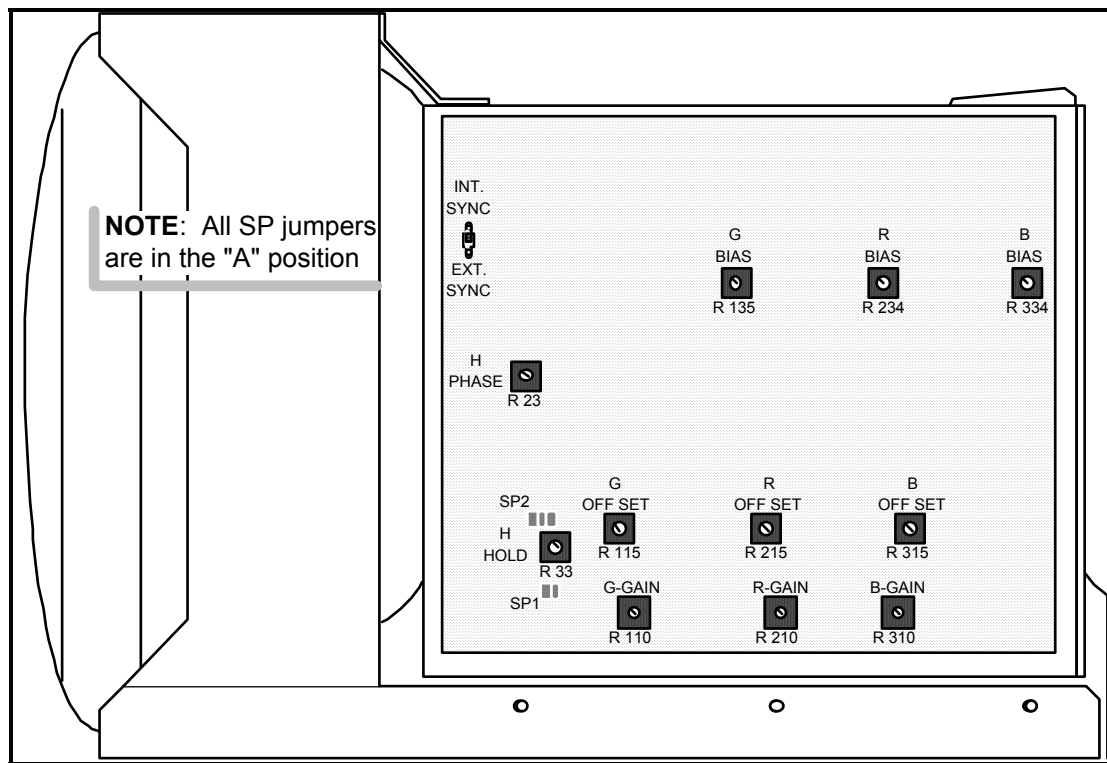


Figure 4-10 — VID-127 Control Locations

3312

Load the Station Diagnostics (HVTS OPIF tests for the PIC/VDG control card combination—HVTS PDGI tests for an EPDG or PDG control card) so that video test patterns are accessible. The green video contains composite sync so the **INT. SYNC/EXT. SYNC** switch (S1) on the video PC card must be in the **INT. SYNC** position.

If there is no sync signal present on the green video input, there will be no visible raster present. If this condition exists, check to be sure S1 is set to the INT SYNC position. If the switch is correctly set, place a scope probe on the sync line and check for proper signal amplitude as shown on the waveforms in Figure 4-8.

4.6.5.1 Horizontal/Vertical Synchronization

Use Figures 4-6 and 4-7 to locate the controls mentioned in this procedure.

1. Apply a known, all white, test pattern signal to the monitor before attempting to adjust it. The video source should be a crosshatch pattern.
2. Turn brightness control knob fully clockwise (up).
3. Adjust horizontal hold **H HOLD** R33 (VID-127) until picture locks in the horizontal direction.
4. Adjust vertical hold **V HOLD** R502 (on DFL-006) until picture locks in the vertical direction.
5. Switch the **INT. SYNC/EXT. SYNC** switch (S1 on VID-127) back and forth to cause the monitor to lose, then regain sync. If necessary, readjust the controls in Steps 3 and 4 to make sure the monitor synchronizes properly and the controls are not at the edge of their locking range.

4.6.5.2 Raster Size, Centering, and Geometric Adjustments

Use Figures 4-6 and 4-7 to locate the controls mentioned in this procedure.

1. Adjust the **WIDTH** control R602 (DFL-006) until the display width measures 36 cm from the left side of the white outline to the right.
2. Adjust the **HEIGHT** control R505 (DFL-006) until the display height measures 27 cm from the top of the white outline to the bottom.
3. Turn the brightness control down until a dim display of the test pattern can be seen.
4. Adjust the **G2 BIAS** control R638 (DFL-006) until a dim display of the background raster can be seen.
5. Adjust the **H PHASE** (horizontal phase) control R23 (VID-127) until the test pattern is centered on the background raster.
6. Adjust the **SIDE PINCUSHION** control R520 (DFL-006) until the sides of the raster are straight.

4. DISPLAY CRT ASSEMBLY 51108308-100

4.6. Setup and Adjustment (Color Video Monitor Qualified)

7. Adjust the **G2 BIAS** control R638 (VID-127) until the background raster disappears.
8. Adjust the horizontal and vertical centering controls **H CENT** R631 and **V CENT** R526 (DFL-006) until the test pattern is centered on the screen.
9. Adjust the vertical linearity control **V LIN** R507 (DFL-006) for the best overall vertical linearity. If necessary, readjust the vertical centering and then readjust the vertical linearity.
10. Adjust the **FOCUS** control VR901 (rear panel—see Figure 4-3) to obtain optimum line sharpness.
11. Check the display width and display height to see that they are 36 cm and 27 cm, respectively. If necessary, repeat steps 1 through 11 again to obtain the correct dimensions.

4.6.5.3 Input Signal Level Compensation/Gray Scale Tracking

Use Figures 4-6 and 4-7 to locate the controls mentioned in this procedure.

1. Signal level compensation and gray scale tracking must be made with no more than 30% video present on the raster.

NOTE

A 10 cm (4 inch) wide video square block in the center of the screen is ideal for this setup.

2. Turn the contrast control knob counterclockwise (down) until the video display is just visible.
3. Adjust the red, green, and blue bias controls, **R BIAS** R235, **G BIAS** R135 and **B BIAS** R335 (VID-127) for a gray display. Only adjust the color control that is needed to make gray. If background raster appears, readjust **G2 BIAS** control R638 (DFL-006).
4. Turn the contrast control knob fully clockwise (up).
5. Adjust the red, green, and blue gain controls, **R GAIN** R210, **G GAIN** R110 and **B GAIN** R310 (VID-127) for a white non-blooming display.
6. Turn the contrast control knob up and down slowly and check gray scale tracking. Repeat Steps 2 through 5 to obtain good gray scale tracking.
7. Adjust the **FOCUS** control for best overall crispness of display.

4.6.5.4 High Voltage/ABL Measurement and Adjustment

WARNING

MEASUREMENT AND ADJUSTMENT OF HIGH VOLTAGE CIRCUITS REQUIRE SPECIAL TOOLS AND TECHNIQUES. DO NOT ATTEMPT THIS TASK UNLESS QUALIFIED!

Only authorized factory personnel should perform any repair or adjustment of the monitor's critical circuits, including the high voltage section. Should unauthorized repair become necessary, use only exact duplicates when critical components are replaced.

Use Figures 4-6 and 4-7 to locate the controls mentioned in this procedure.

If the high voltage section has been repaired, make the following measurements and adjustments:

1. Make sure the monitor is turned OFF.
2. Place a high voltage probe (100 Meg ohms or higher) on the anode button of the CRT.
3. Check that the red lead to the anode button is secure and that it will remain connected throughout this procedure.
4. Connect the high voltage probe ground lead to the monitor chassis.
5. Turn the monitor ON.
6. Turn the brightness control fully counterclockwise (down). The high voltage reading should be $25 \text{ kV} \pm 300 \text{ V}$.
7. If the reading is not within this tolerance, turn the monitor off and break the seal on potentiometer **HV ADJ R408** (DFL-006).
8. Turn monitor on and adjust **HV ADJ R408** (DFL-006) to obtain a reading of 25 kV.
9. Leave the monitor on for 15 minutes. Check to see if the high voltage reading has changed. Readjust **HV ADJ R408** (DFL-006) if the reading is not within $25 \text{ kV} \pm 50 \text{ V}$. Turn the monitor off and reseal **HV ADJ R408** with Glyptol™ or nail polish.

NOTE

Always recheck the high voltage when any critical component or assembly has been repaired or replaced.

The Automatic Brightness Limiter (ABL) circuit limits the average beam current in the CRT and does not require field adjustment. This current has been factory adjusted and the control has been sealed. **DO NOT ATTEMPT TO ADJUST THE ABL CONTROL.**

4.6.5.5 Purity and Convergence Adjustments

The CRT assembly has been adjusted for purity and convergence at the factory and those adjustments have been sealed. If you feel convergence or purity settings need adjustment, first degauss the monitor components using the instructions in subsection 4.6.4. If degaussing does not solve the problem, reject the CRT assembly (51107145-111) and exchange it for a properly-prepared unit.

4.6.5.6 Permanently Sealed Potentiometers

If a card fails which contains a permanently sealed potentiometer, such as R714 in the PSB-004 power supply, Honeywell recommends the card be discarded and replaced.

4.7 Startup

After the CRT monitor has been installed, check that there is power to both the electronics module and the CRT module.

Turn the POWER switch ON at the electronics module (there is no power switch on the CRT monitor). Notice that the red LEDs on the cards in the electronics module light for a few seconds (less than 30), then turn off and the green LEDs turn on. Check the power status display on the electronics module and see that the **PWR OK** green LED is lit and no red LEDs are on. Check the "fan alarm" LED near the top of the 5-card chassis to ensure that its red LED is off. Check the cards and note that all green LEDs are on and the status alphanumeric display indicates the node address.

Adjust CRT brightness and contrast for best resolution on the character that appears in the corner of the screen. Check the *Touch Screen to CRT Alignment* in subsection 6.4.3.

4.8 Spare Parts

Table 4-3 — Spare Parts List

Part Number	Description
* 51108308-100	CRT Color Monitor (includes Remote Cable Assembly)
51107145-101	DFL-006 Deflection Card Assembly
51107145-102	PSB-004 Power Supply Card Assembly
51107145-103	VID-127 Video Card Assembly
51107145-104	DVA-001 Differential Video Card Assembly
51107145-105	Focus Potentiometer
51107145-106	T901 High Voltage Transformer
* 51107145-107	3AG, 2 amp/250 V Fuse (220/240 V units)
* 51107145-108	3AG, 3 amp/250 V Fuse (120 V units)
51107145-109	M901 High Voltage Multiplier
51107145-110	M902 Capacitor Module
51107145-111	19" Slow-Scan HC ARLP Cathode Ray Tube Assembly
51107145-112	Remote Cable Assembly

- ORU Level Replacement Item

4. DISPLAY CRT ASSEMBLY 51108308-100

4.8. Spare Parts

5 DISPLAY CRT ASSEMBLY 51195127-100

NOTE

The 51195127-100 DISPLAY MONITOR described in this section is no longer available from the manufacturer. Spare parts are still available. Use the 51304449-100 monitor described in Section 6 as a functional replacement.

5.1 Cleaning

NOTE

Cleaning of the 51195127-100 CRT, touchscreen, and keyboards should be done with **power off**, and the station off-line; therefore, it is best to do them all at once as described below. For touchscreen cleaning procedure, see subsection 6.1.

1. Open the door below the keyboards and set the main power circuit breaker located in the upper-right corner to the down (OFF) position.
2. If you have a keyboard power supply, set its switch, located just to the left of the main power circuit breaker, to "0" (OFF).
3. Open station rear door (below CRT chassis) and set the Module Power Switch to OFF.
4. Switch all other peripheral equipment associated with the station, such as floppy disk drives, etc. to OFF.
5. Clean the CRT face with a mild detergent solution and soft, lint-free cloth (do not use paper wipes of any kind). It is best to spray the solution on the CRT face to dissipate any static charge before wiping with the cloth.
6. Clean the Operator/Engineer Keyboard by dampening a soft cloth with mild detergent and wiping keys. Do not spray detergent solution on keys as this may contaminate the switch mechanism under the keys.
7. Restore power in accordance with subsection 5.7, *Startup*.

5.2 ORU Checks and Adjustments

5.2.1 Operator Adjustments

The checks and adjustments provided in this subsection are intended for use by the maintenance technician. The only adjustments available to the operator are listed as follows:

- Degauss—With power applied to the station, press the degauss button on the right side of the screen. Hold the button for a few seconds until colors are restored.
- Brightness and Contrast—Adjust brightness and contrast thumb wheels, on the right side of the screen for the desired affect.

5.2.2 Service Checks

There are no built-in tests for the CRT monitor. Fault isolation consists of ensuring that power is applied, and determining whether a fault is caused by the CRT monitor, by the touchscreen, or by some other source. The following list of symptoms and isolation procedures should be consulted before performing any maintenance technician adjustments:

SYMPTOM	FAULT ISOLATION PROCEDURE
Display black	Ensure that the power cord is plugged in. Check the power circuit breaker behind the door under the keyboard. On the electronics module, ensure that all LEDs are green. Set the power circuit breaker to OFF. Remove the rear door to the cabinet by opening and lifting off the hinge pins. Loosen the captive screws at the bottom of the CRT rear cover and pull the cover straight off the unit. Check the fuse located on the rear panel of the CRT chassis and replace, if required, with a fuse of the same type and rating, see Figure 5-5. Ensure that the ribbon cable is present and tight. Replace cover and door, and refer to <i>Startup</i> , subsection 5.7.
No display, just a raster, or "snow"	Ensure that power is applied to the electronics module. Remove the cabinet rear door and CRT rear cover (see Display black—Fault Isolation Procedure). Check the cables on the rear of the CRT and the electronics module to ensure that they are secure. Refer to the <i>Five/Ten-Slot Module Service</i> manual for further trouble shooting instructions. Replace cover and door.
Display just vertical or horizontal line or dot	Refer to the <i>Five/Ten-Slot Module Service</i> manual for module troubleshooting procedures. If the fault cannot be isolated to the module, the CRT is defective.
Colors not right	Press the degauss button on the front of the touchscreen and hold for three to five seconds. Ensure that contrast and brightness are correctly adjusted. Remove the rear door and the CRT rear cover (see Display black—Fault Isolation Procedure). Check the ribbon cable between the CRT monitor and electronics module.

5.3 ORU Removal and Replacement

If the fault with the CRT monitor cannot be found at this point, the monitor must be removed and replaced in its entirety. Removal and replacement of the CRT monitor assembly should be performed only by a qualified technician using the best shop practices.

WARNING

The CRT assembly contains high voltages and a vacuum in the glass CRT envelope. Ensure that the power is removed, by unplugging the power cable, before working on the unit. Use only insulated tools. Handle the CRT assembly only by the chassis; never lift by the neck or strike the glass.

5.3.1 CRT Assembly Removal

1. Open door below keyboards and set the main power circuit breaker located in the upper-right corner to the down (OFF) position.
2. Open station rear door (below CRT chassis) and set the Module Power Switch to OFF.
3. Switch all other peripheral equipment associated with the station, such as floppy disk drives, etc., to OFF.
4. Lower Tier mounted— Gain access to the rear of the CRT chassis by opening the rear door of the console and lifting the door off the hinge pins. Remove the rear cover from the assembly by loosening two captive screws on the bottom of the cover and pulling the cover off.

Upper Tier mounted— Loosen the rear cover from the assembly by removing four screws. Remove the cover by pulling it straight back and away from the CRT chassis. Use care not to drop the cover on the fragile glass cathode ray tube.

5. Label all cables to the CRT chassis (if required) and remove them. These include:
 - Power cable
 - Ribbon cable to I/O card
 - Remote cable assembly

5. DISPLAY CRT ASSEMBLY 51195127-100

5.3. ORU Removal and Replacement

6. Lower Tier mounted— Push the CRT chassis forward to relieve pressure on the retainers, then press on the retainer blocks to rotate the retainers down and away from the rear of the chassis (it may be necessary to unscrew the hex-head bolts slightly so that they clear the chassis rails). The CRT chassis, mounted on a slide tray, will now slide clear of the cabinet on rails and stop at a detent position.

WARNING

On Lower Tier mounted units, when the side buttons are again released in the next step, the CRT CHASSIS WILL BE FREE TO MOVE! Since the chassis is heavy with most of its weight concentrated at the front, you are strongly advised to seek assistance in removing the unit from the cabinet slide rails.

THE CHASSIS CENTER OF GRAVITY IS IN THE AREA OF THE CRT ANODE CONNECTOR. Two people can lift the unit if each of them grasp the rail at the top of the chassis on each side of the anode with one hand while pressing the side button, then steadying the chassis with the other hand while separating the slide rails.

7. Lower Tier mounted— With the assistance of a helper, again release the chassis by pressing the side buttons on the rails. Pull the unit free of the console along the slide rails, and place it on a bench for further disassembly.
8. Remove six bolts (3 on each side) to free the CRT chassis from the slide tray.

NOTE

If a Touchscreen must be removed or replaced, see *Touchscreens*, Section 6, for assistance.

5.3.2 CRT Assembly Replacement

1. With the aid of an assistant, set the CRT chassis on the tray between the mounting brackets, and align it with the holes on the mounting bracket. Install six bolts (3 on each side) to secure the chassis to the tray.
2. Lower Tier— With the aid of your assistant, engage the chassis slide rails with the slide rails on the console and push the assembly beyond the detent position so that both rails lock and hold the assembly in place.
3. Lower Tier— As you move the assembly forward in this step, check the alignment of the CRT face with the touchscreen. There should be a very small gap between them. Press the detent buttons and carefully push the CRT assembly forward. Lift the retainer blocks up behind the CRT chassis to hold the chassis in position. You may need to adjust the hex-headed bolts on the retainers to maintain the proper clearance with the touchscreen. Tighten the lock nuts on the retainers when you are satisfied with the alignment.
4. The CRT should also be centered in the screen. Center it by loosening the touchscreen mounting screws with a long-handled screwdriver, and moving the touchscreen slightly. You may need to readjust this setting again after power has been applied.
5. Connect the ribbon cable and the remote cable assembly to their connectors. Plug in the AC line cable.
6. Lower Tier mounted— Replace the rear cover and secure it with two captive screws. Reinstall the rear door of the console by sliding it back onto its hinges.

Upper Tier mounted— Be sure to engage both tracks inside the bottom of the cover, then slide the cover over the CRT chassis and secure it with four screws.

5.4 PWB Checks and Adjustments

If PWBs have been replaced, refer to Table 5-1 to determine which adjustment procedure needs to be done. **The adjustments are made with a crosshatch display that is generated by running HVTS PDGI Test 18 Version 5.0 or later.**

REQUIRED TEST EQUIPMENT

- Digital Volt Meter (DVM)
- High Voltage Probe (0-40 kV) compatible with DVM
- Minolta Color Analyzer Model II

Table 5-1 — Board Replacement/ Adjustment Procedures

Board Replaced	Adjustment
19" CRT Tube	CRT Bias, Degaussing, Color Temperature
CRT-002	None
DFL-009	High Voltage, Linearity and Geometry
FSP-001	B ⁺
VID-132	CRT Bias, Color Temperature

5.4.1 Horizontal Hold

The (H-HOLD 1-3)* control locks the picture into synchronization horizontally. Although the picture should be presently locked and stable, move the adjustment in both directions finding each edge of the lock. Position control in the approximate center of the lock-in range (or midpoint between the two edges of lock). See Figure 5-1.

5.4.2 Vertical Hold

The (V-HOLD 1-3) control locks the picture into synchronization vertically. Because the vertical deflection circuit seeks lock in only one direction, only one true edge of lock can be found. Move the control CCW (counterclockwise) until lock is broken. Reverse the direction of the control slowly until lock is regained. Move the control a little further into lock to be certain the picture remains stable when switching modes. Avoid adjusting the control too far into lock (CW—Clockwise direction). See Figure 5-1.

5.4.3 Vertical Height

The (HEIGHT 3) control adjusts the height of the display vertically. Adjust the control CW (Clockwise) to make the display taller and CCW (Counter-clockwise) to shrink the size. Using this control, adjust the height of the display to measure 27 cm. from the top of the white outline to the bottom. It may become necessary to adjust the

Vertical Centering (V-CENT) control before achieving the final height setting if the display is not centered. Move the control CW to position the display closer to the top and CCW for the opposite direction. See Figure 5-1.

* This monitor has 4 modes of operation (resolution). We use mode 3 exclusively, e.g., (H-Hold 1-3) is used for modes 1 to 3.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.4. PWB Checks and Adjustments

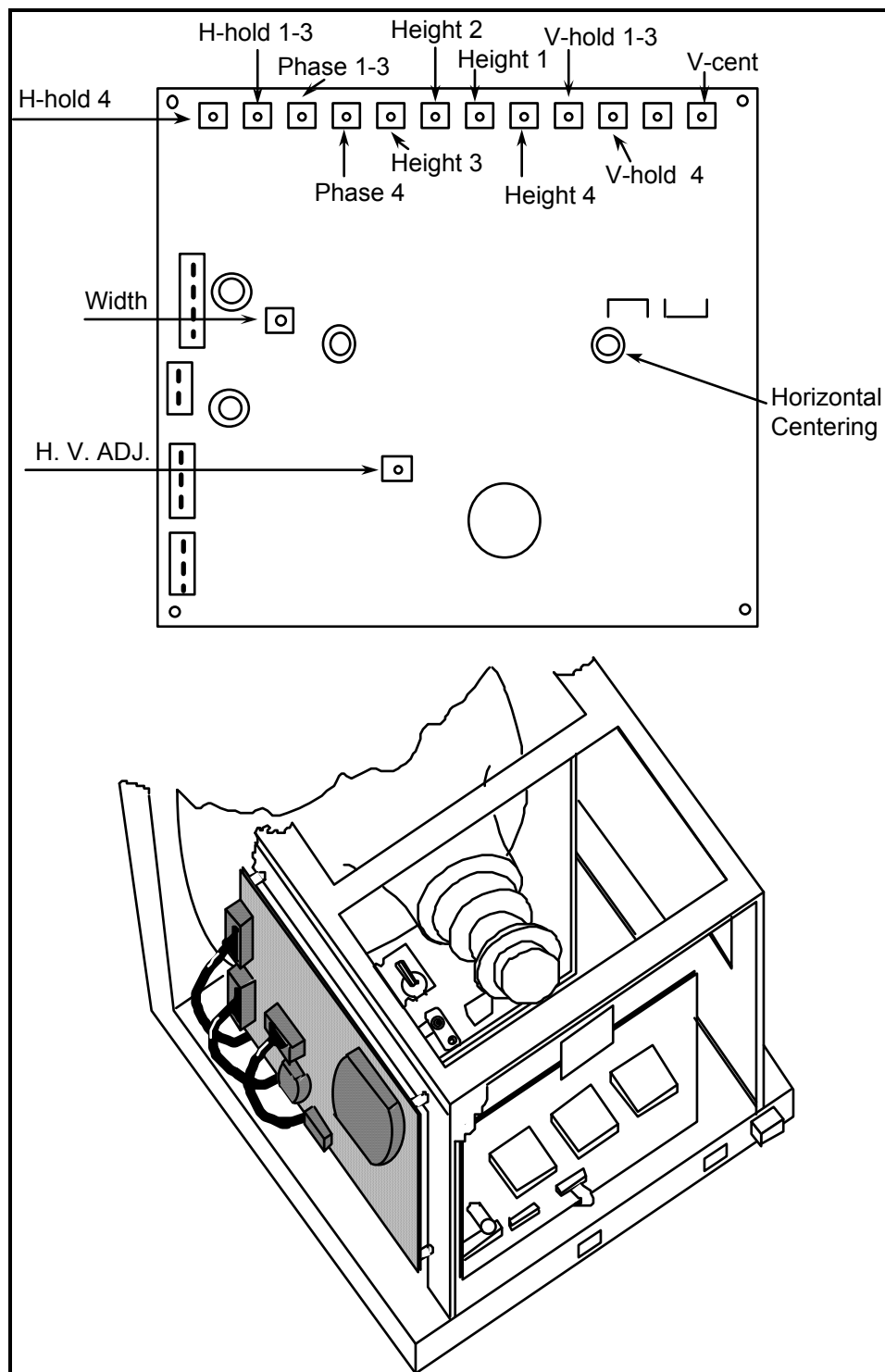


Figure 5-1 — DFL-009 Control Adjustment

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5.4.4 Horizontal Width

The (WIDTH) control adjusts the width of the display (see Figure 5-1.). Adjust the control CW (Clockwise) to make the display wider and CCW (Counter-Clockwise) for the opposite. Using this control, adjust the width of the display to measure 36 cm. from the left side of the white outline to the right. It may become necessary to adjust the Horizontal Centering (H-CENT) and the Horizontal Phase (PHASE 1-3) before achieving the final width setting if the display is not centered either on the background or within the bezel. Adjust the monitor Brightness control to the maximum so that the raster background is visible. If the side edges of the raster (background) are not visible, reduce the width of the display until they become visible. Using the Horizontal Centering control (H-CENT), adjust the raster centering until the raster is centered. Then, using the Horizontal Phase control (PHASE 1-3), adjust the position of the display within the raster until it is centered. Return the monitor Brightness control to the minimum setting and adjust the width to the final setting of 36 cm.

5.4.5 Linearity and Geometry

The controls for Linearity and Geometry are: Vertical Linearity (V-Lin) and Side Pincushion (SIDE PIN). Correction for horizontal linearity is made by adjusting Horizontal Linearity Coil. Top and bottom pincushion correction is built into the CRT yoke. See Figure 5-2 for location of controls on the DFL-009 circuit card.

5.4.5.1 Vertical Linearity

1. While the crosshatch display is shown, adjust the (V-LIN) control for linearity of the display from top-to-bottom with reference to the center. Move the control CW (Clockwise) to make the top lines of the crosshatch farther apart and the bottom lines closer together or CCW (Counter-Clockwise) to make the top lines closer together and the bottom lines further apart. The top lines should be the same distance apart as the bottom lines.
2. If a significant adjustment is made to the V-LIN control, it is also necessary to redo the Vertical Centering using V-CENT.

5.4.5.2 Side Pincushion Adjustment

With the crosshatch pattern displayed, adjust the Side Pincushion as follows:

1. The (SIDE PIN) control adjusts the straightness of the side of the display, correcting a distortion that occurs naturally with the CRT. Adjust so that the sides of the display are as straight as possible. Other distortions that can occur in the display, such as trapezoidal or parallelogram shapes, are a function of the tube performance and cannot be corrected using the SIDE PIN control.
2. If a significant adjustment was made to the SIDE PIN control it is necessary to redo the WIDTH setting to 36 cm.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.4. PWB Checks and Adjustments

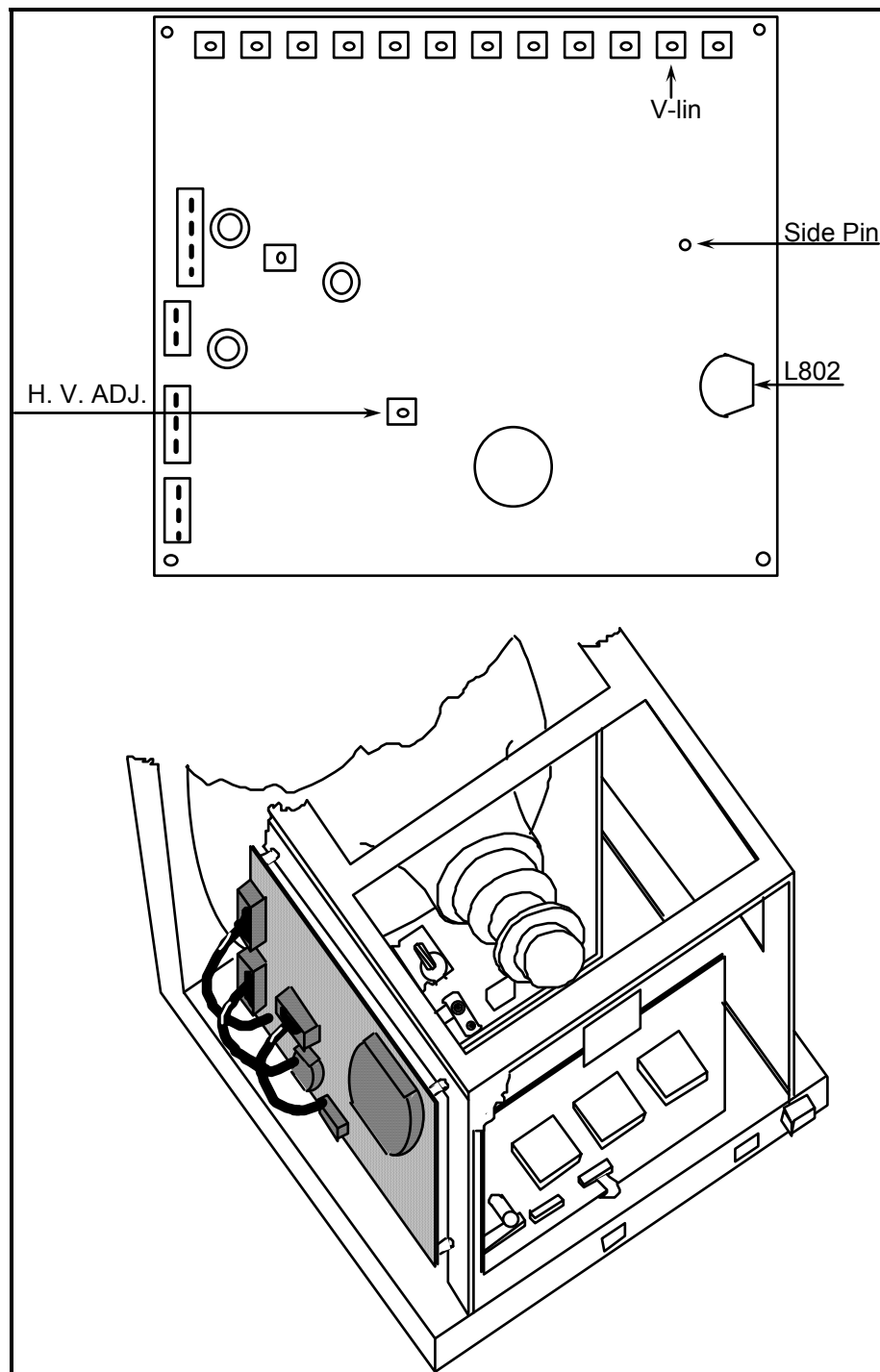


Figure 5-2 — Location of Controls on DFL-009 for Linearity and Geometry Adjustments

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5.4.6 Horizontal Linearity Adjustment

Adjust the Horizontal Linearity as follows:

1. Horizontal Linearity (L802)—This control adjusts the linearity of the display side-to-side with reference to the center. Rotating the thumbwheel on the side of the coil (taking care not to touch exposed terminals), changes the left side of the display, stretching or compressing the width of the blocks. Position the thumbwheel so that the width of the blocks on the left are equal to those on the right. See Figure 5-2.
2. If a significant adjustment is made to the H-LIN control, it may be necessary to redo the width adjustment and the horizontal centering using the H-CENT control.

5.4.7 CRT Bias Adjustment

Using the crosshatch display from HVTS PDGI Test 18, the CRT Bias may be adjusted as follows:

1. Check to see that the monitor Brightness control is at the minimum position.
2. Adjust the Red Bias (R BIAS), Green Bias (G BIAS), and Blue Bias (B BIAS) controls, aligning the adjustment slot to the six o'clock position. See Figure 5-2.
3. Adjust the screen control (G2), located on the Focus control assembly, CW (Clockwise) to produce a dim display, and then CCW (Counter-Clockwise) until the point is reached where the display just disappears.
4. Observing the display with a cupped hand or with an eyepiece, while in a darkened area, adjust each of the cathode bias controls in turn to balance cutoff levels. Using the R BIAS control, adjust CW until a dim red background is seen and then CCW to the point at which the red just disappears. Repeat this procedure using the G BIAS and the B BIAS controls.
5. Adjust the monitors Brightness control to the maximum position and observe that the background is a dim gray. Should the gray be shifted in hue towards any color, repeat the balancing operation in Step 4.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.4. PWB Checks and Adjustments

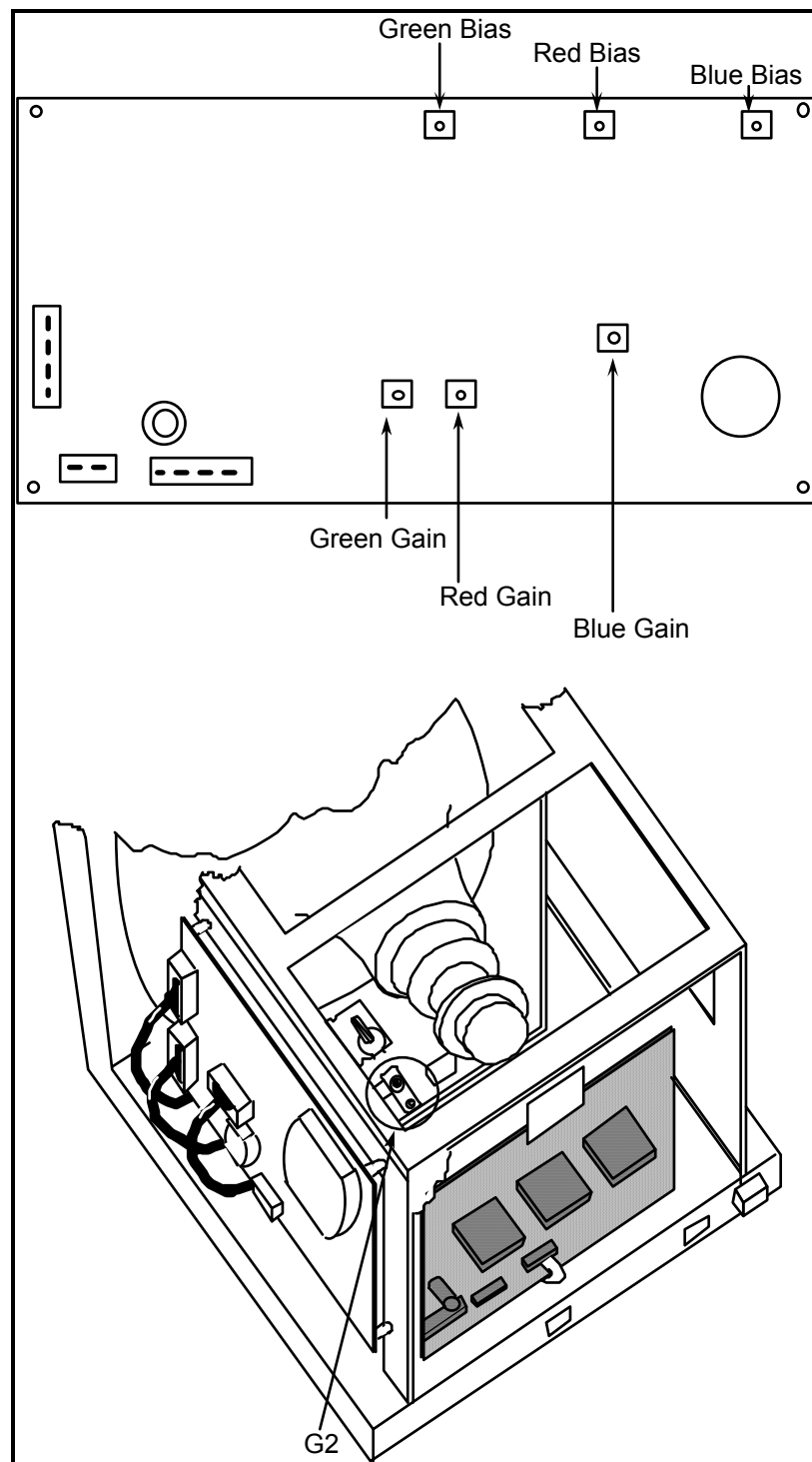


Figure 5-3 — Location of Controls (VID-132) for Color Temperature Adjustment 2767

5.4.8 Color Temperature

NOTE

- *1. This adjustment requires the use of a Minolta Color Analyzer Model II or equivalent and is not recommended unless this equipment is used. If it is not available, only make adjustments if colors are obviously off.
2. The CRT Bias adjustment must be done before starting this test.
3. The controls required for this adjustment are located on the video circuit card, VID-132. Location of the card and controls is shown in Figure 5-3.
4. This adjustment procedure adjusts the color temperature using the display generated by HVTS PDGI Test 17.

* See subsection 5.4.

1. Adjust the monitor's contrast control (upper thumbwheel) to the maximum position and the Brightness control (lower thumbwheel) to the minimum.
2. Observe the white square and the gray square in the display. At this point two determinations can be made: Is the picture bright enough (sufficient light output)? Is the color temperature of white (9300° Kelvin) correct?

In determining the light output level question, it should be considered that excessive drive causes spot growth (fuzzy characters) and shadow mask doming (a spot heating effect which temporarily shifts beam landing). The best adjustment of light output is one that approaches spot growth in level but does not allow brighter operation.

The assessment of color balance is simple when a good mental reference of the desired white is in place. The best description of 9300 degrees Kelvin (white), is a greenish-bluish white. It can be observed, that when white is properly adjusted, the secondary (mixed) colors are also proper in hue.

3. If the light output level requires adjustment, first adjust the R GAIN, G GAIN, and B GAIN by the same amount, either CW to increase or CCW to decrease. This adjustment can be done in a rough fashion for color balance.
4. Measure the color temperature of the white square, if it is not white (9300° - X=281, Y=311) take note of the color or hue it is shifted towards. (The X and Y values are the coordinate values displayed by the Color Analyzer.) If it is one of the primary colors (red, green, or blue), simply reduce that color drive by turning the appropriate GAIN adjust CCW (the three gain controls are somewhat interactive; that is, will change either one or both X/Y readouts). If it is shifted towards a secondary color (yellow, magenta or cyan), adjust the missing color not used to produce the secondary color) CW to increase the drive for that color. In this fashion a good white can be achieved.
5. Observe the gray square. If the gray is not pure, the color balance procedure is in subsection 5.4.7, step 4.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.4. PWB Checks and Adjustments

5.4.9 High Voltage

NOTE

We recommend that only experienced personnel be permitted to perform any repair or adjustment of the VGA 20's critical circuits (high voltage). Should unauthorized (component) repair be attempted, use only exact (part type) duplicate parts when designated critical components are replaced.

1. Turn the monitor's brightness and contrast controls to the minimum position.
2. Turn the main breaker off and wait one minute.
3. Place the tip of the high voltage probe (1000 Megohms or higher) on the anode contact button of the CRT by lifting the boot and sliding the probe in place. It is desirable to have the boot cover as much of the probe tip as possible and to lay flat on the CRT.
4. Connect the ground lead clip from the high voltage probe to the monitor chassis. Make sure this is secure and stays in place during the measurement.
5. Turn on the power at the main break of the Universal Station. If any arcing occurs, turn off the power, correct the connection to the anode of the CRT before proceeding. Allow a 15 minute warmup period.
6. Taking into consideration the attenuation ratio of the probe, check the meter for 26,000 volts (25,970 to 26,030 volts).
7. If the reading is not within this range, check the HIGH VOLTAGE (H.V. ADJ.) control on the deflection card DFL-009. See Figure 5-2. If sealed, remove power, replace the control (use only an exact duplicate part), adjust control to full counterclockwise direction and wait 15 minutes for the voltage to stabilize.
8. Adjust the high voltage control for 26 kV. After 15 minutes recheck the reading.
9. Turn the monitor off and wait one minute before removing the High voltage probe. Check to see that the anode contact and the boot are properly seated.
10. Permanently seal the H.V. ADJ control (R414) using cyanocrylate cement between the rotor and the body of the control.

5.4.10 B⁺ Checks (Low Voltage)

The B⁺ checks should be done with an average picture load — use HVTS PDGI Test 18.

Connect the negative lead of a DVM to the monitor chassis and using the positive lead, measure to see that the following dc voltages are at the locations listed. (The locations given were selected for ease of measurement rather than using the actual pins on the power supply.)

Table 5-2 — B⁺ Voltage Measurements

Power Supply Voltages and Tolerances DO THE MEASUREMENTS IN THE ORDER PRESENTED		
	VDC	TEST POINT
(Set this one with B ⁺ ADJ.)	16.0 $\pm$ 0.16 7.4 $\pm$ 0.3 30.0 $\pm$ 2.0 110.0 $\pm$ 3.0	JV2-Pin 7 (VID-132) JV3-Pin 1 (VID-132) JV5-Pin 4 (DFL-009) JV5-Pin 6 (DFL-009)

NOTE

Each connector is keyed with a plastic pin in one of the pin positions. Do not count these as pin numbers.

Should the +16.0 Vdc measurement be out of limits, it is likely that the other dc voltages are also out of limits.

Adjust the B⁺ ADJ control located on the power supply card (see Figure 5-4) as follows:

1. Turn off monitor and unplug the power cord. Remove the sealed B⁺ potentiometer and replace it with an exact duplicate part. Position the control at the center of its adjustment range.
2. With the DVM connected to the +16.0 Vdc test point, plug in the power cord and turn on the monitor. Carefully adjust the B⁺ ADJ control (R729) so that the DVM reads +16.0 Vdc. Turn off the monitor, unplug power cord, and permanently seal the control using cyanocrylate cement between the rotor and the body of the control so that no further adjustment is possible.
3. Check all voltages.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.4. PWB Checks and Adjustments

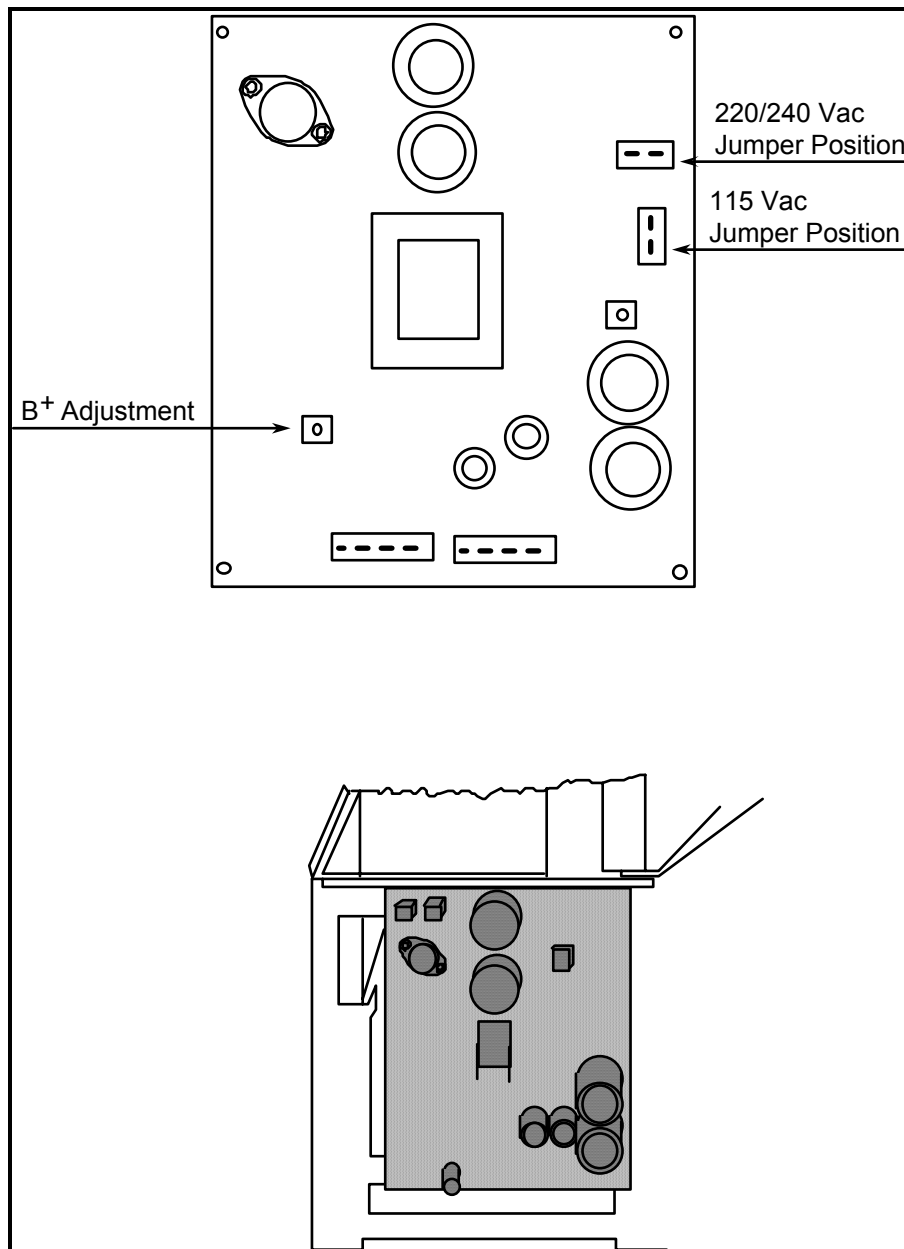


Figure 5-4 — B⁺ Adjustment on FSP-001

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5.5 Test/Troubleshooting

To be able to troubleshoot the monitor, one must understand the partitioning of the circuitry. Figure 5-5 shows a block diagram of the monitor with the circuitry partitioned to board types. Table 5-3 gives a description of the major functions of each board type.

Table 5-3 — Board Type Functional Descriptions

Board Type	Description
CRT-002	The CRT board contains the connector for the CRT and interfaces the video output signals from the VID-132 board, the focus voltage and G2 voltage from the high voltage output section.
DFL-009	The deflection board controls the scanning of the electron beams in the CRT. • Horizontal Deflection Circuit • Vertical Deflection Circuit • Regulated High Voltage Power Supply • Scan Mode Control Switches
FSP-001	Supplies a regulated B ⁺ from an input of 120/240 Vac.
VID-132	Controls the color and brightness of the display. The sync signals that come in on the video input connector are processed on this board and sent to the deflection board, DFL-009.

Figure 5-5 shows a block diagram of the monitor.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.5. Test/Troubleshooting

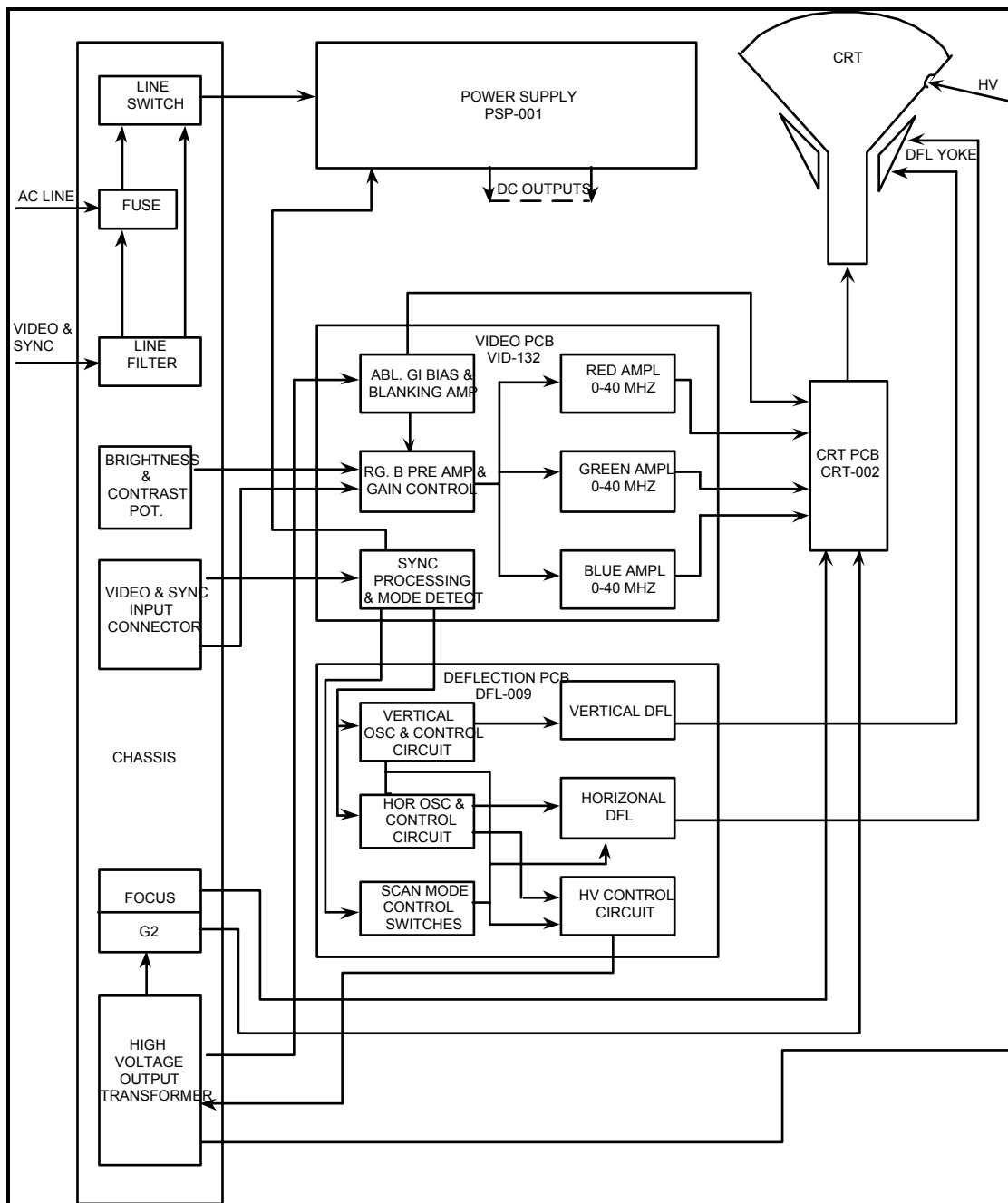


Figure 5-5 — Monitor Block Diagram 2772

5.6 Parts Removal/Replacement

5.6.1 Required Tools

#2 Phillips Screwdriver

1/4" Hex Nut Driver

5/16" Hex Nut Driver

*3/8" Hex Nut Driver

25 W Soldering Pencil and Resin Solder

Diagonal Wire Cutters

Long Nose Pliers

*Safety Goggles

*Shop Apron

* Required for CRT replacement

WARNING

Always turn off the ac power breaker on the Universal Station before attempting to replace any part of the monitor.

5.6.2 FSP-001 (Power Supply Card)

1. Disconnect the ac power cord from the rear of the monitor.
2. Disconnect the CRT Interface Cable from the rear of the monitor.
3. Loosen the two retaining clamps at the rear of the assembly and carefully slide the assembly back on the slide rails. If the assembly must be removed from the rails, press in on the detent on each side and slide the assembly out.
4. Disconnect connectors JP1, JP3 and JP4 thru JP8.
5. Remove the hex fastener (1/4" AF = [across flats]) securing the corner near JP7.
6. Unsnap the other three corner mounts and lift out the circuit card.
7. Install the new Power Supply Card in the reverse order, snapping in the three corners and reattaching the hex fastener that secures the fourth corner.
8. Reinstall the connectors removed in step 4. Check to see that the ac input selector plug SP701 is installed and in the correct position.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.6. Parts Removal/Replacement

9. Slide the assembly carefully forward until the retaining clamps can be swung into the locking position.
10. Reconnect the ac power.
11. Perform the B⁺ check in subsection 5.4.10.

5.6.3 DFL-009 (Deflection and High Voltage Card)

1. Disconnect the ac power cord from the rear of the monitor.
2. Disconnect the CRT Interface Cable from the rear of the monitor.
3. Loosen the two retaining clamps at the rear of the assembly and carefully slide the assembly back on the slide rails. If the assembly must be removed from the rails, press in on the detent on each side and slide the assembly out.
4. Disconnect connector JD1 through JD6.
5. Remove the circuit card by unlatching the four corner mounts and lifting the card out.
6. Install the new circuit card in the same manner, snapping the four corners into the mounts.
7. Reinstall the connectors removed in step 4.
8. Slide the assembly carefully forward until the retaining clamps can be swung into the locking position.
9. Reconnect the ac power and the Interface Cable.
10. Perform the High Voltage adjustment, Linearity and Geometry adjustments in subsections 5.4.5, 5.4.6, and 5.4.9.

5.6.4 VID-132 (Video Card)

1. Disconnect the ac power cord from the rear of the monitor.
2. Disconnect the CRT Interface Cable from the rear of the monitor.
3. Loosen the two retaining clamps at the rear of the assembly and carefully slide the assembly back on the slide rails. If the assembly must be removed from the rails, press in on the detent on each side and carefully slide the assembly out.
4. Disconnect connector JV1 through JV6 plus JV102, JV202, and JV302.
5. Remove the two hex fasteners (1/4" = [across flats]) securing the lower corners of the card.
6. Unlatch the mounts securing the upper corners of the card and lift the circuit card out.
7. Install the new circuit card in the reverse order, installing the lower corners on the standoffs and snapping in the upper corners.
8. Reattach the two hex fasteners securing the lower corners.
9. Reinstall the connectors removed in step 4.
10. Slide the assembly carefully forward until the retaining clamps can be swung into the locking position.
11. Reconnect the ac power and the Interface Cable.
12. Perform the CRT Bias and Color Temperature in subsections 5.4.7 and 5.4.8.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.6. Parts Removal/Replacement

5.6.5 CRT-002 (Tube Socket Connector Card)

1. Disconnect the ac power cord from the rear of the monitor.
2. Disconnect the CRT Interface Cable from the rear of the monitor.
3. Loosen the two retaining clamps at the rear of the assembly and carefully slide the assembly back on the slide rails. If the assembly must be removed from the rails, press in on the detent on each side and carefully slide the assembly out.
4. Disconnect connectors XV1 and JC1 through JC5.
5. Remove the ground strap from the ground screw on the card by removing the hex nut (5/16" AF = [across flats]) and washer.
6. Slide the insulating boot back from the Focus wire (White wire from XV1 to the focus pack Term 2), unsolder the focus lead from the ferrule, and remove the card.
7. Install the new card as follows:
 - a. Place the insulating boot over the Focus wire properly oriented. Slide the boot back to expose the lead end.
 - b. Resolder the Focus wire to the terminal 2 on the Focus pack. Slide the boot up over the ferrule of terminal 2.
 - c. Reinstall the ground strap over the ground screw and secure the nut and washer.
8. Carefully slide the assembly forward until the retaining clamps can be swung into the locking position.
9. Reconnect the ac power and the Interface Cable.

5.6.6 IFT-001 (Integrated Flyback Transformer Card)

1. Disconnect the ac power cord from the rear of the monitor.
2. Disconnect the CRT Interface Cable from the rear of the monitor.
3. Loosen the two retaining clamps at the rear of the assembly and carefully slide the assembly back on the slide rails. If the assembly must be removed from the rails, press in on the detent on each side and slide the assembly out.
4. Disconnect connector JD8 and the CRT anode connector (using a grounded screwdriver to check discharge of the anode). Unsolder the Focus wire:
 - a. Slide back the boot.
 - b. Unsolder the lead from the ferrule.
 - c. Remove the boot from the wire.
5. Remove the two screws securing the card to the chassis and lift out the card.
6. Install the new card in the reverse order, fastening the card to the chassis with the two screws, installing the boot over the Focus lead, and soldering the lead to the ferrule on the Focus terminal. Slide the boot up and over the ferrule to the terminal.
7. Reconnect connector JD8 and the anode cap.
8. Carefully slide the assembly forward until the retaining clamps can be swung into the locking position.
9. Reconnect the ac power and the Interface Cable.
10. Perform the High Voltage adjustment in subsection 5.4.9.

5. DISPLAY CRT ASSEMBLY 51195127-100

5.6. Parts Removal/Replacement

5.6.7 Cathode Ray Tube (CRT)

CAUTION

DO NOT attempt CRT replacement unless you have been trained to do so and know the precautions that must be taken.

1. Disconnect the ac power cord from the rear of the monitor.
2. Disconnect the CRT Interface Cable from the rear of the monitor.
3. Loosen the two retaining clamps at the rear of the assembly and carefully slide the assembly back on the slide rails. If the assembly must be removed from the rails, press in on the detent on each side and slide the assembly out.

WARNING

Do not remove, install, or handle the picture tube in any manner unless you wear shatter-proof goggles and a shop apron. Persons not so equipped should be kept at least 10 feet away and behind the tube face while the picture tube is being handled.

CAUTION

If the picture tube has been ON before removal, wait at least one minute (this is to allow the high voltage to bleed off) before attempting to remove the high voltage anode connector.

4. Unplug the CRT-002 card connector from the tube base.
5. Remove the high voltage anode connector. Use a grounded screwdriver to check the discharge of the Anode.
6. Unplug the yoke harness connector JD6 (DFL-009) and cut the cable tie holding the yoke harness to the chassis. Feed the harness through the card upright and carefully wrap it around the neck of the CRT behind the yoke.
7. When removing the hex nuts that attach the tube mounting ears to the chassis, first remove two nuts on diagonally opposite corners. Then holding the tube in place, remove the remaining two nuts making sure that the tube does not move unexpectedly.
8. Carefully lift out the tube making sure the tube neck does not contact the chassis. Breaking the tube causes an implosion that could result in injury.
9. Install the new CRT by reversing steps 8 through 1.
10. Perform CRT Bias, Focus, Degaussing and Color Temperature adjustments.

5.7 Startup

Ensure that all of the Universal Station equipment is turned off (this prevents power surges that could damage the equipment when the main power switch is turned on). Apply power by setting the main power circuit breaker, located behind the door under the keyboard, to "1" (ON). Set the keyboard power supply switch to "1" (ON), and close the door. On the rear of the unit, open the rear door and set the module power switch to ON. After 1.75 minutes, check the module light emitting diode (LED) annunciators to ensure that they are all green. The alphanumeric indicator displays the module node address when the operating system is running or not running. If any of the above conditions cannot be met, refer to the *Five/Ten-Slot Module Service* manual. Close the Station rear door. Apply power to any other equipment turned off at the beginning of this procedure.

Adjust CRT brightness and contrast for best resolution on the character that appears in the corner of the screen. Check the *Touchscreen to CRT Alignment* in subsection 6.4.3.

5.8 Spare Parts List

Table 5-4 — Monitor Spare Parts

Part Number	Description
* 51304360-100	Replacement Monitor Unit (120 Vac)
* 51304360-200	Replacement Monitor Unit (220 Vac)
51107143-101	Tube, CRT Replacement
51107143-102	PWB, FSP-001
51107143-103	PWB, VID-132
51107143-104	PWB, DFL-009
51107143-105	PWB, CRT-002
51107143-106	Pot, Focus ASM
51107143-107	Potentiometer, Contrast
51107143-108	Potentiometer , Brightness
51107143-109	Switch, Degauss
51107143-110	Card Assembly, IFT-001
51107143-111	Degauss Coil
51107143-112	Remote Cable Assy (includes potentiometers and switch)
51107143-113	Transistor, Plug-in (Q901, Q902)

5. DISPLAY CRT ASSEMBLY 51195127-100

5.8. Spare Parts List

51304366-002	Cable, CRT Interface (2 m./6.6 ft. for lower tier Monitors)
51304366-003	Cable, CRT Interface (3 m./9.8 ft. for upper tier Monitors)

* = Optimum Replaceable Unit (ORU)

6 DISPLAY CRT ASSEMBLY 51304449-100 and 51196696-200

NOTE

The 51195127-100 DISPLAY MONITOR described in this Section is no longer available from the manufacturer. The monitors described in this section (with its remote cable assembly) is a functional replacement.

Assembly 51196696-200 is the same monitor as assembly 51304449-100, but with a different CRT glass tube.

This CRT Monitor is an enhanced-definition, noninterlaced CRT Monitor. This assembly requires the EPDG board, and an EPDG I/O, or EPDGP I/O board for control. The remote control harness (including brightness, contrast, and degauss controls) has different value potentiometers and thus, is unique to this assembly. The monitor is defined as an Optimum Replaceable Unit (ORU) and no spare parts other than a replacement monitor are given.

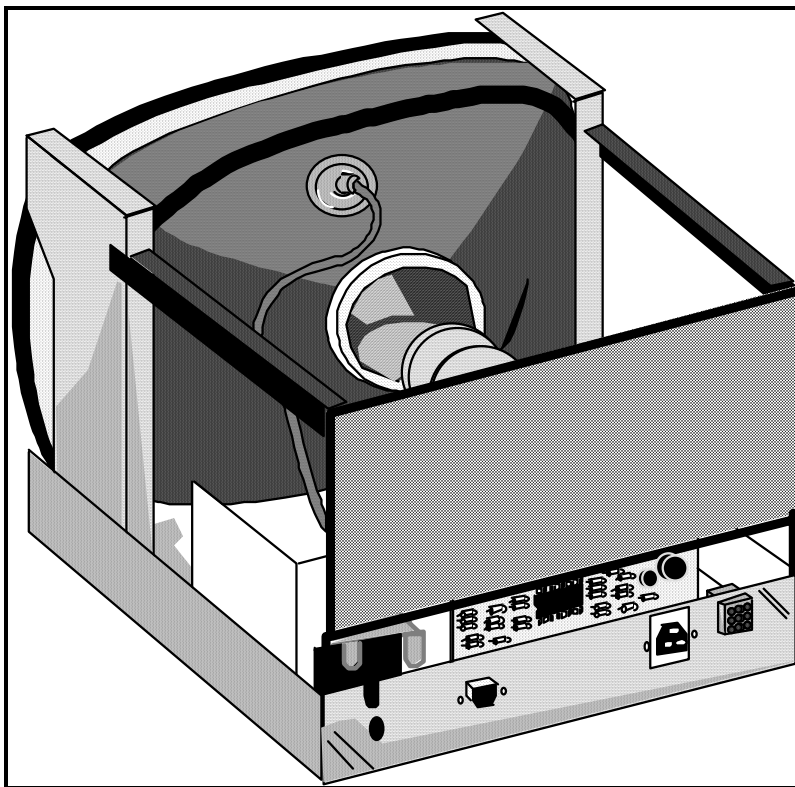


Figure 6-1 — CRT Monitor Assembly 51304449-100 and 51196696-200

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6.1 Cleaning

NOTE

Cleaning of the color monitor CRT, touchscreen, and keyboards should be done with **power off**, and the station off-line; therefore, it is best to do them all at once as described below. For touchscreen cleaning procedures, see subsection 7.1.

1. Open the door below the keyboards and set the main power circuit breaker located in the upper-right corner to the down (OFF) position.
2. If you have a keyboard power supply, set its switch, located just to the left of the main power circuit breaker, to "0" (OFF).
3. Open station rear door (below CRT chassis) and set the Module Power Switch to OFF.
4. Switch all other peripheral equipment associated with the station, such as floppy disk drives, etc., to OFF.
5. Clean the CRT face with a mild detergent solution and soft, lint-free cloth (do not use paper wipes of any kind). It is best to spray the solution on the CRT face to dissipate any static charge before wiping with the cloth.
6. Clean the Operator/Engineer Keyboard by dampening a soft cloth with mild detergent and wiping keys. Do not spray detergent solution on keys as this may contaminate the switch mechanism under the keys.
7. Restore power in accordance with subsection 6.5, *Startup*.

6.2 ORU Checks and Adjustments

6.2.1 Operator Adjustments

The checks and adjustments provided in this subsection are intended for use by the maintenance technician. The only adjustments available to the operator are listed as follows:

- Degauss—With power applied to the station, press the degauss button on the right side of the screen. Hold the button for no less than 15 seconds to ensure proper operation of the degaussing circuit. This will restore correct color purity and sometimes convergence.
- Brightness and Contrast—Adjust brightness and contrast thumbwheel controls on the right side of the screen for the desired effect. The brightness control is the upper thumbwheel and the contrast control is below it.

6.2.2 Service Checks

There are no built-in tests for the CRT monitor. Fault isolation consists of ensuring that power is applied, and determining whether a fault is caused by the CRT monitor, by the touchscreen, or by some other source. Table 6-1 contains a list of symptoms and basic isolation procedures. Consult this table before performing any maintenance technician adjustments.

Table 6-1 — Basic Fault Isolation Procedures

SYMPTOM	FAULT ISOLATION PROCEDURE
Display black	Ensure that the power cord is plugged in. Check the power circuit breaker behind the door under the keyboard. Check the contrast thumbwheel control to be sure it isn't turned down too far. On the electronics module, ensure that all LEDs are green. Set the power circuit breaker to OFF. Remove the rear door to the cabinet by opening and lifting it off the hinge pins. Loosen the captive screws at the bottom of the CRT rear cover and pull the cover straight off the unit. Check the fuse and replace it, if required, with a fuse of the same type and rating (5x20 mm, slow blow, 5 A, 250 V). The fuse is located on the rear panel of the CRT chassis, in the power cord receptacle assembly, immediately below the socket where the power cord is plugged in. To access the fuse, unplug the power cord, and with a small screwdriver, gently pry the fuse holder out of the receptacle assembly. The fuse holder has a picture of a fuse on it. There is a spare fuse in a slide-out compartment of the fuse holder. Ensure that the ribbon cable is present and tight. Replace cover and door, and refer to <i>Startup</i> , subsection 6.5.
No display, just a raster or "snow"	Ensure that power is applied to the electronics module. Remove the cabinet rear door and CRT rear cover (see Display black—Fault Isolation Procedure). Check the cables on the rear of the CRT and the electronics module to ensure that they are secure. Refer to the <i>Five/Ten-Slot Module Service</i> manual for further troubleshooting instructions. Replace cover and door.
Display just vertical or horizontal line or dot	Refer to the <i>Five/Ten-Slot Module Service</i> manual for module troubleshooting procedures. If the fault cannot be isolated to the module, the CRT is defective.

6. DISPLAY CRT ASSEMBLY 51304449-100 and 51196696-200

6.2. ORU Checks and Adjustments

Colors not right	Press the degauss button on the front of the touchscreen and hold for at least 15 seconds. Ensure that contrast and brightness are correctly adjusted. Remove the rear door and the CRT rear cover (see Display black—Fault Isolation Procedure). Check the ribbon cable between the CRT monitor and electronics module.
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6.3 ORU Removal and Replacement

If the fault with the CRT monitor cannot be found at this point, the monitor must be removed and replaced in its entirety. Removal and replacement of the CRT monitor assembly should be performed only by a qualified technician using the best shop practices.

WARNING

The CRT assembly contains high voltages and a vacuum in the glass CRT envelope. Ensure that the power is removed, by unplugging the power cable, before working on the unit. Use only insulated tools. Handle the CRT assembly only by the chassis; never lift by the neck or strike the glass.

6.3.1 Monitor Assembly Removal

1. Open door below keyboards and set the main power circuit breaker located in the upper-right corner to the down (OFF) position.
2. Open station rear door (below CRT chassis) and set the Module Power Switch to OFF.
3. Switch all other peripheral equipment associated with the station, such as floppy disk drives, etc., to OFF.
4. Lower Tier mounted— Gain access to the rear of the CRT chassis by opening the rear door of the console and lifting the door off the hinge pins. Remove the rear cover from the assembly by loosening two captive screws on the bottom of the cover and pulling the cover off.

Upper Tier mounted— Loosen the rear cover from the assembly by removing four screws. Remove the cover by pulling it straight back and away from the CRT chassis. Use care not to drop the cover on the fragile glass cathode ray tube.

5. Disconnect all cables to the CRT chassis. These include:
 - Power cable
 - Ribbon cable to I/O card
 - Remote cable assembly.

6. DISPLAY CRT ASSEMBLY 51304449-100 and 51196696-200

6.3. ORU Removal and Replacement

Place the cables so that they will not interfere with sliding the monitor to the rear.

6. Lower Tier mounted— Push the CRT chassis forward to relieve pressure on the retainers, then press on the retainer blocks to rotate the retainers down and away from the rear of the chassis (it may be necessary to unscrew the hex-head bolts slightly so that they clear the chassis rails). The CRT chassis, mounted on a slide tray, will now slide clear of the cabinet on rails and stop at a detent position.

WARNING

On Lower Tier mounted units, when the side buttons are again released in the next step, the CRT CHASSIS WILL BE FREE TO MOVE! Since the chassis is heavy with most of its weight concentrated at the front, you are strongly advised to seek assistance in removing the unit from the cabinet slide rails.

THE CHASSIS CENTER OF GRAVITY IS IN THE AREA OF THE CRT ANODE CONNECTOR. Two people can lift the unit if each of them grasp the rail at the top of the chassis on each side of the anode with one hand while pressing the side button, then steadying the chassis with the other hand while separating the slide rails.

7. Lower Tier mounted— With the assistance of a helper, again release the chassis by pressing the side buttons on the rails. Slide the unit free of the console along the slide rails, and place it on a bench for further disassembly.
8. Remove six bolts (3 on each side) to free the CRT chassis from the slide tray.

NOTE

If a Touchscreen must be removed or replaced, see Touchscreens, Section 7, for assistance.

6.3.2 Monitor Assembly Replacement

1. With the aid of an assistant, set the CRT Monitor chassis on the tray between the mounting brackets, and align it with the holes on the mounting bracket. Install six bolts (3 on each side) to secure the chassis to the tray.
2. Lower Tier— With the aid of your assistant, engage the chassis slide rails with the slide rails on the console and push the assembly beyond the detent position so that both rails lock and hold the assembly in place.
3. Lower Tier— As you move the assembly forward in this step, check the alignment of the CRT face with the touchscreen. There should be a very small gap between them. Press the detent buttons and carefully push the CRT assembly forward. Lift the retainer blocks up behind the CRT chassis to hold the chassis in position. You may need to adjust the hex-head bolts on the retainers to maintain the proper clearance with the touchscreen. Tighten the lock nuts on the retainers when you are satisfied with the alignment.
4. The CRT should also be centered in the screen. Center it by loosening the touchscreen mounting screws with a long-handled screwdriver, and moving the touchscreen slightly. You may need to readjust this setting again after power has been applied.
5. Connect the ribbon cable and the remote cable assembly to their connectors. Plug in the AC line cable.

6. Lower Tier mounted— Replace the rear cover and secure it with two captive screws. Reinstall the rear door of the console by sliding it back onto its hinges.

Upper Tier mounted— Be sure to engage both tracks inside the bottom of the cover, then slide the cover over the CRT chassis and secure it with four screws.

6.4 PWB Checks and Adjustments

NOTE

The following adjustments are made with a crosshatch display that is generated by running HVTS PDGI Test 18 Version 5.0 or later.

REQUIRED TEST EQUIPMENT

- Digital Volt Meter (DVM)
- High Voltage Probe (0-40 kV) compatible with DVM
- Minolta Color Analyzer Model II

6.4.1 Horizontal Hold

The horizontal control (H-HOLD—R657) locks the picture into synchronization horizontally. This control will normally not require adjustment. If it should become misadjusted, the proper setting is as follows: Turn the pot fully counterclockwise, and then clockwise 1/8th of a turn. If the pot is turned clockwise too far, the X-ray protection circuits will shut down both the high voltage and the horizontal scan circuits. If this occurs, power must be removed and the proper adjustment made before the circuits will operate. See Figure 6-2 for the location of the H-HOLD adjustment pot (potentiometer).

6.4.2 Vertical Hold

The vertical hold control (V-HOLD—R551) locks the picture into synchronization vertically. To perform this adjustment, rotate the pot in one direction until vertical synchronization is lost. Then rotate the pot in the opposite direction until synchronization is restored. Continue rotating the pot in this direction until synchronization is lost again. The correct setting for the pot is the midpoint between the two positions where synchronization is lost. See Figure 6-2 for location of the V-HOLD adjustment pot.

6.4.3 Vertical Height

The vertical height control (HEIGHT—R553) adjusts the height of the display vertically. Adjust the control CW (Clockwise) to make the display taller and CCW (Counter-clockwise) to shrink the size. Using this control, adjust the height of the display to measure 27 cm. from the top of the white outline to the bottom. It may become necessary to adjust the Vertical Centering (V-CENT) control before achieving the final height setting if the display is not centered. Move the control CW to position the display closer to the bottom and CCW for the opposite direction. See Figure 6-2 for the location of the HEIGHT adjustment pot. Note that you are adjusting the height of the display, not the height of the raster. Your measurement is the distance between the left and right borders of the crosshatch pattern.

6.4.4 Horizontal Width

The horizontal width control (WIDTH—R802) adjusts the width of the display (see Figure 6-2). Adjust the control CW (Clockwise) to make the display wider and CCW (Counter-Clockwise) for the opposite. Using this control, adjust the width of the display to measure 36 cm from the left side of the white outline to the right. It may become necessary to adjust the horizontal centering (H-CENT—R815) before achieving the final width setting if the display is not centered either on the background or within the bezel. Adjust the monitor Brightness control to the maximum so that the raster background is visible. If the side edges of the raster (background) are not visible, reduce the width of the display until they become visible. Using the Horizontal Centering control (H-CENT—R815), adjust the raster centering until the raster is centered. Then, using the Horizontal Phase control (H-PHASE—R653), adjust the position of the display within the raster until it is centered. Return the monitor Brightness control to the minimum setting and adjust the width to the final setting of 36 cm. Note that you are adjusting the width of the display, not the width of the raster. Your measurement is the distance between the left and right borders of the crosshatch pattern.

6.4.5 Linearity and Geometry

The controls for Linearity and Geometry are: Vertical Linearity (V-LIN—R563 and Side Pincushion (SIDE PIN—R565). Correction for horizontal linearity is made by adjusting the horizontal linearity coil. Top and bottom pincushion correction is built into the CRT yoke. See Figure 6-2 for location of control.

6.4.5.1 Vertical Linearity

1. While the crosshatch display is shown, adjust the (V-LIN—R563) control for linearity of the display from top-to-bottom with reference to the center (see Figure 6-2). Move the control CW (Clockwise) to make the top lines of the crosshatch farther apart and the bottom lines closer together or CCW (Counter-Clockwise) to make the top lines closer together and the bottom lines further apart. The top lines should be the same distance apart as the bottom lines.
2. If a significant adjustment is made to the V-LIN control, it is also necessary to redo the Vertical Centering using V-CENT.

6.4.5.2 Side Pincushion Adjustment

With the crosshatch pattern displayed, adjust the Side Pincushion as follows:

1. The (SIDE PIN—R565) control adjusts the straightness of the side of the display, correcting a distortion that occurs naturally with the CRT (see Figure 6-2). Adjust so that the sides of the display are as straight as possible. Other distortions that can occur in the display, such as trapezoidal or parallelogram shapes, are a function of the tube performance and cannot be corrected using the SIDE PIN control.
2. If a significant adjustment was made to the SIDE PIN control, it is necessary to redo the WIDTH setting to 36 cm.

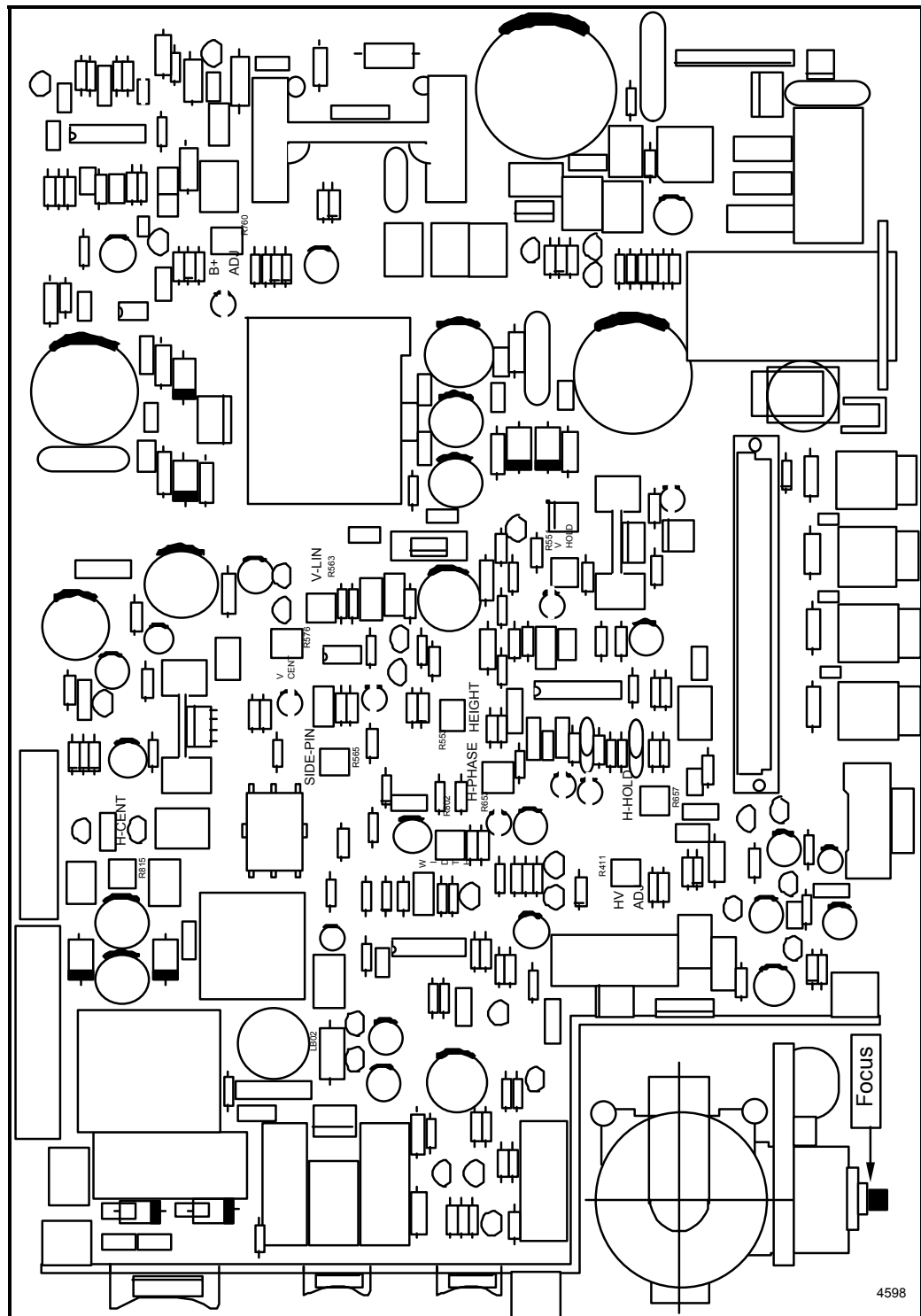


Figure 6-2 — Circuit Board FSM-002 Component Layout

6.4.6 Horizontal Linearity Adjustment

Adjust the Horizontal Linearity as follows:

1. Horizontal Linearity (L802)—This control adjusts the linearity of the display side-to-side, with reference to the center. Rotating the thumbwheel on the side of the coil (taking care not to touch exposed terminals) changes the left side of the display, stretching or compressing the width of the blocks. Position the thumbwheel so that the width of the blocks on the left are equal to those on the right (see Figure 6-2).
2. If a significant adjustment is made to the H-LIN control, it may be necessary to redo the width adjustment and the horizontal centering using the H-CENT control.

6.4.7 CRT Bias Adjustment**WARNING**

This adjustment is sealed in the factory and should not be changed as a matter of course. The procedure is given here for those skilled personnel trained to make such adjustments. Improper adjustment of the CRT bias can drastically affect the lifetime of the CRT.

Using the crosshatch display from HVTS PDGI Test 18, the CRT Bias may be adjusted as follows:

1. Check to see that the monitor Brightness control is at the minimum position.
2. Adjust the Red Bias (R BIAS), Green Bias (G BIAS), and Blue Bias (B BIAS) controls, aligning the adjustment slot to the twelve o'clock position (see Figure 6-3).
3. Adjust the screen control (G2), located on the Focus control assembly (lower control on T401, located in the rear left-hand side facing the rear of the Monitor), CW (Clockwise) to produce a dim display, and then CCW (Counter-Clockwise) until the point is reached where the display just disappears.
4. Observing the display with a cupped hand or with an eyepiece, while in a darkened area, adjust each of the cathode bias controls in turn to balance cutoff levels. Using the R BIAS control (R257 in Figure 6-3), adjust CW until a dim red background is seen and then CCW to the point at which the red just disappears. Repeat this procedure using the G BIAS (R157) and the B BIAS (R357) controls.
5. Adjust the monitors Brightness control to the maximum position and observe that the background is a dim gray. Should the gray be shifted in hue towards any color, repeat the balancing operation in Step 4.

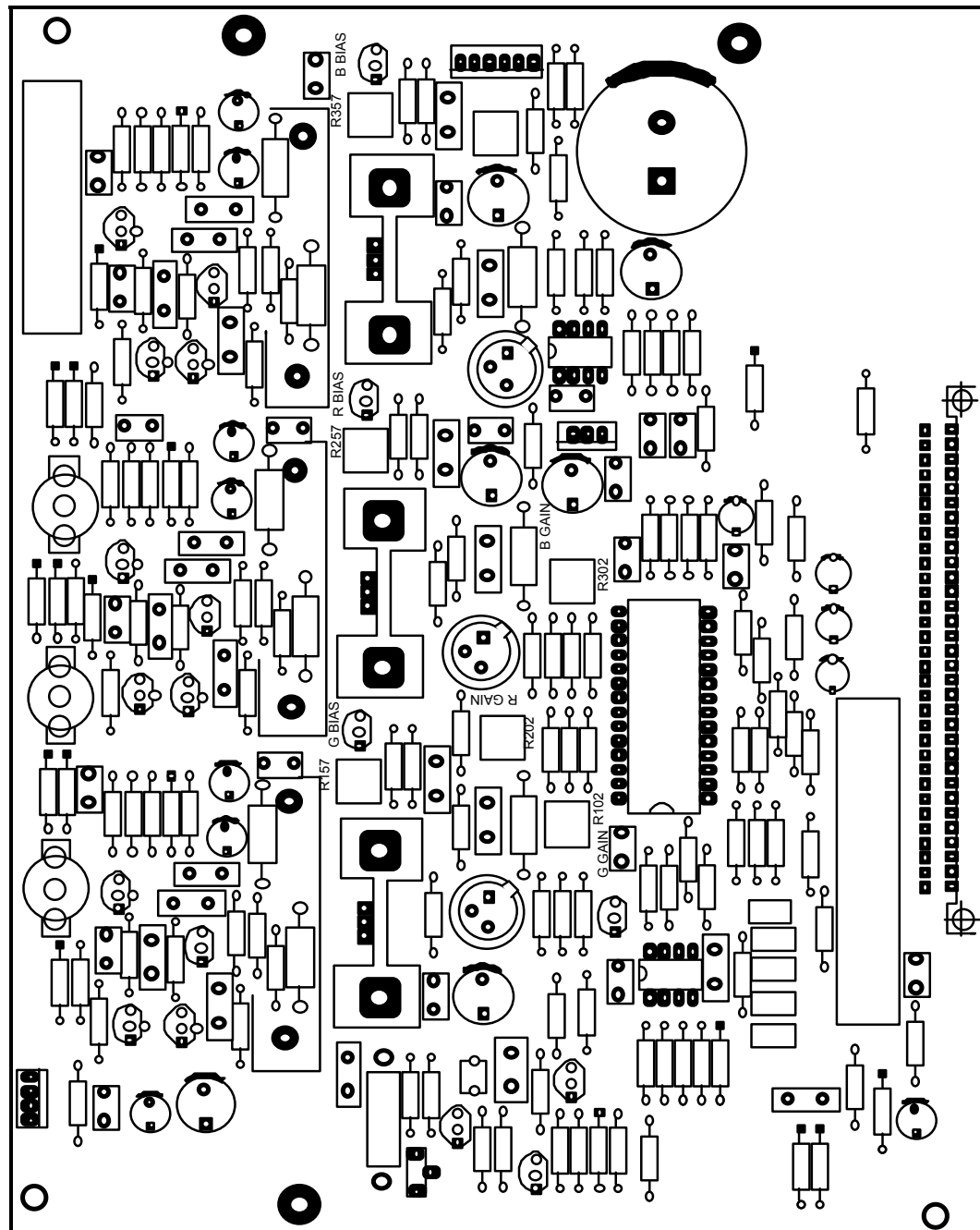


Figure 6-3 — Circuit Board FSV-301 Component Layout 4599

6.4.8 Color Temperature

NOTE

- *1. This adjustment requires the use of a Minolta Color Analyzer Model II, or equivalent, and is not recommended unless this equipment is used. If it is not available, only make adjustments if colors are obviously off.
2. The CRT Bias adjustment must be done before starting this test.
3. The controls required for this adjustment are located on the video circuit card, FSV-301. Location of the controls are shown in Figure 6-3.
4. This adjustment procedure adjusts the color temperature using the display generated by HVTS PDGI Test 17.

* See subsection 6.4.

1. Adjust the monitor's contrast control (upper thumbwheel) to the maximum position and the Brightness control (lower thumbwheel) to the minimum. These controls are located on the bezel on the front of the Universal Station (US).
2. Observe the white square and the gray square in the display. At this point, two determinations can be made: Is the picture bright enough (sufficient light output)? Is the color temperature of white (9300° Kelvin) correct?

In determining the light output level question, it should be considered that excessive drive causes spot growth (fuzzy characters) and shadow mask doming (a spot heating effect which temporarily shifts beam landing). The best adjustment of light output is one that approaches spot growth in level but does not allow brighter operation.

The assessment of color balance is simple when a good mental reference of the desired white is in place. The best description of 9300 degrees Kelvin (white), is a greenish-bluish white. It can be observed, that when white is properly adjusted, the secondary (mixed) colors are also proper in hue.

3. If the light output level requires adjustment, first adjust the G GAIN (R102), R GAIN (R202), and B GAIN (R302) by the same amount, either CW to increase or CCW to decrease. This adjustment can be done in a rough fashion for color balance.
4. Measure the color temperature of the white square, if it is not white (9300° - X=281, Y=311) take note of the color or hue it is shifted towards. (The X and Y values are the coordinate values displayed by the Color Analyzer.) If it is one of the primary colors (red, green, or blue), simply reduce that color drive by turning the appropriate GAIN adjust CCW (the three gain controls are somewhat interactive; that is, will change either one or both X/Y readouts). If it is shifted towards a secondary color (yellow, magenta, or cyan), adjust the missing color not used to produce the secondary color CW to increase the drive for that color. In this fashion, a good white can be achieved.
5. Observe the gray square. If the gray is not pure, the color balance procedure is in subsection 6.4.7, step 4.

6.4.9 High Voltage

WARNING

This adjustment is sealed in the factory and should not be changed. The reason for this is to prevent the generation of X-rays that are harmful to personnel.

6.4.10 B⁺ Checks (Low Voltage)

WARNING

This adjustment is sealed in the factory and should not be changed.

6.5 Startup

Ensure that all of the Universal Station equipment is turned off (this prevents power surges that could damage the equipment when the main power switch is turned on). Apply power by setting the main power circuit breaker, located behind the door under the keyboard, to "1" (ON). Set the keyboard power supply switch to "1" (ON), and close the door. On the rear of the unit, open the rear door and set the module power switch to ON. After 1.75 minutes, check the module light emitting diode (LED) annunciators to ensure that they are all green. The alphanumeric indicator displays the module node address when the operating system is running, or not running. If any of the above conditions cannot be met, refer to the *Five/Ten-Slot Module Service* manual. Close the Station rear door. Apply power to any other equipment turned off at the beginning of this procedure.

Adjust CRT brightness and contrast for best resolution on the character that appears in the corner of the screen. Check the *Touchscreen to CRT Alignment* in subsection 7.4.3.

6. DISPLAY CRT ASSEMBLY 51304449-100 and 51196696-200

6.6. Spare Parts List

6.6 Spare Parts List

Monitor 51196696-200 is the same monitor as 51304449-100, except it has a different CRT glass tube.

Table 6-2 — Monitor Spare Parts

Part Number	Description
* 51196696-200	Replacement Monitor Unit (replaces 51304449-100)
N/A	Fuse, 5x20 mm, slow blow, 5 A, 250 V

* = Optimum Replaceable Unit (ORU)

7 TOUCHSCREENS

7.1 Cleaning

The Universal Station can withstand adverse environmental conditions, often encountered in process control applications. Forced airflow has been provided over the CRT and touchscreen components. In dirty (especially tobacco smoke) environments, however, the touchscreen will require periodic cleaning.

CAUTION

Touchscreen cleaning can be done with system power ON to minimize down time. Be aware of potentially high voltages on the CRT chassis. If down time is scheduled at frequent periods, touchscreen cleaning should be done on the power-down schedule.

There are two different (physical) versions of the touchscreen. A cleaning procedure for each configuration is provided as follows:

7.1.1 Cleaning Touchscreen Windows with Dark Red Plastic Arc

(approximate cleaning time: 15 minutes)

NOTE

If tobacco smoke is present in the Universal Station environment, the touchscreen windows should be wiped with isopropyl alcohol before proceeding to step 5. A small foam-type paint brush is ideal for this use and can also be used to support the lint-free cloth in step 5.

1. Clear the screen (this function is on the System Menu).
2. Open rear door of CRT cabinet.
3. Note stop bars on each side of the CRT chassis. Loosen the stop bars and swing them out of the way (requires 12 mm and 8 mm wrenches).
4. Carefully slide the CRT assembly back until it stops against the CRT rear cover—about 10 cm (4 inches).
5. Using a mild window cleaner, such as Windex® and a clean lint-free cloth (not paper), clean the inside of the transparent (red) window and the tops of the LEDs (emitters and receivers, 48X54 array) located just behind the plastic strip.

CAUTION

DO NOT USE PAPER WIPES OF ANY KIND

7. TOUCHSCREENS

7.2. Adjustments

6. Test for proper touchscreen operation.
7. Push the CRT back in place and secure stop bars. Close rear door and resume operation.

7.1.2 Cleaning Touchscreen with an Arc Formed by an Open Slot

(approximate cleaning time: 5 minutes)

NOTE

For moderate to heavy tobacco smoke environments it may be necessary to periodically clean the tops of the LED components. Use a cotton swab moistened with isopropyl, follow by wiping with a dry cotton swab.

1. Clear the screen (this function is on the System Menu).
2. Clean the touchscreen components by inserting an insulated brush through the open slot. The brush should be equivalent to Honeywell part number 51109399-100. The components can also be cleaned by blowing clean, dry, compressed air (25 PSI maximum) through the open slots.
3. Test for proper touchscreen operation.
4. Continue operations.

7.2 Adjustments

None

7.3 Test/Troubleshooting

Refer to the *System Maintenance Guide*. If you have a keyboard power supply, check that it is on. If you don't have a keyboard power supply, touchscreen power is supplied from the PDG I/O or EPDG I/O board. In any case, check the cabling which feed the touchscreen to verify power and signals are being delivered before attempting to replace parts. If cabling and power are OK, the touchscreen is probably defective. Touchscreen cabling information is provided in subsection 7.6 and in the *System Installation Manual*. Touchscreen replacement parts are listed in subsection 7.5.

7.3.1 Touchscreen Alignment Verification

When a US node is first powered up or is reset, the firmware executes a self-test. The results of the self-test can be displayed on the screen by pressing the CONS STATS button on the Operator's Keyboard.

Table 7-1 Procedure for Verifying Touchscreen Alignment

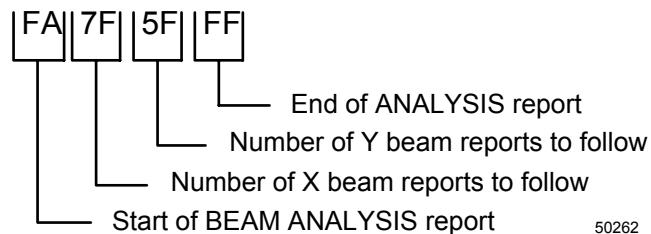
Step	Action
1	Turn the power switch to the US node to the ON position or, if power is already on, press the RESET button on the Operator's Keyboard.
2	After the > and the "box" symbol both appear in the upper left-hand side of the display, press the CONS STATS button on the Operator' Keyboard. Now analyze the display.

What follows is an excerpt of the explanation of the data fields, found in the *Customer Resource Manual*, Tab 15, Page B05. Most of the description in the *Customer Resource Manual* is for engineering debug. The BEAM ANALYSIS and SIGNAL VALUE are the two fields useful to service personnel.

NOTE

There are two different report formats; one reports a numeric value of beam strength, the other reports a go/no go value. The go/no go value is report in the same format as the other value except that the value is 00 for good and 77 for bad.

This value is reported by the firmware in newer touchscreen units. The change was made because units with a low value reported were being replaced when they were still functional, but dusty.



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- Each failed beam report will be 2 bytes long. The first byte (of the two center bytes) is the byte address and the second byte is the fault found with the beam.

7. TOUCHSCREENS

7.3. Test/Troubleshooting

- Addressing is from top down and left to right.

The fault codes are:

00 Firmware could not determine the problem

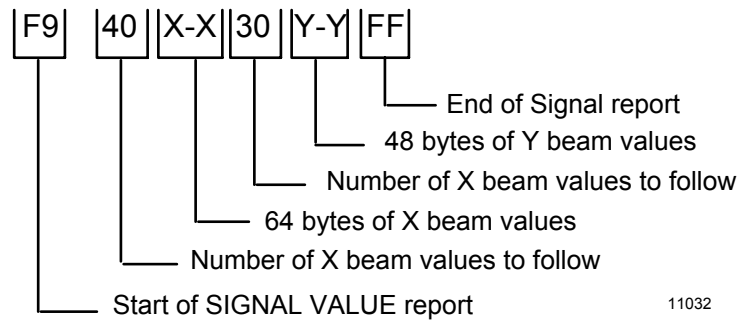
01 Failed photo-transistor

02 Failed LED

03 Failed photo-transistor and LED

04 Individual components appear to work separately, but not together; failure cannot be isolated.

EXAMPLE: FA 01 03 02 00 FF shows 1 failed beam X beam (01), failed address of 3 (03), and is a failed LED. No failed Y beams (00).



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The Signal Value is the differential value between the ambient reading and the reading with the beam on.

- Any reading above 00 should work, but the factory doesn't ship any unit that has a value less than 05.
- A reading of 01 or 02 may be reported as failed.
- A reading of 00 is failed.
- The screen will continue to work as beams fail, but the resolution will suffer.
- If any values are found to be below 05, then the screen should be cleaned and tested again.
- The screen doesn't need to be replaced until the BEAM ANALYSIS reports one or more failed beams.

Example:

SIGNAL VALUE F9 40 [49 51 0D 4B 38 47 28 40 44 20 10 18 2E 4D 37
1B 27 3B 10 23 32 11 0D 17 35 44 4C 33 10 18 2E 28 58 58 11 31 35 4B
1B 1C 3E 56 48 11 44 21 57 4F 15 3B 4D 56 53 42 55 52 59 21 4E 02 59
59 50 42] 30 [3A 31 04 25 3B 45 2B 25 40 2B 27 4C 30 2E 28 37 44 26
1A 47 1D 2D 1E 2C 40 2F 25 15 28 33 14 2B 1B 0A 37 17 30 16 0A 1B
2A 32 27 2F 19 23 22 2B] FF

NOTE

The warrantee on units is such that only units with a reading of less than 02 will be replaced.

7.4 Removal/Replacement

7.4.1 Touchscreen Removal

NOTE

The following instructions apply to replacing a touchscreen with the new Smartframe touchscreen. (Field Spares Kit 51195192-100). This kit is the only available replacement.

The Smartframe touchscreen only interfaces PDG I/O or EPDG I/O boards. If you have the older PIC/VDG controller (and I/O) boards, you must install Field Spares Kit 51195190 as a prerequisite. This kit will supply a PDG I/O board.

If you have an EPDG I/O board, you are interfacing the new noninterlaced CRT monitor. Field Spares Kit 51195192-100 and these instructions are still correct—just read “EPDG I/O” in place of “PDG I/O” in the following instructions.

1. Turn OFF the ac Power Switch on the front of the Universal Station.
2. Disconnect all cables from the CRT and remove the CRT assembly from the station. **If you don’t remove it now, you’ll have to later.** Slide the CRT rails out of the way.
3. Disconnect the cable from the touchscreen.
4. Remove the touchscreen by removing four screws from the sides of the touchscreen (from the inside).
5. Remove the touchscreen window:
 - If slotted type, slide the window out.
 - If solid with red window, break the glue joint, remove the touchscreen window pieces, and remove all adhesive residue.

7.4.2 Touchscreen Replacement

1. Install the new touchscreen windows sections by sliding in the new pieces.
2. Install the new Smartframe touchscreen with four M5 screws provided in the kit.

WARNING

Use extreme care when installing the touchscreen, to avoid contact between the touchscreen and the touchscreen window, which would damage the end diodes.

3. Install the Smartframe power cable.
 - Connect the Smartframe power cable, 51304031-150, to the Smartframe touchscreen as shown in Figure 7-1.
 - Install the cable in the cabinet to the back of the electronics module.
 - Connect the cable to J4 on the PDG I/O (EPDG I/O or EPDGP I/O) board as shown in Figure 7-2.

7. TOUCHSCREENS
7.4. Removal/Replacement

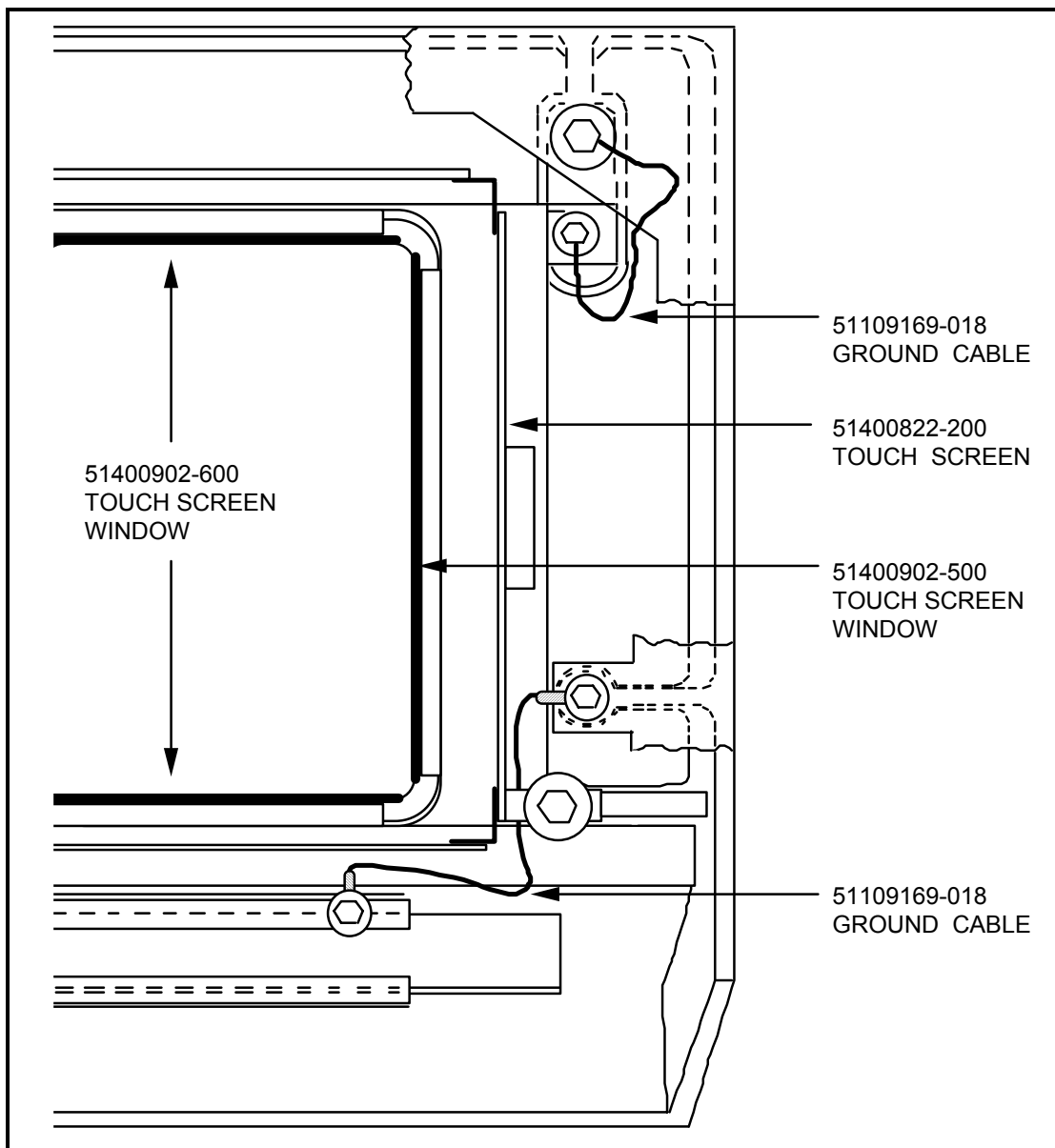


Figure 7-1 — Smart Frame Touchscreen Cabling

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4. Install the Smartframe data cable.
- Connect the Smartframe data cable, 51304030-150, to the Smartframe touchscreen as shown in Figure 7-1.
- Install the cable in the cabinet to the back of the electronics module.
- Connect the cable to J1 on the PDG I/O board as shown in Figure 7-2. The EPDG I/O or EPDGP I/O board connections are the same.
5. Connect the ground cable, 51109169, as shown in Figure 7-1.
6. Reinstall the CRT assembly in the Universal Station. Reconnect all cables to the CRT assembly.

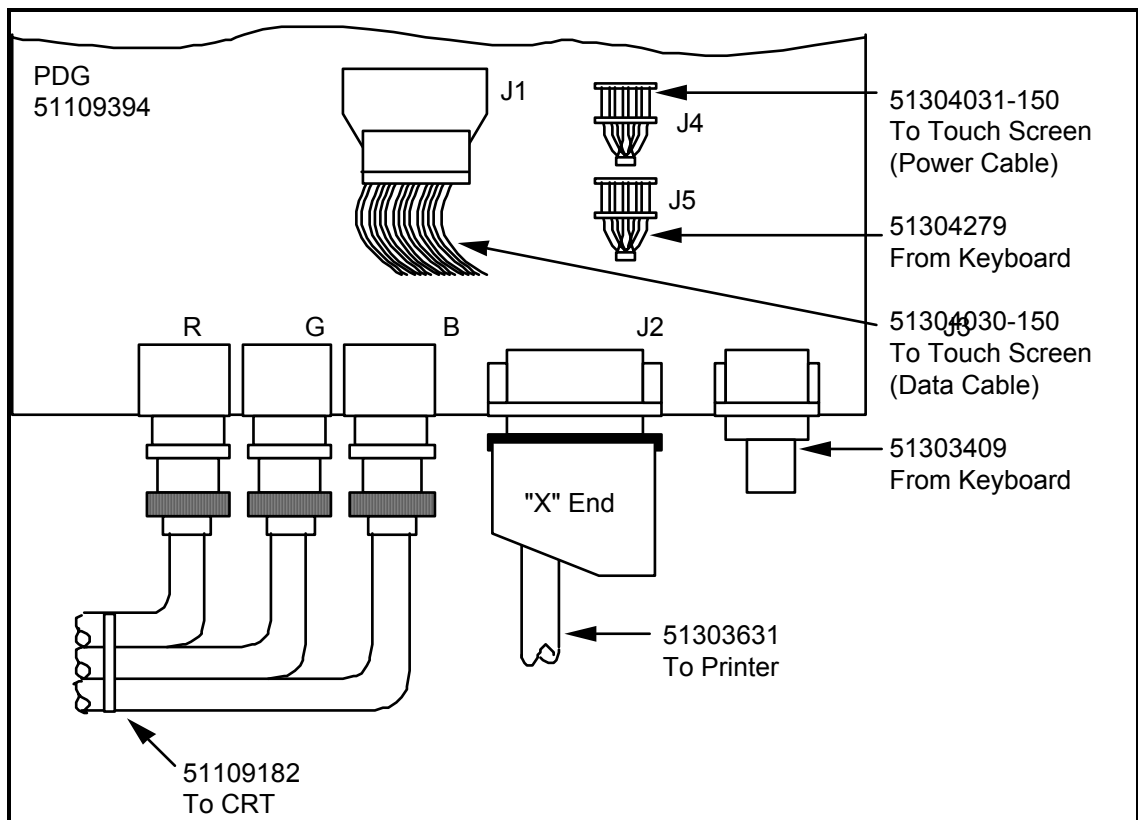


Figure 7-2 — PDG I/O Board Cable Connections

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7.4.3 Touchscreen to CRT Alignment

This procedure determines if the touchscreen is in alignment. It only works if your Universal Station is equipped with either a PDG or an EPDG controller board.

Calibration is done by adjusting the width, height, and centers of the test image on the CRT to match the position of the touchscreen. You will be using four places on the face of the CRT as calibration points. They are the middle right and left edges, and the middle vertical top and bottom edges. Figure 7-3 shows the pattern.

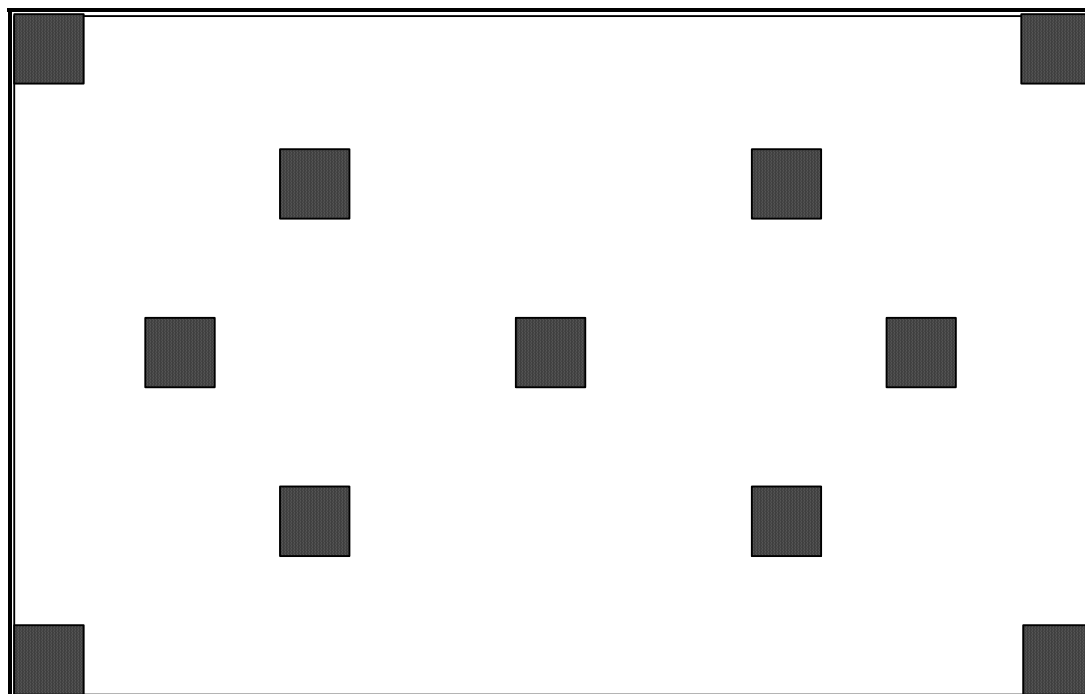


Figure 7-3 — HVTS PDGI Test 15 Test Pattern⁴⁶⁰³

1. Power-up or reset the Universal Station.
2. Using HVTS, run PDGI Test 15. After the pattern appears on the screen, abort PDGI.
3. Place your finger on the square located at the middle left edge of the pattern. Notice where the cursor appears. Place your finger on the square at the middle right edge of the pattern. After considering the results of the two tests, adjust the horizontal centering and horizontal width controls to make the cursor appear directly under your finger each time you touch the CRT surface. Repeat this adjustment until the cursor is centered in the pattern square for each side.

Place your finger on the square located at the middle top edge of the pattern. Notice where the cursor appears. Place your finger on the square at the middle bottom edge of the pattern. After considering the results of the two tests, adjust the vertical centering and vertical width controls to make the cursor appear directly under your finger each time you touch the CRT surface. Repeat this adjustment until the cursor is centered in the pattern square for each side. Note that the cursor will always be slightly below the finger when testing the top target. This is due to

the need to have the touchscreen high in the bezel so that the bottom beam will not be partially blocked off by the side slot's bottom edge.

You must be careful not to spread the pattern too wide or too tall, such that the corners of the pattern are too wide or too tall for the screen.

7.5 Spare Parts

Table 7-2 — Smartframe Touchscreen Spare Parts List

Part Number	Description
* 51400988-100	Smartframe Touchscreen
* 51400988-200	Smartframe Touchscreen, CE Compliant
51304030-150	Cable, Data, Touchscreen to PDG I/O or EPDG I/O
51304031-150	Cable, Power, PDG I/O or EPDG I/O to Touchscreen

* ORU Level Replacement Item

7.6 Schematics

Cabling schematics of the various keyboard/touchscreen arrangements follow.

- Figure 7-4 illustrates the older (standard) keyboard and touchscreen. A separate keyboard power supply is required—which also drives the touchscreen.
- Figure 7-5 illustrates the new low power keyboard and Smartframe touchscreen. Note the Smartframe receives its power from J4 and supplies data to J1 of the PDG I/O board. Also note the PDG I/O board may be replaced with the EPDG I/O board (which drives the new non-interlaced CRT monitor)—J1 and J4 on the PDG I/O board are identical to J1 and J4 on the EPDG I/O board.

7. TOUCHSCREENS

7.6. Schematics

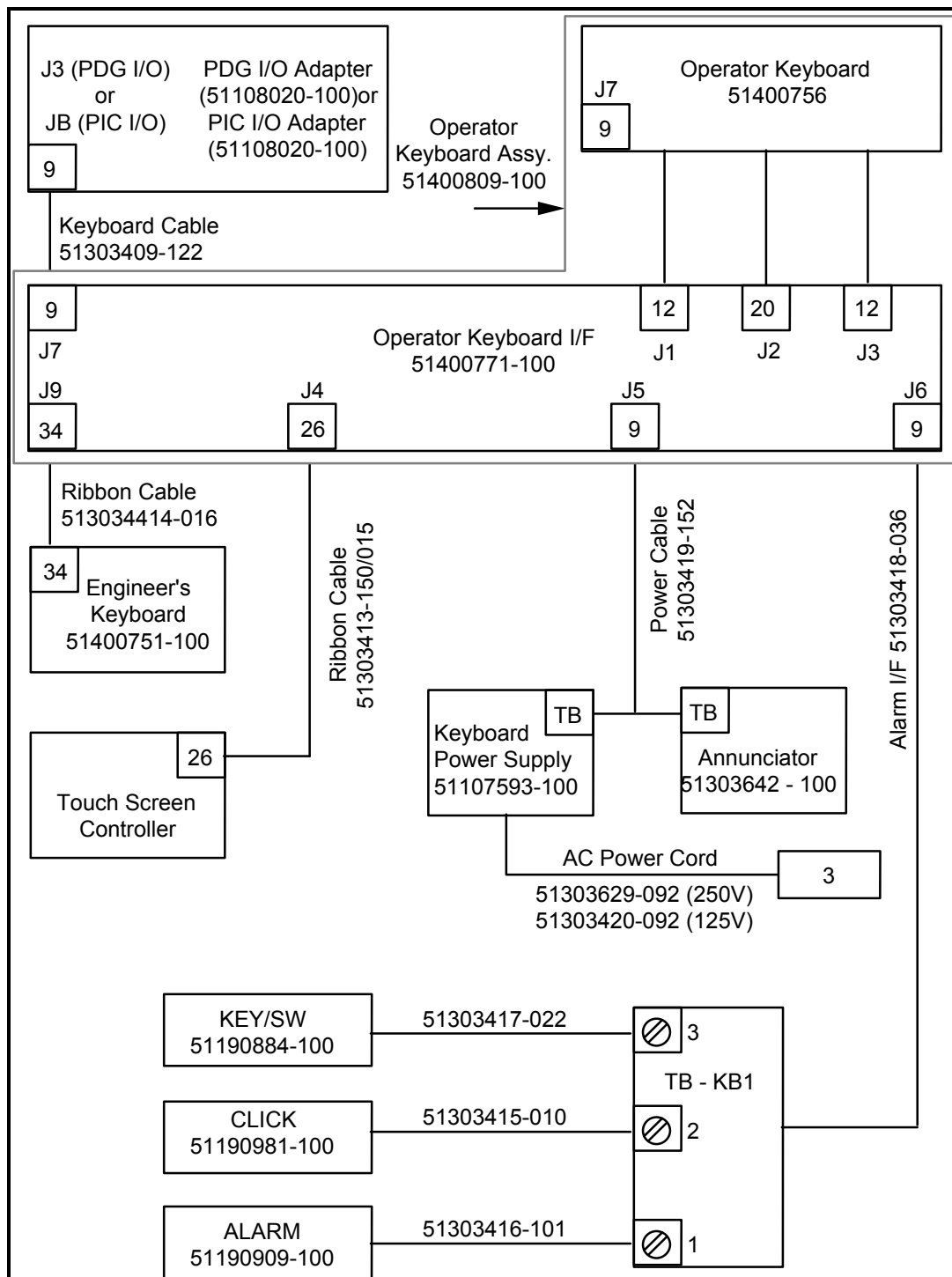


Figure 7-4— Keyboard/Touchscreen Cabling (Standard) for Early Production Units 1262

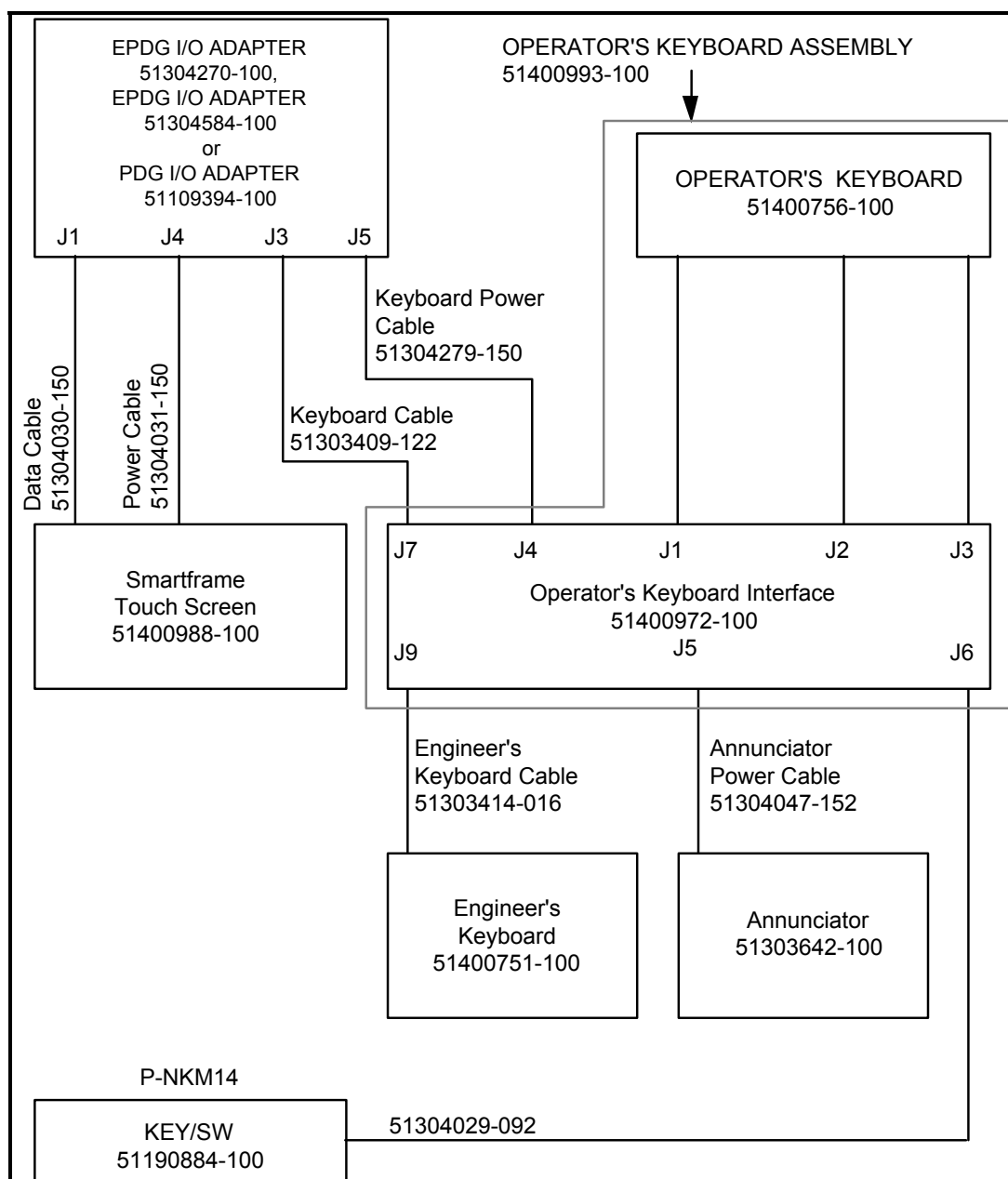


Figure 7-5 — Low Power Keyboard/Touchscreen Cabling for Newer Production Units 2771

For CE Compliant cabling schematic see subsection 2.4.

7.7 Startup

Refer to the *Startup* subsection in any CRT monitor section in this manual.

7. TOUCHSCREENS

7.7. Startup

8 KEYBOARDS

8.1 Cleaning

Clean the keyboards at the same time you clean the CRT. Refer to the *Cleaning* subsection in any CRT monitor section in this manual for keyboard cleaning instructions.

8.2 Adjustments

None.

8.3 Test/Troubleshooting

Refer to the *System Maintenance Guide*. In general, if one or a few keys don't work, the keyboard is defective. If no keys work, check the Five/Ten-Slot Module (see subsection 2.2.10). Replace the PDG I/O, EPDG I/O, EPDGP, or EPDGC I/O card (return the original card if it is not faulty).

Check cabling between the module and the keyboard to ensure power is being supplied and signals are being passed. Keyboard cabling information is provided in subsection 8.6. If you have a keyboard power supply, check its output and replace it if faulty (refer to 8.4.3).

8.4 Removal/Replacement

Open the console front door and the set main power circuit breaker to OFF.

8.4.1 Operator's Keyboard

1. Remove the Engineer's Keyboard cover and remove four (Phillips-head) screws from the front edge of the keyboard tray. Remove four screws and washers from under the keyboard tray, and lift the bezel off. At the rear of the keyboard tray, rotate the panel back to expose the keyboard-mounting screws. Remove the Operator's Keyboard by removing two screws at rear of the keyboard and lifting the front of the keyboard to expose the connectors. Disconnect all five connectors and lift the keyboard off.
2. Replace the Operator's Keyboard by connecting five connectors to the bottom of the board and setting it on the keyboard tray. Secure to tray with two screws. Assemble the bezel to the keyboard tray. Secure with four screws in the front edge, and four screws and washers through the bottom of the tray. At the rear of the keyboard tray, rotate panel into position.

8.4.2 ROVING ENGINEER'S KEYBOARD

1. Replacement or removal of a Roving Engineer's Keyboard is accomplished by disconnecting the keyboard cable from the right-hand side of the gooseneck that supports the Operator keyboard.

8. KEYBOARDS

8.5. Startup

8.4.3 Engineer's Keyboard

1. Remove the Engineer's Keyboard cover and remove four (Phillips-head) screws from the front edge of the keyboard tray. Remove four screws and washers from the bottom of the keyboard tray, and lift the bezel off. Remove four screws and washers from the keyboard, and lift the rear edge to expose the connector. Remove the connector, and lift the keyboard off.
2. Replace the Engineer's Keyboard by connecting the ribbon cable into the keyboard. Assemble the keyboard to the keyboard tray, and secure with four screws and washers. Assemble the bezel to the keyboard tray. Secure with four screws in the front edge, and four screws and washers through the bottom of the tray.

8.4.4 Keyboard Power Supply

NOTE

YOUR EQUIPMENT MAY NOT HAVE A KEYBOARD POWER SUPPLY.

If your equipment has been upgraded to the Low Power CMOS keyboard (or if the CMOS keyboard was originally supplied), power for the keyboard and Smartframe touchscreen is obtained from J4 and J5 on the PDG I/O or EPDG I/O board.

1. Open the front door of the cabinet to gain access to the power supply, and set the power-supply switch to "0" (OFF). Remove the power plug, then label (if required) and remove seven wires from the power supply (note how these wires are connected). Remove the power supply by removing four slot-head screws.
2. Assemble the power supply to the cabinet and secure in place with four slot-head screws. Secure seven wires to the power supply, and insert the power plug. Set the power switch to "1" (ON) and close the cabinet door.

8.5 Startup

Refer to the *Startup* subsection in any CRT monitor section in this manual.

8.6 Keyboard Cabling

There are two types of Operator's Keyboards found in the Universal Station:

1. Original Operator's Keyboard. See Figure 8-1.
 - Interfaces the PIC/VDG control boards and PIC/VDG I/O boards (PDG or EPDG and PDG I/O are functional replacements).
 - Requires a Keyboard Power Supply. If a Field Spares Kit (51195191) is installed, the Keyboard Power Supply is removed.
2. Low Power CMOS Operator's Keyboard. See Figures 8-2 and 8-3.
 - Interfaces the PDG or EPDG control boards and the PDG I/O (for RGB monitors). Also interfaces the EPDG control board with EPDG I/O, for enhanced monitors.
 - Receives power from the PDG I/O (or EPDG I/O) card.

8. KEYBOARDS

8.6. Keyboard Cabling

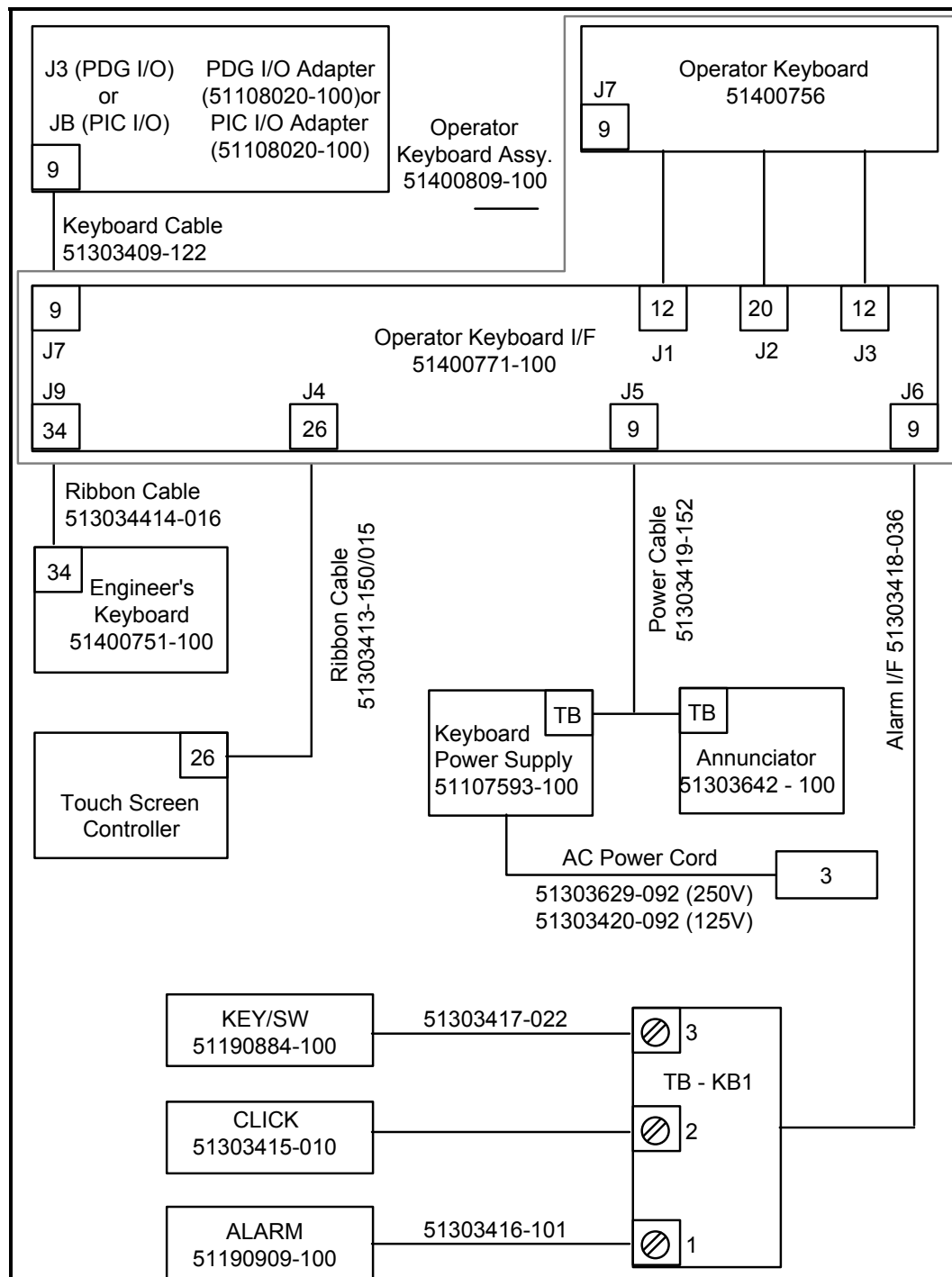


Figure 8-1 — Keyboard Cabling (Standard)

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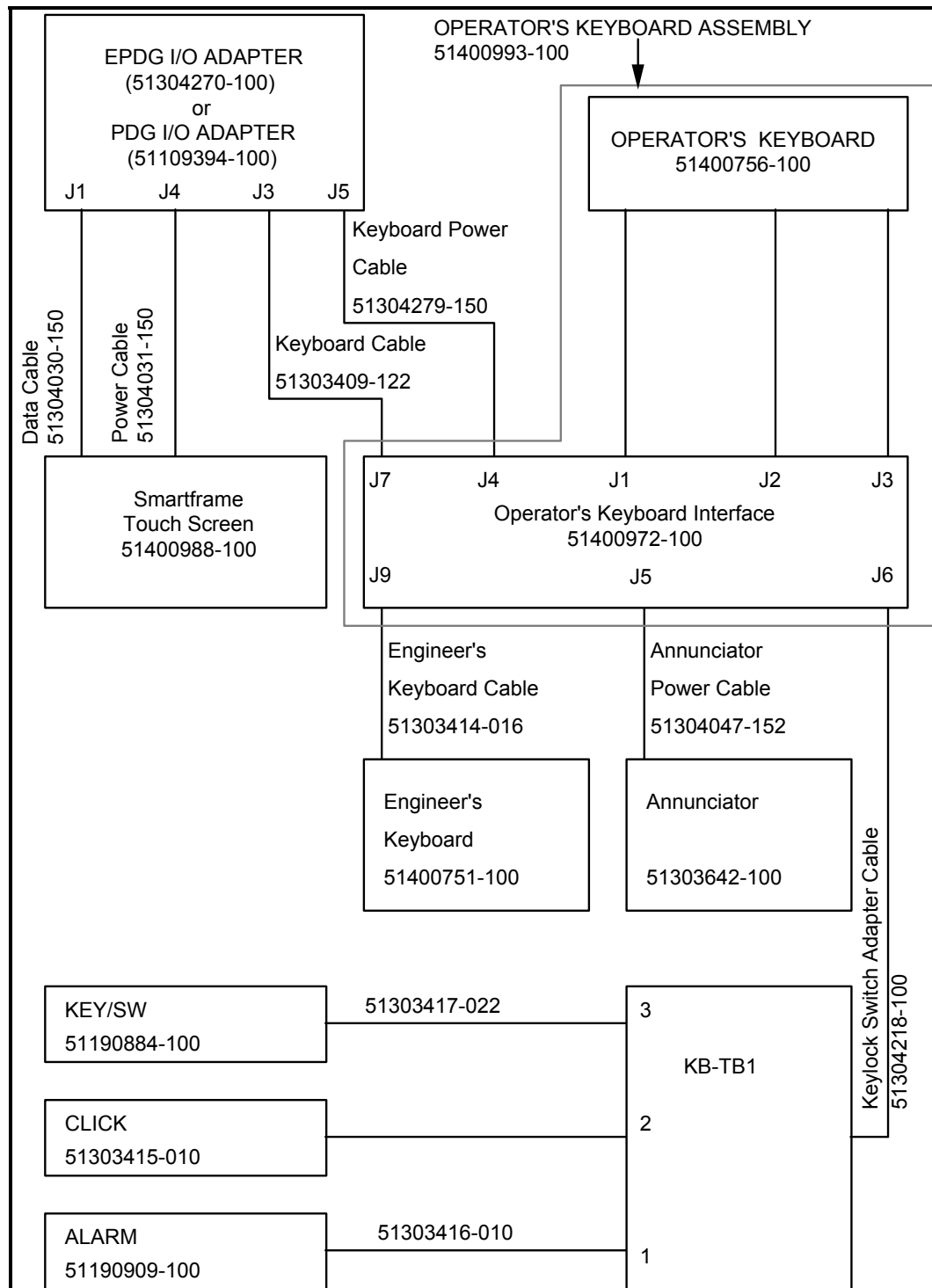


Figure 8-2 — Low Power Keyboard Cabling (Field Spare Kit - 51195191)

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8. KEYBOARDS

8.6. Keyboard Cabling

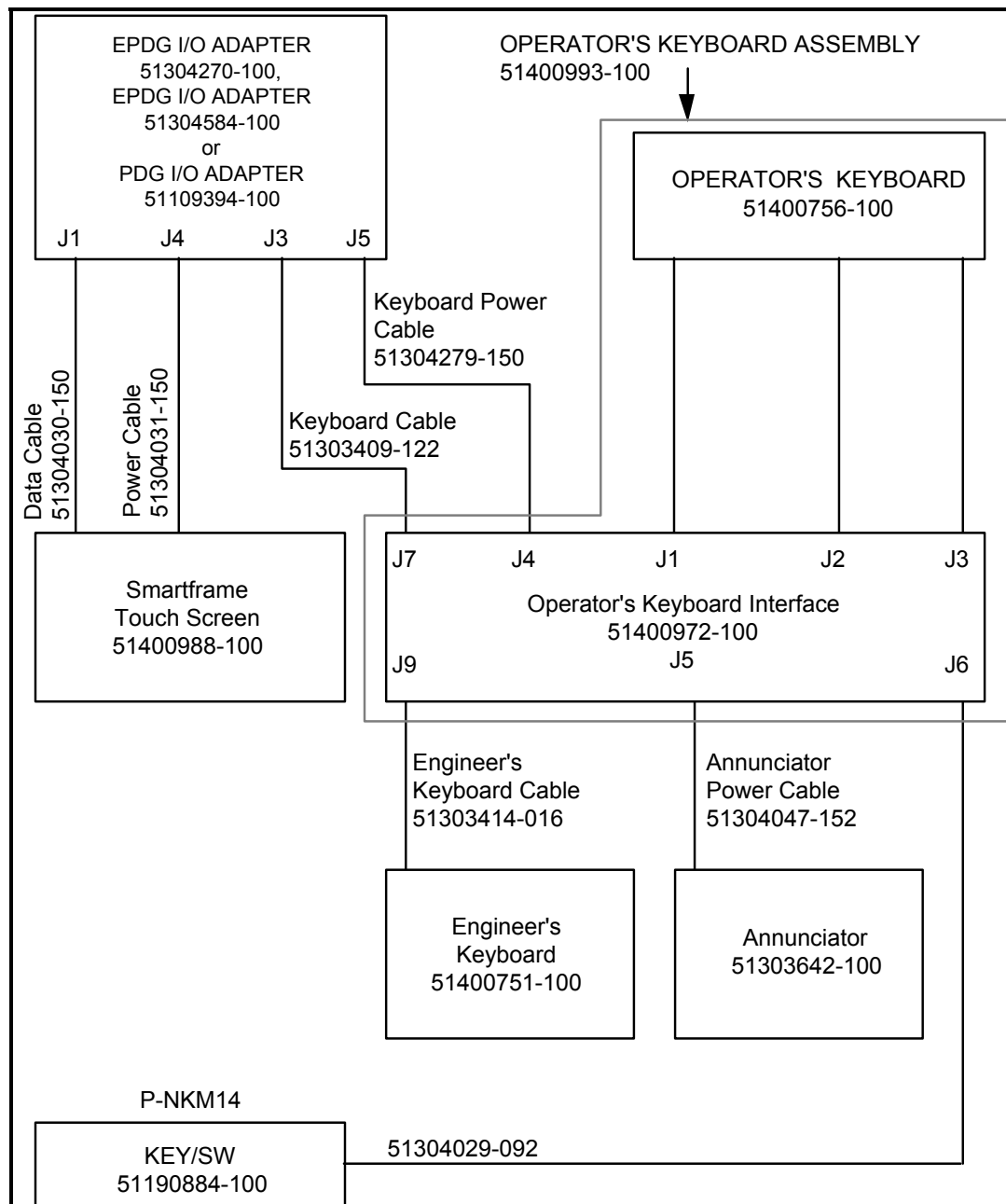


Figure 8-3 — Low Power Keyboard Cabling

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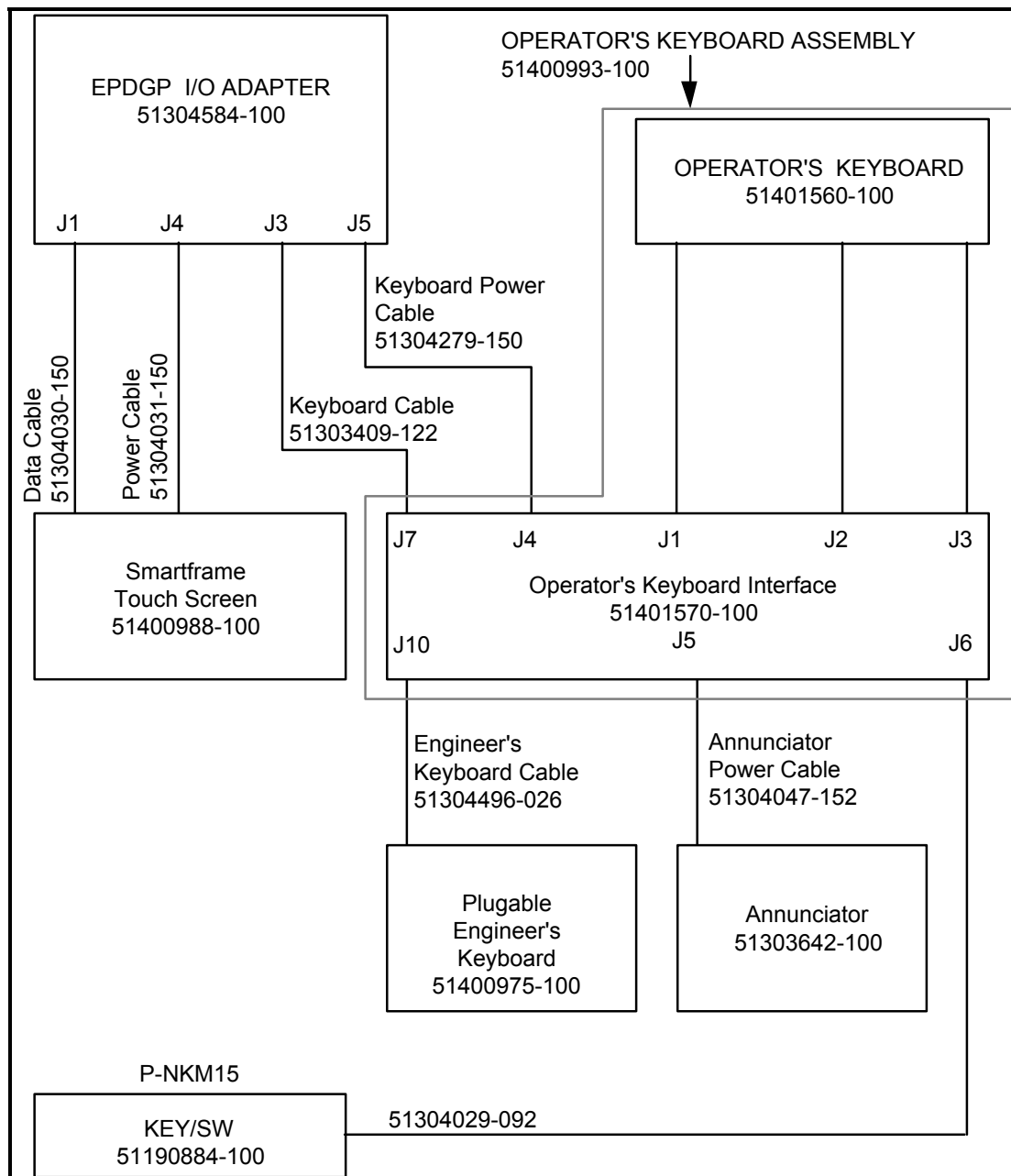


Figure 8-4 — Keyboard Cabling for Pluggable Engineer's Keyboard 4468

The CE Compliant Station cabling is found in Section 17.

8. KEYBOARDS

8.7. Spare Parts

8.7 Spare Parts

Because there are two keyboard versions presently in use, complete Spare Parts Lists for each are presented here. CE Compliant parts are designated in the far right column as “CE Compliant.”

The CE Compliant station cabling schematic is located in Section 17.

Table 8-1 — Original Keyboard Spare Parts

Part Number	Description
* 51400809-100	Operator's Keyboard Assembly
51400756-100	Operator's Keyboard
*51401577-500	QWERTY Operator Keyboard – CE Compliant
*51400993-400	ABCDEF Operator Keyboard – CE Compliant
51400771-100	Keyboard Interface
* 51400751-100	Engineer's Keyboard
51308126-100	Keyboard Cable – CE Compliant
* 51107593-100	Keyboard Subsystem Power Supply
* 51190516-100	Relay, Solid State, Annunciator Assembly
51303409-122	Cable, Keyboard Cable
51303413-150	Cable, Keyboard to Touchscreen
51303414-016	Cable, Engineer K/B to Operator K/B
51303415-010	Cable, Keyboard Click (and clicker)
51303416-010	Cable, Keyboard Alarm
51303417-022	Cable, Key Lock Switch
51304029-100	Keylock Switch Cable – CE Compliant
51303418-036	Cable, Alarm Interface
51303419-152	Cable, Keyboard DC Power Interface
51303642-200	Annunciator TB Assembly
51304047-652	Annunciator Power Cable – CE Compliant

* ORU Level Replacement Item

Table 8-2 — CMOS Low Power Keyboard Spare Parts

Part Number	Description
* 51400993-100	Operator's Keyboard Assembly
51400756-100	Operator's Keyboard
* 51400751-100	Engineer's Keyboard
51400972-100	Operator's Keyboard Interface
* 51190516-100	Relay, Solid State, Annunciator Assembly
51304029-092	Cable, Keylock Switch
51303409-122	Cable, Keyboard Cable
51303414-016	Cable, Engineer K/B to Operator K/B
51303415-010	Cable, Keyboard Click
51303416-010	Cable, Keyboard Alarm
51303417-022	Cable, Key Lock Switch

* ORU Level Replacement Item

Table 8-3 — Roving Engineer's Keyboard

Part Number	Description
51400975-100	Auxiliary Station Engineer's Keyboard (Roving Engineer's Keyboard)

8. KEYBOARDS

8.7. Spare Parts

9 TRACKBALL

9.1 Cleaning

To clean the ball in the trackball assembly, wipe the surface of the ball with a slightly damp cloth that has been dipped in a mild soap solution.

9.2 Adjustments

None.

9.3 Test/Troubleshooting

1. If the cursor does not move on the monitor when the trackball is scrolled, check the cable connections at the EPDGP I/O board in the rear of the Five-Slot Module.
2. If this is correctly connected, replace the EPDGP I/O board.
3. Replace Trackball Interface Card, 51108312-200.

9.4 Replacement of Trackball Interface Card

1. Open the rear door of the Universal Station and turn off the US node power switch on the Dual Node Module.
2. Remove the two Phillips-head screws located under the keyboard tray, 2.5 inches from the front, on each side.
3. Remove four hex-head screws located on the two sides near the edge of the bottom of the keyboard tray.
4. Remove two Phillips-head screws located on the top, at the rear of the Operator's Keyboard.
5. Pivot the keyboard bezel/trackball assembly, raising it toward yourself.
6. Tilt the operator towards the CRT.
7. Disconnect the trackball cable at the nine-pin connector attached to the cable connecting to the Trackball Interface Card.
8. Remove the cable connecting the Trackball Interface Card to the EPDGP I/O card (through the gooseneck). Release the tie-downs as necessary.
9. Replace the Trackball Interface Card after removing the two slotted head screws.
10. Reconnect the Trackball Interface Cable to the EPDGP I/O card connector J1.
11. Retie the tie-downs previously release in step 8.
12. Reconnect the Trackball cable to the nine-pin connector that is attached to the Trackball Interface Card.

9. TRACKBALL

9.5. Startup

13. Pivot the Trackball/bezel assembly up toward yourself.
14. Place the Operator's Keyboard in the keyboard tray.
15. Place the Trackball/bezel assembly on the keyboard tray.
16. Replace the four hex-head screws removed in step 4.
17. Replace the two Phillips-head screws removed in step 3.

9.5 Startup

Turn on the ac power to the US node in the Dual Node Module. Reload the station with the appropriate personality.

9.6 Spare Parts

Table 9-1 — Trackball Spare Parts

Part Number	Description
*51108313-100	Trackball Assembly
*51108313-300	Console Mount Trackball – CE Compliant
51108313-200	Interface Card and attached Interface Cables
51304673-100	Left trackball switch with wires*
51304673-200	Right trackball switch with wires

* Has longer wires to reach terminal strip.

10 FIVE/TEN-SLOT MODULE - DUAL NODE MODULE

10.1 Cleaning

There is no cleaning of the module as such; however, periodically the air filter on the rear station door should be removed, washed in detergent, rinsed, and dried.

10.2 Adjustments

None.

10.3 Test/Troubleshooting

Refer to the *Five/Ten-Slot Module Service* or *Dual Node Module Service* manual. These publications also specify the required complement of electronic boards for each available configuration of the Universal Station and its resident modules.

10.4 Removal/Replacement

10.4.1 Module Removal

1. Open the rear door on the console and place the module's power switch in the OFF (0) position.
2. Open the front door on the console. Label (if required) and remove all cables connected to the module I/O boards. Taking care not to cut or nick the wire insulation, cut the tie wraps that secure the wires to the rear of the module. Unplug the module power cord(s). Remove grounding wire between the module rear panel and the power cord by removing the screw and washer from the grounding lug.
3. At the rear of the console, remove the bolts and washers that secure the module to the console, and slide the module out of the console.

10.4.2 Module Replacement

1. Slide the module into the rear of the console, and secure with eight bolts and washers.
2. At the front of the console, secure the grounding-wire lug to the rear of the module chassis with a screw and washer. Connect the power cable(s) to the module. Bundle the wires and secure them to the rear of the module with tie wraps. Install the cables on the I/O boards.

10.5 Startup

At the rear of the console, place the module's power switch in the ON (1) position. After 2 minutes, check the module LED indicators; all red ones should be off, all green ones should be on, and the alphanumeric display should show the module node address. For further information, refer to the *Five/Ten-Slot Module Service* manual or *Dual Node Module Service* manual.

10.6 Spare Parts

Spare parts information is found in the *Five/Ten-Slot Module Service* or *Dual Node Module Service* manual.

11 FLEXIBLE DISK DRIVE

11.1 Cleaning

The read/write heads in the Flexible Diskette Drive (FDD) unit(s) should be periodically cleaned. Any commercially available 5 1/4" disk drive head-cleaning kit can be used. Follow the cleaning instructions provided with the kit. In general, place a few drops of alcohol on the cleaning disk, insert the disk, and close the latch. After the diskette spins for 10 to 30 seconds the heads should be clean. Open the latch and remove the cleaning diskette.

If the operator station is located in a dirty environment or where smoking is permitted, head cleaning must be scheduled more frequently.

NOTE

DO NOT use Q-TIPS to clean the heads. Also, the drive may be damaged by attempting to close the latch lever without having a diskette or the shipping insert in place.

11.2 Adjustments

None.

11.3 Test/Troubleshooting

Equipment Required:

- Digital Multimeter (DVM)

In general, if one disk drive fails, the disk drive is at fault. If both disk drives are inoperative, open the rear door of the cabinet and ensure that the power supply DC Status LED is on. (Note that a few installations have a power supply for each disk drive.) Next, check the FDD Controller card and FDD I/O card in the US electronics module (refer to the *Five/Ten-Slot Module Service* manual in the *Service* binder).

NOTE

New versions of the US have the Enhanced Power Supply in the Five/Ten-Slot Module that supplies power to the floppy drives by means of a cable, 51304490-300. See Figure 11-3.

11.3.1 DC Voltage Check

Input power should be +4.75 Vdc to +5.25 Vdc and +11.4 Vdc to +12.6 Vdc measured at the input to the FDD (see Figure 11-1).

11.4 Unit Removal/Replacement

11.4.1 Removal

1. Open the door on the rear of the disk drive cabinet, press up on the drive chassis to relieve the force on the chassis retainers, and turn the knurled knobs to rotate the retainers up, and out of the way. The chassis will slide out on the rails to a detent position.
2. Set the FDD Power Supply Switch to "0" (OFF), (see Figure 11-2). If there is no FDD Power Supply, disconnecting the power cable from the J2 on the FDD will remove power. See Figure 11-1 for the location of J2.
3. Remove the floppy disk drive by unplugging two connectors (J1 and J2); then remove four hex-headed screws (two on each bracket) holding the brackets to the base pan. Slide the assembly back and up. Remove the bracket screws from the FDD, two on each side, and transfer the brackets to the replacement unit.
4. Remove the FDD power supply by labeling (if necessary) and disconnecting the four wires from the terminal board. Disconnect the ac status plug and the ac power plug. Remove four screws and washers securing the power supply to the chassis and lift the power supply free.

11.4.2 Replacement

1. To replace the FDD, attach the mounting brackets on each side, attach the front plate, position the assembly on the chassis, and slide it into position. Attach with hex-head screws, two on each side. Connect the ribbon and the power cables. Note that the red stripe is to your left, facing the rear of the FDD.
2. Place the power supply on the chassis and secure with four screws and washers. Connect four wires to the power terminal and plug in the ac status cable (red stripe to the right). Finally, install the ac power cable.
3. Several different device pinning configurations are shown in Figure 11-4. Note that your replacement device may not be of the same manufacture or model number as your original. Locate the correct configuration by checking the sketches in Figure 11-4, then pin the new drive according to the device address used in the device being replaced.
4. Press the detent/release buttons on the sides of the rails and push the disk drive assembly into the cabinet. Rotate the knurled knobs to move the retainers down behind the locking tabs. Set the power switch on the power supply to "1" (ON).

CAUTION

Before operating the floppy disk drive, open the latch lever and remove the shipping insert. Do not attempt to close the latch lever without first inserting a diskette or the shipping insert.

Also be aware that to write-protect the five and a quarter inch diskette, the write tab must be taped over.

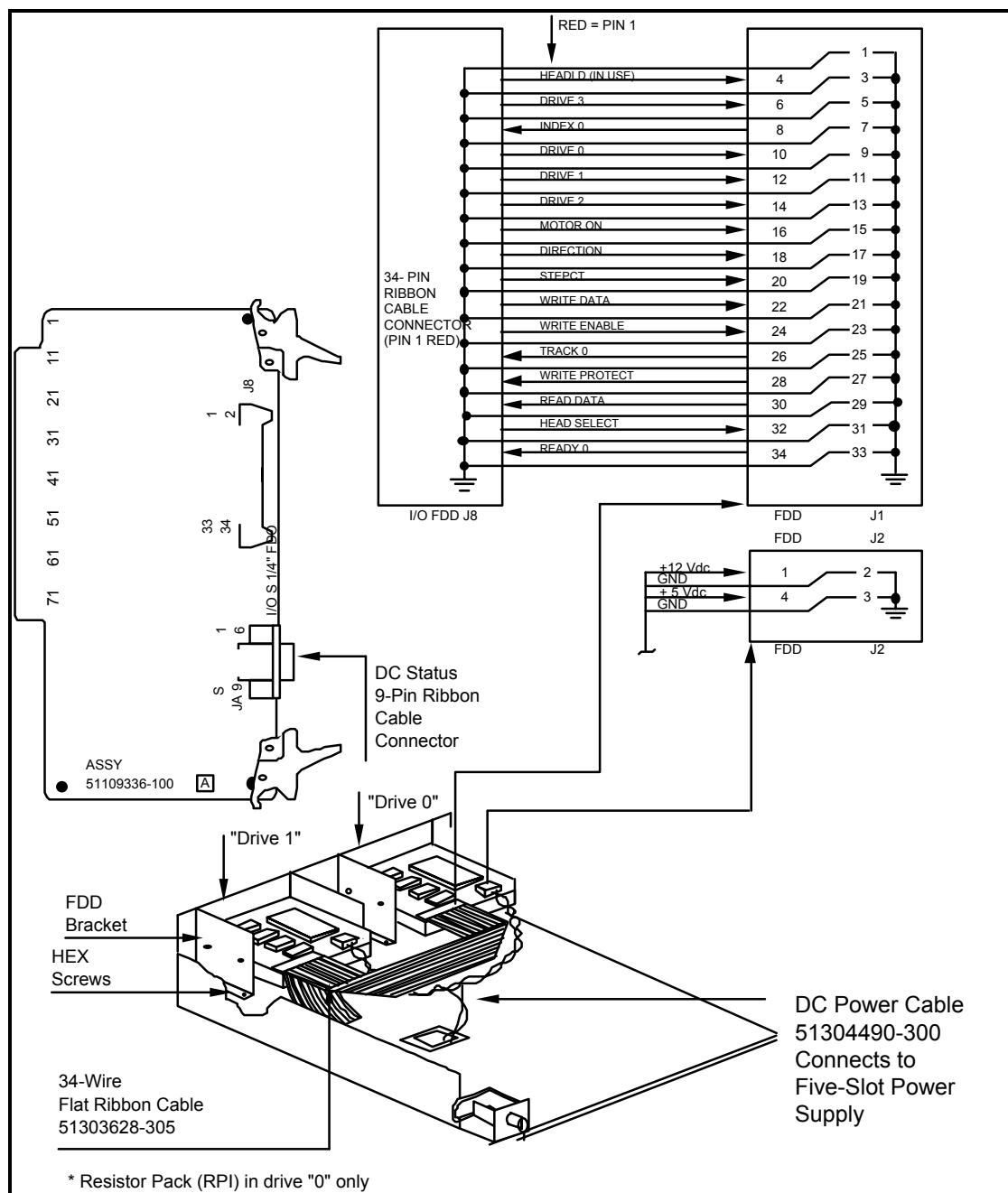


Figure 11-1 — FDD Interface Connections (Showing DC Power Cable That Connects to Five-Slot Power Supply) 4 139

11.4. Unit Removal/Replacement

Before operating the floppy disk drive, open the latch lever and remove the shipping insert. Do not attempt to close the latch lever without first inserting a diskette or the shipping insert.

Also be aware that to write-protect the five and a quarter inch diskette, the write tab must be taped over.



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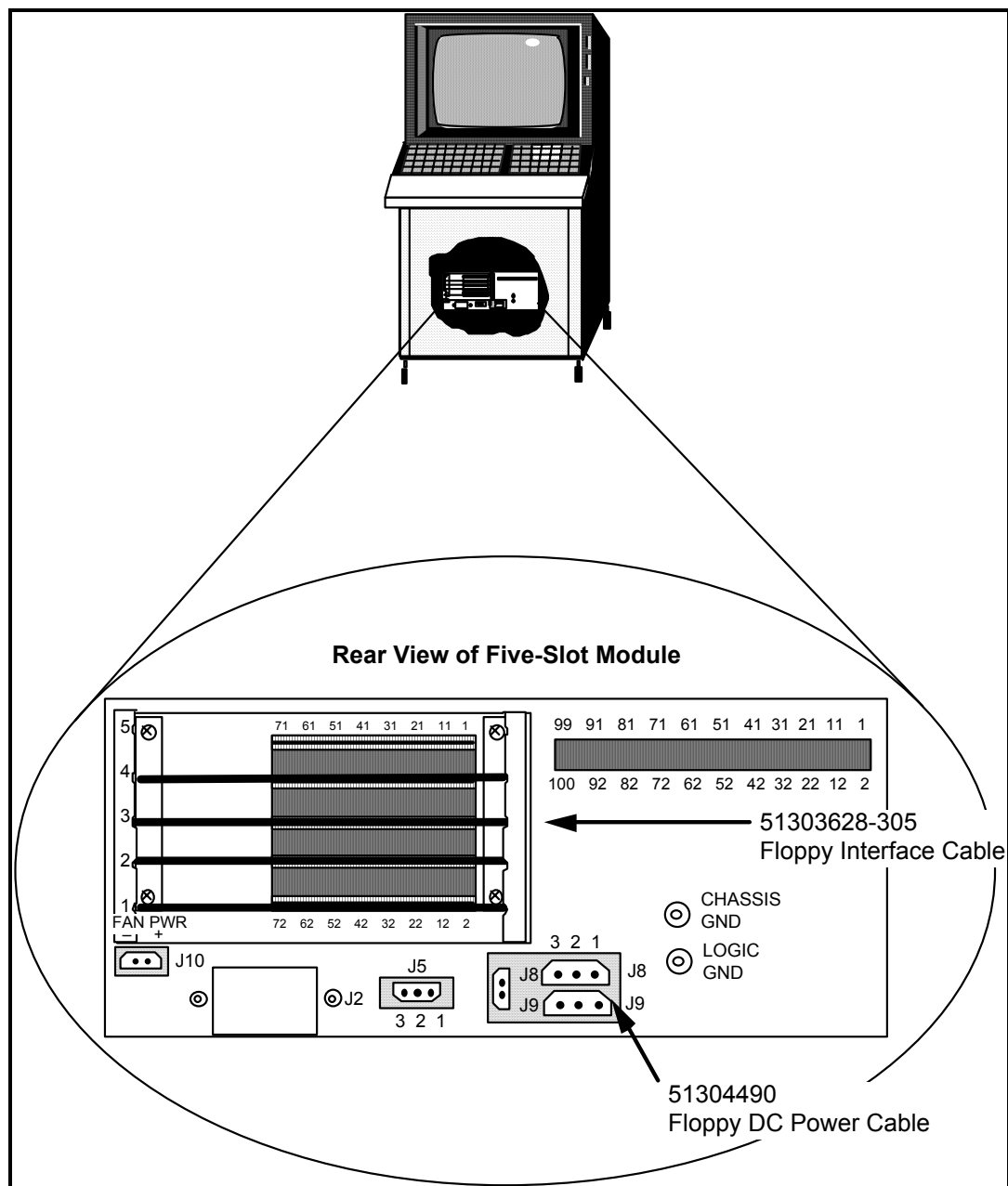


Figure 11-3 — Floppy Drive Power Cabling Using Enhanced Power Supply 4713

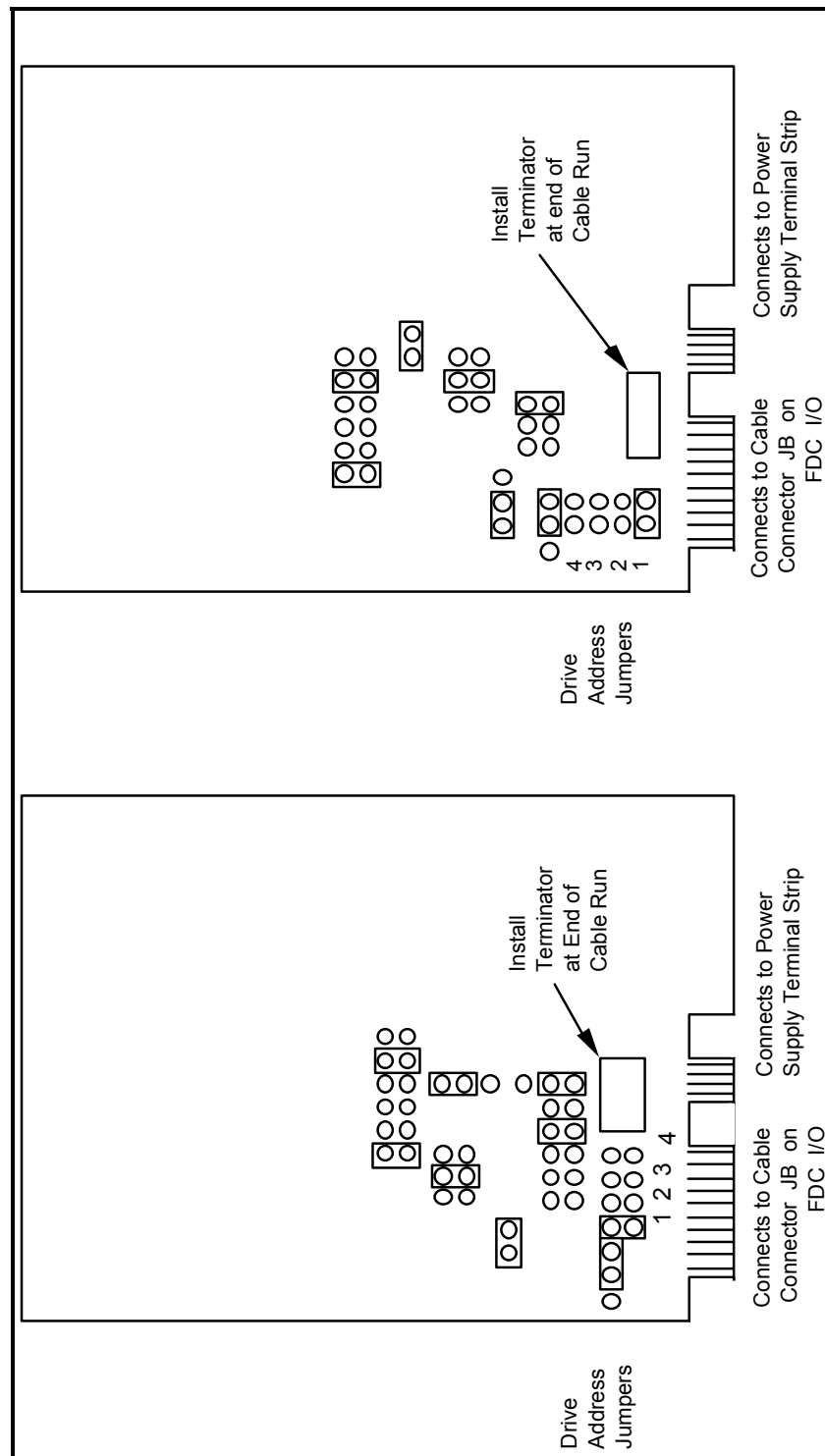
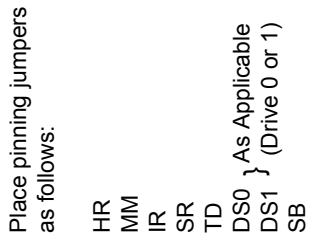


Figure 11-4 — Floppy Disk Drive Address Pinning (Sheet 1 of 3)

2096



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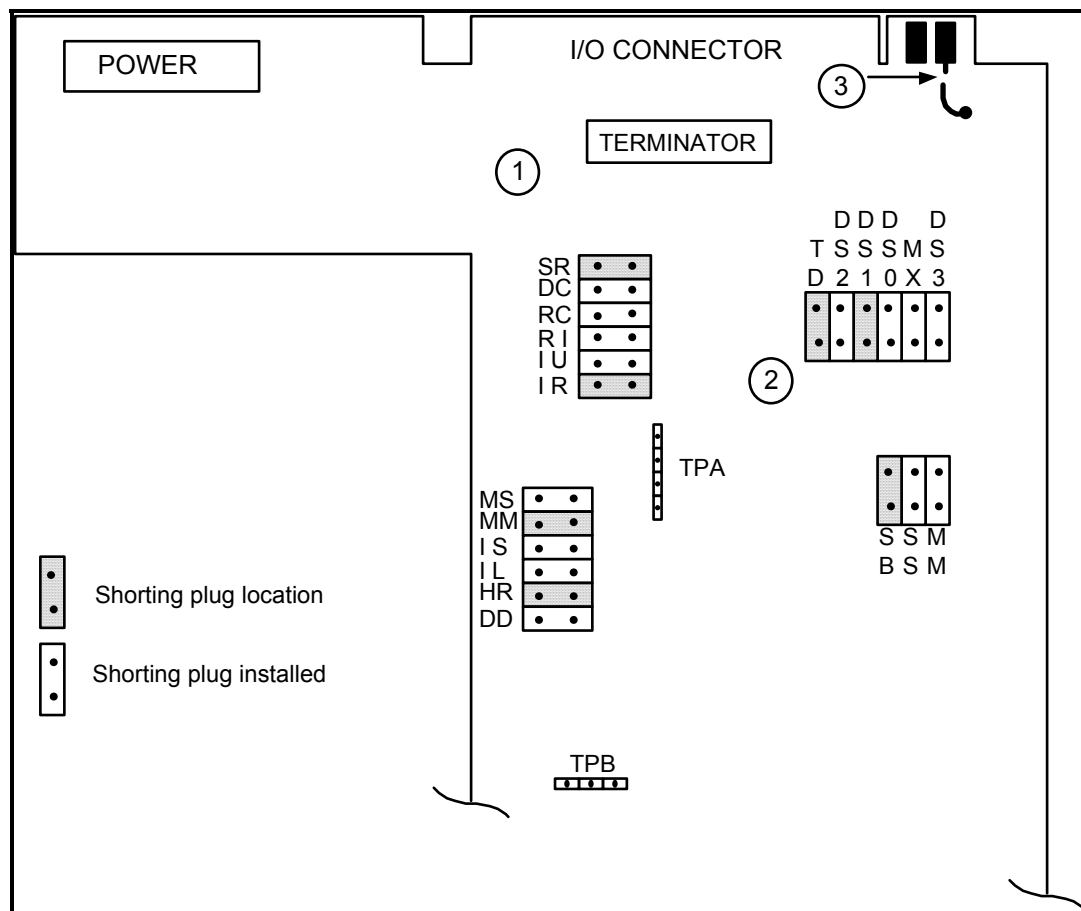


Figure 11-4 — Floppy Disk Drive Address Pinning (Sheet 3 of 3)

4601

11.5 Component Removal and Replacement Procedures

None. The unit must be returned to the factory for repair and adjustment.

11.6 Startup

None.

11.7 Spare Parts

Table 11-1 — Floppy Disk Spare Parts

Part Number	Description
* 51109701-100	5 1/4 Inch Floppy Disk Drive
* 51109818-100	Purchase Spec. (Floppy Power Supply)
51303409-100	Keyboard Cable (DC Power Status)
51303536-061	Winchester DC Power (Floppy DC Power)
51303628-305	Winchester Command Cable (Floppy Command Cable)
51304490-300	Cartridge DC Power Cable (used in Floppy)

- ORU Level Replacement Item

11. FLEXIBLE DISK DRIVE

11.7. Spare Parts

12 CARTRIDGE DISK DRIVE

THE CARTRIDGE SHOULD NOT REMAIN IN THE CARTRIDGE DRIVE FOR EXTENDED PERIODS OF TIME WHEN IT IS NOT IN USE. The drive automatically retracts the heads after a period of time without use, but the media is still spinning. This shortens the life of the cartridge and could result in lost data.

The 150 MB Iomega Bernoulli cartridge drive automatically spins down after 30 minutes of inactivity.

To load a cartridge, insert it into the drive in a smooth fashion until it clicks (with about 3 cm [1 inch] still sticking out) and starts to spin. The green LED will start flashing, then will remain steadily on when the drive is ready.

WARNING

If the green LED starts to flash before the cartridge is in all the way, stop and wait till the LED stops flashing. If the LED does not stop flashing, press the STOP button. When the LED stops flashing, remove the cartridge.

12.1 Cleaning

Periodic cleaning of the read/write head is important for prolonging head and cartridge life. The head should be cleaned after every 50 hours of use (more frequently in dusty environments). We recommend immediate head cleaning whenever abnormal drive performance, such as frequent errors, occurs. Continued operation with a dirty head causes damage to the data surface of the cartridge disk. If the head is not cleaned, eventually contaminants will not be completely removed from the head and the drive becomes inoperative.

If a cartridge performs abnormally, we recommend head cleaning before use of another cartridge. Continued use of a defective or worn cartridge results in head contamination.

A special cleaning kit is available for this purpose—see subsection 2.2. It includes a head cleaning cartridge, wiper pads, and an approved head-cleaner solution. The head cleaning cartridge inserts into the drive as does a standard data cartridge. Instructions for the cleaning operation are included with the head cleaning kit.

Turn the drive power on to position the heads for cleaning. Remove the data cartridge from the drive. Press the front panel button on the disk drive and hold until the green LED starts flashing (about two seconds). This places the drive in the cleaning mode to accept the cleaning cartridge. When the cleaning cartridge is inserted into the drive, the green LED glows steadily to signal that the heads are in position and ready to be cleaned.

12. CARTRIDGE DISK DRIVE

12.2. Adjustments

CAUTION

Follow instructions on the container that holds the disk drive head cleaner solution to avoid vapor inhalation or skin exposure.

The cleaning cartridge contains a head wiper pad that is mounted on the pad-arm assembly. Change the pad after five uses. Apply the cleaner by squeezing the solution from the container onto the pad through the cartridge window and center slot. The cleaner should be applied to the pad just before inserting the cartridge into the drive. The pad should be dampened but not saturated with the solution. The wiper lever automatically moves back and forth for 30 cleaning cycles.

After cleaning is completed, press the STOP button. When the green light goes out, remove the cartridge. A label is provided to record the drive identification and the date of cleaning. Place the label on the outside of the storage sleeve.

12.2 Adjustments

None.

12.3 Test/Troubleshooting

In general, if only one disk drive fails, the disk drive is most likely at fault. If both disk drives are inoperative, open the rear door of the cabinet and ensure that the power supply DC STATUS LED is on. Next, check the associated Smart Peripheral Controller (SPC) board (or EPDG board—refer to the *Five/Ten Slot Module Service* manual, in the *LCN Service* binder). If the cartridge drive power comes from the Five/Ten-Slot Module Power Supply, check its indicators (refer to *Five/Ten-Slot Service* manual).

12.3.1 DC Voltage Check

Equipment Required:

- Digital Multimeter (DVM)

Input 5 V power should be +4.75 to +5.25 Vdc, measured on pin 4 of the dc power cable, 51303536-061, at the input of the cartridge drive.

Input 12 V power should be +12.4 to +12.6 Vdc, measured on pin 1 of the dc power cable, 51303536-061, at the input of the cartridge drive.

12.4 Removal/Replacement

12.4.1 Removal of Cartridge Disk Drive (CDD) and DC Power Supply

1. Open the door on the rear of the disk drive cabinet, press up on the drive chassis to relieve the force on the chassis retainers, and turn the knurled knobs to rotate the retainers up, out of the way. The chassis will slide out on the rails to a detent position.

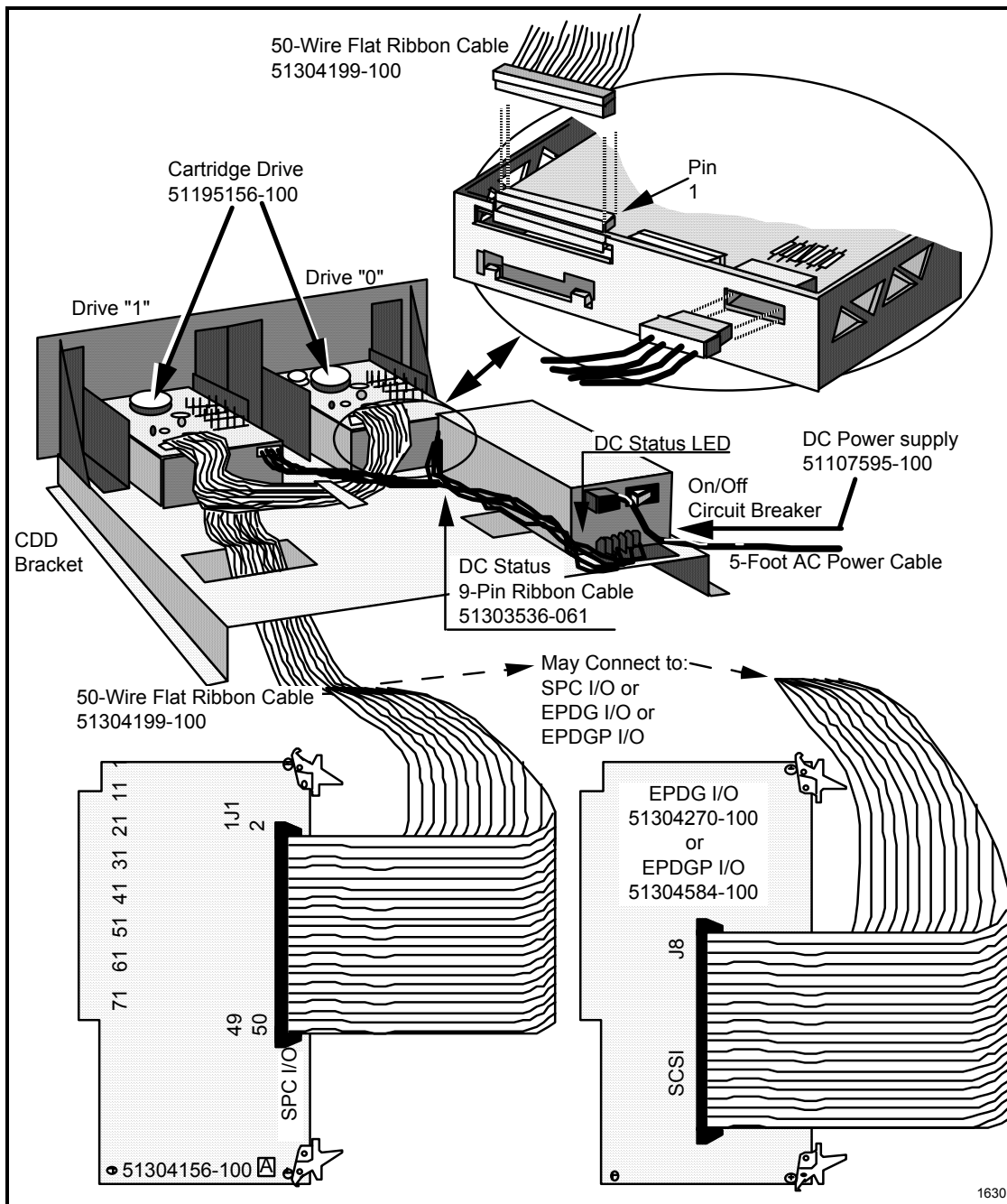


Figure 12-1 — CDD Interface Connections

12. CARTRIDGE DISK DRIVE

12.4. Removal/Replacement

2. Set the CDD power supply switch to "0" (OFF), see Figure 12-1. If there is no CDD power supply, the power comes from the Five-Slot Module Power Supply. In this case turn off the power supply in the Five-Slot Module—**be aware of the fact that this will shut down the US node.**

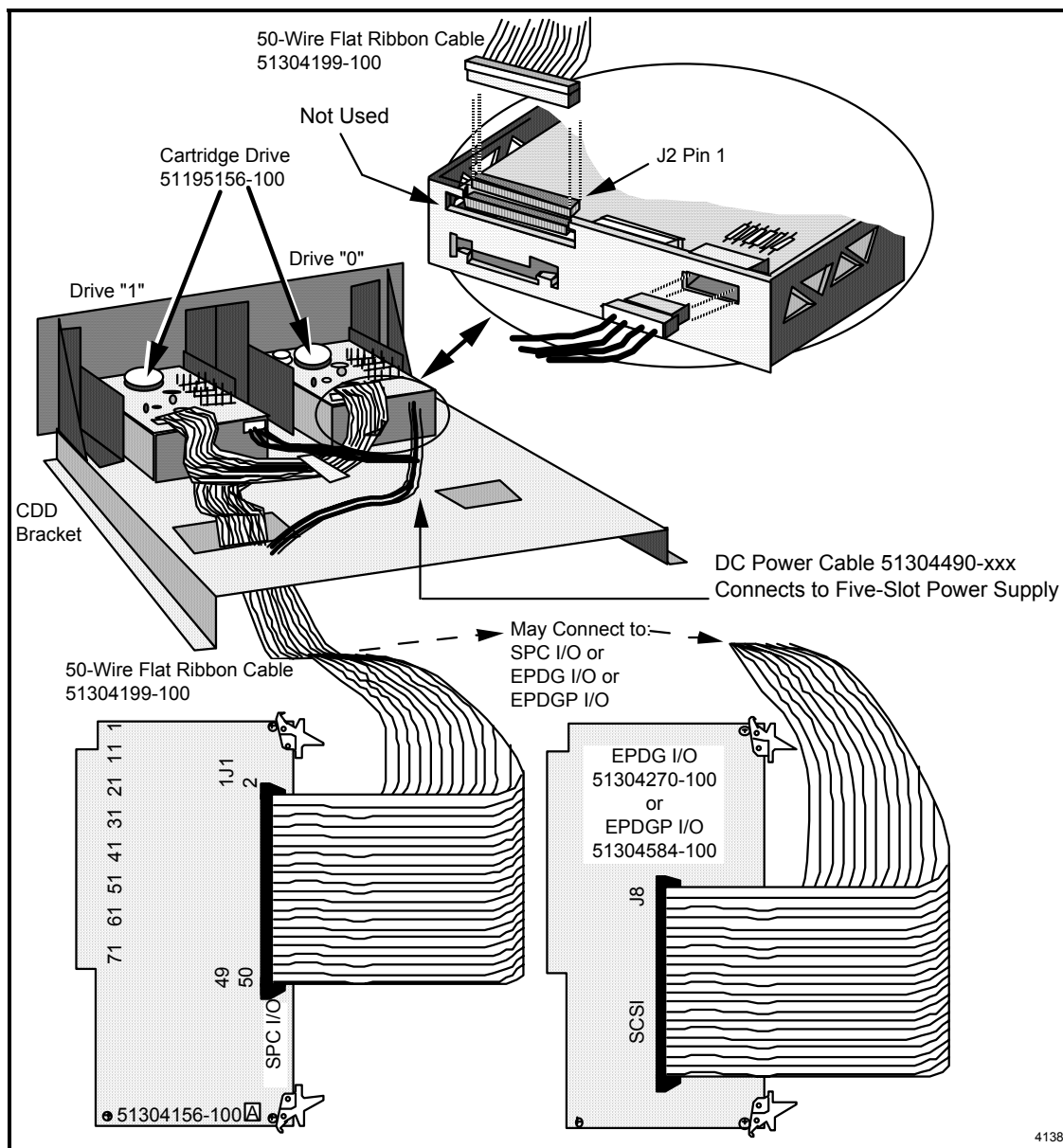


Figure 12-2 — CDD Assembly Without Power Supply , Using Power Cable

3. Remove the cartridge disk drive by unplugging the two cables connected to J1 and J2, then removing four hex-headed screws (two on each bracket) that hold the brackets to the slide assembly. Slide the assembly back and

up. Remove the bracket screws from the CDD, two on each side and transfer the brackets to the replacement unit.

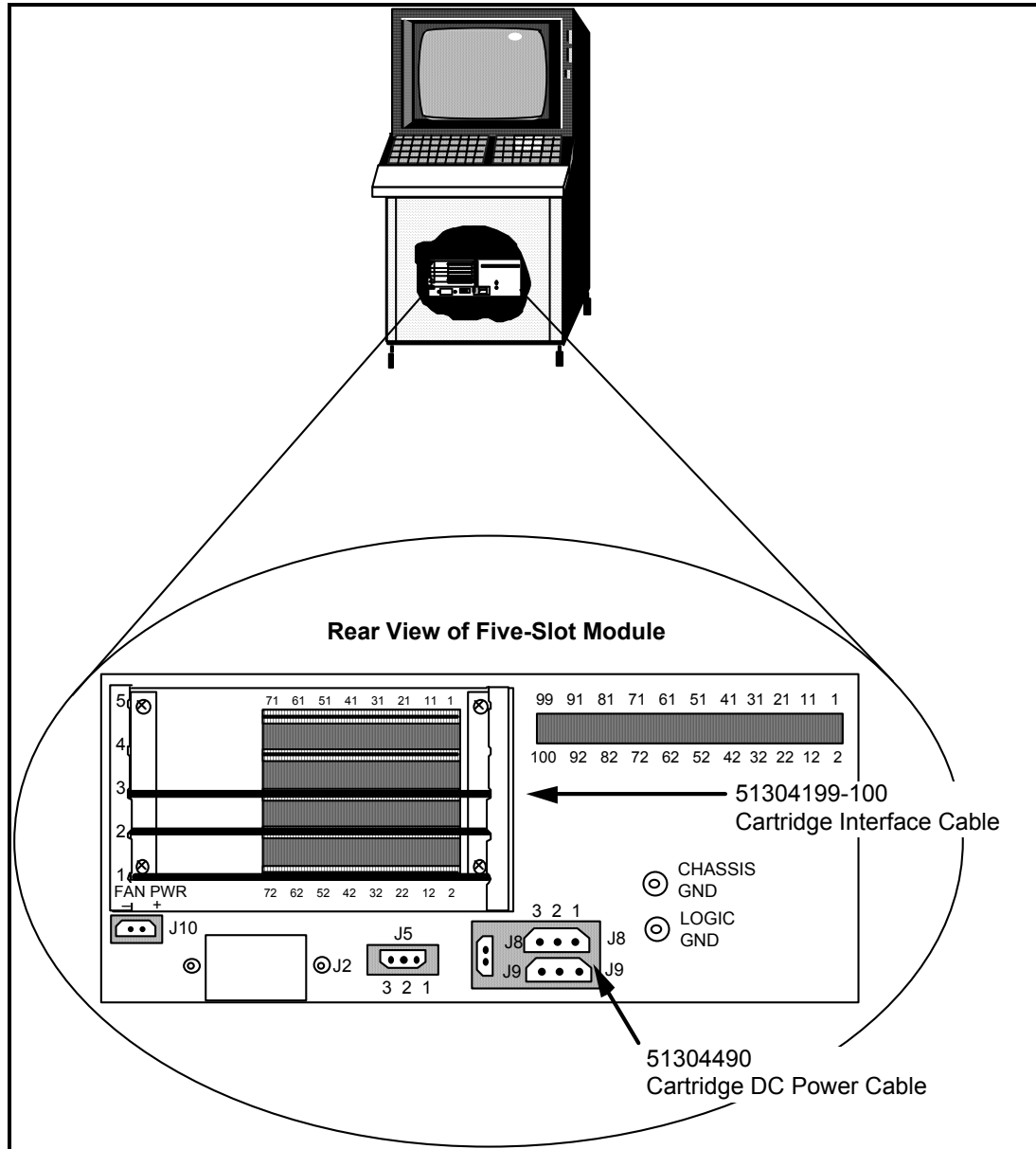


Figure 12-3 — Cartridge Drive Power Cabling Using Enhanced Power Supply

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12. CARTRIDGE DISK DRIVE

12.4. Removal/Replacement

4. Remove the CDD power supply by disconnecting the four wires of the dc power cable, 51303536-061, from the terminal board. Disconnect the ac status cable connector and the ac power plug. Remove four screws and washers securing the power supply to the chassis, and lift the power supply free. If there is no CDD power supply—skip this step.

If there is no CDD power supply, the power to operate the CDD comes from the Five-Slot Module power supply. Power cabling, using the Enhanced Power Supply, is shown in Figure 12-3.

5. Reconnect the cables by reversing the above procedures.

12.4.2 Replacement of Cartridge Disk Drive (CDD) and DC Power Supply

1. Configure the drive pinning as shown in Figure 12-4, 12-5, or 12-6. The following parameters must be configured:

If this is Drive 0,

Drive Address = 0 S1 = Off

S2 = Off

S3 = Off

Parity Checking S4 = On

Diagnostic Operations S5 = Off

Manual Power-On S6 = Off

Termination Resistors RP1 = In

RP2 = In

RP3 = In

If this is Drive 1,

Drive Address = 1 S1 = On

S2 = Off

S3 = Off

Parity Checking S4 = On

Diagnostic Operations S5 = Off

Manual Power-On S6 = Off

Termination Resistors RP1 = Out

RP2 = Out

RP3 = Out

2. To replace the cartridge drive, attach the mounting brackets on each side, attach the front plate, position the assembly on the chassis, and slide it into position. Attach with hex-head screws, two on each side. Connect the ribbon and the power cables. Note that the red stripe on the cable is to your right, facing the rear of the cartridge drive.
3. If there is a separate power supply, place it on the chassis and secure with four screws and washers. Connect four wires to the power terminal, according to cable marking labels, and plug in the ac status cable (red stripe on the cable to the right). Finally, install the ac power cable.
4. Ensure that the device end of the flat ribbon cable, 51304199-100, is connected to the host interface connector, **not to the drive interface connector**.
5. Press the detent/release buttons on the sides of the rails, and push the disk drive assembly into the cabinet. Rotate the knurled knobs to move the retainers down behind the locking tabs. Set the power switch on the power supply to "1" (on).
6. Remove and retain the polystyrene insert. If the drive or console must be transported in the future, insert it into the cartridge slot to protect the heads from being damaged.

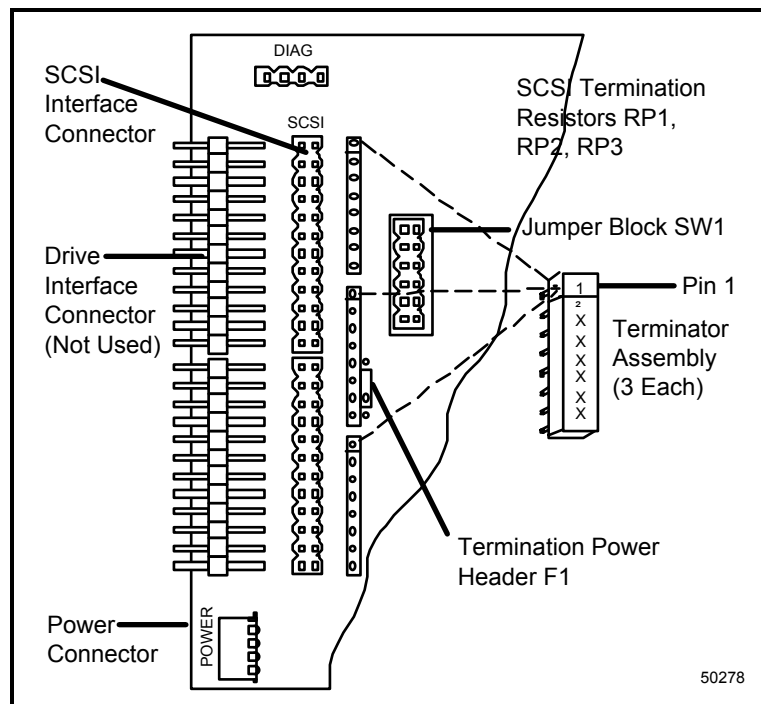


Figure 12-4 — 20 MB Iomega Cartridge Disk Termination

12. CARTRIDGE DISK DRIVE

12.4. Removal/Replacement

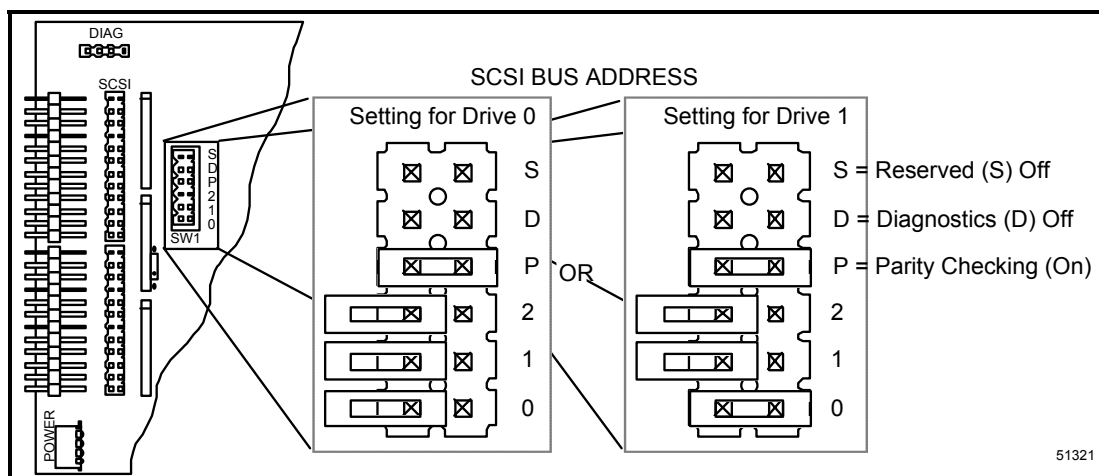


Figure 12-5 — 20 MB Cartridge Disk Configuration

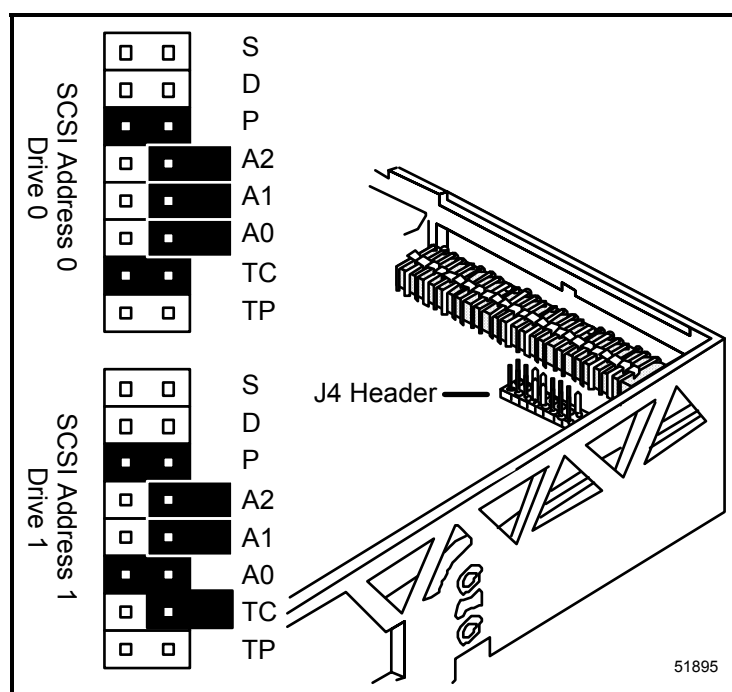


Figure 12-6 — 150 MB Cartridge Disk Configuration and Termination

12.4.3 Replacement of SPC Controller Boards

If the SPC Controller board is replaced, the pinning on the new board must be verified according to Figure 12-7. The SPC board has two applications—Cartridge Disk and WREN III disk drives (History Module). To distinguish between the applications a pin is placed in location 11D on the board.

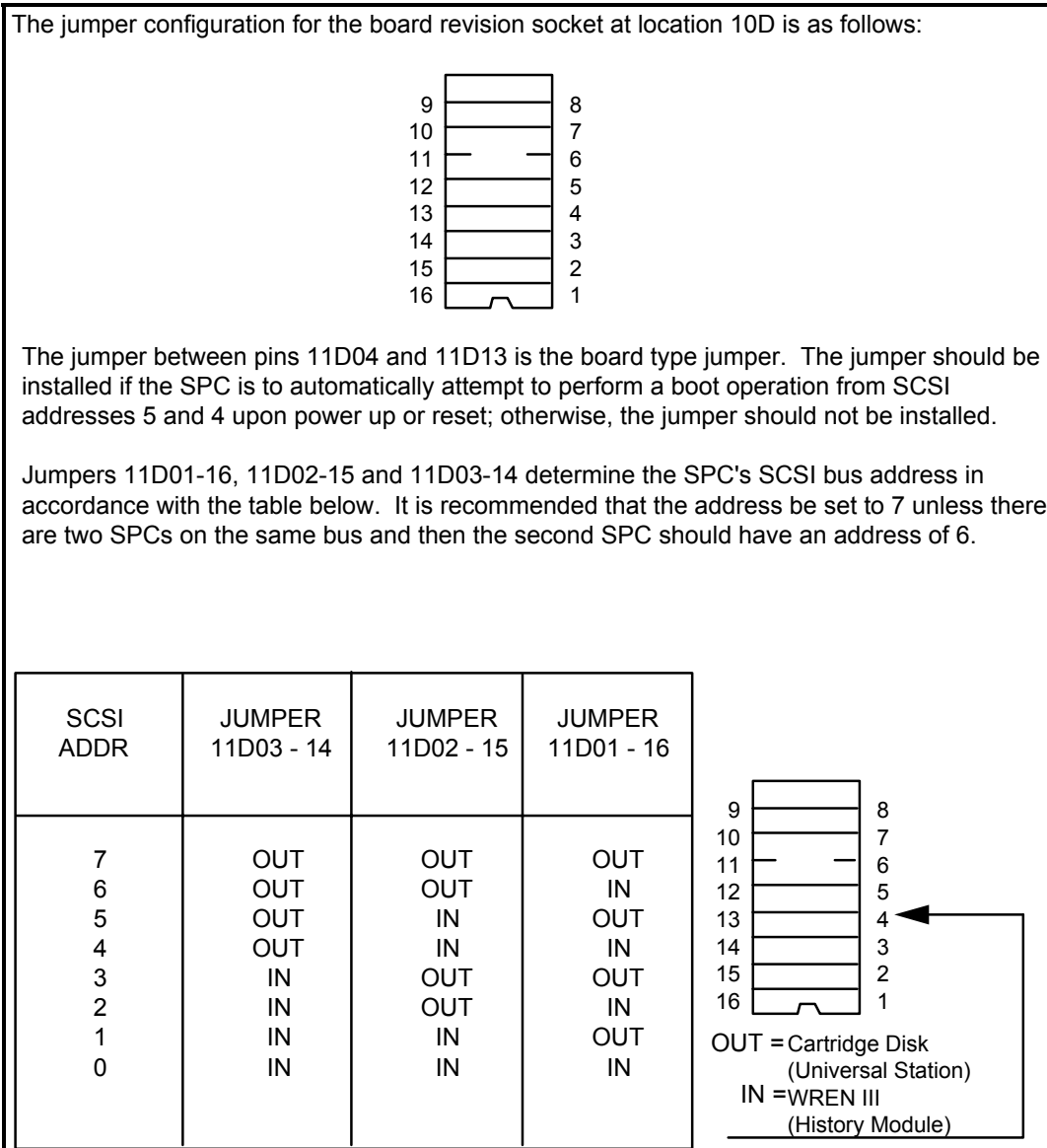


Figure 12-7 — SPC Board Address Pinning for LCN and SCSI Bus Addresses

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12.5 Cartridge Removal Without Power

Use the following steps to remove a cartridge when power is not available; see Figure 12-8.

1. Power to the drive must be turned off. To guard against power being restored during the cartridge removal procedure, set the Power Supply dc power switch to the "0" position (if there isn't a separate power supply — turn of the power supply switch on the Five-Slot Module).
2. Use a small, flat-blade screwdriver to remove the motor access panel from the front panel of the drive.
3. Insert the eraser end of a pencil into the space behind the motor access panel and press the eraser against the motor housing on the right side of the space. Rotate the motor until the dot on the motor housing is aligned with the dot on the pencil shield.
4. With the dots aligned with each other in the motor access space, insert the end of a straightened paper clip in the small hole below the stop button on the front of the drive. Press the paper clip firmly into the hole; simultaneously, pull the cartridge from the drive. Pressure must be maintained with the paper clip until the cartridge is fully removed.
5. Place the motor access panel back in the front panel.
6. Set the power supply dc power switch to the "1" position.

12.6 Component Removal and Replacement

None. Unit must be returned to factory for repair.

12.7 Startup

IMPORTANT

A. The position of the Write-Protect switch on the cartridge disk should not be changed while the disk is in a drive. Unload the cartridge to change the switch.

B. Inserting and removing a cartridge:

When inserting a cartridge into the drive, wait for the following events to occur before proceeding:

1. Green light flashes as drive motor accelerates to operating speed
2. Yellow light blinks as data is read several times
3. Green light remains illuminated

To remove a cartridge disk from the drive:

1. Push the STOP button.
2. Wait until the green light stops flashing (is extinguished) and an audible "click" is heard (approximately 5 seconds). If the cartridge resists removal, push the Eject button again.

WARNING: **ATTEMPTED REMOVAL OF THE DISK WITHOUT FOLLOWING
THE ABOVE PROCEDURE CAN RESULT IN SERIOUS DAMAGE
TO BOTH THE DISK AND DISK DRIVE.**

12. CARTRIDGE DISK DRIVE

12.7. Startup

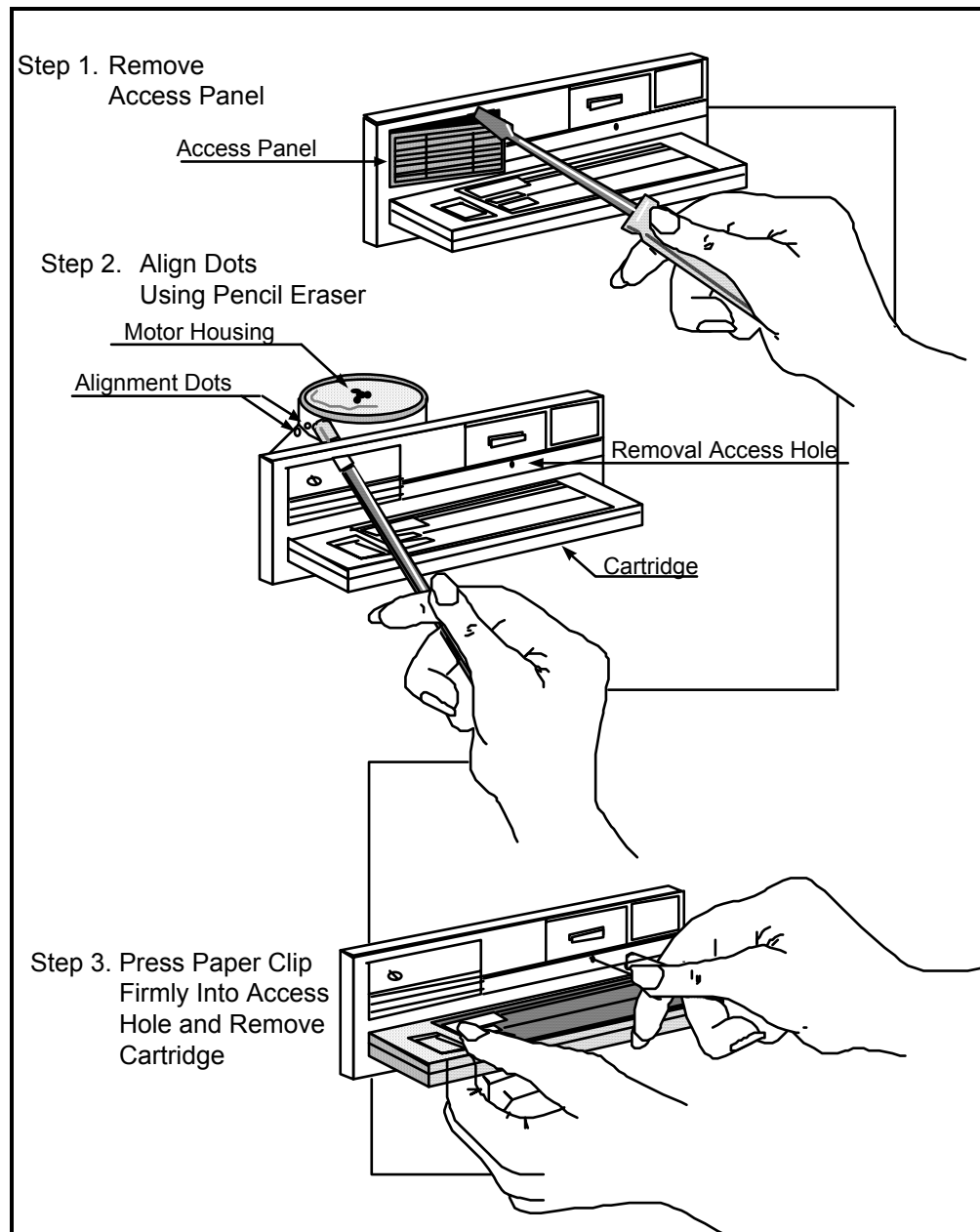


Figure 12-8 — Power Down Cartridge Removal

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12.8 Spare Parts

Table 12-1 — 20 MB Cartridge Disk Drive Spare Parts

Part Number	Description
*51195156-100	BETA 20 (Iomega Cartridge Drive)
51304199-100	Cable Cartridge Disk Interface
51303536-061	Winchester DC Power (DC Power to Cartridge)
*51107595-100	Winchester Power Supply (Used for DC Power)
51304490-300	Cartridge DC Power Cable

* ORU Level Replacement Item

Table 12-2 — 150 MB Cartridge Disk Drive Spare Parts

Part Number	Description
*51196483-100	Iomega's MultiDisk 150 Cartridge Drive
51304199-100	Cable Cartridge Disk Interface
51304490-300	Cartridge DC Power Cable
51196485-100	35 MB MultiDisk 150 MB compatible 5.25" cartridge
51196485-200	65 MB MultiDisk 150 MB compatible 5.25" cartridge
51196485-300	105 MB MultiDisk 150 MB compatible 5.25" cartridge
51196485-400	150 MB MultiDisk 150 MB compatible 5.25" cartridge
51196485-500	5.25" Cleaning Care Kit—includes • cleaning cartridge • one 0.25 oz bottle of cleaning solution • 2 extra cleaning pads
51196485-600	Multi-pak Cleaning Solution—Refill Kit • three 0.25 oz bottles of cleaning solution
51196485-700	Wiper Arm Pads (Quantity of 2 pads)

* ORU = Optimum Replaceable Unit

12. CARTRIDGE DISK DRIVE

12.8. Spare Parts

13 CE COMPLIANT CARTRIDGE DISK DRIVE

13.1 Overview

The cartridge drive is located in an EMI isolation enclosure and is powered by the power supply in the enclosure. The enclosure will accommodate single or dual drives.

The cartridge drive is a 150 MB Iomega Bernoulli drive that has multidisk capability for 35, 65, 105, 150 MB disks.

WARNING

THE CARTRIDGE SHOULD NOT REMAIN IN THE CARTRIDGE DRIVE FOR EXTENDED PERIODS OF TIME WHEN IT IS NOT IN USE. The drive automatically retracts the heads after a period of time without use, but the media is still spinning. This shortens the life of the cartridge and could result in lost data.

The 150 MB cartridge drive has an "auto sleep" feature that spins down the drive after approximately 30 minutes. The drive will spin up again when accessed.

13.2 Loading a Cartridge

Use the following procedure to load a cartridge in the drive.

1. Insert the cartridge into the drive with a smooth motion until it clicks in place (with about 3 cm [1 inch] still protruding out) and starts to spin.
2. The green LED should start to flash and then will remain steadily on when the drive is ready. If the LED starts to flash before the cartridge is fully inserted, stop and wait until the LED stops flashing.

13.3 Unloading a Cartridge

3. If the LED does not stop flashing, press the STOP button. When the LED stops flashing, remove the cartridge.

Use the following procedure to unload a cartridge from the drive.

1. Press the button on the front and release it. The LED indicator should flash on and off. Do not pull on the cartridge until the LED stops flashing.
2. When the LED indicator stops flashing, remove the cartridge by pulling firmly on the center of the cartridge.

13.4 Cleaning a Cartridge Drive

Periodic cleaning of the read/write head is important to prolong head and cartridge life. The head should be cleaned:

- every 50 hours of use (more frequently in dusty environments).

13. CE COMPLIANT CARTRIDGE DISK DRIVE

13.4. Cleaning a Cartridge Drive

- whenever frequent errors occur.

NOTE

Each drive should have its own cleaning cartridge.

The cleaning cartridges are available from Honeywell. See Table 13-1 for part numbers. The kits include:

- a head-cleaning cartridge,
- wiper pads, and
- an approved head-cleaner solution.

The head-cleaning cartridge inserts into the drive in the same manner as a standard data cartridge. Instructions are included with the head-cleaning kit. The cleaning cartridge has a pad that is mounted on the pad-arm assembly. Change the pad after five uses.

Use the following procedure to clean the cartridge drive.

1. Remove any cartridge that is in the drive.
2. Press the front panel button on the disk drive and hold until the green LED starts flashing (about two seconds). This places the drive in the mode to accept the cleaning cartridge.
3. Follow the instructions on the container that holds the disk drive cleaner solution to avoid vapor inhalation or skin exposure.
4. Apply the cleaner by squeezing the solution from the container onto the pad through the cartridge window and center slot. The cleaner should be applied to the pad just before inserting the cartridge into the drive. The pad should be dampened, but not saturated with the solution.
5. Place the cleaning cartridge in the drive. The LED will illuminate steadily.
6. Allow the wiper lever on the cartridge to move back and forth for 30 cleaning cycles.
7. After the 30 cleaning cycles are complete, press the button on the front panel.
8. When the green light extinguishes, remove the cartridge.
9. Record the cleaning date on the label provided with the cartridge. Place the label on the outside of the storage sleeve.

13.5 Troubleshooting an Inoperative Drive

In general, if only one drive fails, the drive is most likely at fault. If both drives are inoperative, check the power source. Consider the following:

- If the node is in a Five-Slot Module, the module power supply supplies the power.
- If the node is in a Dual Node Module, the peripheral power supply supplies the power.

Use the following procedure to isolate and troubleshoot a failed drive.

1. If the node is in a Five-Slot Module, check the indicators on the module power supply. Refer to the *Five/Ten Slot Module Service* manual for guidance.
2. Check the TPDGX board error LED (red).
3. Check the power cable by disconnecting it from the cartridge drive.

Pins 1 and 2 = $12 \pm .1$ Vdc

Pins 3 and 4 = $5.0 \pm .25$ Vdc

13.6 Removing a Cartridge Drive

Use the following procedure to remove a cartridge drive.

1. Remove the rear cover from the upper pod after removing the four retaining screws.
2. Disconnect the cables from the back of the drive enclosure.
3. Remove the two hex-head screws (from each side) that attach the enclosure to the mounting brackets.
4. Lift the enclosure off the support tray and place it on an ESD-safe work surface.
5. Remove the 16 screws from the top cover of the enclosure.
6. Remove the top cover and set it aside.
7. Disconnect the 4-prong dc power cable from the rear of the cartridge drive.
8. Disconnect the 50-pin ribbon cable from the rear of the cartridge drive.
9. Remove the four hex-head machine screws holding the cartridge drive to the removable media drive disk tray assembly.
10. Pin the new drive according to the instructions in subsection 13.7. Observe ESD precautions while handling the cartridge drive.

13.7 Pinning a Cartridge Drive

Pin the cartridge drive as illustrated in Figure 13-1.

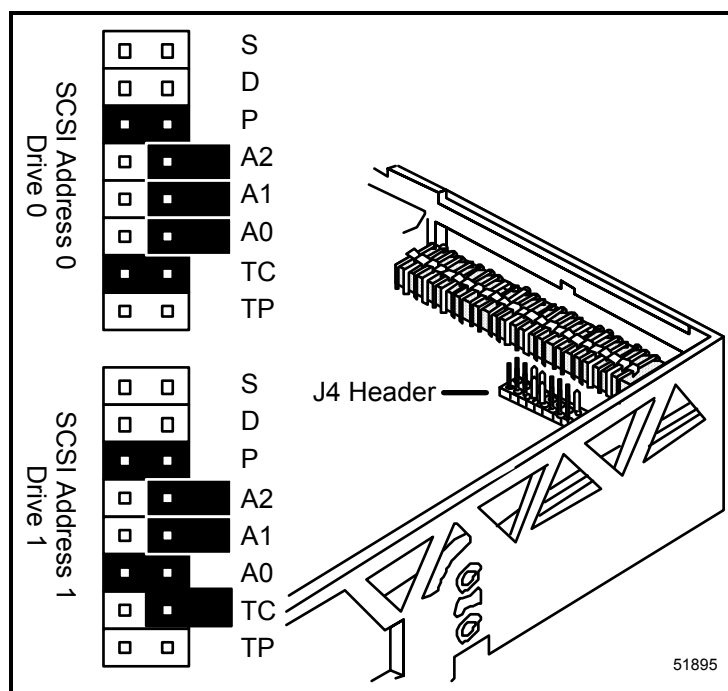


Figure 13-1 — 150 MB Cartridge Drive Pinning

13.8 Replacing a Cartridge Drive

Use the following procedure to replace a cartridge drive. Figure 13-2 illustrates the cartridge drive cabling.

1. Place the cartridge drive on the removable media drive tray and fasten it in place with four screws.
2. Reconnect the dc power cable to the rear of the drive.
3. Reconnect the SCSI 50-pin ribbon cable to the rear of the drive.
4. Place top cover on the enclosure and replace the 16 screws to ensure a good ground.
5. Place the drive enclosure on the mounting tray and replace the four hex-head screws that attach the enclosure.
6. Reconnect the power and interface cables to the enclosure.
7. Replace the half upper tier rear cover or return the slide tray fully forward to the front of the station. In either case, check to see if the faceplate is in proper contact with the bezel.

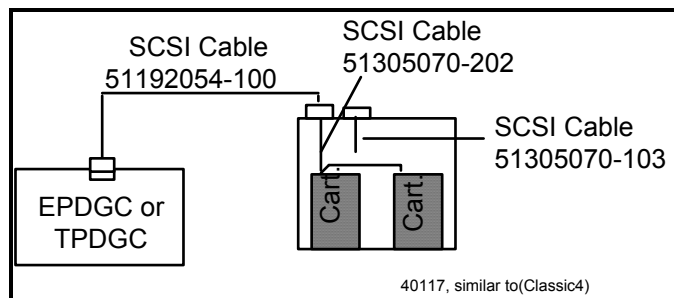


Figure 13-2 — 150 MB Cartridge Drive Cabling

The SCSI bus ribbon cable **connects to the back connector (the only one)** on a 150 MB cartridge drive.

13.9 Spare Parts

Table 13-1 — 150 MB Cartridge Disk Drive Spare Parts

Part Number	Description
*51196403-100	Beta 0 (Iomega Cartridge Drive)
51192054-102	SCSI Interface Cable TPDGC I/O Board to Drive Enclosure
51305070-202	SCSI Interface Cable for Drive Enclosure (3 connector)
51305070-103	SCSI Interface Cable for Drive Enclosure (2 connector)
51305341-100	Drive Cable DC Power
51196485-100	35 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-200	65 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-300	105 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-400	150 MB Multidisk 150 MB compatible 5.25" cartridge
51196485-500	Cartridge Drive Cleaning Kit—includes: • cleaning cartridge • three 0.25 oz bottles of cleaning solution • two extra cleaning pads
51196485-600	Cartridge Drive Cleaning Solution—refill kit includes: • three 0.25 oz bottles of cleaning solution
51196485-700	Wiper Arm Pads (quantity of 2)

* ORU = Optimum Replaceable Unit

14 ZIP DISK DRIVE

14.1 Overview

The Iomega Zip disk drive uses a 100-megabyte (MB) disk for loading or backing up data.

Zip drive configurations can consist of:

- dual drives (if there is no Bernoulli cartridge drive) in a half upper pod or on a table top slide-tray.
- a single drive that coexists with a Bernoulli cartridge drive in a half upper pod or on a table top slide-tray.

The Universal Station node is resident in a Five-Slot Module or Dual Node Module and the Zip drive(s) is powered by the Acme power supply in the Five-Slot Module or a peripheral power supply in the half upper pod or table top cabinet.

Figure 14-1 illustrates the front panel of the Zip disk drive and identifies its features. Depressing the eject pushbutton normally ejects the disk. If the disk cannot be ejected by this method because of an electronic failure, the release hook that is located in the upper left corner of the front panel can be used to mechanically eject the disk. Use a small paper clip or similar tool to release the disk by pulling forward on the hook.

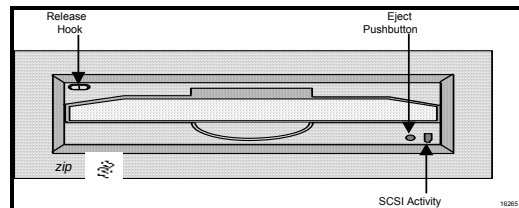


Figure 14-1 — Iomega Zip Disk Drive Front Panel

14. ZIP DISK DRIVE

14.2. Loading/Unloading the Drive

14.2 Loading/Unloading the Drive

NOTE

Avoid leaving the disk in the drive when it is not being used to reduce wear. The drive automatically retracts its heads after a period of time when not in use, but the disk still rotates. Drive rotation stops completely if the disk is not used during a 15-minute time period.

Use the procedure in Table 14-1 to load a disk into the Zip drive.

Table 14-1 — Zip Drive Disk Loading Procedure

Step	Action
1	Insert the disk into the drive with a smooth motion until a click is felt. The disk should start to rotate and the activity LED indicator should flash and then extinguish. The edge of the disk should be flush with the front surface of the drive.
2	The LED indicator is essentially a disk activity monitor. The intensity of the indicator represents the rate the data transfer on the SCSI bus.

Use the procedure in Table 14-2 to unload a disk from the Zip drive.

Table 14-2 — Zip Drive Disk Unloading Procedure

Step	Action
1	Press the eject pushbutton once and release it. The activity LED indicator should flash briefly. The disk will then be ejected.
2	Remove the disk by grasping the center of the disk.

14.3 Troubleshooting an Inoperative Zip Drive

In general, if two drives are installed and only one drive fails, the drive is most likely defective. If both drives are inoperative, check the power supply in the Five-Slot Module or the peripheral power supply.

Use the troubleshooting procedure in Table 14-3 to isolate a drive problem. Refer to the *Five/Ten-Slot Module Service* or *Dual Node Service* manual as needed.

Table 14-3 — Troubleshooting Procedure

Step	Action
1	Check the error (red) LED indicator on EPDG I/O or EPDG2 I/O board.
2	Check the indicators on the Five-Slot Module's power supply or the peripheral power supply.
3	If two drives are installed and both fail, and if applicable, disconnect the power cable from module's power supply and check for the presence of +5 Vdc and +12 Vdc at the power supply connector. Refer to the <i>Five/Ten-Slot Module Service</i> manual as needed. If the voltages are present, reconnect the power cable and go to Step 4. If the voltages are not present, check the module's power supply. If a peripheral power supply is present, check for the presence of +5 Vdc and +12 Vdc at its terminal strip.
4	Disconnect the power cable at the drive and check for the presence of +5 Vdc and +12 Vdc in the cable. • Pins 1 and 2 = 12 Vdc $\pm$.1 V • Pins 3 and 4 = 5.0 Vdc $\pm$.25 V If present, reconnect the power cable and go to Step 5.
5	If the power cable connection(s) is secure, replace the drive.
6	If the problem still exists, check the SCSI bus cable connections.

14. ZIP DISK DRIVE

14.4. Zip Drive Removal Procedure

14.4 Zip Drive Removal Procedure

CAUTION

Observe ESD precautions while handling the Zip drive.

Use the procedure in Table 14-4 to remove a Zip drive.

Table 14-4 — Zip Drive Removal Procedure

Step	Action
1	If the Zip drive is located in an upper pod, remove the pod's rear cover after removing the four screws that secure the cover. If the Zip drive is located in a desk top cabinet, open the cabinet's rear door, loosen the tray's fasteners, and extend the tray out of the cabinet.
2	Place the peripheral power supply's power switch in the OFF (0) position. If there is no peripheral power supply, power is provided by a Five-Slot Module's power supply. Place the Five-Slot Module's power supply's power switch in the OFF (0) position.
3	Remove the four hex-head retaining screws from each side of the slide-tray on which the Zip drive(s) is mounted.
4	Move the slide-tray backward enough to provide access to the Zip drive. Be careful not to move the slide-tray too far back on the support tray. Hint: Temporarily replace one of the slide-tray's attachment screws to hold the slide-tray in place while replacing the Zip drive.
5	Disconnect the dc power cable from the rear of the Zip drive. An adapter cable is used between the dc power harness connector and the drive.
6	Disconnect the 50-pin SCSI bus interface ribbon cable from the rear of the Zip drive.
7	Remove the four pan-head machine screws that attach the Zip drive assembly to the the vertical brackets on the slide-tray.

14.5 Zip Drive SCSI Bus Address Pinning Procedure

Figure 14-2 illustrates the method of pinning the Zip drive. Pinning for SCSI bus addresses of one (1) and zero (0) are illustrated in the figure. By normal convention, the drive that is pinned for a SCSI bus address of zero (0) is the end device on the bus, and the termination jumper must be in place. Duplicate the pinning of the drive that is being replaced.

If a RULA is installed, see the RULA section of this publication for special drive pinning instructions.

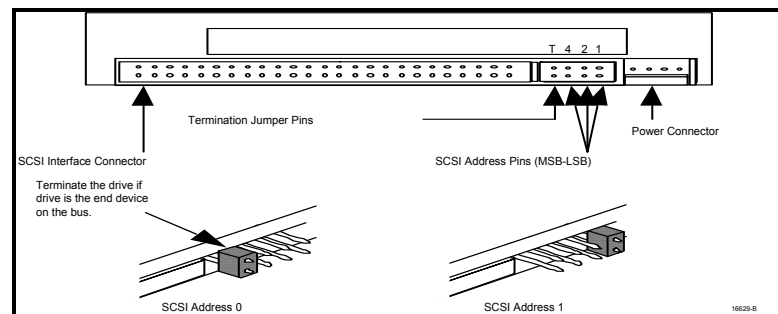


Figure 14-2 — Zip Drive SCSI Bus Pinning

14. ZIP DISK DRIVE

14.6. Zip Drive Installation Procedure

14.6 Zip Drive Installation Procedure

CAUTION

Observe ESD precautions while handling the Zip drive.

Use the procedure in Table 14-5 to install a replacement Zip drive.

Table 14-5 — Zip Drive Installation Procedure

Step	Action
1	Verify that the replacement drive is correctly pinned for its application. See Figure 14-2.
2	Attach the drive assembly to the vertical brackets on the slide-tray and secure it in place with the four screws that were previously removed.
3	Reconnect the dc power adapter cable at the rear of the drive.
4	Reconnect the 50-pin SCSI bus interface ribbon cable at the rear of the drive.
5	Move the slide-tray forward to the front of the station. Check to see that the faceplate is properly in contact with the bezel.
6	Install the four retaining screws at each side of the slide-tray. The screw slots in the slide-tray are used to align the tray with the faceplate/bezel.
7	Fold the SCSI bus interface ribbon cable as necessary. Fasten it in place with the cable clamp(s).
8	Place the peripheral power supply or Five-Slot Module power supply switch in the ON (1) position. Check the indicators for proper operation.
9	If the Zip drive is located in an upper pod, replace the pod's rear cover and secure the cover with the four screws that were previously removed. If the Zip drive is located in a desk top cabinet, return the tray to its normal position, secure it with its fasteners, and close the cabinet's rear door.

14.7 Spare Parts

Table 14-6 is a list of Zip drive related parts.

Table 14-6 — Zip Drive Replacement Parts

Part Number	Description
*51196933-100	Iomega 100-Megabyte Zip Drive Assembly
51304842-200	SCSI Bus Interface Cable (single drive)
51304199-201	SCSI Bus Interface Cable (dual drives) 3m
51304199-300	SCSI Bus Interface Cable (dual drives) 1m
51308055-100	DC Power Cable

* ORU = Optimum Replaceable Unit

14. ZIP DISK DRIVE

14.7. Spare Parts

15 CE COMPLIANT ZIP DISK DRIVE

15.1 Overview

The Iomega Zip disk drive uses a 100-megabyte (MB) disk for loading or backing up data.

Zip drive configurations can consist of:

- dual drives (if there is no Bernoulli cartridge drive) in a half upper pod or on a table top slide-tray.
- a single drive that coexists with a Bernoulli cartridge drive in a half upper pod or on a table top slide-tray.

The Universal Station node is resident in a Five-Slot Module or Dual Node Module and the Zip drive(s) is powered by the peripheral power supply inside the drive enclosure.

Figure 15-1 illustrates the front panel of the Zip disk drive and identifies its features. Depressing the eject pushbutton normally ejects the disk. If the disk cannot be ejected by this method because of an electronic failure, the release hook that is located in the upper left corner of the front panel can be used to mechanically eject the disk. Use a small paper clip or similar tool to release the disk by pulling forward on the hook.

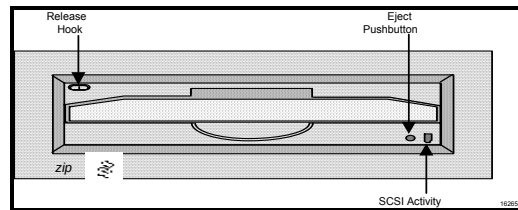


Figure 15-1 — Iomega Zip Disk Drive Front Panel

15.2 Loading/Unloading the Drive

NOTE

Avoid leaving the disk in the drive when it is not being used to reduce wear. The drive automatically retracts its heads after a period of time when not in use, but the disk still rotates. Drive rotation stops completely if the disk is not used during a 15-minute time period.

Use the procedure in Table 15-1 to load a disk into the Zip drive.

Table 15-1 — Zip Drive Disk Loading Procedure

Step	Action
1	Insert the disk into the drive with a smooth motion until a click is felt. The disk should start to rotate and the activity LED indicator should flash and then extinguish. The edge of the disk should be flush with the front surface of the drive.
2	The LED indicator is essentially a disk activity monitor. The intensity of the indicator represents the rate the data transfer on the SCSI bus.

Use the procedure in Table 15-2 to unload a disk from the Zip drive.

Table 15-2 — Zip Drive Disk Unloading Procedure

Step	Action
1	Press the eject pushbutton once and release it. The activity LED indicator should flash briefly. The disk will then be ejected.
2	Remove the disk by grasping the center of the disk.

15.3 Troubleshooting an Inoperative Zip Drive

In general, if only one drive fails, the drive is most likely at fault. If both drives are inoperative, check the drive enclosure's peripheral power supply.

Use the procedure in Table 15-3 to isolate the Zip drive problem. Refer to the *Five/Ten-Slot Module Service* or *Dual Node Service* manual as needed.

Table 15-3 — Troubleshooting Procedure

Step	Action
1	Check the error (red) LED indicator on EPDG I/O or EPDG2 I/O board.
2	Remove the drive enclosure's top cover.
3	Check to be sure that the power cable connection(s) inside the drive enclosure is proper.
4	Disconnect the adapter power cable from the Zip drive and check for the presence of +5 Vdc and +12 Vdc in the cable. • Pins 1 and 2 = 12 Vdc $\pm$.1 V • Pins 3 and 4 = 5.0 Vdc $\pm$.25 V If present, replace the drive. If not present, replace the peripheral power supply.
5	If the problem still exists, check the SCSI bus interface cable connections.

15. CE COMPLIANT ZIP DISK DRIVE

15.4. Zip Drive Removal Procedure

15.4 Zip Drive Removal Procedure

CAUTION

Observe ESD precautions while handling the Zip drive.

Use the procedure in Table 15-4 to remove a Zip drive.

Table 15-4 — Zip Drive Removal Procedure

Step	Action
1	If the Zip drive is located in an upper pod, remove the rear cover from the pod after removing the four screws that secure it in place. If the Zip drive is located in a table top cabinet, open the rear door, release the latches at each side of the slide-tray, and fully extend the tray.
2	If the Zip drive is located in an upper pod, disconnect the power and interface cables from the back of the drive enclosure. Label them, if necessary, to ensure that they will be properly reconnected.
3	If the Zip drive is located in an upper pod, remove the eight hex-head screws that attach the drive tray to the cabinet structure. Move the tray backward enough to provide access to the drive enclosure's top cover.
4	Remove the 16 screws that attach the drive enclosure's top cover.
5	Remove the top cover.
6	Disconnect the 4-pin dc power adapter cable from the rear of the Zip drive.
7	Disconnect the 50-pin SCSI bus ribbon cable from the rear of the Zip drive.
8	Remove the four machine screws that attach the Zip drive assembly to the vertical mounting brackets.
9	Remove the Zip drive assembly from the drive enclosure.

15.5 Zip Drive SCSI Bus Address Pinning Procedure

Figure 14-2 illustrates the method of pinning the Zip drive. Pinning for SCSI bus addresses of one (1) and zero (0) are illustrated in the figure. By normal convention, the drive that is pinned for a SCSI bus address of zero (0) is the end device on the bus, and the termination jumper must be in place. Duplicate the pinning of the drive that is being replaced.

If a RULA is installed, see the RULA section of this publication for special drive pinning instructions.

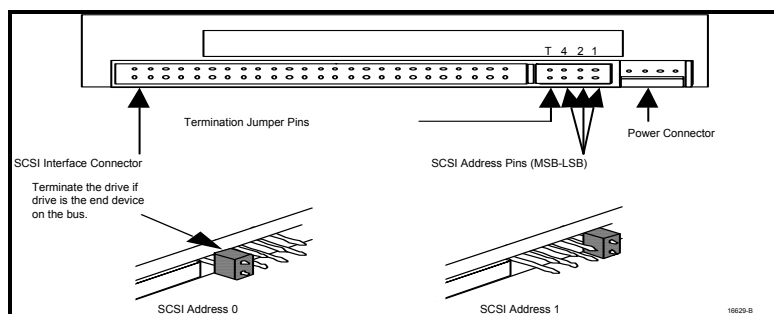


Figure 15-2 — Zip Drive SCSI Bus Pinning

15.6 Zip Drive Installation Procedure

CAUTION

Observe ESD precautions while handling the Zip drive.

Use the procedure in Table 15-5 to install a replacement Zip drive.

Table 15-5 — Zip Drive Installation Procedure

Step	Action
1	Be sure the replacement drive is pinned correctly. Attach the drive to the vertical mounting brackets in the drive enclosure with the four screws that were previously removed.
2	Reconnect the dc power adapter cable to the rear of the drive.
3	Reconnect the 50-pin SCSI bus interface ribbon cable to the rear of the drive.
4	Place the cover on the drive enclosure and secure it with the sixteen screws that were previously removed. All screws must be in place for EMI containment.
5	If the Zip drive is located in an upper pod, move the slide-tray forward against the front bezel. Attach the drive-tray to the cabinet structure with the eight hex-head screws that were previously removed.
6	Reconnect the power and SCSI bus interface cable to the drive enclosure.
7	If the Zip drive is located in an upper pod, replace the rear cover on the pod and secure it with the four screws that were previously removed. If the Zip drive is located in a table top cabinet, release the slide-tray stops and slide it fully into the cabinet. Secure it in place with its latches. Close the rear door.

Figure 15-3 illustrates the drive enclosure's internal and external SCSI bus cabling.

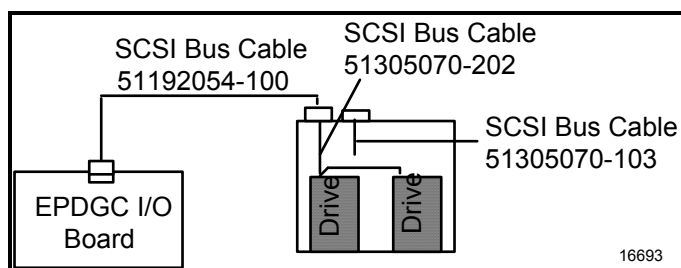


Figure 15-3 — Drive Enclosure Internal/External SCSI Bus Cabling

15.7 Spare Parts

Table 15-6 is a list of Zip drive related parts.

Table 15-6 — Zip Drive Replacement Parts

Part Number	Description
*51196933-100	Iomega 100-Megabyte Zip Drive Assembly
51305070-202	Drive Enclosure Interior SCSI Bus Interface Cable (3-connector)
51305070-103	Drive Enclosure Interior SCSI Bus Interface Cable (2-connector)
51192054-102	EPDGC I/O Board to Drive Enclosure SCSI Bus Cable

* ORU = Optimum Replaceable Unit

15. CE COMPLIANT ZIP DISK DRIVE

15.7. Spare Parts

16 ASPI-32 PRINTER

16.1 Cleaning

Clean the Printer using a soft-bristled brush and vacuum. Set the power switch (see Figure 16-1) to OFF and unplug the power cord. Lift the top-half of the cover and remove the paper, if present. Vacuum the dirt/dust from inside of the unit. Use a brush to loosen dirt accumulations, but take care not to damage or misalign delicate items, such as the print head. Close the cover, and wipe the outside clean with a cloth and mild detergent solution. Plug the power cord back into the unit.

16.2 Adjustments

(See subsection 16.4, *Removal/Replacement*)

16.3 Test/Troubleshooting

The matrix printer includes a built-in diagnostic test, intended for use by the operator. See Figure 16-1 for power switch and control panel location. To initiate the test:

1. Set the power switch to OFF. Ensure that there is paper in the printer. Press the START button while setting the power switch to ON, then release the START button. The printer prints the full character-set. Press the STOP button to stop printing. The print should be clear and complete; if not, see Table 16-1 Troubleshooting Guide.

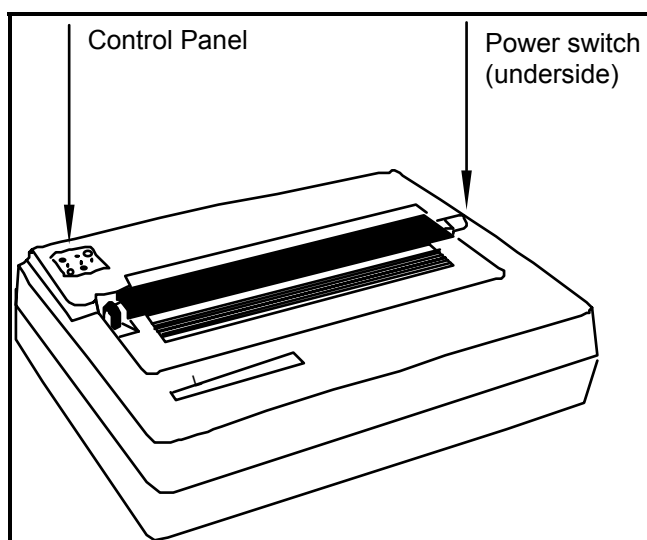


Figure 16-1 — ASPI-32 Printer

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2. Set the power switch to OFF. Press and hold the BREAK button while setting the power switch to ON, then release the button. The ERROR, STAND BY, ON LINE, and PAPER OUT light indicators blink sequentially, then flash in unison for about five seconds before the ON LINE indicator steadily lights while the other indicators are off. If a fault is indicated, see Table 16-2 for test display description.

Table 16-1 — Troubleshooting Guide

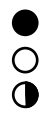
	Self Test Cause	Symptom
NO	Unit does not turn ON.	1. AC fuse 2. PWA BROSY 3. Bulk PS
NO	Unit not initialized.	1. Fuses, PWA BROSY 2. PWA BROSY

		3. Optical Sensor
NO	Unit initialized, plus PAPER OUT and ERROR lit.	1. No paper 2. Paper switch
YES/NO	One or more needles do not print.	1. PWA BROS Y 2. Print head
YES	Inked ribbon does not turn. Printing correct.	1. Ribbon cords 2. Print head
YES/NO	Jammed on first line of print after first few lines and is repetitive; shift to left or right of the line of print.	1. PWA BROS Y 2. Carriage rods dirty 3. Ribbon jammed 4. Optical reader 5. Carriage motor
YES	First printed characters expanded.	1. Belt slackened 2. Pulley slackened
YES	Line feed not executed or badly accomplished. Page set not executed.	1. PWA BROS Y 2. Paper knob 3. Paper motor
YES	Tabulations not executed only line feed executed.	1. Thermistor
YES	Does not make connection with console module (PIC).	1. PWA ROS-A 2. Line trouble
NO	Print quality is poor on left and/or right side, even after ribbon replacement.	1. Penetration Adj. may be set back too far.

Table 16-2 — Test Display Description

Printer Panel Indicators	Fault Analysis	
	1.	Test on RAM of DA. If there is a fault display, replace PWA ROS-A.
	2.	Test on EPROM of DA (U13). If there are fault displays, replace it (wired on PWA ROS-A).
	3.	Test on EPROM of DA (U11). If a fault is displayed, replace it (wired on PWA ROS-A).
	4.	Test on series I/O gate of DA. When a fault is displayed, replace PWA ROS-A.
	5.	Test on DA parallel I/O gate. When a fault is displayed, replace PWA ROS-A.
	6.	Test on interface bus between DA and DOE. When a fault is displayed it will have occurred on PWA BROSY or PWA ROS-A.
	7.	Test on RAM of DOE. When a fault is displayed, replace PWA BROSY.
	8.	Test on the interface of DOE towards the print head and the carriage. When a fault is displayed, replace PWA BROSY.

Table 16-2 — Test Display Description

Printer Panel Indicators		Fault Analysis	
		9.	Test on EPROM of DOE. When a fault is displayed, replace it (wired on PWA BROSY).
		10.	Test on EPROM of DOE. When a fault is displayed, replace PWA BROSY.
		END	Display of end of diagnostic test.
Indicator Legend		Note	
		Displays other than those shown, indicate major errors beyond the scope of the diagnostic test. Check the indicator lamps and panel.	
LIT			
OFF			
BLINKING			

3740

16.4 Removal/Replacement

The following parts replacement and adjustment procedures are to be performed by the maintenance technician, as called for in the Troubleshooting Guide, Table 16-1, and the Test Display Description, Table 16-2.

CAUTION

Before performing any maintenance or parts replacement, disconnect the printer power cord from the power entry box at the rear of the printer.

To gain access to the printer, turn off the power switch, remove paper and raise top cover by pushing up on the front of the cover. Before replacing the OPRO1, BROSY, or ROS-A boards, set the configurator switches and jumpers to the same configuration as the removed PWA, see Figures 16-2 through 16-8.

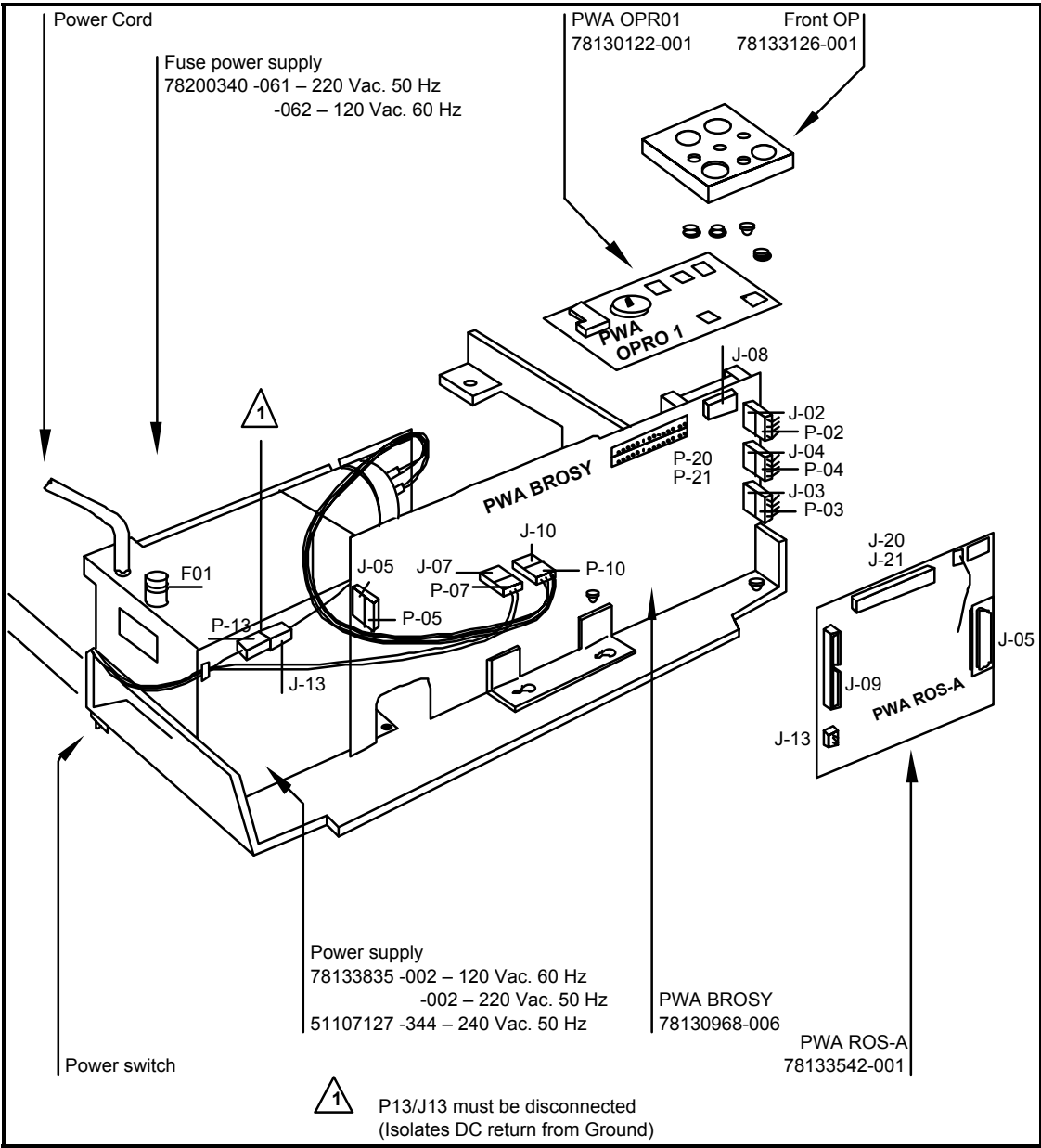


Figure 16-2 — PWA and Bulk Power Assembly

1653

16. ASPI-32 PRINTER

16.4. Removal/Replacement

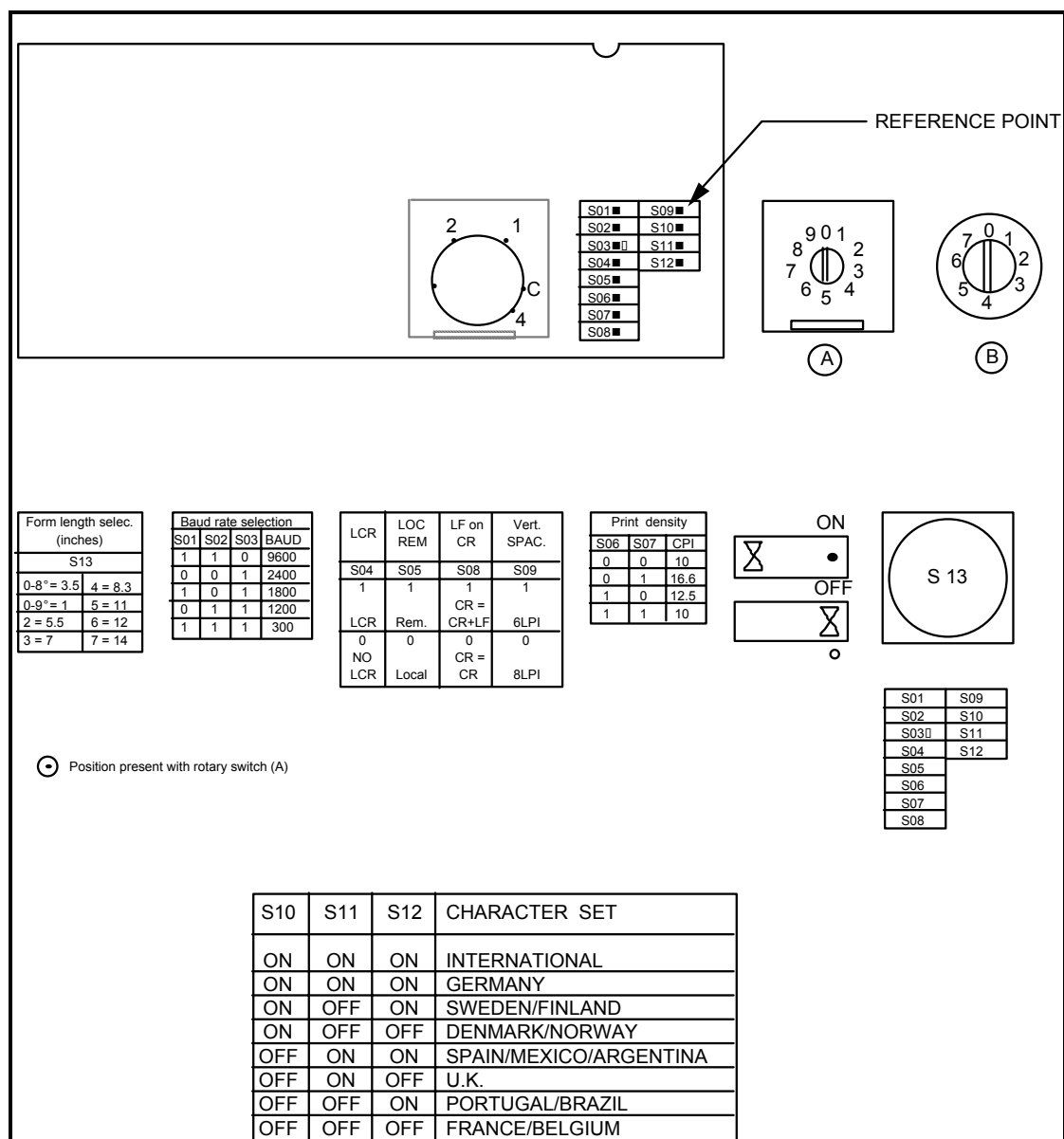


Figure 16-3 — Print/Function Selection

1654

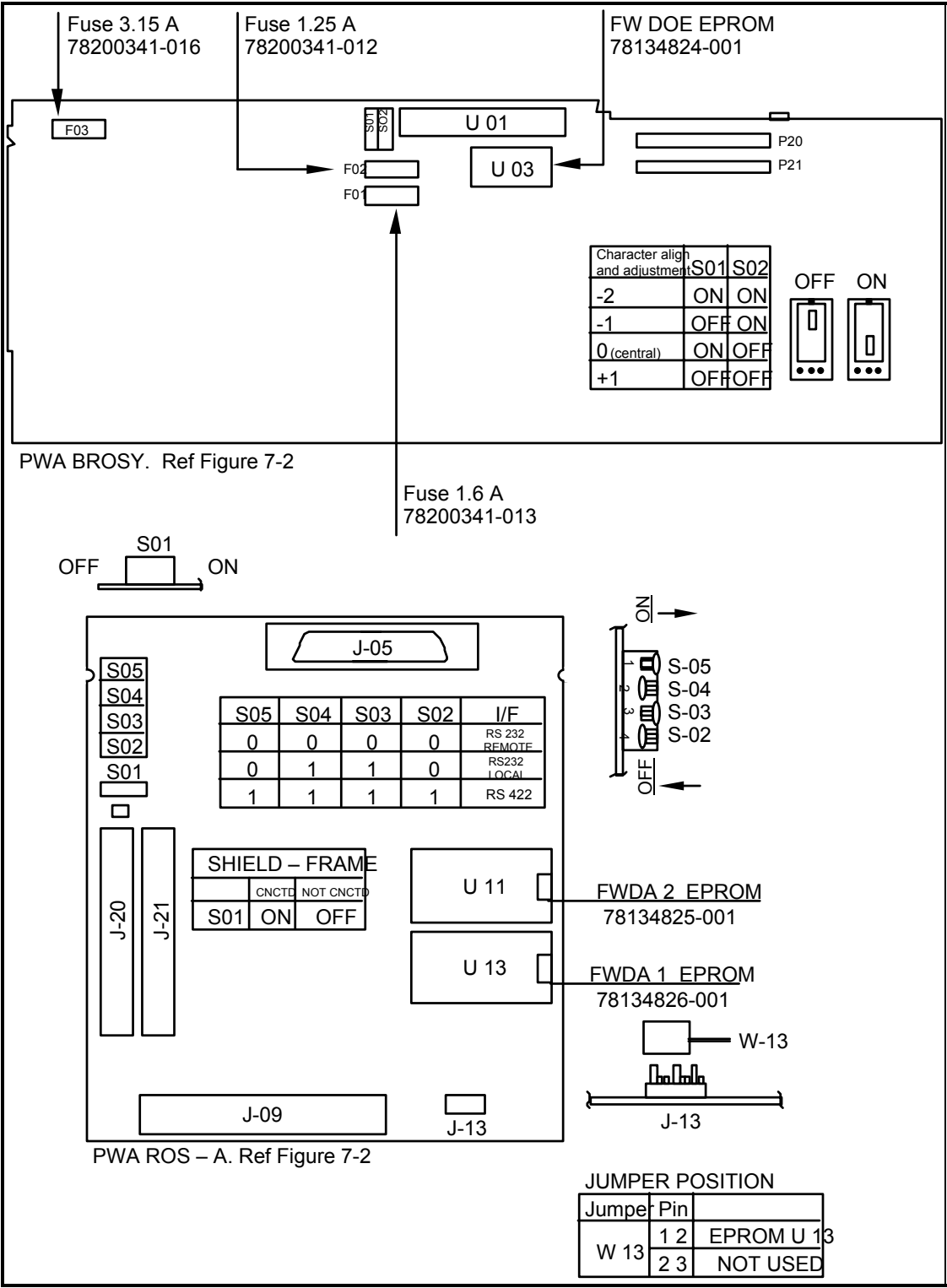


Figure 16-4 — PWA Pinning Options 1655

16.4. Removal/Replacement

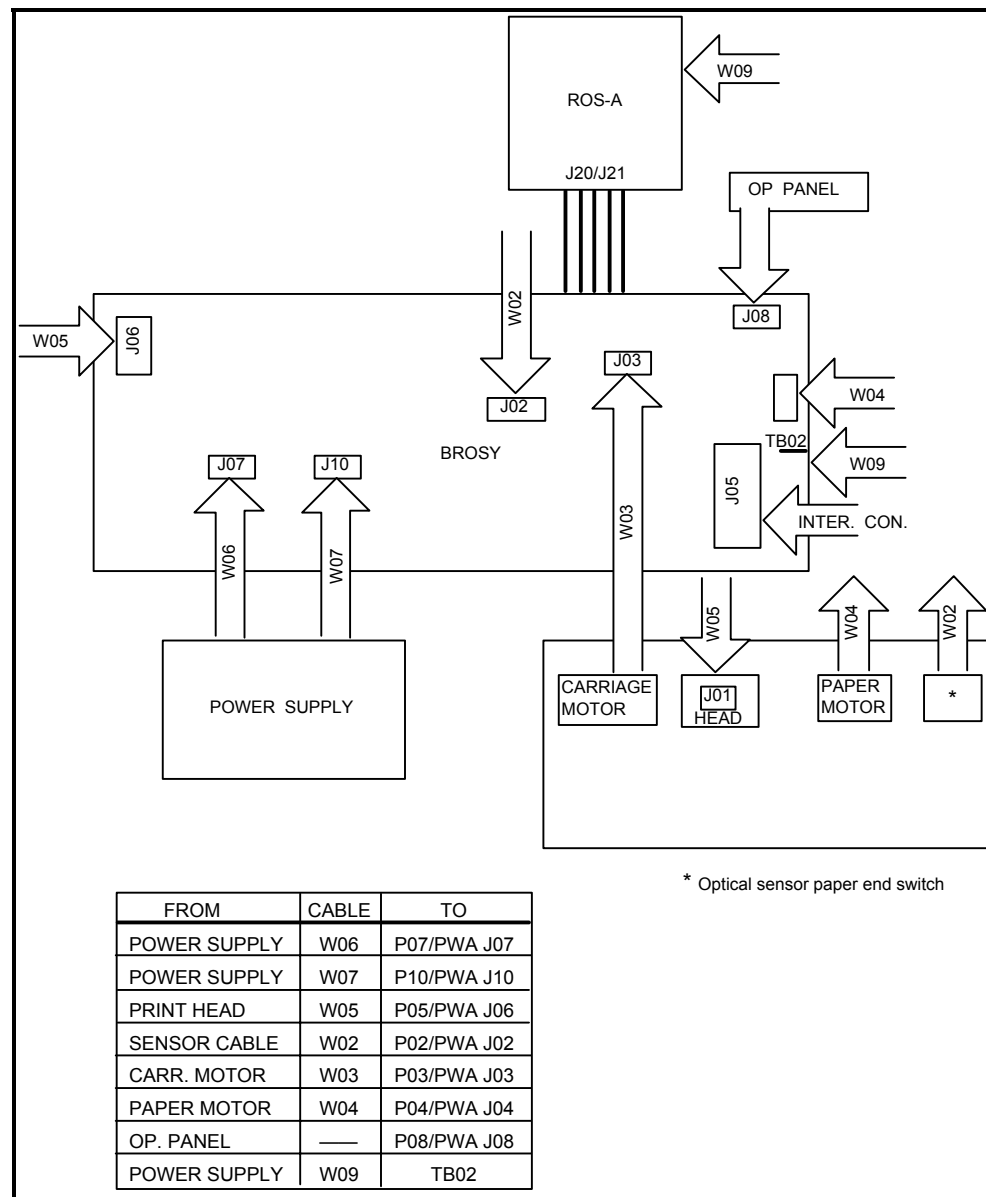


Figure 16-5 — Printer Intercabling

1656

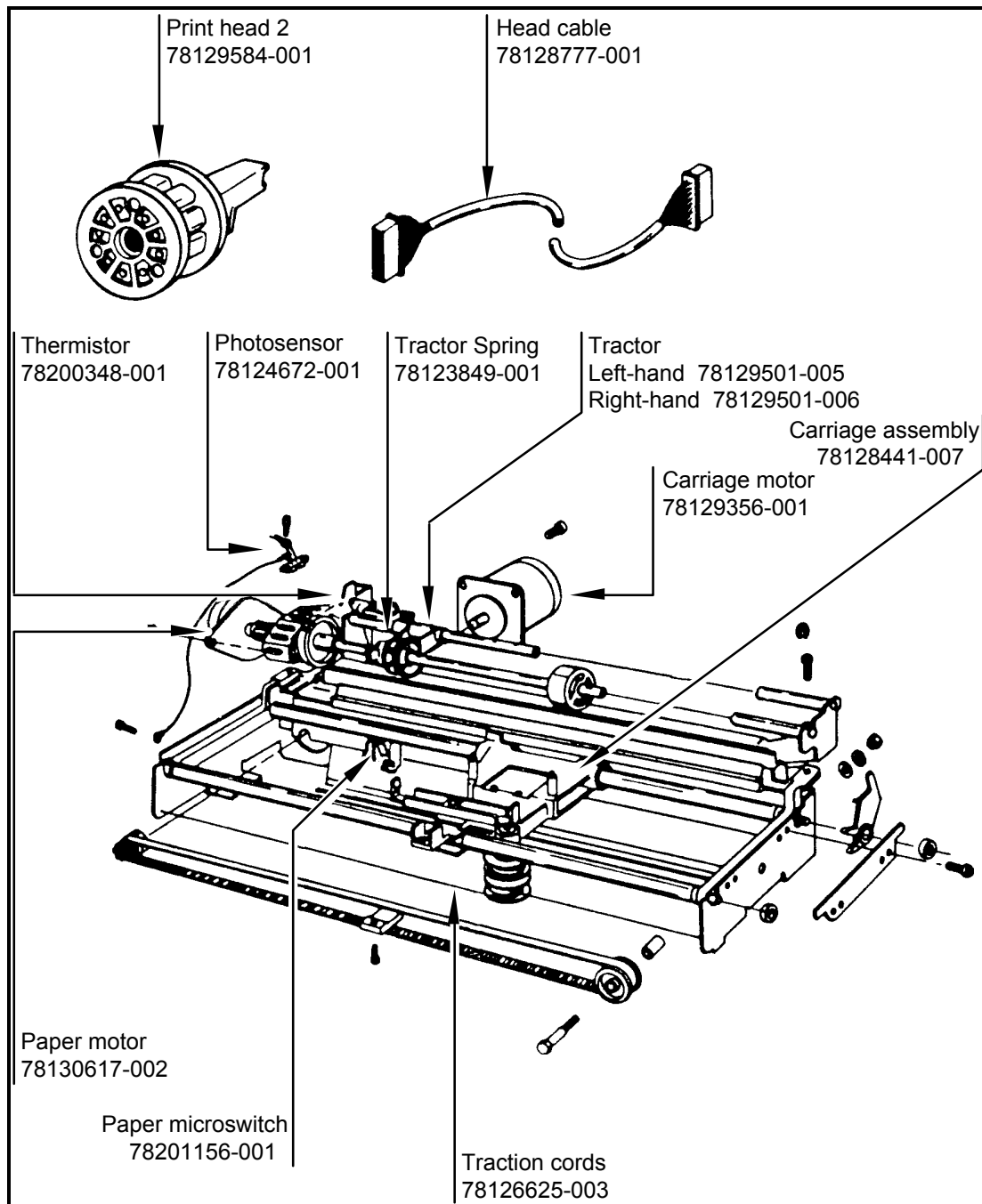


Figure 16-6 — MIRA Mechanical Assembly

1657

16. ASPI-32 PRINTER

16.4. Removal/Replacement

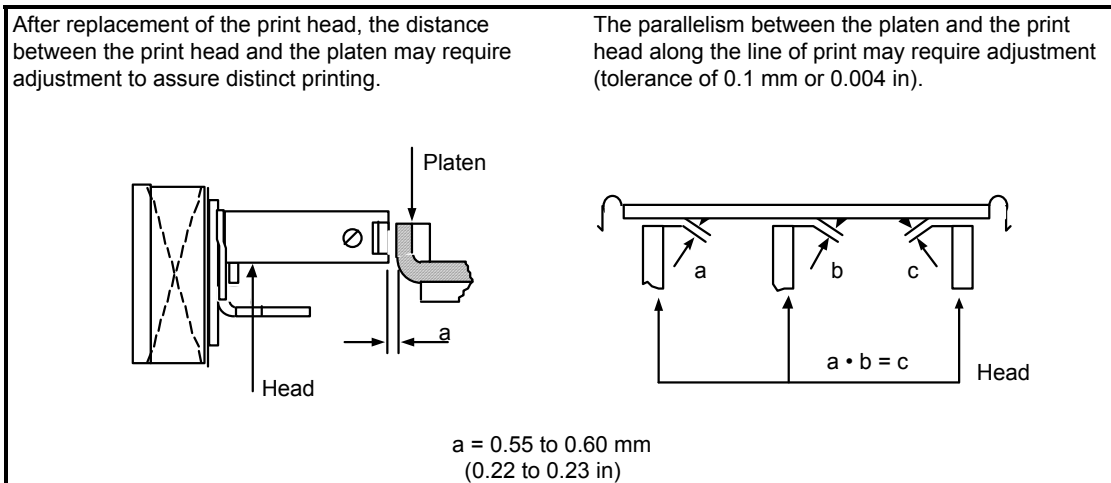


Figure 16-7 — Print Head Adjustment 1658

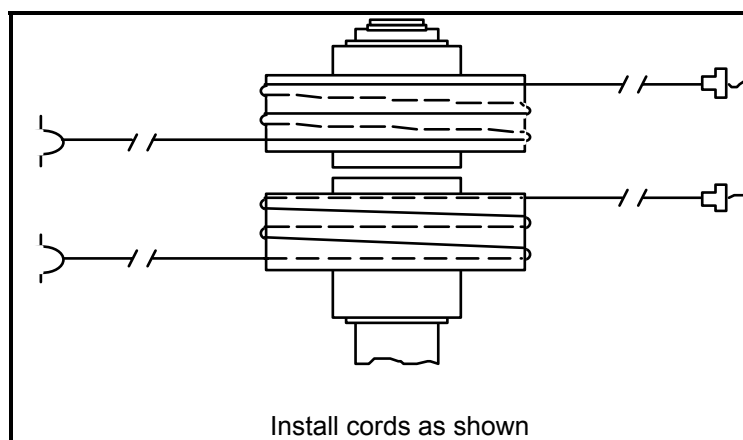


Figure 16-8 — Inked Ribbon Traction Cords Positioning

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16.5 Spare Parts

Table 16-3 — ASPI-32 Printer Spare Parts

Part Number	Description
* 51108254-400	Printer, 120 Vac, 47-63 Hz
* 51108254-500	Printer, 220 Vac, 47-63 Hz
* 51108254-700	Basket, Paper
* 78130968-009	PWA BROSY
* 78133542-001	PWA ROS-A
* 78130222-001	PWA OPR01
51303631-003	Cable, RS-232 Direct Connect, 3 m (PIC, PDG, and EPDG(P) I/O board)
51303631-004	Cable, RS-232 Direct Connect, 4 m (PIC, PDG, and EPDG(P) I/O board)
51303631-008	Cable, RS-232 Direct Connect, 8 m (PIC, PDG, and EPDG(P) I/O board)
51303631-010	Cable, RS-232 Direct Connect, 10 m (PIC, PDG, and EPDG(P) I/O board)
51303631-015	Cable, RS-232 Direct Connect, 15 m (PIC, PDG, and EPDG(P) I/O board)
51308103-003	Cable, RS-232 Direct Connect, 3 m (EPDGC I/O board)
51308103-008	Cable, RS-232 Direct Connect, 8 m (EPDGC I/O board)
51308103-015	Cable, RS-232 Direct Connect, 15 m (EPDGC I/O board)
78133126-001	Front OP
* 78200340-061	FUSE, Power Supply —220 Vac 50 Hz
* 78200340-062	FUSE, Power Supply —120 Vac 60 Hz
* 78200341-013	FUSE, PWA F01 1.6 A
* 78200341-012	FUSE, PWA F02 1.25 A
* 78200341-016	FUSE, PWA F03 3.15 A
78134824-001	FW DOE PROM
78134825-001	FW DA 2 PROM
78134826-001	FW DA 1 PROM

16. ASPI-32 PRINTER

16.5. Spare Parts

Table 16-3 — ASPI-32 Printer Spare Parts

Part Number	Description
78140531-001	Carriage Assembly
78129356-001	Carriage Motor
78128777-001	Head Cable
78130617-002	Motor, Paper
78201156-001	Microswitch, Paper
78124672-001	Photosensor
* 78133835-002	Power Supply, 120 Vac 60 Hz
* 78133835-001	Power Supply, 220 Vac 50 Hz
* 78129584-001	Print Head 2 (NFXT use)
* 78123849-001	Spring, Tractor
78200348-001	Thermistor
78126625-003	Traction Cords
* 78129501-005	Tractor, Left-hand
* 78129501-006	Tractor, Right-hand

* ORU Level Replacement Item

17 ASPI-41 PRINTER

NOTE

The ASPI-32 printer (described elsewhere in this manual) is no longer available from the manufacturer. This printer, the ASPI-41, is a functional replacement for the ASPI-32 printer.

17.1 Cleaning

Clean the exterior of the printer, using a soft-bristled brush and vacuum. Turn off the ac power switch located on the rear of the printer, just above the power cord, then unplug the power cord. Lift the top transparent cover from the rear and remove any paper, if present. Vacuum the dirt and dust from inside of the unit. Use a brush to loosen dirt accumulations, but take care not to damage or misalign delicate items, such as the print head. Close the cover and wipe the outside clean with a cloth and mild detergent solution. Plug the power cord back into the unit and turn power on if required.

17.2 Adjustments

There are no adjustments to this printer at the ORU (Optimum Replaceable Unit) level.

A skilled user, however, may choose to replace parts and make adjustments beyond the ORU level. See subsections 17.5 and 17.6 for assistance in removing and replacing those parts and making the necessary adjustments.

17.3 Test/Troubleshooting

To help you isolate and correct minor printer problems, the topics covered here include problem solving, the printer self-tests, the built-in printer diagnostic procedure, and the error messages resulting from the diagnostic procedure.

When your printer has a problem, follow this troubleshooting sequence:

1. Use the Problem Solving Guide (Table 17-1) to identify and correct problems.
2. Run the Printer Self-Test. The instructions are in subsection 17.3.2.
3. Run the Diagnostic Procedure. The instructions are in subsection 17.3.3.
4. Call the Honeywell Technical Assistance Center (TAC). In the U.S.A. use our toll-free number 1*800-822-7673 (available in the contiguous states except Arizona; in Arizona dial 863-5558).

17.3.1 Problem Solving

Use the guide in Table 17-1 to help identify and correct printer problems.

Table 17-1 — Problem Solving Guide

Problem	Possible Cause	Solution
POWER indicator does not light.	Power Switch off.	Turn printer on.
	Power cable not connected properly.	Verify that the cable is connected properly to the printer and the ac power outlet.
Printer does not print.	Display window shows WAIT .	Press the ON LINE button to enable printing.
	Interface cable is not properly connected.	Push cable firmly into connectors at both ends, and make sure the screws are secure.
Continuous forms paper is not advancing.	Paper holes are torn.	Remove torn paper and replace with the next good sheet. Check tension of paper between left and right sprocket wheels. If too tight, loosen by moving the right tractor slightly to the left.

Table 17-1 — Problem Solving Guide

Problem	Possible Cause	Solution
	Paper is not correctly aligned on sprocket wheels.	Reload paper, ensuring that the corresponding holes at each side of the paper are correctly aligned on the sprocket wheels.
Print fading.	Ribbon not feeding.	Check that ribbon is correctly loaded (see subsections 17.4.4 and 17.4.5). Turn ribbon advance knob to ensure that ribbon is not jammed.
	Ribbon worn or damaged.	Replace ribbon (see subsections 17.4.4 and 17.4.5).
	Print head too far from paper.	Push paper thickness lever towards the back of the printer to move print head closer to the paper. Run self-test (see subsection 17.3.2) as a check.
	Print head is worn or damaged.	Replace print head (see subsection 17.4.6) and run self-test (see subsection 17.3.2).
Dark, smudgy printing.	Print head too close to paper.	Pull paper thickness lever towards the front of the printer to move print head away from the paper.
Self-test not printed.	ON LINE not pressed while turning the printer on.	Repeat the sequence (see subsection 17.3.2).
	Carriage fault.	Press ON LINE and repeat the sequence (see subsection 17.3.2).

Table 17-1 — Problem Solving Guide

Problem	Possible Cause	Solution
Display window shows JAM.	Paper jammed or torn.	Check paper. If it is jammed or torn, try to move paper forwards or backwards using the or keys. Otherwise, try to extract the paper manually. Note: Clear JAM indicator by turning power off, then on.
The display shows FLT.	A carriage synchronization error has occurred. The font cartridge has not been inserted or removed correctly.	Turn the printer off, and then on. Check cartridge. Turn the printer off and then on.
Single sheet is not loaded correctly.	You have not inserted the single sheet correctly.	Insert single sheet well into the front paper entry slot, making sure that it reaches the platen. The sheet must be pushed a little further in after P.E. disappears from the display.

17.3.2 Printer Self-Tests

After installing the printer (see subsection 17.4.2), test its operation by using the self-test procedure. Before beginning the self-test, verify that:

1. The paper and ribbon cartridge are inserted correctly.
2. The power cable is connected and the power switch (right rear) is off.
3. The printer cover is closed.

Perform the following steps:

1. Press and hold the **ON LINE** button while powering the printer on.

The display window shows **TEST**, and a pattern is printed on the loaded paper as shown in Figure 17-1. The printer continues to print until you press the **ON LINE** button.

2. The display window shows **WAIT**.

You may run self-test with the interface cable disconnected. If the interface cable is not connected to a operating host, the printer will show **OFFD** rather than **WAIT** when stopped.

```

ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
BCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
CDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
DEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
EFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
FGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
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PQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
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WXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
XYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£

```

Figure 17-1 — Self-Test Print Pattern 3659

NOTE

1. The self-test also ends if the printer runs out of paper and the window shows **P.E.**
2. The self-test pattern can be printed in Quality print by pressing the **QUALITY** button.
3. You can also select different character fonts for self-test by inserting optional font cartridges and pressing **FONT**.

17.3.3 Diagnostic Procedure

This diagnostic procedure helps identify printer problems so that when a unit fails, you know for certain it is the printer at fault and not some other part of the subsystem.

The diagnostic procedure checks the control panel indicators, the display window, and internal printer functions, then prints out its hardware and firmware configuration. At the end of the diagnostic, it either displays **?END** or an error number if an error is found.

Before starting, verify that:

- The printer power is turned off.
- The printer's top and rear covers are closed.
- Paper is loaded.
- The LCN-based system is not sending data.

As you perform the diagnostic procedure, note any problems you find. To run the diagnostic, follow these steps:

1. Press the  button while turning on the power to the printer. The indicators on the control panel light up.
2. The display window shows, in a sequence of a to e, the following patterns to check its display capability. As each pattern is displayed, the printer "beeps." Verify that the display window shows:

- | | | | | |
|----|---|---|---|--|
| a. |  |  |  |  |
| b. | * |  |  |  |
| c. | * | * |  |  |
| d. | * | * | * |  |
| e. | * | * | * | * |

After displaying the above sequence, the diagnostic procedure automatically enters test execution.

3. The display window shows **T&Dx**, where **x** is an alphanumeric number in sequence as the test continues. Some of the T&D tests may take several seconds while some are finished in an instant.
4. When the display window shows **T&DG**, the printer prints configuration information similar to that shown on Figure 17-2. The configuration information is:
 - epr Indicates firmware release
 - sio Indicates serial interface
 - ocg Indicates if the font cartridge is present
 - asf Indicates if the automatic sheet feeder has been connected to the printer

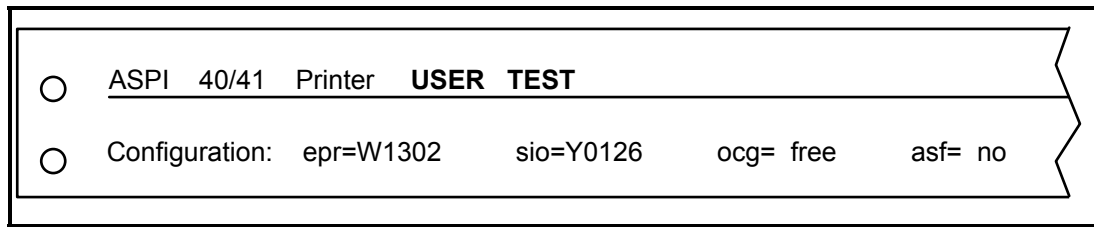


Figure 17-2 — Printer User Diagnostic Test Message 3011

5. The display reports it is an ASPI-41 by flashing **A 41**, then the test terminates by either:

- Flashing **?END** for 5 seconds, or
- Displaying an error message in the format **ORUx** where **x** is the error number. The printer will also "beep" three times and the indicators will blink. Table 17-2 lists the possible error messages.

6. After displaying **?END** for 5 seconds, the printer is ready to print.

If you press the **PROG** button while either **?END** or the error message is displayed, the diagnostic procedure will be repeated. It can be stopped by pressing **PROG** within 5 seconds of the subsequent **?END** display.

Table 17-2 — Diagnostic Procedure Error Messages

Error Number (X)	Description
ORU0	General fault
ORU1	Fault on main board
ORU2	Fault on interface board
ORU3	Fault on chip U03 (interface board)
ORU4	(Reserved for future use)
ORU5	Fault on operator panel
ORU6	Character generator fault
ORU7	Paper load lever fault

17.4 ORU Removal and Replacement

17.4.1 Printer Removal

Turn off the ac power switch located on the rear of the printer, just above the power cord. Disconnect the power cord. Disconnect the data cable from the left rear of the printer.

17.4.2 Printer Replacement or Installation

Check that the line voltage is the same as the voltage requirement displayed on the back of the printer. If the line voltage does not match the printer voltage, do not use the printer; instead, seek advice from your supplier.

1. Make sure that the power switch on the back of the printer is in the off position (0), then insert the power cable plug into the socket on the back of the printer.
2. Insert the other end of the power cable in a convenient outlet.
3. Remove the transparent cover as shown in Figure 17-4, illustration 1. Remove any shipping ties or blocks in the printer. If your printer was shipped from the factory, the paper chute was packed separately in the carton. See subsection 17.4.3 for the procedure for attaching the chute. If not already installed, install a ribbon cartridge as covered in subsection 17.4.5.
4. Connect one end of the 25-pin interface cable firmly to the printer I/O interface on the LCN-based host equipment (check the specific Honeywell manual for details), then connect the other end to the socket at the rear of the printer. Tighten all screws on the cable ends to insure the cable will remain in place.
5. To power the printer on, press the part of the power switch marked 1 on the back of the printer. Note that the print head moves and stops near the left side of the printer.

17.4.3 Paper Chute Installation

When the printer was packed for shipment, the paper chute was placed separately in the carton. To install it correctly, stand in front of the printer holding the chute in the position shown in Figure 17-3, illustration 4.

Numbers on illustrations in Figure 17-3 refer to the steps which follow.

- Step 1. Insert the hook on the left side of the chute into the hole near the paper entry slot.
- Step 2. Bend the chute towards the front of the printer and insert the hook on the right side into the corresponding hole in the printer cabinet.
- Step 3. The paper chute assumes two positions depending upon the type of paper used:
- Raised (**A** position) for single sheet use.
 - Lowered (**B** position) for continuous forms paper (common usage).
- Step 4. To position the chute down, use both hands on either side of the chute to lift it up, then lower it gently until it stops. To position the chute up, hold the chute with a hand on either side and place it in the **A** position. Now gently, but firmly push it toward the printer body and note the chute moves about a centimeter and locks.

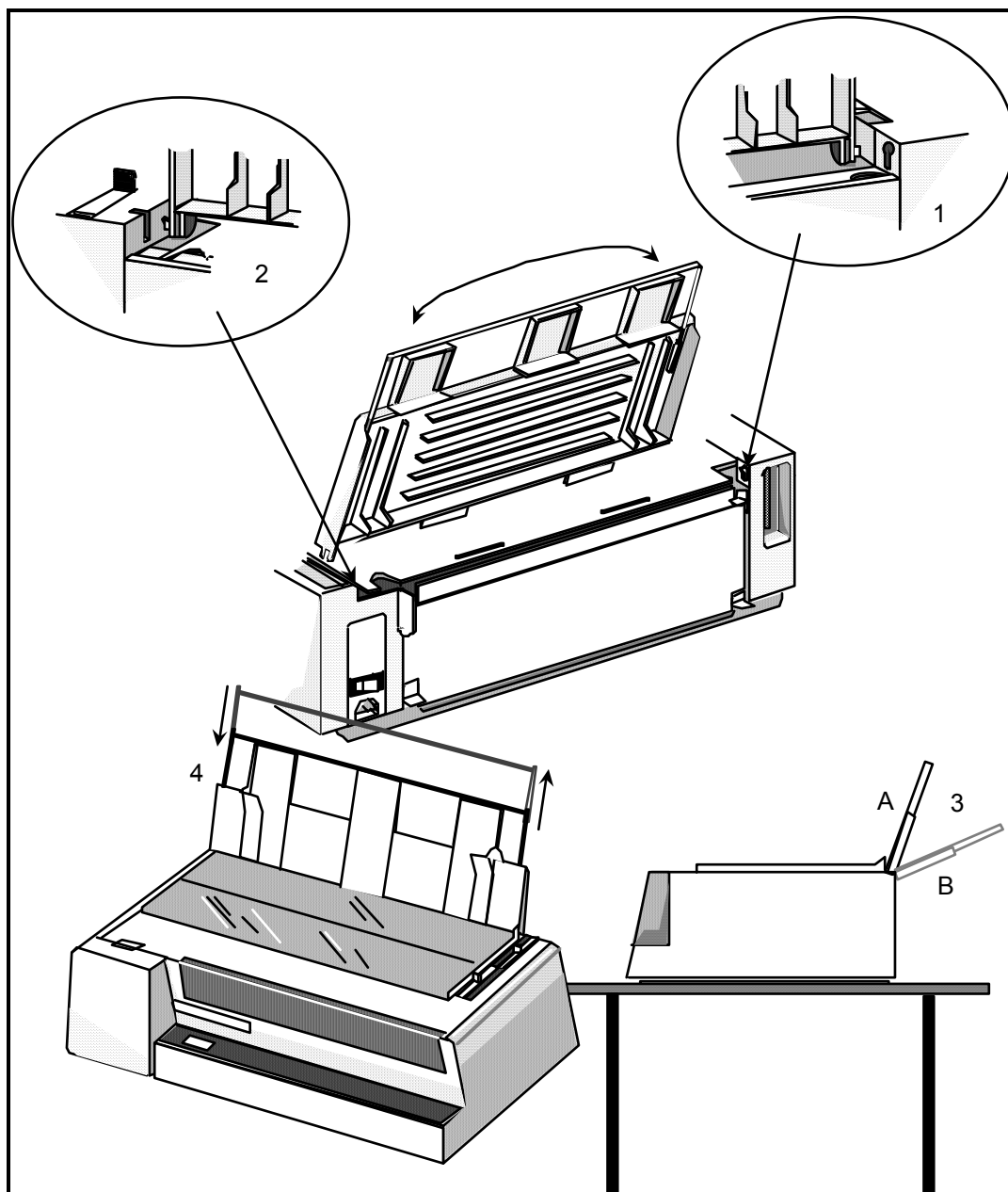


Figure 17-3— Paper Chute Installation

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17. ASPI-41 PRINTER

17.4. ORU Removal and Replacement

17.4.4 Ribbon Removal

NOTE

If power is turned off with data in the input buffer, the data will be lost.

The ribbon cartridge can be removed and replaced either with the printer on (the display window showing **P.E.** or **WAIT**) or with the printer off.

Numbers on illustrations in Figure 17-4 refer to the steps which follow.

- Step 1. Remove the printer cover.
- Step 2. Move the printer carriage to the center of the printer. **DON'T** perform this step if the printer is on.
- Step 3. Grasp the cartridge firmly by its sides and lift the ribbon cartridge straight up until the side clips release it.

Now you are ready to insert a new ribbon cartridge by continuing with subsection 17.4.5.

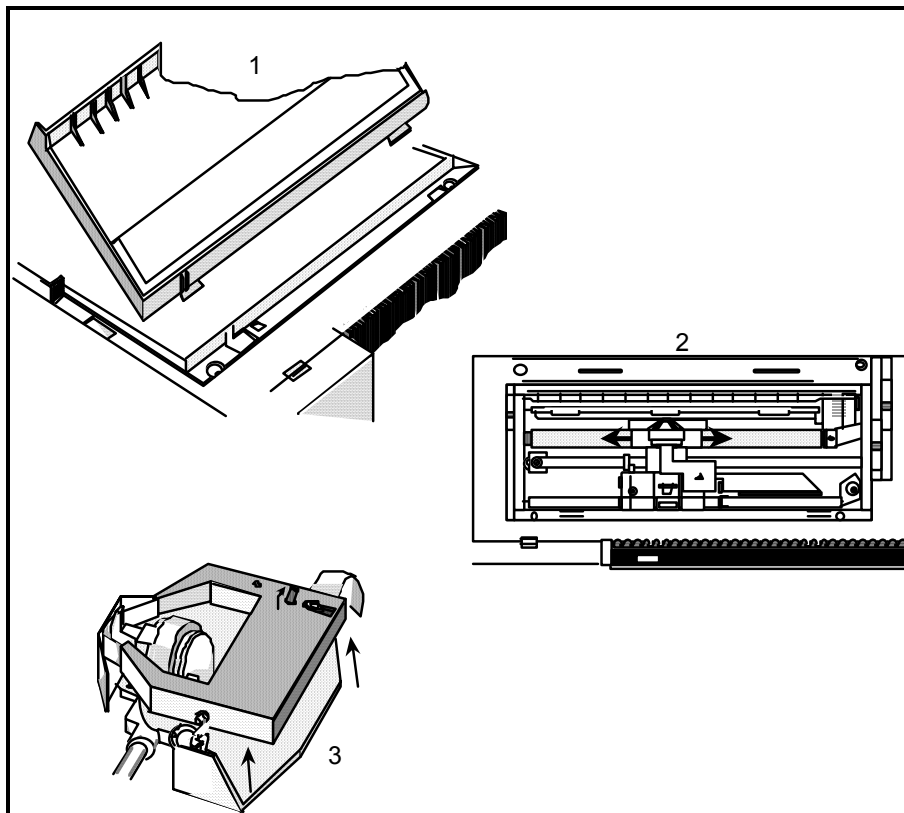


Figure 17-4 — Ribbon Removal

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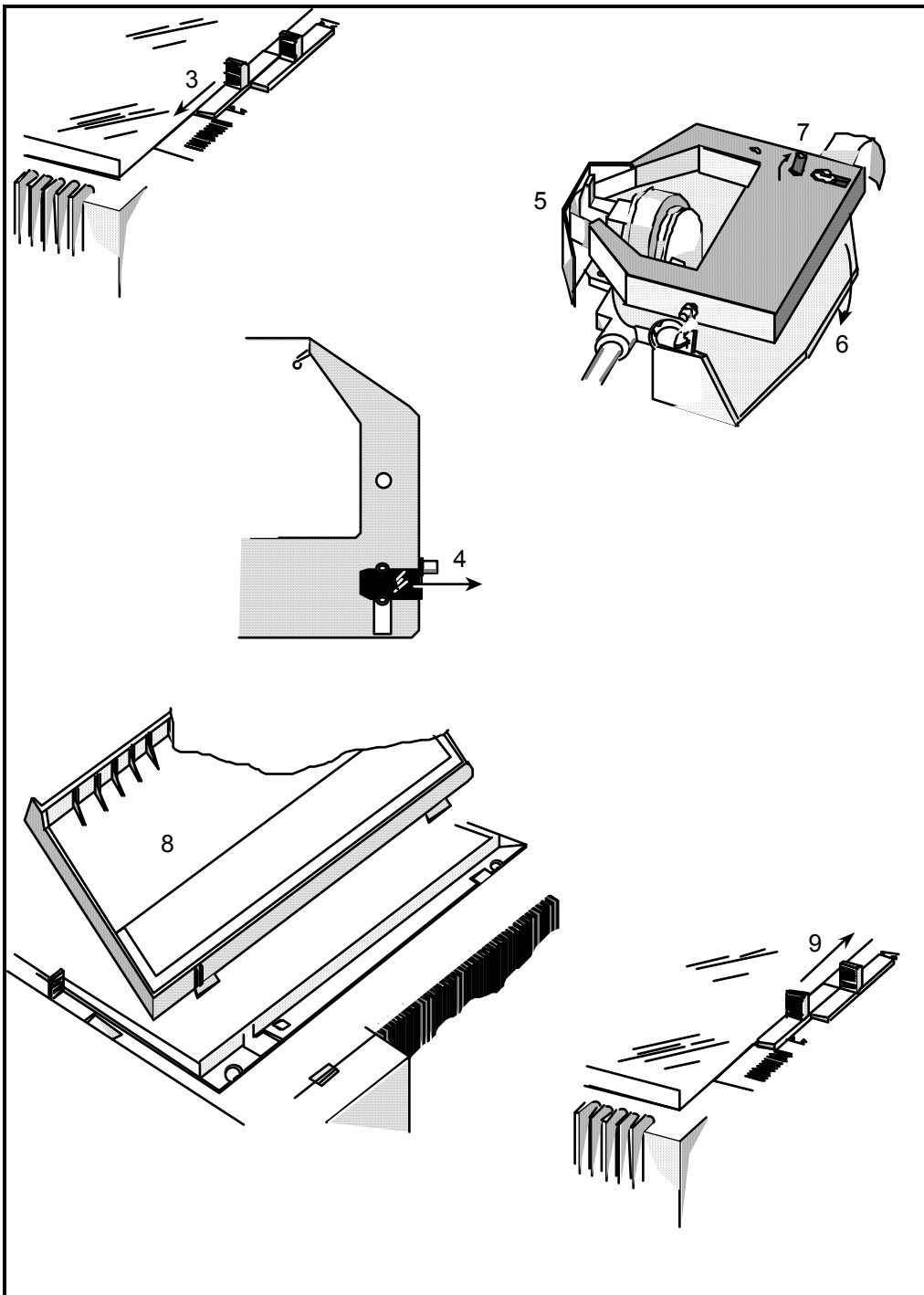


Figure 17-5 — Ribbon Replacement

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17. ASPI-41 PRINTER

17.4. ORU Removal and Replacement

17.4.5 Installing the Ribbon Cartridge

Numbers on illustrations in Figure 17-5 refer to the steps which follow.

- Step 1. Remove the printer cover (Figure 17-4, step 1).
- Step 2. Move the printer carriage to the center of the printer. **DON'T** perform this step if the printer is on (Figure 17-4, step 2).
- Step 3. Pull the paper thickness lever towards the front of the printer to facilitate ribbon insertion.
- Step 4. Remove the new ribbon cartridge from its bag. Remove and discard the holdfast that blocks the ribbon.
- Step 5. Insert the ribbon between the print head and the print head mask. Lay the cartridge over the printer carriage. Make sure the two pins on each side of the cartridge are positioned over the white retaining clips of the printer carriage.
- Step 6. Push the cartridge gently down while turning the tension knob. Make sure that the cartridge clips into place.
- Step 7. To tighten the ribbon, turn the tension knob in the direction shown by the arrow on the ribbon cartridge.
- Step 8. Replace the printer cover.
- Step 9. Move the paper thickness lever towards the back of the printer and adjust it to obtain the best quality printing. Check the quality of print with the self-test procedure in subsection 17.3.2. If the pattern does not satisfy your expectations, adjust the paper thickness lever accordingly.

17.4.6 Installing a New Print Head

- 1. Turn the power switch off and remove the power cord. Remove the ribbon cartridge per subsection 17.4.4. Make sure the print head is not too hot to touch. Do not replace the printer cover.
- 2. Disconnect the color connector (the smallest one, vertically oriented) from the print head and unscrew the two screws on the print head. Lift the print head. Remove the stirrup that secures the other (horizontal) connector, then unplug and discard the used print head.
- 3. Install a new print head by performing the above steps in reverse order. Replace the printer cover.

Run the self-test (subsection 17.3.2) to verify that the new print head is properly installed.



17.5 Printer Parts Replacement

A skilled user may replace other parts in the printer beyond the ORU level (see the Spare Parts subsection 17.8). Use this subsection as a guide in removing and replacing those parts.

The printer is composed of the following basic assemblies:

1. The cabinet.
2. An interface board (GX2P4X parallel interface or GX2S41 serial interface).
3. A print head.
4. The printer mechanism composed of the mechanism group and the carriage group.
5. The main board BAS4X PWA.
6. The power supply and line filter (ALI4X PWA).
7. An operator panel.

17.5.1 Cabinet Removal/Replacement

The cabinet must be removed to access the main PWA (BAS4X) and you must remove the print mechanism to access the power supply PWA (ALI4X).

Numbers in parentheses refer to parts shown in Figure 17-6.

1. Unplug the power cable and disconnect the interface cable.
2. Open the font cartridge cover (7) by pulling its top forward. Remove the interface board cover (10) by pulling up on its front latch and sliding the cover to the rear. Release the interface board by pulling out on the locking knob at the rear of the board. The board can then be unplugged and removed.
3. Remove the paper chute (1) and the transparent cover (4).
4. Remove drive selection (2), paper thickness (3) and paper bail (8) knobs by pulling them straight up.
5. Lift off the panel caps (11, 12).
6. Unscrew four screws (13) holding the corners of the cabinet (5) and one screw (14) near the operators panel.
7. Remove font cartridges (6), if provided. Close the font cartridge cover (7).
8. Unlatch the rear cover (9) by pulling out on both lower corners, then lift the cabinet straight up off its base.

To replace the cabinet, reverse the above procedure.

17.5.2 Replacement of Main PWA (BAS4X)

The printer cabinet must be removed before you can gain access to the main PWA board. Remove the cabinet (see subsection 17.5.1) before proceeding.

Numbers on illustrations in Figure 17-7 refer to the steps which follow.

- Step 1. Using a screwdriver, release the cable to the print head. Carefully remove the cable.
- Step 2. Disconnect the operators panel cable and the sensors cable.
- Step 3. Release the board from the printer base by pulling up on the locking knob at the base of the board.
- Step 4. Disconnect the motor connectors P05 and P06.
- Step 5. The operators panel is now loose. Use care to position it properly during reassembly.

Reverse the above procedure to reassemble the printer.

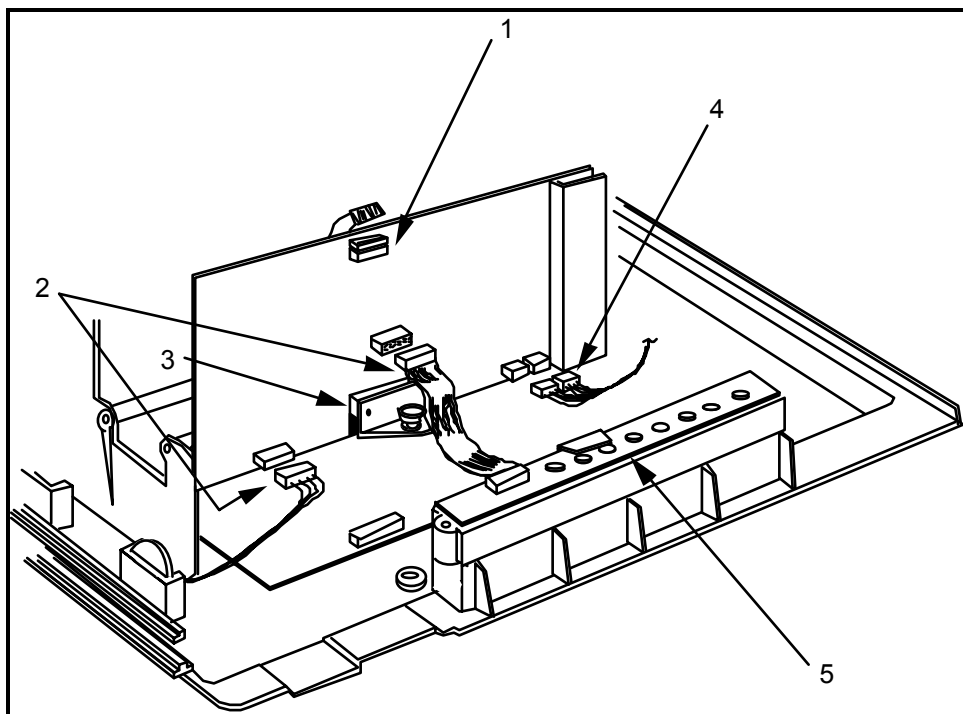


Figure 17-7 — Replacement of Main PWA (BAS4X) 3249

17.5.3 Removal/Replacement of Print Mechanism

The print mechanism must be removed to gain access to the power supply PWA.

Numbers on illustrations in Figure 17-8 refer to the steps which follow.

- Step 1. Slip off the ground connector.
- Step 2. Unscrew and remove the line filter cover.
- Step 3. Remove the screws that secure the mechanism to the printer base.
- Step 4. Lift the mechanism straight up and off the base.

To replace the printer mechanism, reverse the above procedure.

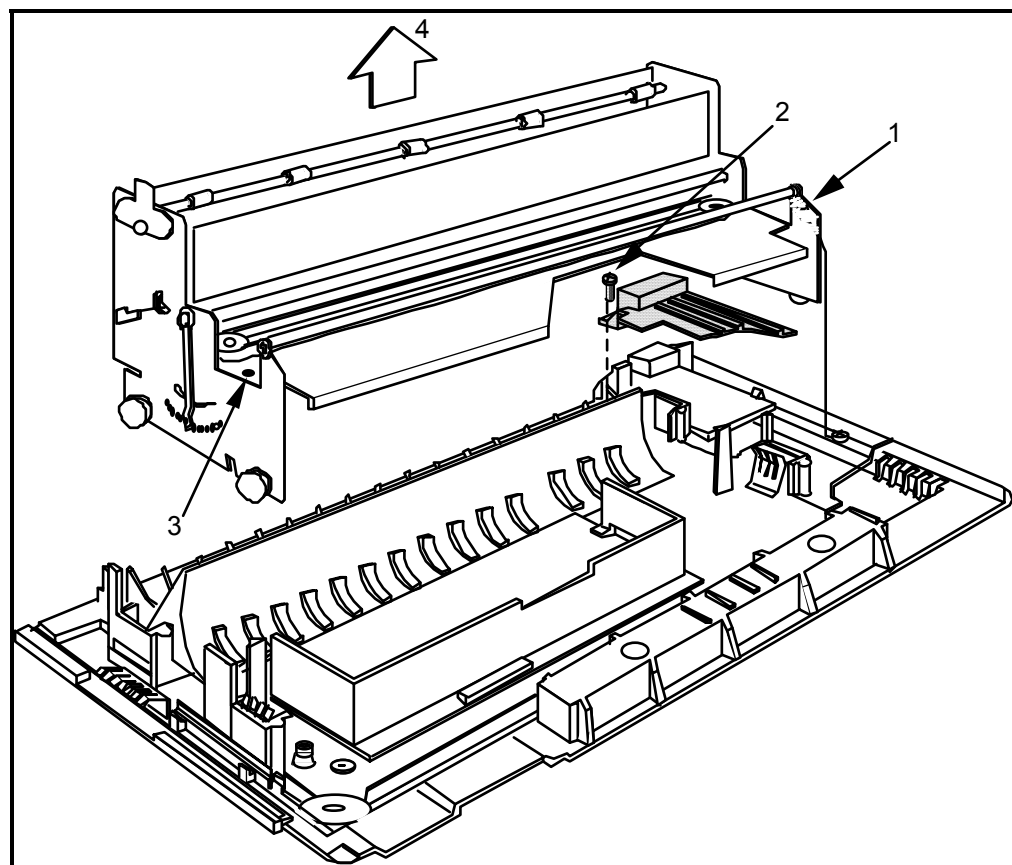


Figure 17-8 — Removal/Replacement of Print Mechanism 3253

17.5.4 Replacement of Power Supply PWA (ALI4X)

The print mechanism must be removed before you can gain access to the power supply PWA. Remove the print mechanism (see subsection 17.5.3) before proceeding.

Numbers on illustrations in Figure 17-9 refer to the steps which follow.

- Step 1. Remove three screws holding the power supply board.
- Step 2. Remove two screws holding the line filter assembly.
- Step 3. Release the power supply board from pins and catches in the printer base as shown in Figure 17-9. Remove the power supply and line filter boards together.

To replace the power supply and line filter boards, reverse the above procedure.

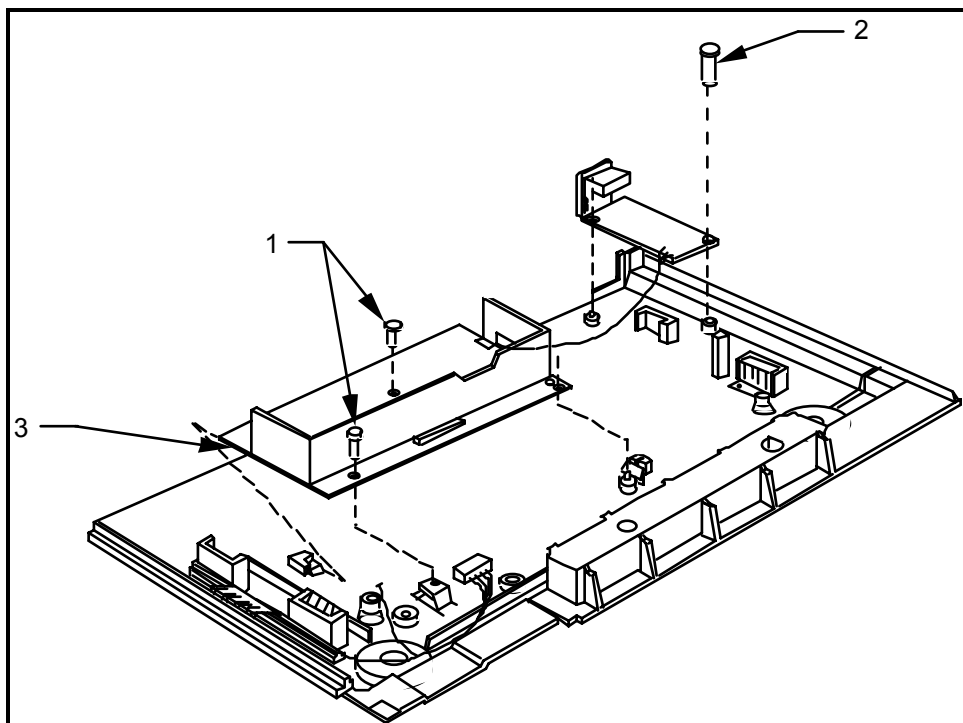


Figure 17-9 — Replacement of Power Supply PWA (ALI4X) 3315

17.5.5 Carriage Group Disassembly

Numbers on illustrations in Figure 17-10 refer to the steps which follow.

Disassembling the left side:

- Step 1. Remove screws from the front and rear drive bars.
- Step 2. Unscrew and remove the inner copy lever.
- Step 3. Remove the front bar lockwasher.
- Step 4. Unscrew and remove the carriage belt idle gear.

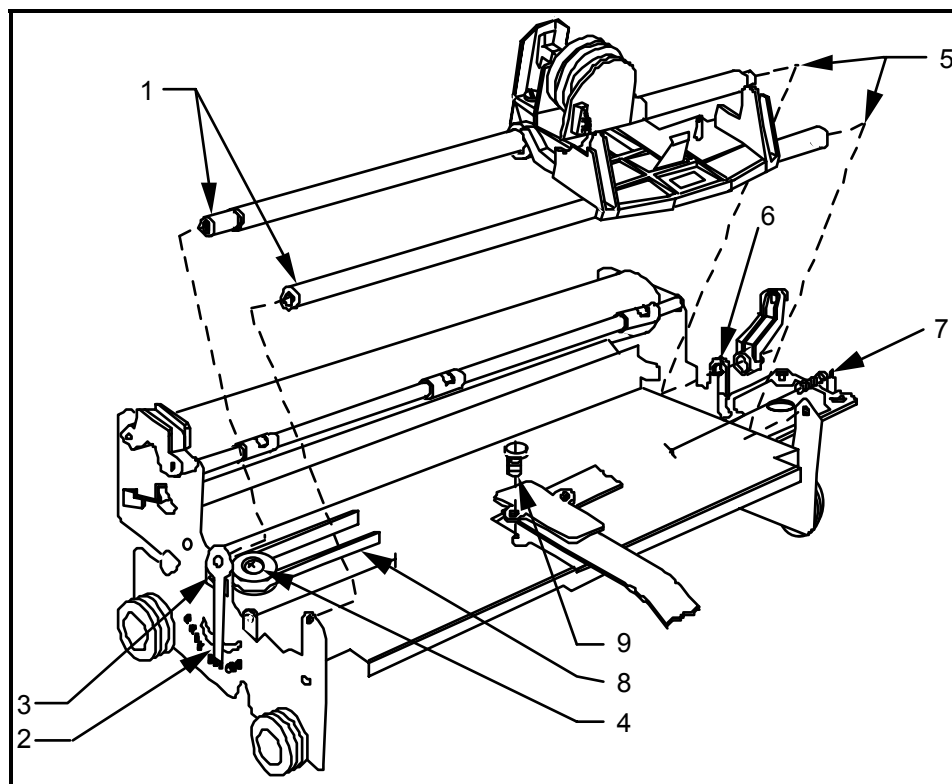


Figure 17-10 — Carriage Group Disassembly

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Disassembling the right side:

- Step 5. Remove screws from the front and rear drive bars.
- Step 6. Remove the front bar lockwasher.
- Step 7. Remove the ribbon take-up tension spring.
- Step 8. Remove the carriage belt.
- Step 9. Remove the print head cable clamp.

The carriage group and drive bars are now free to be removed together.

To replace the carriage group and drive bars, reverse the above procedure.

17.5.6 Motor Group Disassembly

There are three motors in the printer. First, consider removing the paper motor and carriage motor. Numbers on illustrations in Figure 17-11 refer to the following steps.

Removing the Paper Motor:

- Step 1. Remove the screws holding the motor and remove it as shown in Figure 17-11.

Removing the Carriage Motor:

- Step 2. Unscrew and remove the carriage belt idle gear.
- Step 3. Remove the carriage belt.
- Step 4. Remove the screws holding the motor and remove it as shown in Figure 17-11.

Now consider removing the Color Motor. Numbers on illustrations in Figure 17-12 refer to the following steps.

- Step 1. Disconnect the color connector and cable from the print head.
- Step 2. Remove the pulley.
- Step 3. Remove the screws holding the motor and carefully remove it as shown in the illustration.

To replace the motors, reverse the above procedure.

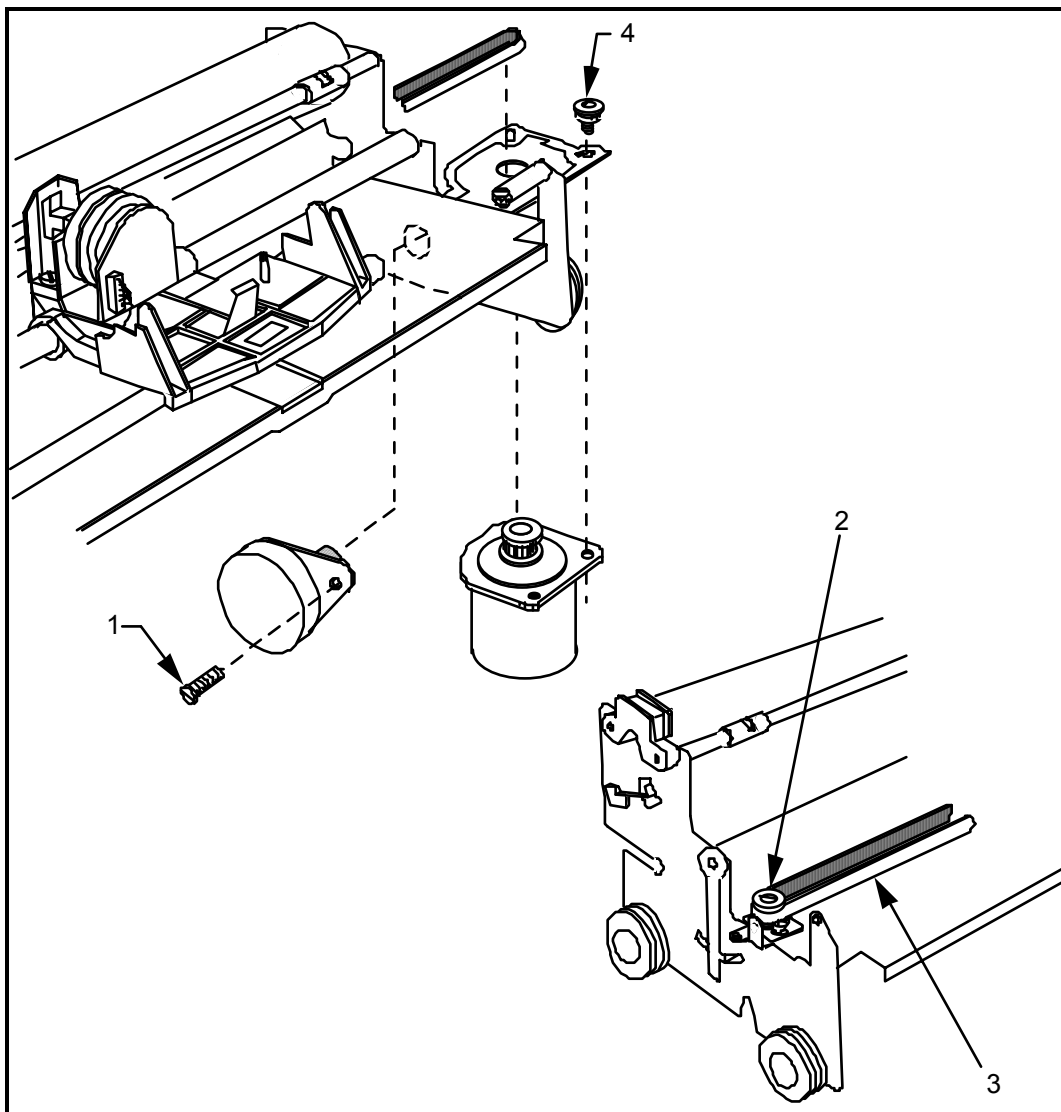


Figure 17-11 — Carriage Motor and Paper Motor Removal 3317

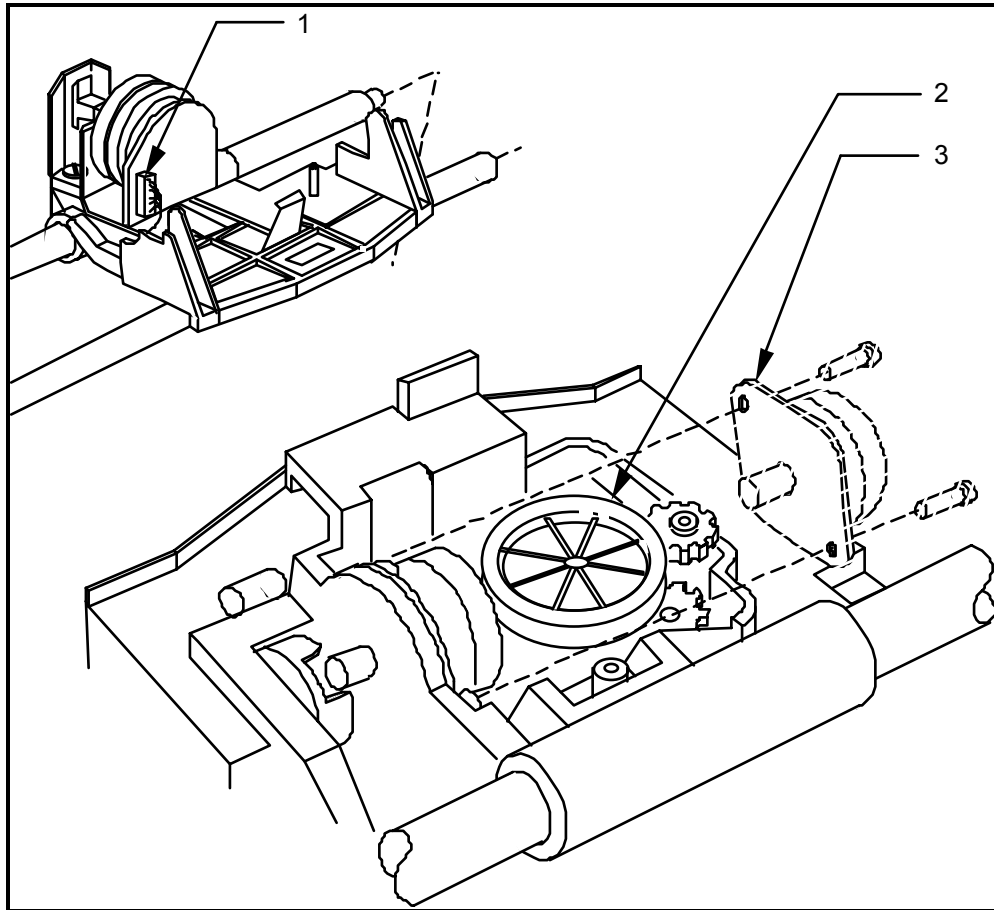


Figure 17-12 — Color Motor Removal

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17.6 Printer Adjustments

A skilled user may replace other parts in the printer beyond the ORU level (see *Spare Parts*, subsection 17.8). Use this subsection as a guide in adjusting those replaced parts.

17.6.1 Carriage Belt Adjustment

Numbers on illustrations in Figure 17-13 refer to the steps which follow.

- Step 1. Loosen (don't remove) the screws holding the cable pulley plate.
- Step 2. Using a tension gauge, apply a force of 3.8 kg + 0.3 kg (8.36 lb + 0.66 lb) to the cable pulley plate as shown.
- Step 3. Secure the cable pulley plate.

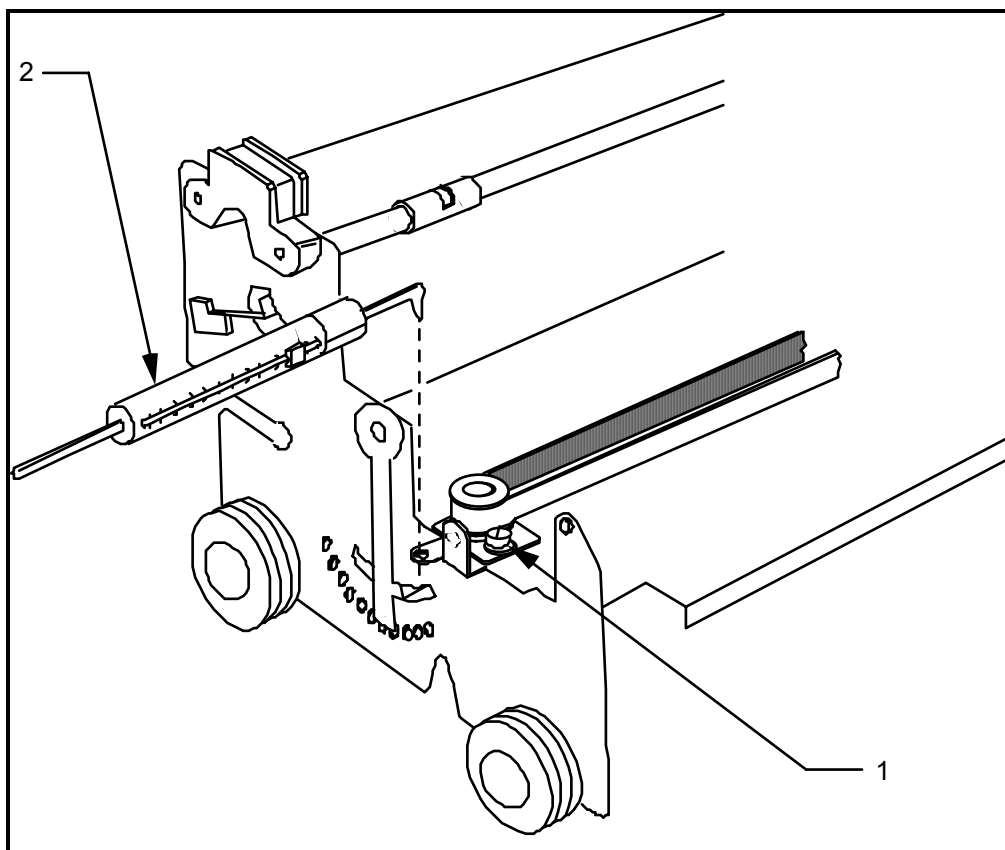


Figure 17-13 — Carriage Belt Adjustment

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17.6.2 Paper Belt Tension Adjustment

Numbers on illustrations in Figure 17-14 refer to the steps which follow.

- Step 1. Loosen (don't remove) the two screws holding the belt idler assembly.
- Step 2. Using a tension gauge, apply a force of 0.5 kg (1.10 lb) downward on the belt idler assembly arm at point 2.
- Step 3. Secure the belt idler assembly.

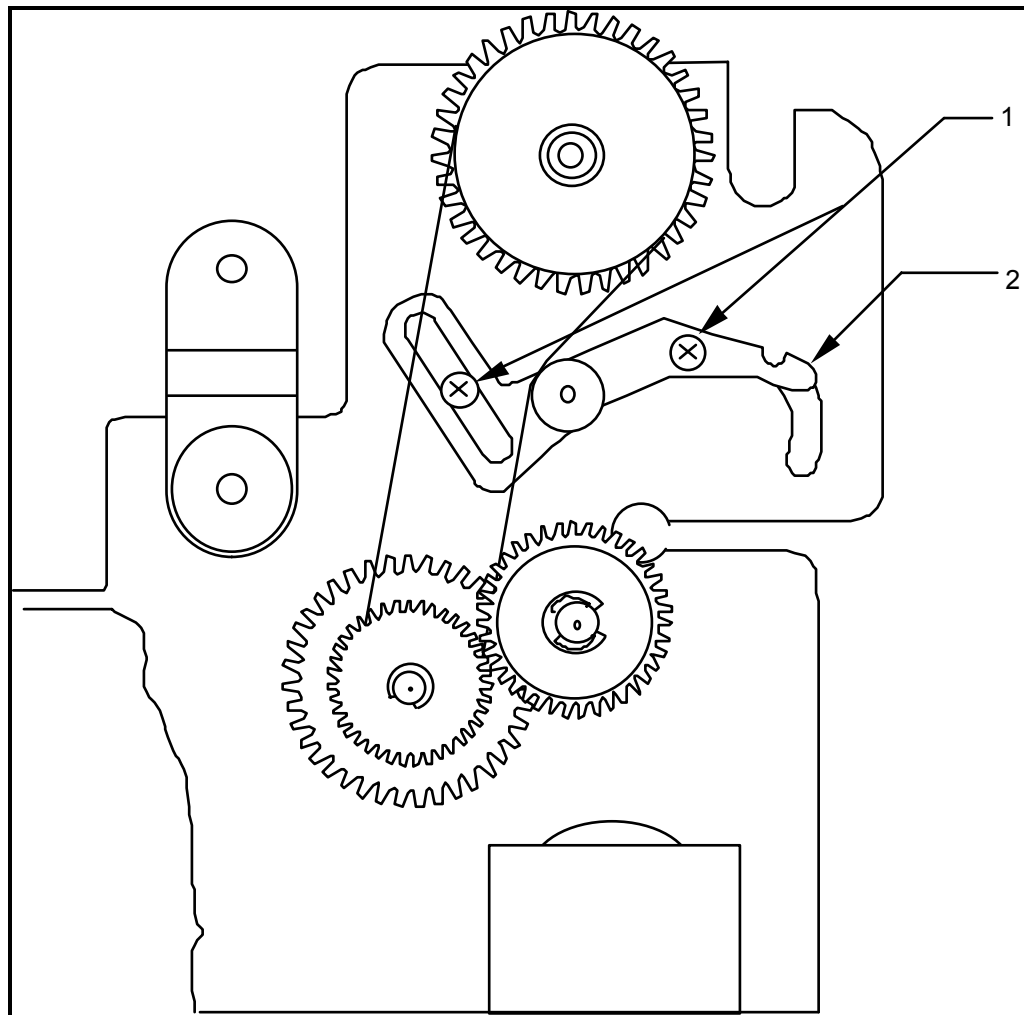


Figure 17-14 — Paper Belt Adjustment

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17.6.3 Print Head to Platen Adjustment

Numbers on illustrations in Figure 17-15 refer to the steps which follow.

- Step 1. Remove the ribbon mask.
- Step 2. Move the carriage to the center of the printer. Loosen the screw holding the front bar positioning lever.
- Step 3. Use the copy lever on the right side to move the carriage in the next step.
- Step 4. Use a feeler gauge to adjust the distance between the print head and the platen to 0.35 mm (0.14 inch).
- Step 5. Move the front bar positioning lever to the right-most position and tighten the screw which holds it.

Recheck your work now that all screws have been tightened. Replace the ribbon mask.

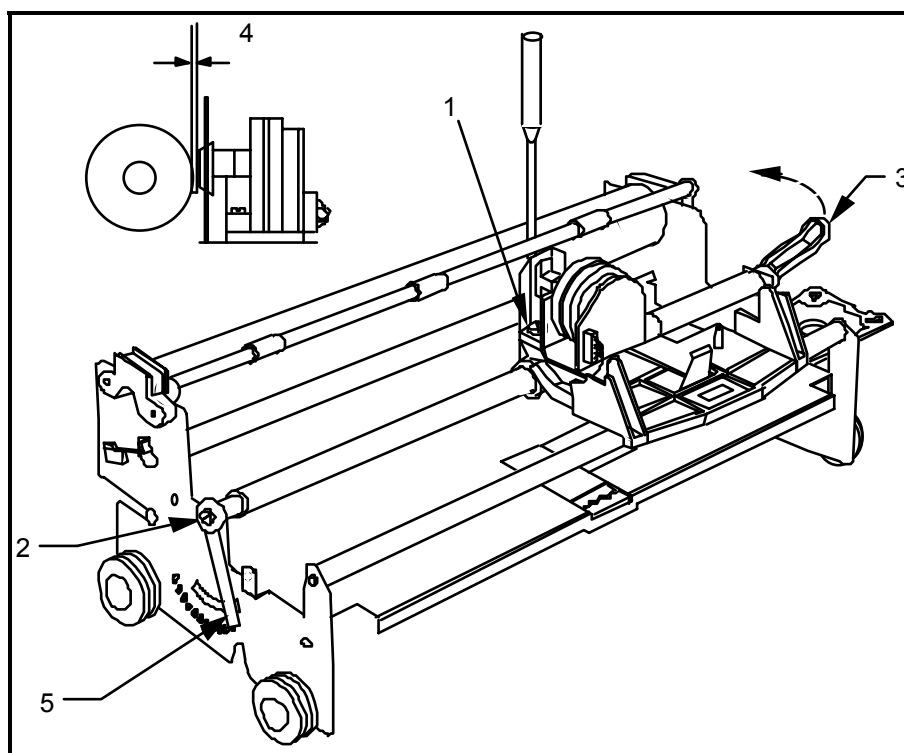


Figure 17-15 — Print Head to Platen Adjustment

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17.7 Startup

Refer to the *Process Operations Manual*, in the *Process Operations* binder, for operational procedures.

17.8 Spare Parts

Table 17-3 — Spare Parts List

Part Number	Description
51107144-114	Bar Tractor Spring
* 51108305-700	Black Ribbon
51107144-129	Cabinet
51303631-003	Cable, RS-232 Direct Connect, 3 m (PIC, PDG, and EPDG(P) I/O board)
51303631-004	Cable, RS-232 Direct Connect, 4 m (PIC, PDG, and EPDG(P) I/O board)
51303631-008	Cable, RS-232 Direct Connect, 8 m (PIC, PDG, and EPDG(P) I/O board)
51303631-010	Cable, RS-232 Direct Connect, 10 m (PIC, PDG, and EPDG(P) I/O board)
51303631-015	Cable, RS-232 Direct Connect, 15 m (PIC, PDG, and EPDG(P) I/O board)
51308103-003	Cable, RS-232 Direct Connect, 3 m (EPDGC I/O board)
51308103-008	Cable, RS-232 Direct Connect, 8 m (EPDGC I/O board)
51308103-015	Cable, RS-232 Direct Connect, 15 m (EPDGC I/O board)
51107144-143	Carriage Belt
51107144-122	Carriage Group
51107144-117	Carriage Motor Group
51107144-115	Color Motor Group
51107144-105	Copy Lever
51107144-107	Front Cartridge Flap
* 51107144-101	Fuse, 3 A 250 V
51107144-119	Idle Pulley
51107144-108	Interface Cover
51107144-111	Internal Copy Lever
51107144-112	Internal F/T
51107144-134	Mylar
51107144-113	Nylon String

17. ASPI-41 PRINTER**17.8. Spare Parts**

Table 17-3 — Spare Parts List

Part Number	Description
51107144-136	Optical Reflection Sensor
* 51108305-900	Optional Pedestal for ASP-41 Printer
51107144-138	P. E. Switch/Color Switch
51107144-139	Power Supply Switch
51107144-123	Paper Bail Spring
51107144-142	Paper Belt
51107144-127	Paper Chute
51107144-106	Paper Load Lever
51107144-116	Paper Motor Group
* 51107144-102	Print Head
51107144-133	Print Head Cable
51107144-120	Print Head Shield
* 51108305-100	Printer, Replacement
51107144-110	Pulley
51107144-121	Push Button O. P.
51107144-125	PWA ALI4X (Power Supply)
51107144-109	PWA BAS4X (Main Board)
51107144-131	PWA GX2P4X (O. P. Group)
51107144-126	PWA GX2S41 (Serial I/F Group)
51107144-130	Rear Cover Group
51107144-103	Ribbon Cartridge Coupler
51107144-118	Roller Assembly
51107144-124	Roller Sensor Assembly
51107144-135	Sensor Cable Group
51107144-132	Spring Roll
51107144-137	TL P802 Optical Interrupter
51107144-140	Tractor Left
51107144-141	Tractor Right
51107144-104	Tractor/Friction Lever

Table 17-3 — Spare Parts List

Part Number	Description
51107144-128	Transparent Cover

* ORU Level Replacement Item

17. ASPI-41 PRINTER

17.8. Spare Parts

18 ASPI-46 PRINTER

18.1 Cleaning

Clean the exterior of the printer, using a soft-bristled brush and vacuum. Set the power switch (on the rear, above the power cord) to OFF and unplug the power cord. Lift the top-half of the cover and remove the paper, if present. Vacuum the dirt/dust from inside of the unit. Use a brush to loosen dirt accumulations, but take care not to damage or misalign delicate items, such as the print head. Close the cover and wipe the outside clean with a cloth and mild detergent solution. Plug the power cord back into the unit.

18.2 Adjustments

None.

18.3 Test/Troubleshooting

To help you isolate and correct minor printer problems, the topics covered here include problem solving, the printer self-tests, printer diagnostic procedure, and the hexadecimal dump procedure.

When your printer has a problem, follow this troubleshooting sequence:

1. Use the Problem Solving Guide (Table 18-1) to identify and correct problems.
2. Run the Printer Self-Test. The instructions are in subsection 18.3.2.
3. Run the Diagnostic Procedure. The instructions are in subsection 18.3.3.
4. Call the Technical Assistance Center (TAC). In the U.S.A. use our toll-free number 1*800-822-7673 (available in the contiguous states except Arizona; in Arizona dial 1*602-863-5558).

18.3.1 Problem Solving

Table 18-1 — Problem Solving Guide

Problem	Possible Cause	Solution
POWER indicator does not light	Power Switch off	Turn printer on
	Power cable not connected properly	Verify that the cable is connected properly to the printer and the ac power outlet.
	AC outlet not working	Test outlet. Use another outlet until the first outlet is restored to service.
	Power supply fuse has blown	Locate power supply fuse on the right side of the printer. Turn printer power off. Unplug both ends of the power cord. Insert a screwdriver under the upper edge of the power fuse holder and pry the holder off the printer. Remove the fuse, check it, and if it has blown, replace it with a new fuse that matches the printer's voltage and current requirements. Replace the fuse holder so the small arrow on the case is pointing to the correct voltage. Turn the printer on.
	After power off, power was turned on within three seconds	Turn printer power off, wait 1.5 minutes, then turn power on again.
Printer does not print	Display window shows "WAIT"	Press the ON LINE button to enable printing.
	Interface cable is not properly connected	Push cable firmly into connectors at both ends, and make sure the screws are secure.
	Incorrect Interface cable	Replace with correct cable.

Table 18-1 — Problem Solving Guide

Problem	Possible Cause	Solution
	Configuration settings don't match host requirements	Correct configuration settings to match host requirements.
Continuous forms paper is not advancing	Friction/Tractor selector is in the wrong position	Set the selector to the "Tractor" position.
	Paper holes are torn	Remove torn paper and replace with the next good sheet. Check tension tractors. If too tight, loosen by moving the right tractor slightly to the left. See paper loading instructions in the <i>Process Operations Manual</i> , in the <i>Process Operations</i> binder.
	Paper is not correctly aligned on tractors	Reload paper, ensuring that corresponding holes at each side of the paper are correctly aligned on the tractors.
Single sheet does not load correctly	The single sheet has not been inserted correctly.	Insert single sheet well into the front paper-entry slot making sure that it reaches the roller. The sheet must be pushed slightly further in after "P.E." disappears from the display window.
Single sheet starts to load, then backs up	Friction/Tractor selector is set to wrong position	Set selector to "Friction" position.
	Forms overlay (OVRL) not selected	Change OVRL to OV Y to select forms overlay.
Display window shows "JAM" while loading a sheet from the ASF, and ejects the sheet or continuous form*	Friction/Tractor selector set to "Tractor"	Set selector to the "Friction" position.
Printing partial, incomplete characters, or light printing	Ribbon not feeding	Check that ribbon is correctly loaded. Turn ribbon advance knob to ensure that the ribbon is not jammed.

Table 18-1 — Problem Solving Guide

Problem	Possible Cause	Solution
	Ribbon worn or damaged	Replace ribbon.
	Print head too far from paper	Push paper thickness lever towards back of printer to move print head closer to paper.
	Print head is worn or damaged	Print head needs replacement.
Dark, smudgy printing	Foreign matter on print head	Remove ribbon, wipe off print head, then replace ribbon.
	Print head too close to paper	Pull paper thickness lever towards front of the paper to move the print head away from paper.
	Ribbon loaded improperly	Remove, then replace properly. See <i>Process Operations Manual</i> , in the <i>Process Operations</i> binder.
Display Window shows "JAM" *	Paper jammed or torn	Remove and reload paper.
	Printer withdrew more than 17 inches of continuous form while inserting single sheet	Press the down-arrow button until "P.E." displays, then continue operation.
	Single sheet is too long (longer than 17 inches) or is jammed	Press the up-arrow button continuously to clear the sheet. If necessary, clear the sheet manually.
Display window shows "FLT"	A carriage synchronization error has occurred	Turn the printer power off, then on. If "FLT" is still displayed, report the problem to the TAC.

Table 18-1 — Problem Solving Guide

Problem	Possible Cause	Solution
	One or both font cartridges have not been inserted or removed correctly	Turn the printer power off. Check the font cartridges. Then turn the power on again.

* NOTE: To clear "JAM" from the display window, try turning the power off, then on again.

18.3.2 Printer Self-Tests

After installing the printer, you can test its operation by using the self-test procedure. Before beginning the self-test, verify that:

1. The paper and ribbon cartridge are inserted correctly.
2. The power cable is connected but the power is turned off.
3. The front cover is closed.

Perform the following steps:

1. Press and hold the ON LINE button while you turn the printer on.

The display window shows "TEST," and a pattern is printed on the loaded paper as shown in Figure 18-1. Note that the printer continues to print until you press the ON LINE button.

2. The display window shows "WAIT."

```

ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
BCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
CDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
DEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
EFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
FGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
GHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
HIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
IJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
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STUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
TUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
UVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
VWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
WXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
XYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£

```

Figure 18-1 — Self-Test Print Pattern

3659

NOTE

1. The self-test also ends if the printer runs out of paper and the window shows "P.E."
2. The self-test pattern can be printed in quality print by pressing the QUALITY button.
3. You can also select different character fonts for self-test by inserting optional font cartridges and pressing FONT.

18.3.3 Diagnostic Procedure

This diagnostic procedure helps to identify printer faults so that when a unit fails, you will know for sure that it is the printer that has failed and not some other part of the subsystem that is at fault.

The diagnostic procedure checks the control panel indicators, the display window, and internal printer functions, then does a printing test. At the end of the diagnostic, the display window shows END or displays an error number if an error is found.

Before starting, verify that:

- The printer power is turned off.
- The printer's front and rear covers are closed.
- Paper is loaded.
- The LCN-based system is not sending data.

As you perform the diagnostic procedure, note any problems you find. To run the diagnostic, follow these steps:

1. Press the PROG button while turning on the power to the printer. The printer beeps five times, and all the indicators on the control panel and on the font cartridges (if inserted) light up.
2. The display window shows, in a sequence of a to e, the following patterns to check its display capability. As each pattern is displayed, the printer "beeps." Verify that the display window shows:

- a.

--

--

--

--
- b. *

--

--

--
- c. * *

--

--
- d. * * *

--
- e. * * * *

The diagnostic procedure automatically enters test execution.

3. The display window shows T&Dx, where x is the hexadecimal number in sequence as the test continues.
4. When the display window shows T&DB, the printer prints a test pattern. See Figure 18-2.
5. The test terminates by:
 - Flashing END? several times, or
 - Displaying an error message in the format ORUx where x is the error number. The printer will also "beep" three times and the indicators will blink. Table 18-2 lists the possible error messages.
6. If you press the PROG button while END? or the error message is displayed, the diagnostic procedure repeats continuously until you stop it by pressing the PROG button.

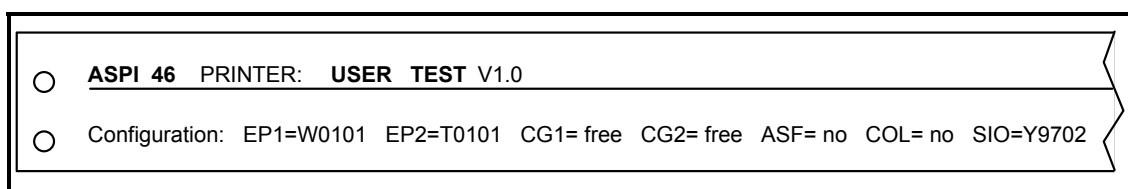


Figure 18-2 — Printer Diagnostic Test Pattern³⁴³⁴

Table 18-2 — Diagnostic Procedure Error Messages

Error Number (X)	Description
ORU0	General fault
ORU1	Fault on main board
ORU2	Fault on firmware board
ORU3	Fault on chip U03 (firmware board)
ORU4	Fault on chip U04 (firmware board)
ORU5	Fault on operator panel
ORU6	Character generator 1 fault
ORU7	Character generator 2 fault
ORU8	Interlock fault
ORU9	Paper load lever fault
ORUA	Color fuse fault
ORUB	Automatic sheet feeder fuse fault
ORUC	Serial interface fault
ORUD	Fault on chip B09 (serial interface board)
ORUE	Fault on chip D10 (serial interface board)

18.4 Removal/Replacement

18.4.1 Printer Removal

Turn off the ac power switch located on the rear of the printer, just above the power cord. Disconnect the power cord. Disconnect the data cable from the left rear of the printer.

18.4.2 Printer Replacement

1. Remove the plastic ties from the print head.
2. Verify that the power fuse holder, located under the power socket on the rear of the printer, has the proper voltage selected (i.e., 120 V) is immediately above the triangular arrow, if 120 Vac power is to be used.
3. Install a ribbon cartridge as covered in subsection 18.4.4.
4. Connect data cable to connector located at the left rear of the printer.
5. Connect ac power cord to the rear of the printer. Turn on power switch located just above the power cord.

18.4.3 Ribbon Replacement

NOTE

Turning off the power with data in the input buffer causes the data to be lost.

The ribbon cartridge can be removed when the printer is on and the display window shows P.E. or WAIT, or when the printer is off.

- Step 1. Raise the front cover (Figure 18-3).
- Step 2. Using both hands, pull the paper bail towards the front of the printer, then lift it up (Figure 18-4).
- Step 3. Pull the left release arm toward the front of the printer (Figure 18-4), then lift the left side of the ribbon cartridge free. Now pull the right release arm toward the front of the printer and lift the right side of the ribbon cartridge free. Lift the ribbon cartridge out of the printer.

18.4.4 Installing The Ribbon Cartridge

- Step 1. Raise the front cover (Figure 18-3).
- Step 2. Using both hands, pull the paper bail (Figure 18-4) towards the front of the printer, then lift it up.
- Step 3. Move the print head to the center of the printer (Figure 18-4).
- Step 4. Pull the paper thickness lever towards the front of the printer to make ribbon insertion easier (Figure 18-4).
- Step 5. Remove the ribbon cartridge from its bag.
- Step 6. Remove and discard the holding pin from the cartridge (Figure 18-5).
- Step 7. Turn the ribbon tension knob (Figure 18-5) in the direction indicated by the arrow, until the ribbon is taut.
- Step 8. Place the ribbon cartridge inside, so it rests between the release arms of the retaining clips (Figure 18-4).
- Step 9. Slide the white plastic mask on the ribbon until the mask is aligned with the print head. Slip the mask onto the pins at either side of the print head (Figure 18-6).
- Step 10. Push the cartridge gently down until it snaps into place (Figure 18-7).
- Step 11. Slide the print head along the print carriage to check that the mask moves freely along the ribbon.
- Step 12. Using two hands, lift the paper bail towards the front of the printer, lower it, then release it gently. This action repositions the paper bail (Figure 18-4).

CAUTION

Before closing the front cover, make sure that the paper bail is correctly positioned; otherwise, the cover could be damaged.

- Step 13. Close the front cover.
- Step 14. Move the paper thickness lever (Figure 18-4) towards the back of the printer. Align the lever with the last mark on the cover.

18. ASPI-46 PRINTER
18.4. Removal/Replacement

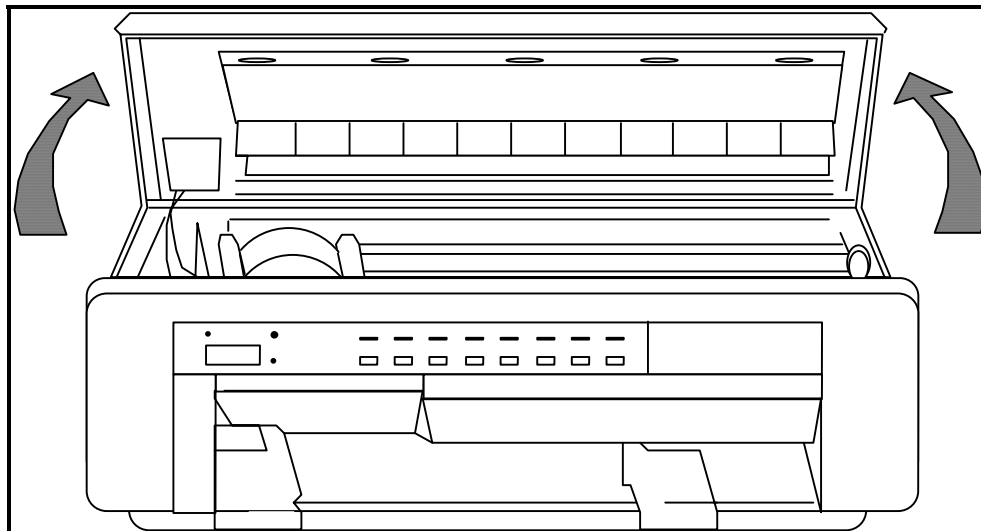


Figure 18-3 — Printer Front Cover 1737

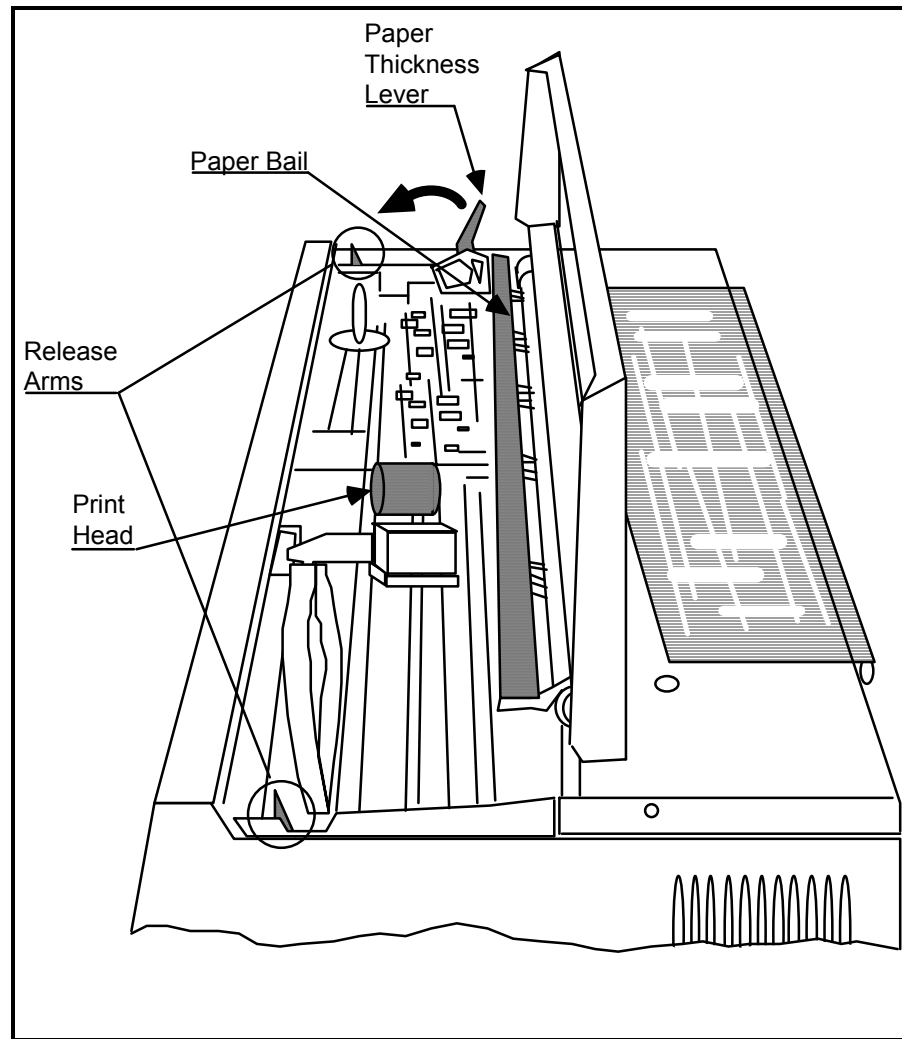


Figure 18-4 — Paper Bail, Print Head, and Paper Thickness Lever 1738

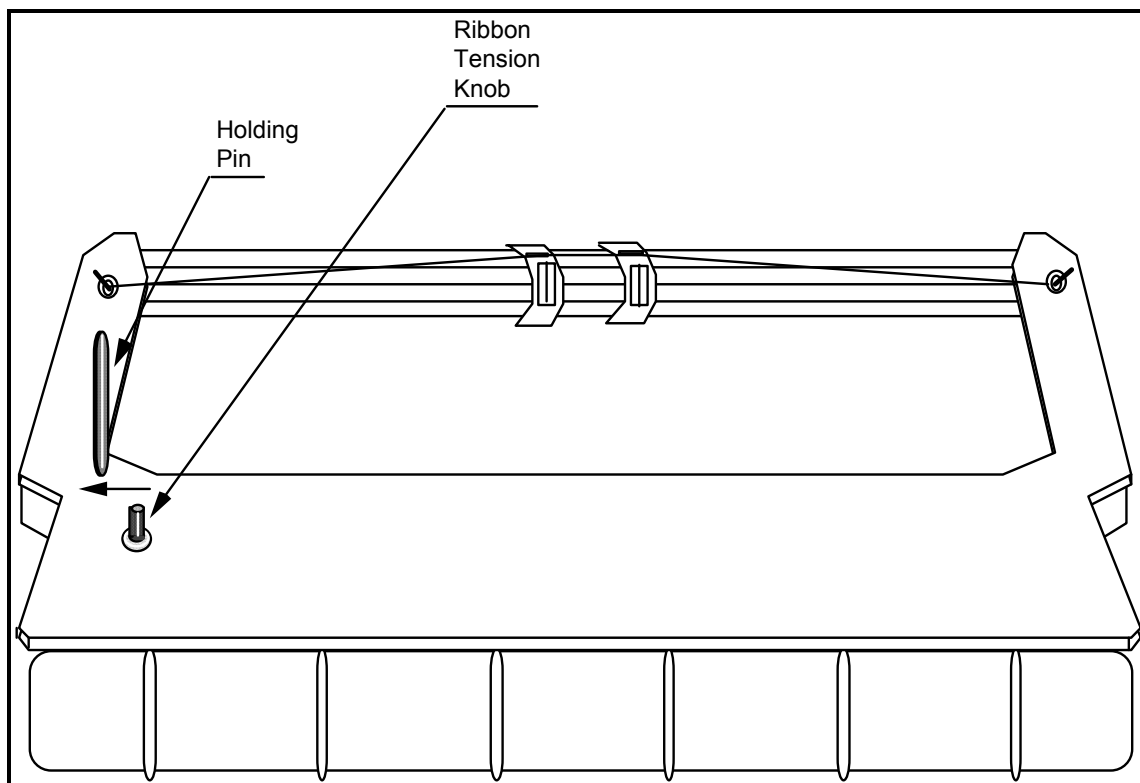


Figure 18-5 — Ribbon Cartridge

1739

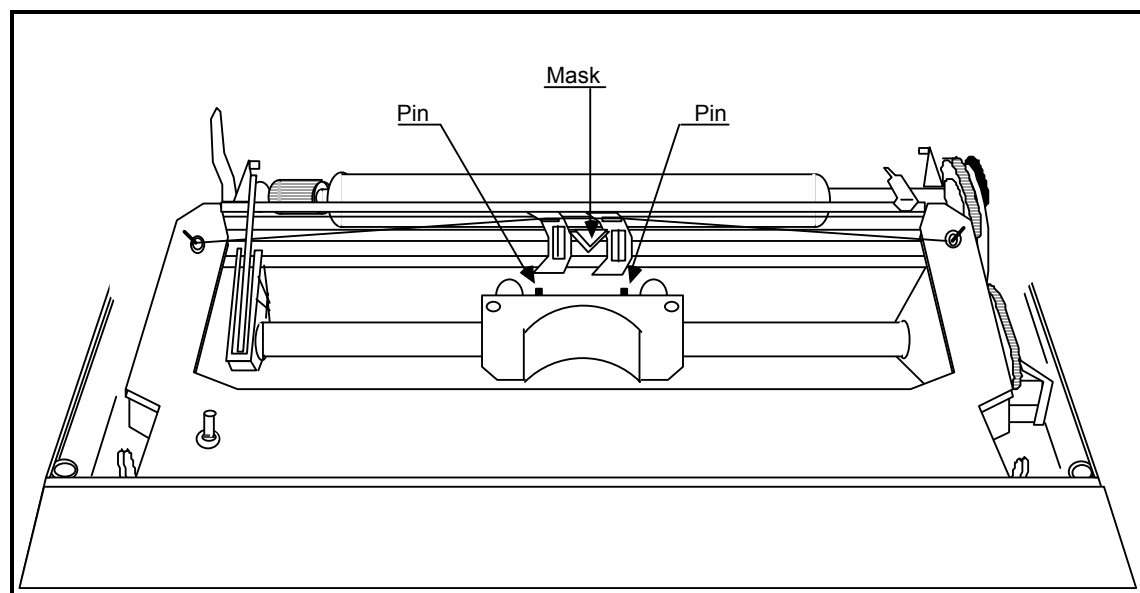


Figure 18-6 — Plastic Mask Attachment

1740

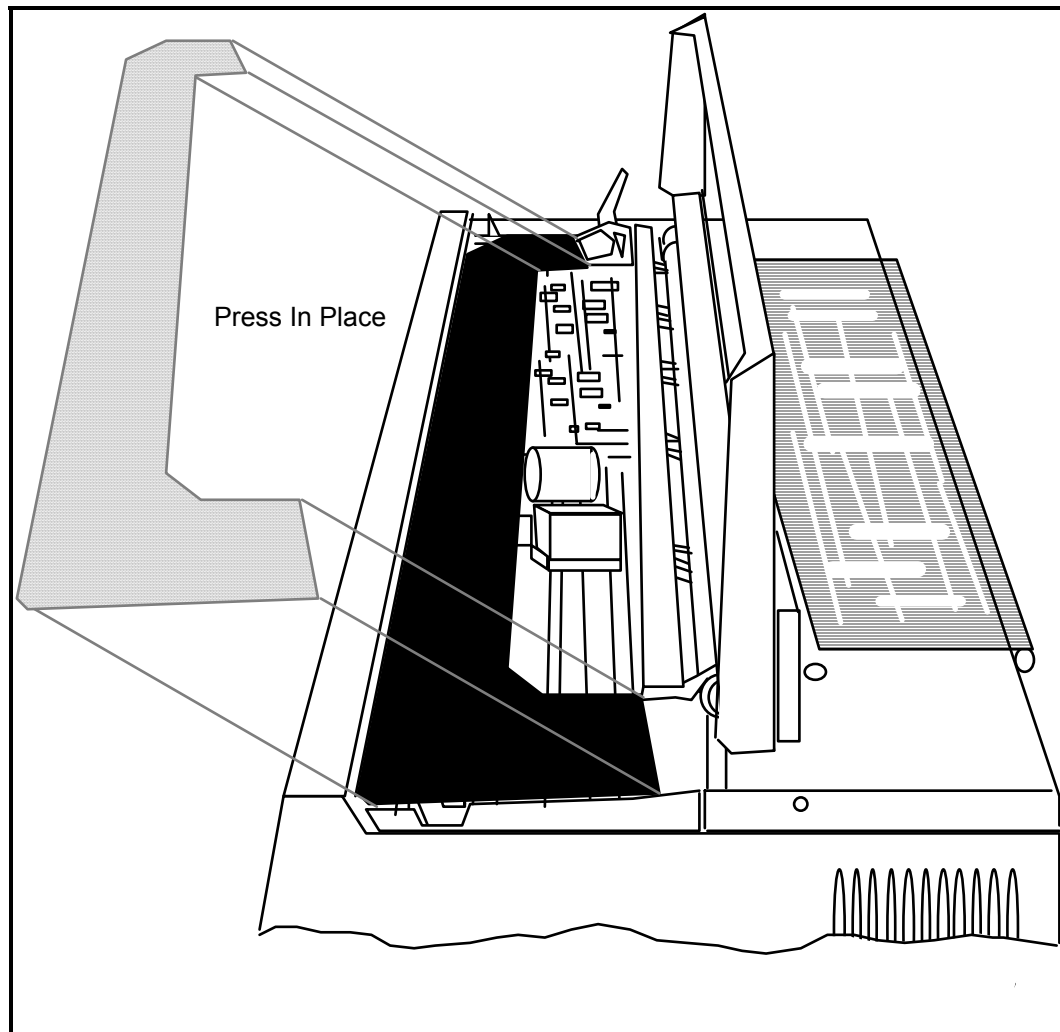


Figure 18-7 — Ribbon Cartridge Alignment 1741

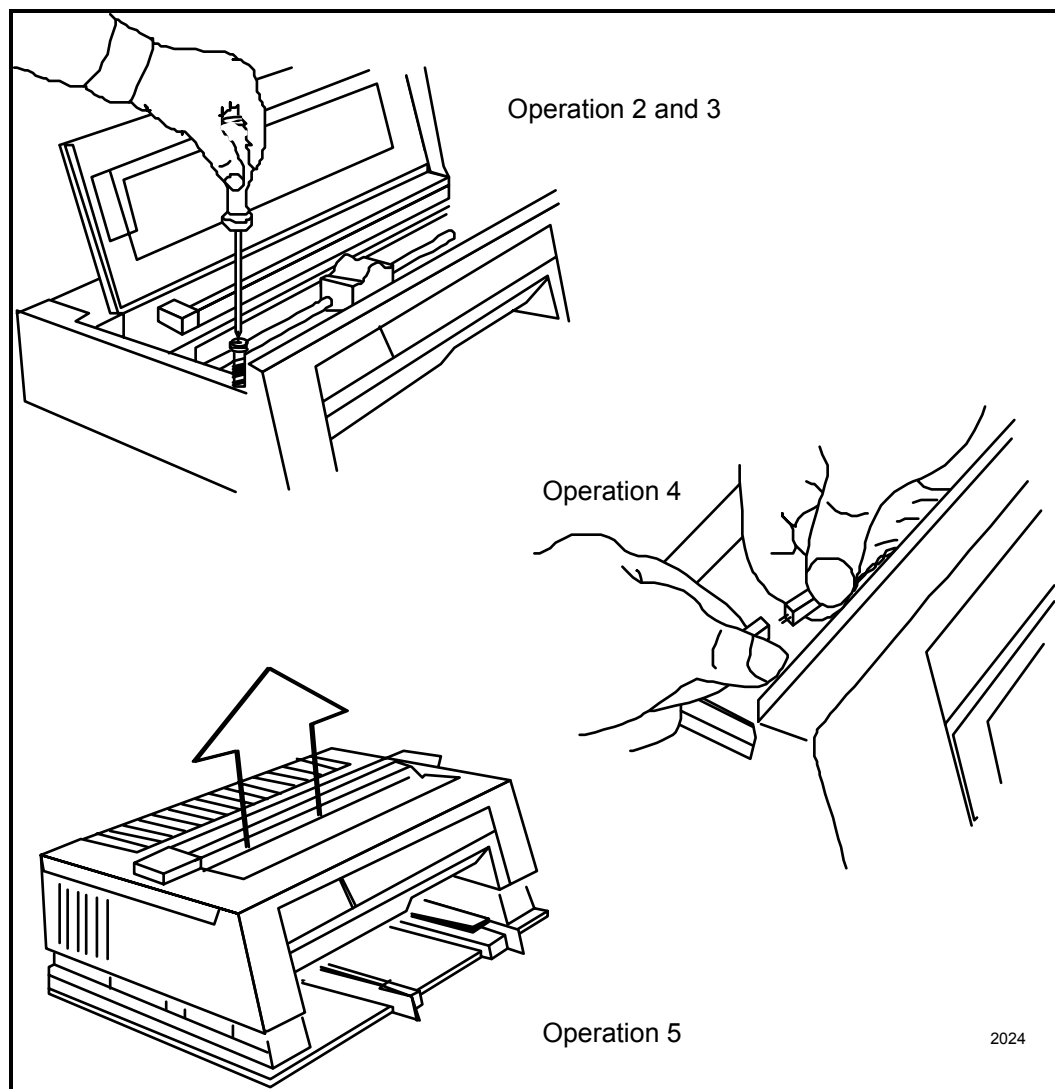


Figure 18-8 — Upper Cabinet Removal

18.5 Printer Parts Replacement

18.5.1 Upper Cabinet Removal/Replacement

1. Unplug the power cable and disconnect the interface cable. Remove the tractor assembly by pulling up on each side of the back of the tractors until the assembly unsnaps from the printer.
2. Remove the two screws located under the front cover (see Figure 18-8).
3. Remove the two screws located under the rear cover.
4. Disconnect the left-side interlock cable connector (note wire-color polarity).
5. Lift off the upper cabinet and set it aside.

To replace the upper cabinet, reverse the above procedure.

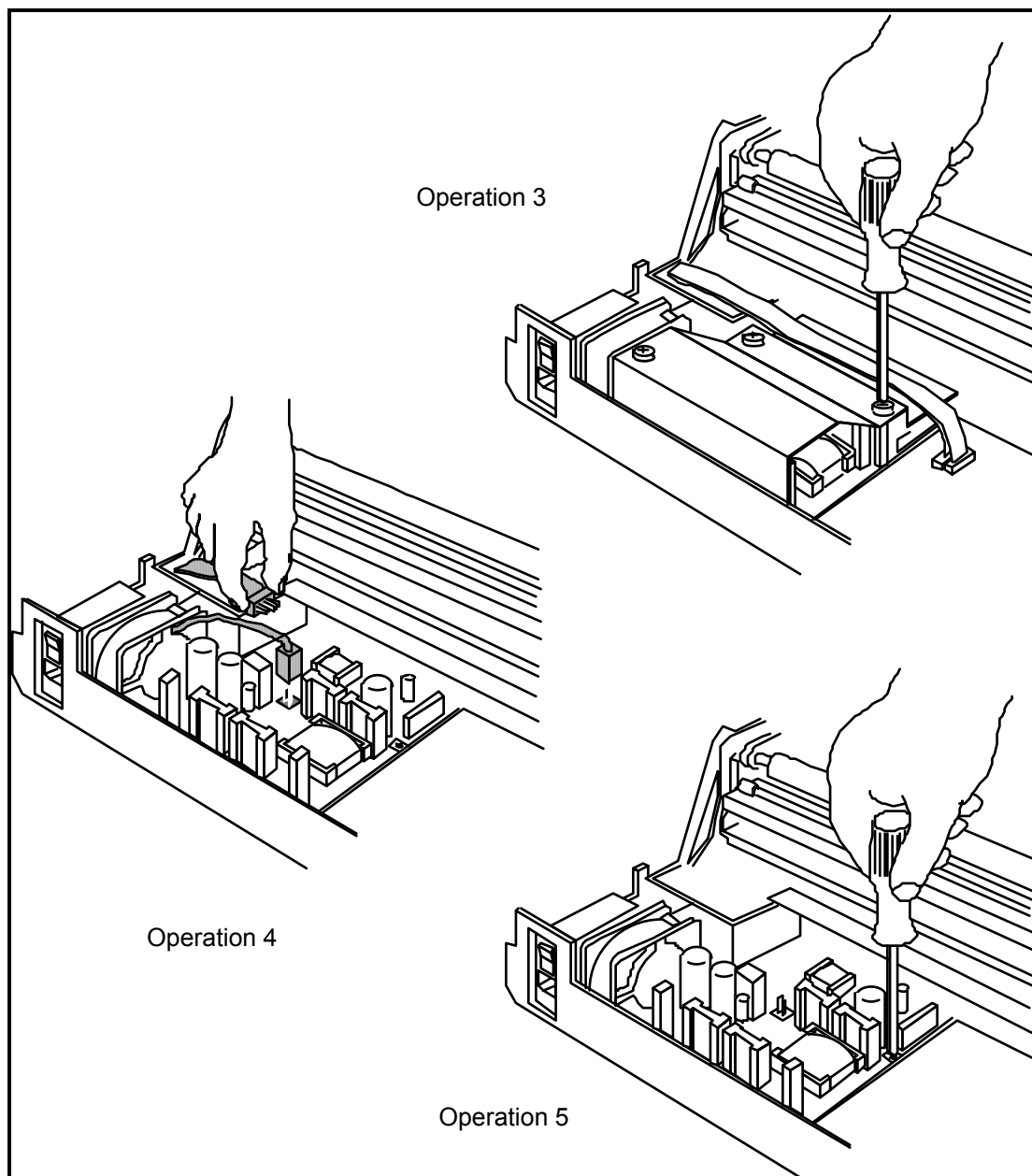


Figure 18-9 — Power Supply PWA Replacement

2112

18.5.2 Replacement of Power Supply PWA

1. Unplug the power cable and disconnect the interface cable.
2. Remove tractor assembly. Remove the upper cabinet. See subsection 18.5.1.
3. Slip all the cables off their clamps and remove the metal protection by unscrewing the three cross-head screws placed on the upper side as shown in Figure 18-9.
4. Carefully disconnect all the cables plugged onto the power supply board.
5. Remove the power supply PWA by unscrewing the four cross-head screws.

To replace the power supply board, reverse the above procedure.

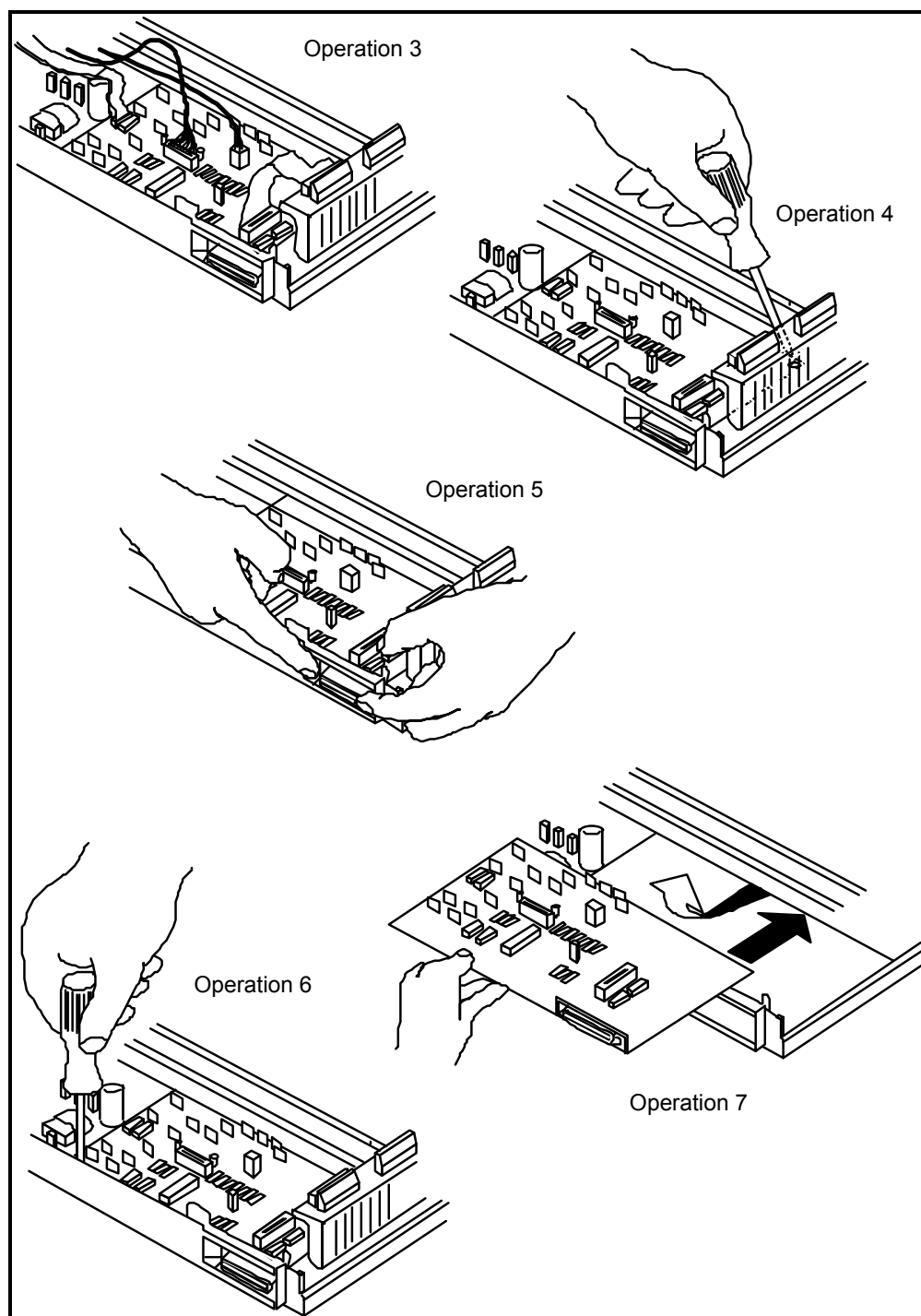


Figure 18-10 — Main Processing PWA Replacement

2113

18.5.3 Replacement of Main Processing Board

1. Unplug the power cable and disconnect the interface cable. Remove ribbon cartridge (subsection 18.4.3).
2. Remove tractor assembly. Remove the upper cabinet. See subsection 18.5.1.
3. Carefully disconnect all the cables plugged onto the main processing board as shown in Figure 18-10. Remove the serial interface board (not shown) by partially removing two cross-head screws which hold it to the base of the printer. One screw is located approximately 4 cm (1.5 inches) to the left of the DB-25 printer connector and the other is just around the corner from the DB-25 connector. When the screws have been loosened, pull straight up on the interface board to disconnect it from the main processing board.
4. Remove the inner side cross-head screw and washer (located on the left-side socket of the lower base), which secures the ASF 15-pin Cannon connector (RS-232) to the main processing board.
5. Slightly bend inward the two side connector clips of the 36-pin connector, located on the left-side socket of the lower base. This allows the connector to clear the cabinet in Step 7.
6. To release the main processing board, unscrew four cross-head screws on the sides of the board and another screw placed behind the power connector (5 screws in all).
7. Carefully remove the processing board by gently pushing it towards the front of the printer until the 36-pin connector clears the back of the printer, then lifting the rear of the board up and out of the printer.

To replace the main processing board, reverse this procedure.

18.6 Startup

Refer to the *Process Operations Manual*, in the *Process Operations* binder, for paper loading, ribbon changing, and other operational procedures.

18.7 Spare Parts

Table 18-3 — Spare Parts List

Part Number	Description
* 51195091-100	Printer, Replacement
51107142-100	PWA GH4B46
51107142-101	PWA GD2SUP
51107142-102	PWA Power Supply
51107142-103	PWA Operator Panel
51107142-104	PWA Protocol FW
* 51107142-105	Print Head
51107142-106	Cable, Print Head
51107142-107	Cartridge, Protocol
51107142-108	Paper Guide Assembly
51107142-109	Tractor, Right
51107142-110	Tractor, Left
51107142-111	Lever, Tractor Unlock
51107142-112	Spring, Tractor Lock
51107142-113	Lever, Roller Arm
51107142-114	Lever, Copy
51107142-115	Cartridge, Motor Assembly
51107142-116	Motor, Carriage
51107142-117	Shaft Assembly
51107142-118	Pulley, Inked Ribbon
51107142-119	Belt, Cartridge
51107142-120	Belt, Paper
51107142-121	Operator Panel Assembly
51107142-122	Fan
51107142-123	Uni-Block

Table 18-3 — Spare Parts List

Part Number	Description
51107142-124	Cover, Transparent
* 51195091-700	Ribbon, Black
* 51107142-125	Fuse, 1.6 A
51107142-126	Color Motor Assembly
51303631-003	Cable, RS-232 Direct Connect, 3 m (PIC, PDG, and EPDG(P) I/O board)
51303631-004	Cable, RS-232 Direct Connect, 4 m (PIC, PDG, and EPDG(P) I/O board)
51303631-008	Cable, RS-232 Direct Connect, 8 m (PIC, PDG, and EPDG(P) I/O board)
51303631-010	Cable, RS-232 Direct Connect, 10 m (PIC, PDG, and EPDG(P) I/O board)
51303631-015	Cable, RS-232 Direct Connect, 15 m (PIC, PDG, and EPDG(P) I/O board)
51308103-003	Cable, RS-232 Direct Connect, 3 m (EPDGC I/O board)
51308103-008	Cable, RS-232 Direct Connect, 8 m (EPDGC I/O board)
51308103-015	Cable, RS-232 Direct Connect, 15 m (EPDGC I/O board)

* ORU Level Replacement Item

18. ASPI-46 PRINTER

18.7. Spare Parts

19 SIGNUM 2043 PRINTER

19.1 Overview

The Signum 2043 printer is a replacement for the ASPI-41 and ASPI-46 printers. It is a 136 column printer that is capable of 400 characters per second with an ASPI protocol. Seven printable colors are available when a color ribbon is installed in the printer.

Two models of the printer are available, a 120 Vac model and a 240 Vac model. Only the 240 Vac model is CE Compliant.

Figures 19-1 and 19-2 are front and rear views of the Signum 2043 printer. Figure 19-3 is a pictorial of the printer's components.

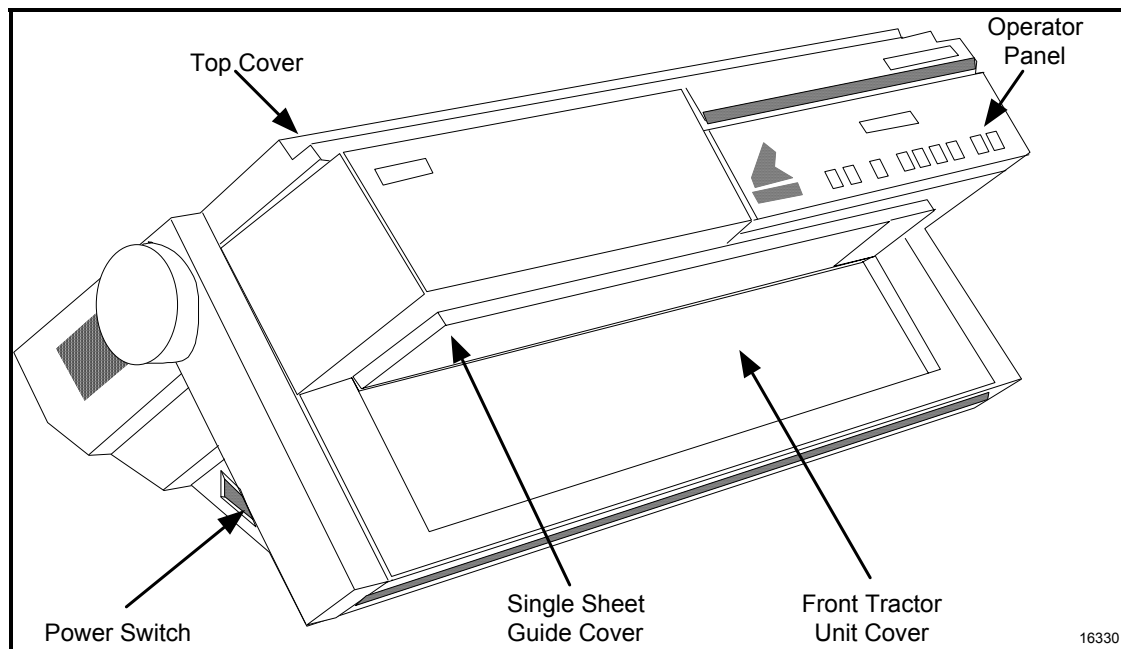


Figure 19-1 — Front View of Signum 2043 Printer

19. SIGNUM 2043 PRINTER

19.1. Overview

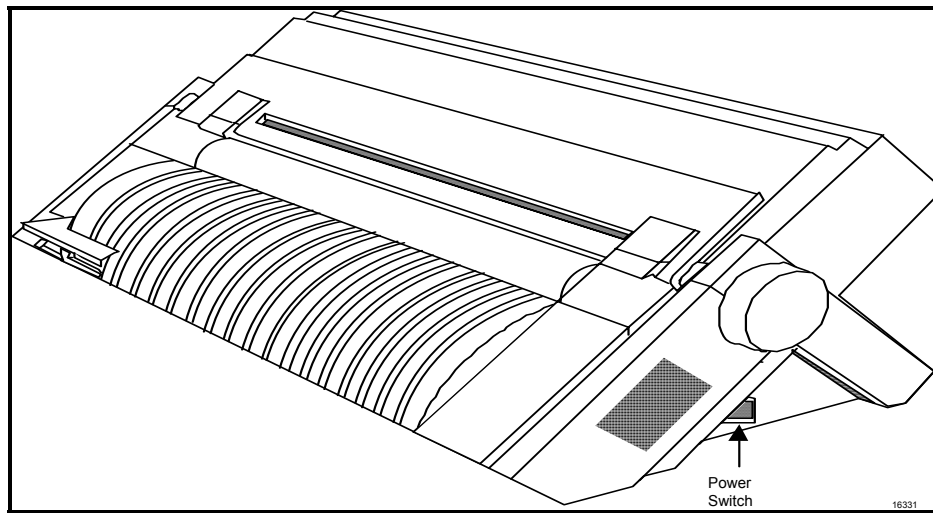


Figure 19-2 — Rear View of Signum 2043 Printer

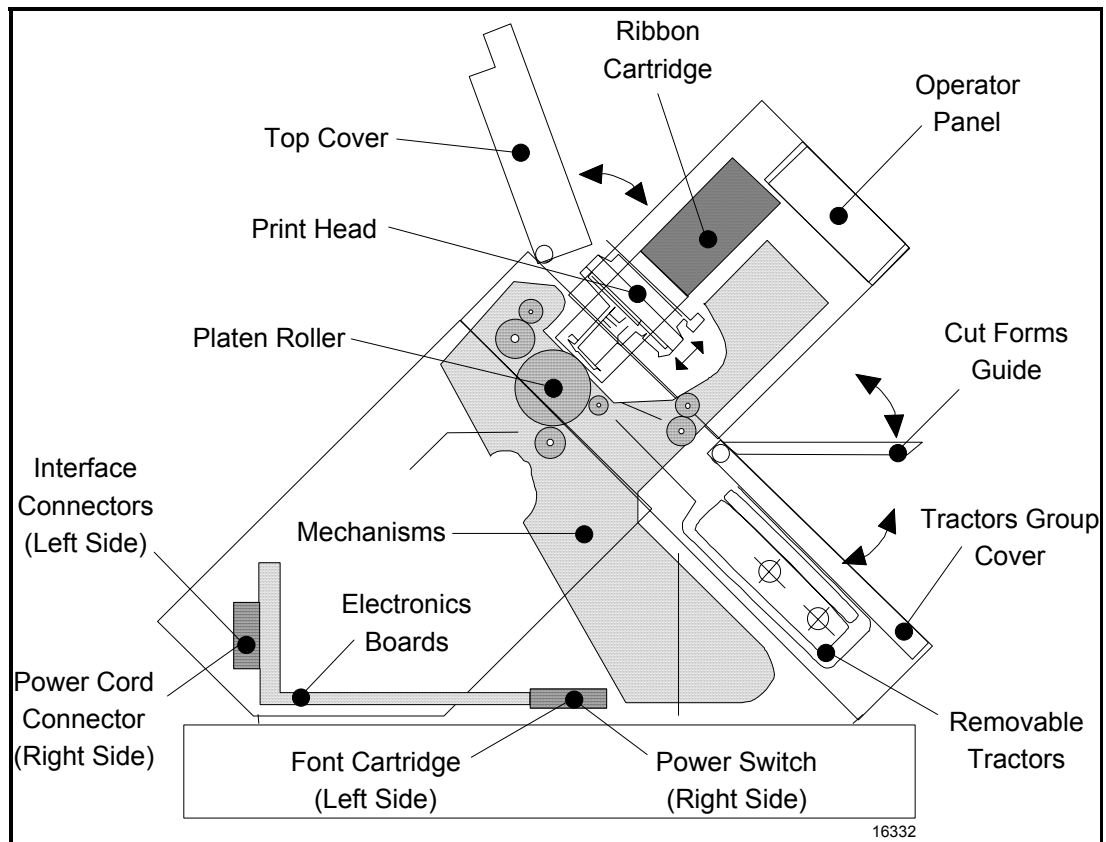


Figure 19-3 — Signum 2043 Printer Pictorial

19.2 Printer Cleaning

CAUTION

Due to the high temperature of some mechanical parts during prolonged operation (print head, carriage motor, and paper motor), be sure power has been removed from the printer for at least 5 minutes before beginning any cleaning.

There is no scheduled preventive maintenance other than periodic cleaning. Periodic cleaning of the printer will help provide optimal printer performance. Use the printer cleaning procedure in Table 19-1.

Table 19-1 — Printer Cleaning Procedure

Step	Action
1	Place the printer's power switch that is located at the left side of printer in the OFF position and unplug the power cord.
2	Open the top cover. If paper is loaded into the printer, remove the paper.
3	With a vacuum cleaner, remove any dirt or dust inside the printer. If necessary, use a brush to loosen any dirt accumulation, but be careful not to damage or misalign delicate components, such as the print head.
4	Close the cover and clean the exterior of the printer by using a soft-bristled brush or vacuum cleaner. Use a neutral detergent or water solution on a soft cloth to remove dirt and grease from the printer cabinet. Do not use an abrasive cloth, alcohol, paint thinner, or similar agents because they may discolor or scratch the cabinet surface.
5	Reconnect the power cord to the printer and place the printer's power switch in the ON position.

19.3 Printer Adjustments

There are no printer adjustments at the ORU (Optimum Replaceable Unit) level.

19.4 Troubleshooting/Testing

To help you isolate and correct minor printer problems, the following problem solving checks and procedures are recommended.

- Initializing procedure with Operator Panel messages
- Power malfunction checks
- Self-test procedure
- Test and diagnostic (T&D) procedure with error messages

When your printer has a problem, use the general troubleshooting/testing procedure in Table 19-2.

Table 19-2 — General Troubleshooting Procedure

Step	Action
1	If appropriate, perform the power malfunction checks.
2	If appropriate, observe the initialization procedure and Operator Panel messages.
3	Perform the printer self-test procedure. The instructions are in subsection 19.5.
4	Perform the diagnostic procedure. The instructions are in subsection 19.8.
5	If unable to diagnose and correct a problem, call the Honeywell Technical Assistance Center (TAC) for assistance. • In the U.S.A., use the toll-free number 1*800-822-7673 (available in the contiguous states except Arizona). • In Arizona, use 602-313-5558.

19.4.1 Power Malfunctions

CAUTION

The printer power supply is a switching type with some parts exposed to high voltage. The power supply cover must be in place before performing any operation while the printer's rear cover is removed.

Use the suggestions in Table 19-3 to help identify and correct a power malfunction.

Table 19-3 — Power Malfunction Guide

Problem	Possible Cause	Solution
POWER indicator extinguished	Power switch not in the on (1) position.	Place the printer's power switch in the on (1) position.
	Power cable not connected.	Verify that the cable is properly connected to the printer and the ac power outlet.
	Power absent at ac power outlet.	Verify that power is present at the ac power outlet.
	Primary power fuse blown.	Replace the primary power fuse. 120 Vac – 5 A (1 fuse) 240 Vac – 3.15 A (2 fuses)
No display information	The Operator Panel assembly is faulty or has a bad connection.	Check the connection to the main circuit board assembly.

19.5 Power Initialization

When power is applied to the printer by placing the Power switch in the on (1) position, it enters the initialization mode (bootstrap) before entering the normal operational mode. While in the initialization mode, the printer firmware initializes the hardware resources for the correction functionality.

The printer display will momentarily display: **STARTING UP** and then perform a full-matrix display check from the leftmost to the rightmost character position.

The printer display will momentarily display: ● ● ● ● ● ● ● ● ● ●

If initialization is successful, the printer enters the normal operational mode, moves the carriage assembly to the leftmost position to check the home position sensor, and then enters either the READY (paper installed) or WAIT (paper not installed) state.

If the printer does not initialize correctly, an error message is displayed on the printer display. See subsection 19.6 for the list of error messages.

19.6 Printer Display Errors

Some errors can be displayed by the printer. The errors are divided into three main groups.

- Status errors
- Recoverable errors
- Nonrecoverable errors

19.6.1 Status Errors

When a Status error occurs, the printer enters the WAIT state and displays a primary message that indicates the error. If appropriate, a secondary error message is displayed that provides more specific information about the primary error.

Pressing the ON LINE key resets the error condition.

Table 19-4 describes the printer status error messages.

Table 19-4 — Printer Status Errors

First Message	Second Message	Cause/Solution
PAPER JAM	CHECK ALL PATHS	Cause: A paper jam error condition exists in the paper path. Solution: Check all the paper paths and remove the jammed paper. Press the ON LINE key to reset the error condition.
JAM ON PATH2	CHECK PAPER	Cause: A paper jam error condition exists in paper path 2. Solution: Check the front push tractor unit and remove the jammed paper. Press the ON LINE key to reset the error condition.
JAM ON PATH3	CHECK PAPER	Cause: A paper jam error condition exists in paper path 3. Solution: Check the front push tractor unit and remove the jammed paper. Press the ON LINE key to reset the error condition.
AGA NOT OPER	ADJUST THE GAP	Cause: The automatic gap adjustment (A.G.A.) is not enabled. Solution: Manually adjust the gap. Press the ON LINE key to reset the error condition.
BUFFER OVERFLOW	CHARACTER LOST	Cause: An overflow condition exists in the buffer. Solution: Place the printer in the OFF position, and the buffer will be cleared. Press the ON LINE key to reset the error condition.

Table 19-4 — Printer Status Errors

First Message	Second Message	Cause/Solution
DATA SET OFF	N/A	Cause: The DSR signal is off, and the printer is not ready for data transfer. Solution: Press the ON LINE key to reset the error condition.
RIBBON BLOCKED	CHECK RIBBON	Cause: The cartridge ribbon is blocked. Solution: Check to be sure that the ribbon cartridge is inserted correctly. Turn the tension knob to verify that the ribbon is not jammed. Press the ON LINE key to reset the error condition.
PATH2 TRIM ERROR	N/A	Cause: A trimmer sensor error exists in paper path 2. Solution: Press the ON LINE key to reset the error condition.
PATH3 TRIM ERROR	N/A	Cause: A trimmer sensor error exists in paper path 3. Solution: Press the ON LINE key to reset the error condition.
PRINT INTEGRITY	N/A	Cause: Abnormal print out because of a possible print carriage blocking. Solution: Press the ON LINE key to reset the error condition.

19.6.2 Recoverable Errors

When a recoverable error occurs, the printer enters the WAIT state and displays a primary error message. If appropriate, a secondary error message is displayed that provides more specific information about the primary error. An intermittent buzzer may sound.

All error displays are reset by either pressing the ON LINE key or by a specific action.

Table 19-5 describes the printer status error messages.

Table 19-5 — Printer Status Errors

First Message	Second Message	Cause/Solution
# COVER OPEN	N/A (Continuous buzzer)	Cause: The top cover was open during the Initialization mode. Solution: Close the cover. Check to see if a magnet is installed on the cover. Cause: The Operator Panel may be faulty. Solution: Replace the Operator Panel.
CLOSE COVER	N/A	Cause: The top cover was opened during operation. Solution: Close the top cover.
PAPER JAM	CHECK ALL PATHS	Cause: A paper jam error condition existed in a paper path during a print head parking or paper ejection operation. Solution: Check to see that the paper is correctly installed and remove the paper if it is jammed.
RIBBON BLOCKED	CHECK RIBBON	Cause: The cartridge ribbon is blocked and the relative sensors detect the fault. The problem occurs when some characters are printed with poor ink density. Solution: Check that the ribbon is properly inserted. To be sure that the ribbon is not jammed, remove the ribbon and turn the tension knob .

Table 19-5 — Printer Status Errors

First Message	Second Message	Cause/Solution
PRINT INTEGRITY	N/A	Cause: The carriage assembly is blocked and the relative home position sensor detected the fault. The problem occurs when the printout is shifted to the right or left side. Solution: Check to see that the carriage assembly is free.
BUFFER OVERFLOW	CHARACTER LOST	Cause: An overflow condition occurred in the input buffer. The EPDG I/O, EPDGC I/O, TPDG I/O, or TPDGC I/O board is transmitting data to the printer even though it was signaled to stop. Solution: Check the connection between the printer and the EPDG I/O, EPDGC I/O, TPDG I/O, or TPDGC I/O board. Press the ON LINE key to reset the error. If the buffer must be cleared, cycle the printer power off, then back on.
DATA SET OFF	N/A	Cause: The DSR signal to the printer is not active, and the printer is not ready for data transfer. Solution: Check the connection between the printer and the EPDG I/O, EPDGC I/O, TPDG I/O, or TPDGC I/O board.

19.6.3 Nonrecoverable Errors

When a nonrecoverable error occurs, the printer is halted and enters an Error state. A nonrecoverable error cannot be reset by a specific action, and therefore the printer power must be cycled off and back on.

Typically, a nonrecoverable error is detected while the printer is in the Bootstrap or normal operational state.

If the nonrecoverable error repeats after cycling the printer power off and back on, run the T&D program that is described in subsection 19.10 to identify the problem. If after correction of the problem, the nonrecoverable error reoccurs, contact your nearest Honeywell Technical Assistance Center.

In the Bootstrap state, the following error messages could be displayed. If one of the messages does appear, the Bootstrap will not complete, and consequently the printer hardware is not properly initialized.

- STARTING UP (fixed)
- SOFTWARE ERROR
- ENGINE DRIVER
- PM SET-UP ERR
- # VOID EXEC LIST
- # MFILE CHECKSUM
- UNK MFILE TYPE
- TOO MANY FILES
- # CARD 0 REMOVED

Generally, all the above error messages are related to either an EPROM, microcontroller, or data bus failure.

In the normal operational mode, the following messages may be displayed. If one of the messages does appear, the Bootstrap was completed, but a handshaking problem occurred between two of the firmware modules.

- SPV FLT n1 n2 n3 (Supervisor fault)
- INT FLT n1 n2 n3 (Interpreter fault)
- P_M FLT n1 n2 n3 (Print Manager fault)
- ENG FLT n1 n2 n3 (Engine Device fault)

19.7 Printer Self-Test

Self-test is an internal printer test that checks the printer's logic and electromechanical performance at the user level. A test pattern will be produced on a sheet of printer paper.

Before beginning the self-test, verify that:

- the paper and ribbon cartridge are installed correctly.
- the power cable is connected and the printer's power switch is in the off (0) position.
- The printer's top cover is closed.

Use the procedure in Table 19-6 to initiate and execute the printer's self-test. Figure 19-4 is an illustration of the printout.

Table 19-6 — Signum 2043 Self-Test Procedure

Step	Action
1	Press and hold the ON LINE key while placing the printer's power switch in the on (1) position. Release the ON LINE key. The display shows TEST, and a pattern is printed on the paper as illustrated in Figure 19-4. The printer continues to print until the ON LINE key is pressed a second time. Self-test will end if the printer runs out of paper.
2	The display shows WAIT. You can run self-test with the interface cable disconnected. If the interface cable is not connected to the EPDG I/O, EPDGC I/O, TPDG I/O, or TPDGC I/O board, the printer will display OFFD rather than WAIT when stopped.

19.7. Printer Self-Test

- Equal character density
- Equal character darkness
- Character sharpness
- Character completeness
- Character underline and descenders
- Character smudges or strikes
- Equal line density
- Character composition (graphics)

```

ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
BCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
CDEFGHIJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
DEFGHIJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
EFGHIJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
FGHIJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
GHIJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
HIJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
IJKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
JKLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
KLMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
LMNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
MNOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
NOPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
OPQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
PQRSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
RSTUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
STUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
TUVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
UVWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
VWXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
WXYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$
XYZ[\]^_ 'abcdefghijklmnopqrstuvwxyz{|}~ ;ç£ø¥|$

```

Figure 19-4 — Signum 2043 Printer Self-Test Print Pattern

19.8 Hex Dump Test

The hex dump test should be performed when a protocol line error is detected, and other internal tests do not detect an error. The hex dump test provides a printout of all the characters that were previously transmitted to the printer by the host hardware. Control codes are also included such as ESC, DC4, CR, LF, etc.

The procedure is executed by entering PROGRAM SETUP, selecting HEX-DUMP PRNT, and choosing the value YES. After initiating the procedure, any character sent to the printer will be printed as its hexadecimal value.

The following is an illustration of a text example in the normal mode.

ASCII: ESC C H
HEX: 1B 43 48
BASIC: CHR\$(27);CHR\$(67);CHR\$(72)

The same text is illustrated in the hex-dump mode.

41 53 43 49 49 3A 20 45 53 43 20 43 20 48 0D 0A 48 45 58 20 20 3A 20 31
42 20 34 33 20 34 38 0D 0A 42 41 53 49 43 3A 20 43 48 52 24 28 32 37 29
3B 43 48 52 24 28 36 37 29 3B 43 48 52 24 28 37 32 29 0D 0A

19.9 Fault Probability Guide

Sometimes a fault is detected, but there is no error display.

Usually, if a fault is associated with an electromechanical assembly, no error is detected by the printer firmware because there is no electronic feedback for the fault condition. Specifically, except for the home position and ribbon sensors, the other mechanical parts do not have a return feedback because they are part of an open loop circuit. The home position and ribbon sensors have a feedback circuit to return the status of the part.

An example of an electromechanical fault is the paper not advancing upon command. The paper motor, driver circuit, and logic circuitry are involved, but there is no way to identify the part that actually failed. For this fault, only paper movement verification is possible during the T&D printout.

Table 19-7 is a list of causes and solutions for inconsistent character printing. The character printing may be dark, light, have gaps or spaces, or be incomplete.

Table 19-7 — Inconsistent Character Printing

Cause	Solution
The ribbon cartridge installed improperly.	Reinstall the cartridge.
The ribbon cartridge may be faulty.	Install a new cartridge.
The ribbon does not perform correctly (RIBBON BLOCKED error message).	Verify that the carriage unit movement does not interfere with the ribbon movement. If it does, the left support group pivot assembly must be replaced.
The print head cable is installed improperly.	Check the mounting of the print head cable. The print head cable must be reinstalled or replaced.
The print head may be faulty.	The print head must be replaced.
The main electronics board may be faulty.	The main electronics board must be replaced.
The power supply electronics board may be faulty.	The power supply electronics board must be replaced.
The gap between the print head and platen is incorrect.	The gap between the platen and print head must be adjusted.

Table 19-8 is a list of causes and solutions when the printer stops printing or prints slowly.

Table 19-8 — Printing Stops or Slow Printing

Cause	Solution
The thermosensor in the print head is damaged.	It may be necessary to replace the print head if it is not to specifications.
The connection between the print head and the main electronics board is faulty.	Check the connection. It may be necessary to replace the cable.
The main electronics board may be faulty.	It may be necessary to replace the main electronics board.

Table 19-9 is a list of causes and solutions when the quality of the color printing is poor.

Table 19-9 — Color Printing Poor

Cause	Solution
The color ribbon cartridge is installed incorrectly.	Reinstall the color ribbon cartridge.
The color motor is not mounted correctly or is faulty.	Check the mounting of the motor. It may be necessary to replace the motor.
The connection between the print head and the main electronics board is faulty.	Check the connections. It may be necessary to replace the cable.
The main electronics board may be faulty.	It may be necessary to replace the main electronics board.

Table 19-10 is a list of causes and solutions when the character printing is misaligned vertically.

Table 19-10 — Character Printing Misaligned Vertically

Cause	Solution
The printer is misaligned.	Run T&D to check vertical printer alignment.
The carriage belt is not adjusted correctly.	The carriage belt may require adjustment.

Table 19-11 is a list of causes and solutions when the tractor/friction solenoid is not operable.

Table 19-11 — Tractor/Friction Solenoid Not Operable

Cause	Solution
The tractor/friction solenoid may be faulty.	It may be necessary to replace the tractor/friction solenoid.
The connection between the tractor/friction solenoid and the main electronics board may be faulty.	Check the connection. It may be necessary to replace the cable.
The main electronics board may be faulty.	It may be necessary to replace the main electronics board.

Table 19-12 is a list of causes and solutions when the paper does not advance correctly.

Table 19-12 — Paper Not Advancing Correctly

Cause	Solution
The connection between the paper motor and the main electronics board is faulty.	Check the connection on the board. It may be necessary to replace the motor.
The paper motor may be faulty.	It may be necessary to replace the motor.
The paper movement does not provide equal line spacing.	The belt tension may require adjustment.
The main electronics board may be faulty.	It may be necessary to replace the main electronics board.
The holes of the paper fanfold are torn.	Remove the torn paper and reload the paper. Adjust the tension of the paper between the left and right tractors.
The paper is not aligned correctly on the sprocket wheels.	Reload the paper and ensure that the corresponding holes on each side of the paper are properly aligned on the tractors

Table 19-13 is a list of causes and solutions when the printer initialized, but the display and LEDs are blank. The Operator keys are available.

Table 19-13 — Display and LEDs Blank

Cause	Solution
The connection between the Operator Panel and the main electronics board is faulty.	Check the connections. It may be necessary to replace the Operator Panel or main electronics board.

Table 19-14 is a list of causes and solutions when the printer initialized, but the display is blank. The Operator keys are not available.

Table 19-14 — Display Blank

Cause	Solution
The connection between the Operator Panel and the main electronics board is faulty.	Check the connections. It may be necessary to replace the Operator Panel or main electronics board.
The ribbon sensor is faulty.	Check the ribbon sensor action through the pivot assembly. It may be necessary to replace the pivot assembly, the sensors cable, or the ribbon motor assembly.

Table 19-15 is a list of causes and solutions when the printer initialized, but the carriage assembly moved 1 inch to the right.

Table 19-15 — Carriage Assembly Moves 1 Inch Right

Cause	Solution
The connection to the home position sensor or the carriage motor is faulty.	Check the home position sensor connection and carriage motor connection on the main electronics board.
The home position sensor or carriage motor assembly are faulty.	Check the ribbon sensor action through the carriage assembly and the impedance of the motor. It may be necessary to replace the home position sensor cable or the carriage motor assembly.

Table 19-16 is a list of causes and solutions when the printer did not initialize and the buzzer sounds continuously.

Table 19-16 — Printer Not Initialized and Buzzer Sounds

Cause	Solution
The connection to the home position sensor or the carriage motor is faulty.	Check the home position sensor connection and carriage motor connection on the main electronics board.
The home position sensor or carriage motor assembly are faulty.	Check the ribbon sensor action through the carriage assembly and the impedance of the motor. It may be necessary to replace the home position sensor cable or the carriage motor assembly.

19.10 Test and Diagnostic Procedure

This section was deleted due to the unavailability of the software.

19.11 Printer Replacement

CAUTION

When replacing a printer, be sure that the power source line voltage is the same as the voltage requirement displayed at the back of the printer. If the line voltage does not match the printer voltage, do not use the printer; instead, consult your Honeywell representative.

The printer is the Optimal Replaceable Unit (ORU). A few lower level replacement parts are also currently (1996) available by factory order.

Use the procedure in Table 19-17 to disconnect and remove a printer.

Table 19-17 — Printer Removal Procedure

Step	Action
1	Place the printer's power switch that is located at the left side of the printer in the off (0) position.
2	Disconnect the ac power cord.
3	Disconnect the data cable from its connector at rear of the printer.
4	Remove the printer.

Use the procedure in Table 19-18 to install a printer.

Table 19-18 — Printer Installation Procedure

Step	Action
1	Lift the printer from its shipping box by firmly grasping both sides of the printer. Temporarily place the printer on a flat work surface.
2	Remove the printer accessories that are included in the box. The accessories should include the following items. • ribbon cartridge • tractor unit • power cable • driver diskette An optional second tractor unit and color ribbon cartridge in separate boxes may be included with the printer.
3	If a ribbon cartridge is not installed in the printer, use the procedure in subsection 19.12 to install a ribbon cartridge.
4	If the printer must operate with one or two tractors, use the procedure in subsection 19.13 to install the tractor(s).
5	Choose a suitable location for the printer. Consider the following: • The distance between the printer and the US or U^XS module must not exceed the length of the interface cable. • The location must be sturdy, horizontal, and stable. • The printer must not be exposed to direct sunlight, extreme heat, cold, dust, or high humidity. • The ac power outlet must be near the printer and compatible with the printer's power cord plug. Ground must be provided through the plug. The voltage at the outlet must match the voltage shown on the printer's name plate. Place the printer in its operating location.
6	Connect one end of the 25-pin interface cable to the I/O interface board (EPDG I/O, EPDGC I/O, TPDG I/O, or TPDGC I/O) in the US or U^XS module that will control the printer and the other end of the cable to the printer. To secure the connections, tighten the connector screws at both ends of the cable.
7	Verify that the printer's power switch that is located at the left side of the printer is in the off (0) position. Connect the power cord to the power connector at the rear of the printer.
8	Place the power switch in the on (1) position. The print head should move to the left side of the printer and stop.

19.12 Ribbon Cartridge Replacement

CAUTION

The print head can become hot during extended operation. Be sure it is cool enough to touch before starting to remove the printer cartridge.

Use the procedure in Table 19-19 to replace a printer cartridge.

Table 19-19 — Ribbon Cartridge Replacement Procedure

Step	Action
1	Place the printer's power switch in the off (0) position.
2	Open the printer cover by pressing on the release buttons.
3	Move the printer carriage to the center of the printer.
4	Pull the paper thickness lever towards the front of the printer to facilitate the removal of the ribbon cartridge.
5	Grasp the ribbon cartridge firmly at its sides and lift it straight up until the side clips release it.
6	Remove the new ribbon cartridge from its bag. Remove and discard the hold fast that blocks the ribbon.
7	Before installing the ribbon cartridge, turn the ribbon winding knob on the cartridge in the direction of the arrow on the cartridge to take up any ribbon slack.
8	Insert the ribbon between the print head and the print head mask. Lay the cartridge over the printer carriage. Make sure the two pins on each side of the cartridge are positioned over the white retaining clips of the printer carriage.
9	Push the cartridge gently down while turning the tension knob. Make sure that the cartridge clips into place.
10	To tighten the ribbon, turn the tension knob in the direction of the arrow on the ribbon cartridge.
11	To ensure that the ribbon guide is free, manually move the print carriage from right to left and back.

19. SIGNUM 2043 PRINTER

19.12. Ribbon Cartridge Replacement

12	Move the paper thickness lever towards the back of the printer and adjust it to obtain the best quality printing.
13	Check the quality of print with the self-test procedure in subsection 19.5. If the pattern is not satisfactory, adjust the paper thickness lever accordingly.
14	Close the front printer cover.
15	Place the printer's power switch in the on (1) position.

19.13 Tractor Installation/Removal

A paper tractor can be installed in three positions. One position is in the front of the printer, and two positions are at the rear of the printer.

The front position provides push paper feeding, and the rear positions are for either push or pull paper feeding.

CAUTION

Be sure the printer's power switch is in the off (0) position before installing or removing a tractor.

Use the procedure in Table 19-20 to install or remove a tractor in the front position.

Table 19-20 — Tractor Front Position Installation/Removal Procedure

Step	Action
1	Place the printer's power switch in the off (0) position.
2	To install the tractor, lift the front tractor cover and fasten it to the single sheet support cover.
3	Lift the tractor connector protector at the right side of the printer. Insert the tractor with its connector on the right by pushing it upward until it clicks.
4	Close the front tractor cover.
5	To remove the tractor, press the two small levers at both sides of the tractor that secure the tractor in place and pull the tractor downward. Close the tractor connector protector.
6	Close the front tractor cover.
7	Place the printer's power switch in the on (1) position.

Figure 19-5 illustrates front tractor push operation.

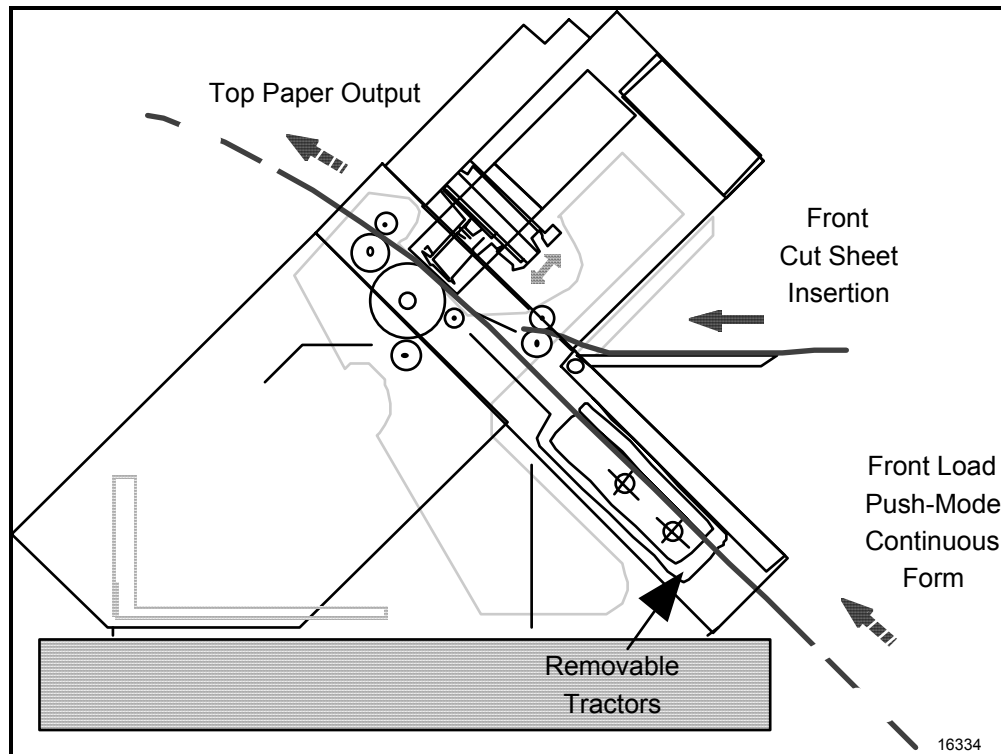


Figure 19-5 — Front Tractor Push Operation

Use the procedure in Table 19-21 to install or remove a tractor in the rear push position.

Table 19-21 — Tractor Rear Push Position Installation/Removal Procedure

Step	Action
1	Place the printer's power switch in the off (0) position.
2	Remove the rear tractor cover.
3	Set the slider on the tractor in the push position (down position).
4	Orient the tractor with its grooves facing downward and its connector on the left side.
5	Insert the left and right tractor grooves into the printer pins until the tractor clicks in place.
6	To remove the tractor, press the two small levers that secure the tractor in place and pull upward.
7	Replace the rear tractor cover.
8	Place the printer's power switch in the on (1) position.

Figure 19-6 illustrates rear tractor push operation.

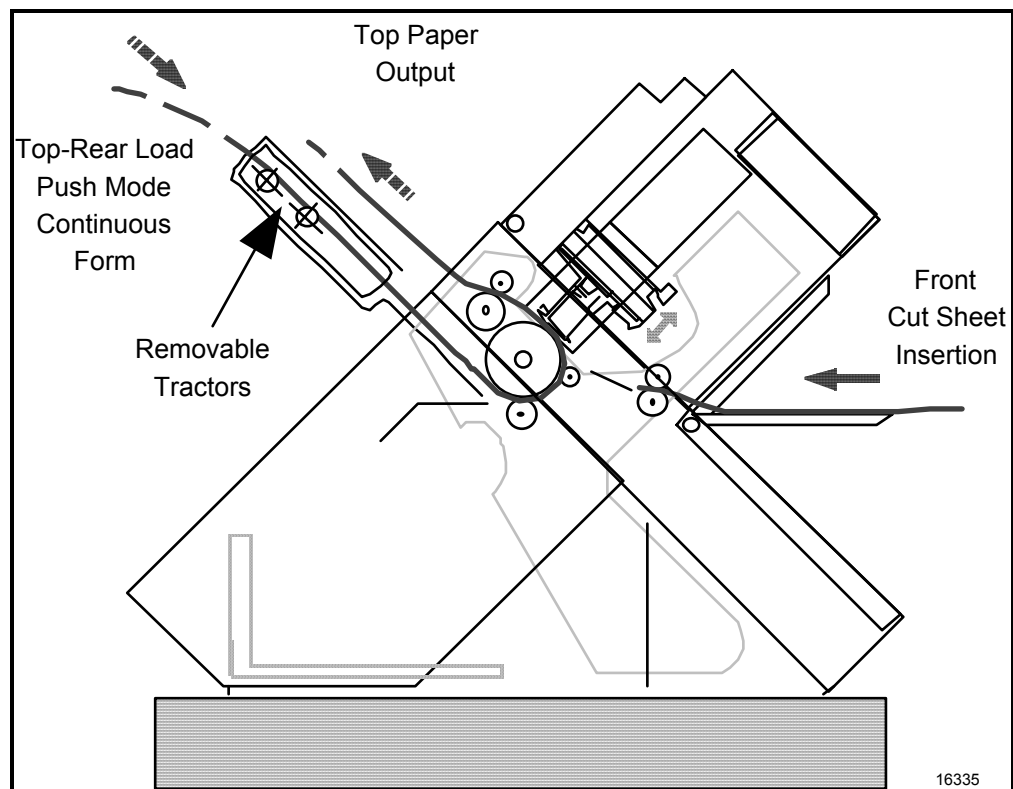


Figure 19-6 — Rear Tractor Push Operation

Use the procedure in Table 19-22 to install or remove a tractor in the rear pull position.

Table 19-22 — Tractor Rear Pull Position Installation/Removal Procedure

Step	Action
1	Place the printer's power switch in the off (0) position.
2	Remove the rear tractor cover.
3	Set the slider on the tractor in the pull position (up position).
4	Move the paper bail to the forward position.
5	Orient the tractor with its grooves facing downward and its connector on the left side.
6	Insert the left and right tractor grooves into the printer pins until the tractor clicks in place.
7	Position the levers under the tractor down to unlock them and gently press the tractor downward until it clicks in place.
8	To remove the tractor, lift the tractor, press the two small levers that secure the tractor, and pull upward.
9	Replace the rear tractor cover.
10	Place the printer's power switch in the on (1) position.

Figure 19-7 illustrates rear tractor pull operation.

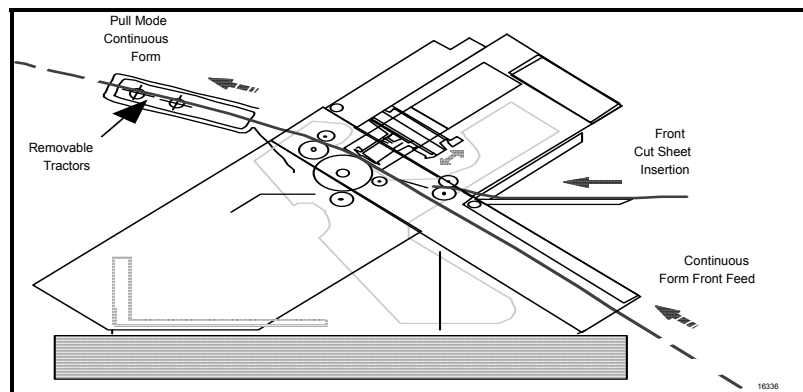


Figure 19-7 — Rear Tractor Pull Operation

Dual tractors can be installed in both the front and rear positions simultaneously. Continuous paper can be loaded at either the front or rear of the printer.

Figure 19-8 illustrates either front tractor push operation or rear tractor pull operation.

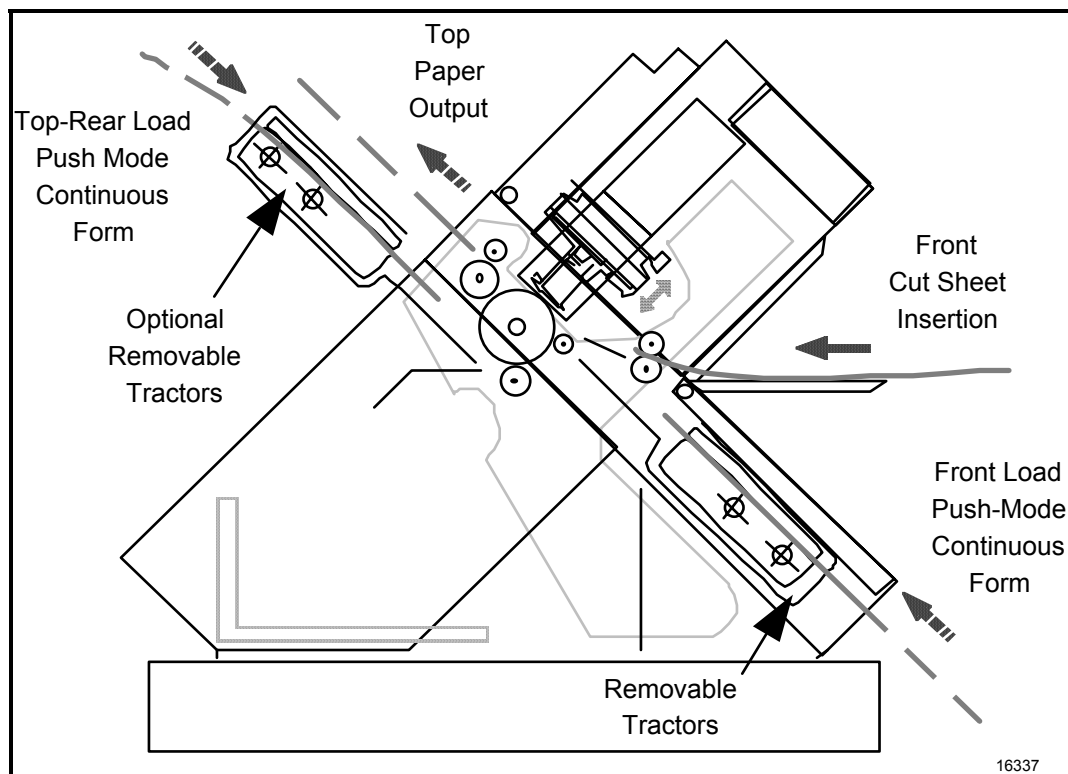


Figure 19-8 — Front Tractor Push or Rear Tractor Pull Operation

Figure 19-9 illustrates front tractor push operation and rear tractor pull operation.

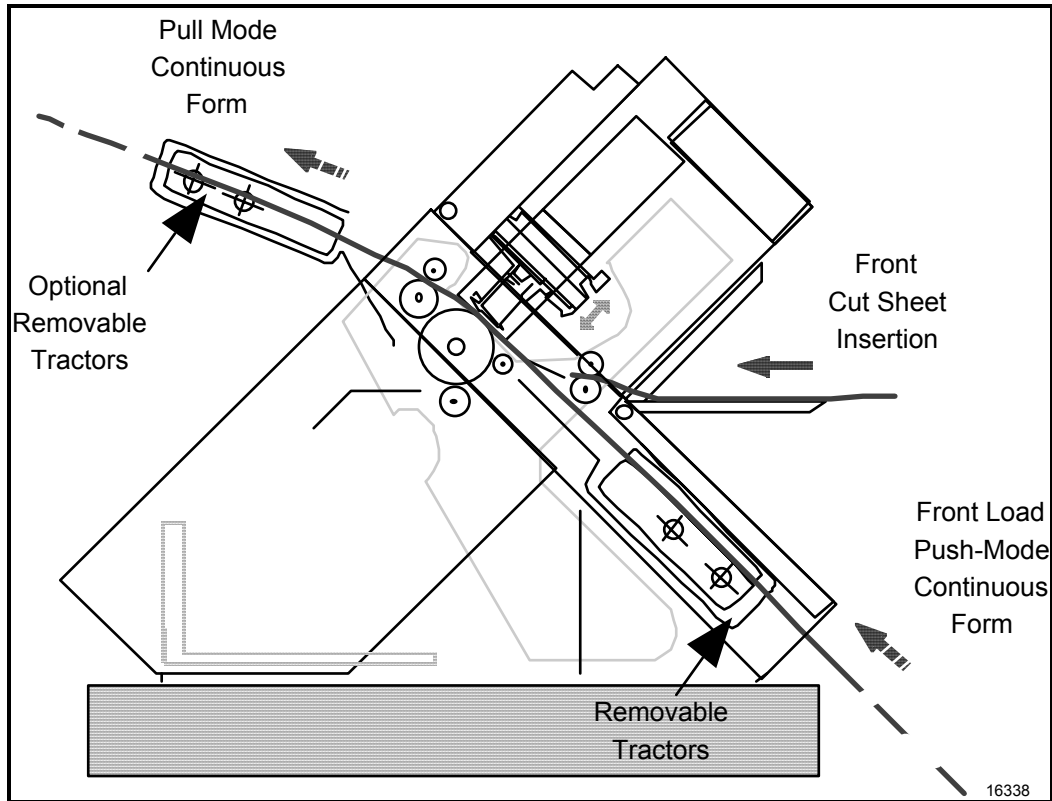


Figure 19-9 — Front Tractor Push and Rear Tractor Pull Operation

19.14 Print Head Replacement

Use the procedure in Table 19-23 to remove and replace a print head.

Table 19-23 — Print Head Removal/Replacement Procedure

Step	Action
1	Place the printer's power switch in the off (0) position. Be sure the print head is not too hot to touch.
2	Depress the push-buttons on each side of the top cover and open the cover.
3	Use the procedure in subsection 19.12 to remove the ribbon cartridge.
4	Loosen and remove the screw that secures the print head cable to the carriage assembly.
5	Disconnect the print head cable from the print head.
6	Loosen and remove the two screws that secure the print head to the carriage assembly.
7	Remove the print head from the carriage assembly.
8	Install a new print head by performing the above steps in reverse order.
9	Run the self-test (subsection 19.7) to verify that the new print head is properly installed.

Figure 19-10 illustrates replacement of the print head.

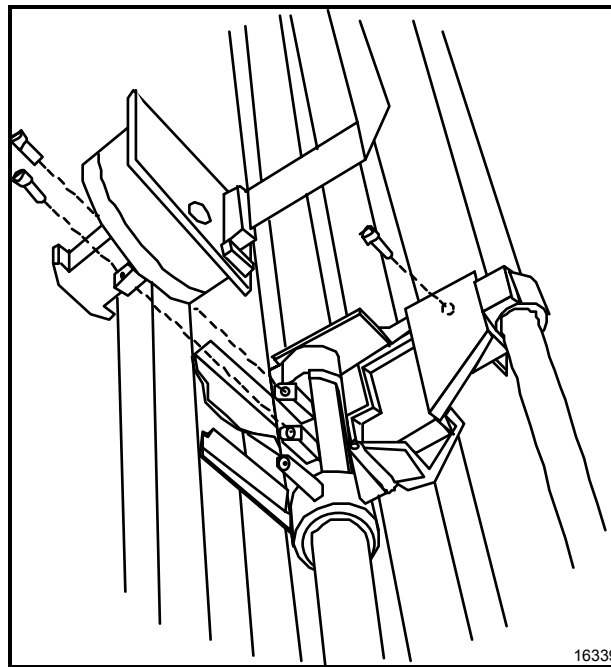


Figure 19-10 — Print Head Replacement

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19.15. Replacement Parts

19.15 Replacement Parts

Table 19-24 is a list of Optimal Replaceable Units (ORU) for the Signum 2043 printer that can be purchased from Honeywell. Other than a replacement print head and a tractor, no other electromechanical assemblies or electronic board assemblies are considered ORU parts.

Table 19-24 — Optimal Replaceable Units (ORU) Parts List

Description	Part Number
Signum 2043 Printer, 120 Vac	51196690-100
Signum 2043 Printer, 240 Vac	51196690-200
Ribbon Cartridge, Black	51196690-400
Ribbon Cartridge, Color	51196690-501
Paper Tractor Unit	51196690-800
Printer Floor Pedestal	51196690-900
Print Head	51196690-920

20 TREND PEN RECORDER

20.1 Cleaning

Clean the recorder face glass with glass cleaner and a soft lint-free cloth. With the front door to the recorder open, grasp the chassis latch (see Figure 20-1) and pull the chassis straight out. Carefully clean the vertical scales with detergent and a cotton-tipped swab or soft cloth.

20.2 Adjustments

Refer to the *Recorder Service Data* and *Recorder Parts List* in the Addendum of this manual.

20.3 Test/Troubleshooting

Refer to the *Recorder Service Data* in the Addendum of this manual. Recorder cabling information is provided in subsection 20.6.

20.4 Removal/Replacement

1. Remove the rear cover from the recorders by loosening four captive screws and sliding the cover straight off the console. Set the switch on the power supply to "0" (off). With the front door open, remove the recorder chassis by pulling straight out on the chassis latch (see Figure 20-1) until the stop is reached. Support the unit with one hand, press the service latch, and withdraw the unit the rest of the way from the console. Disconnect the cable connector to free the unit.

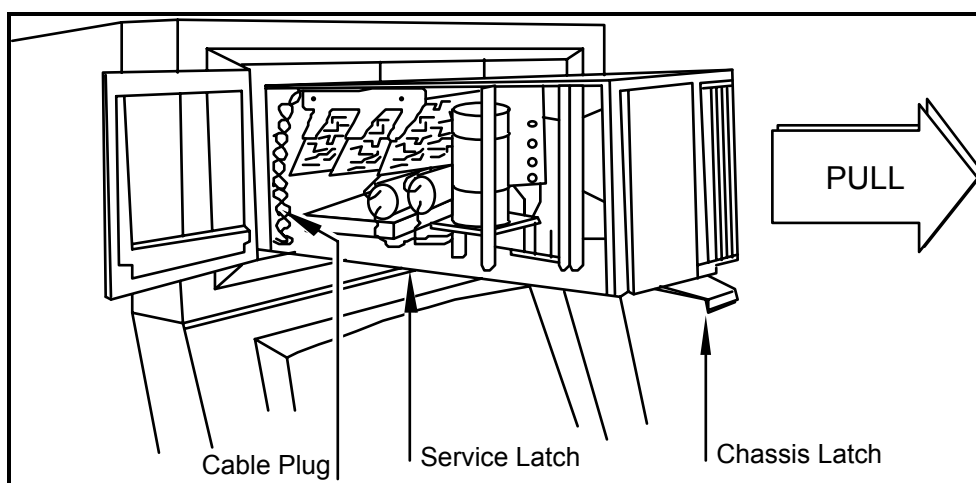


Figure 20-1 — Trend Pen Recorder

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20. TREND PEN RECORDER

20.5. Startup

2. To remove the power supply, tag if required, then remove two wires on the power supply terminal board. Remove the three connectors from the power supply. Remove the four screws, flat washers, and lock washers that secure the power supply to the chassis and lift the power supply off the chassis.
3. Replace the recorder by connecting the cable to the connector, pressing the service latch, and sliding the chassis all the way into the console. Ensure that the wires are not bound or cut and that the chassis latch is secure before closing the front door.
4. Secure the replacement power supply to the chassis with four screws, flat washers, and lock washers. Connect the three cables to the power supply, and secure the two wires to the power supply terminal board with screws and lock washers. Ensure that the wires are installed as indicated on the wire tags.

20.5 Startup

Set the power-supply switch to "1" (on), and ensure that the green LED is on. Slide the rear cover onto the console, and secure with the captive fasteners.

20.6 Cabling

Trend Pen recorder cabling is shown in Figure 20-2. Figure 20-3 shows the Trend Pen recorder cabling with the Motor Stop option installed.

20.7 Spare Parts

Refer to *Recorder Service Data* and *Recorder Parts List* in the Addendum of this manual.

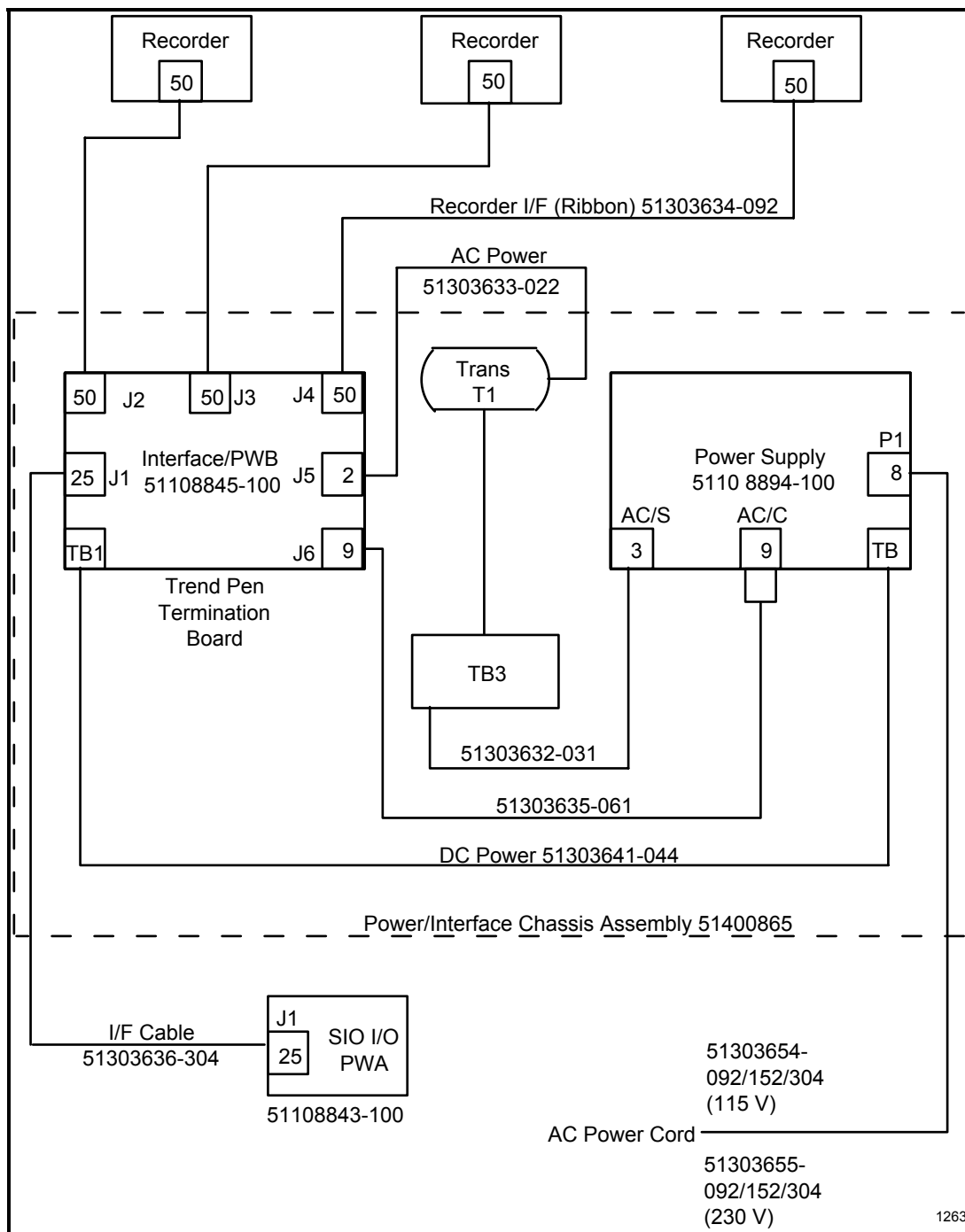


Figure 20-2 — Trend Pen Recorder Cabling

20. TREND PEN RECORDER

20.7. Spare Parts

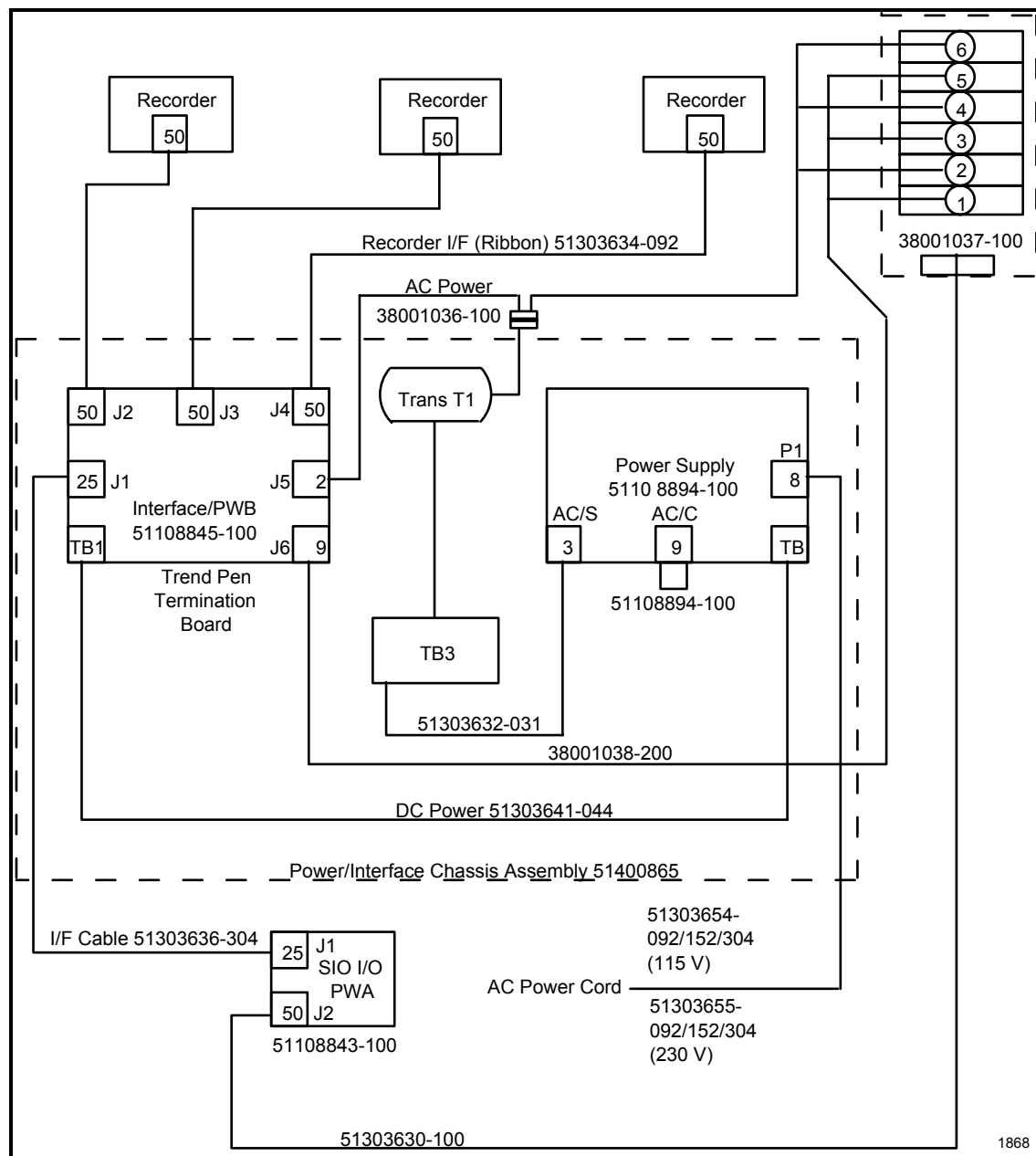


Figure 20-3 — Trend Pen Recorder Cabling with Motor Stop Option

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21.1 Overview

The DPR100 Trend Pen recorder has three pens. Three DPR100 Trend Pen recorders are mounted side by side (arranged in a 3/3/2 configuration) in a pod above the pod that houses a single CRT monitor. The upper pod can accommodate two rows of three recorders each.

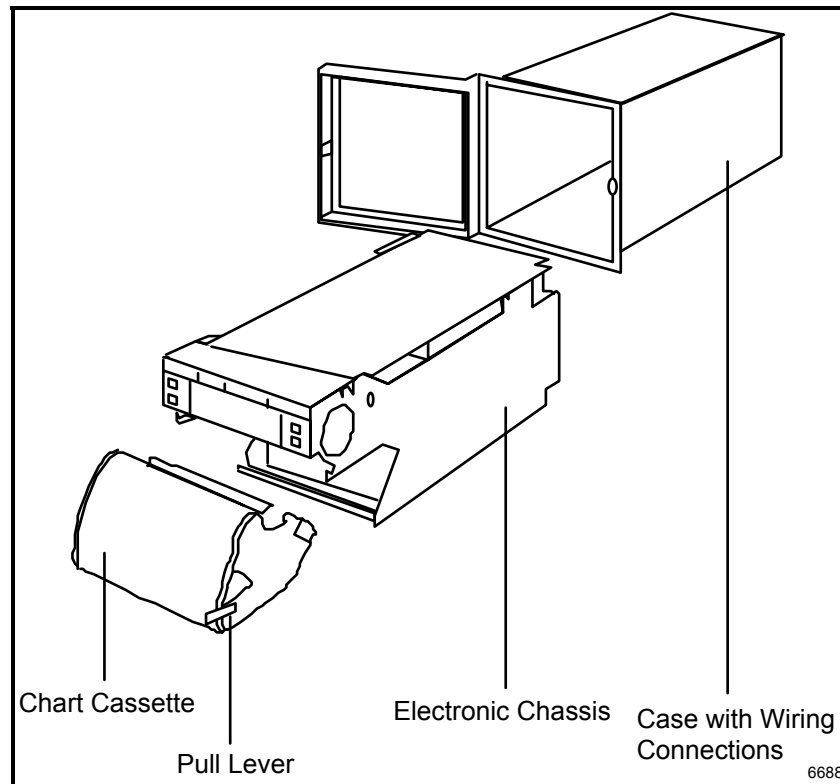


Figure 21-1 — Trend Pen Recorder with Chart Cassette

The DPR100 recorder is only available as a 3-pen unit. The third channel must be turned off when not in use. Refer to the *DPR100 Product manual*, US11-6137, Section 7, *Detailed Configuration Analog Inputs, Sensor*. There should be no entry for Channel 3.

The Trend Pen recorder attaches to the front plate of an upper Universal Station pod. There are two sets of terminals on the recorder.

- ac power terminals
- signal terminals for each recorder channel

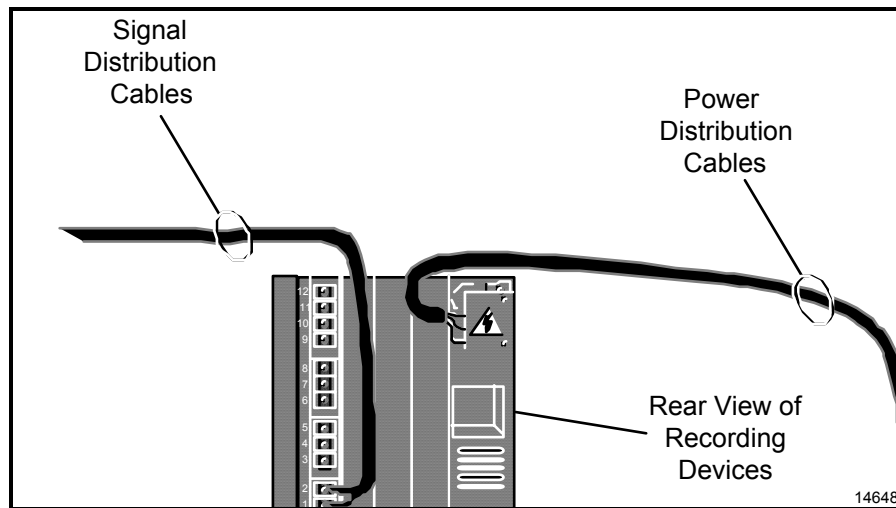


Figure 21-2 — Rear View of Trend Pen Recorder Illustrating Wiring Connections

21.2 Cleaning

The only cleaning recommendation is to clean the front cover with a mild solution of detergent, rinse it thoroughly and dry it with a soft lint-free cloth.

21.3 Loading Paper

Refer to the *DPR100 Product Manual* for instructions. The manual is included with each DPR100 Trend Pen recorder and can be inserted in this binder.

21.4 Cartridge Replacement

Six different colored pens are contained in a print wheel in the print cartridge. When one color is depleted, the cartridge must be replaced.

Use the procedure in Table 21-1 to replace the print cartridge.

Table 21-1 — Print Cartridge Replacement Procedure

Step	Action
1	Remove the paper from the recorder. If the recorder has power applied, the print carriage will move to the center and the display will indicate NO PAP. If the recorder does not have power applied, gently move the print carriage to the center of its travel.
2	If replacing a used print cartridge, secure the print carriage by holding the lever (H) with the left hand and withdraw the print cartridge (F) to the right.
3	Continue to secure the print carriage and fit a new print cartridge , ensuring that the pin on the left engages with the corresponding hole on the print carriage gear wheel. A "click" will be heard when the print cartridge engages correctly.

21.5 Troubleshooting

To service a Trend Pen Recorder subsystem:

- Refer to the DPR100 C/DPR100 D Product manual, US1I-6137, to service the DPR100 Trend Pen recorder.
- Run the HVTS test SIOS for the SIO board.
- Examine the cables and refer to the cabling diagram in Figure 21-3.

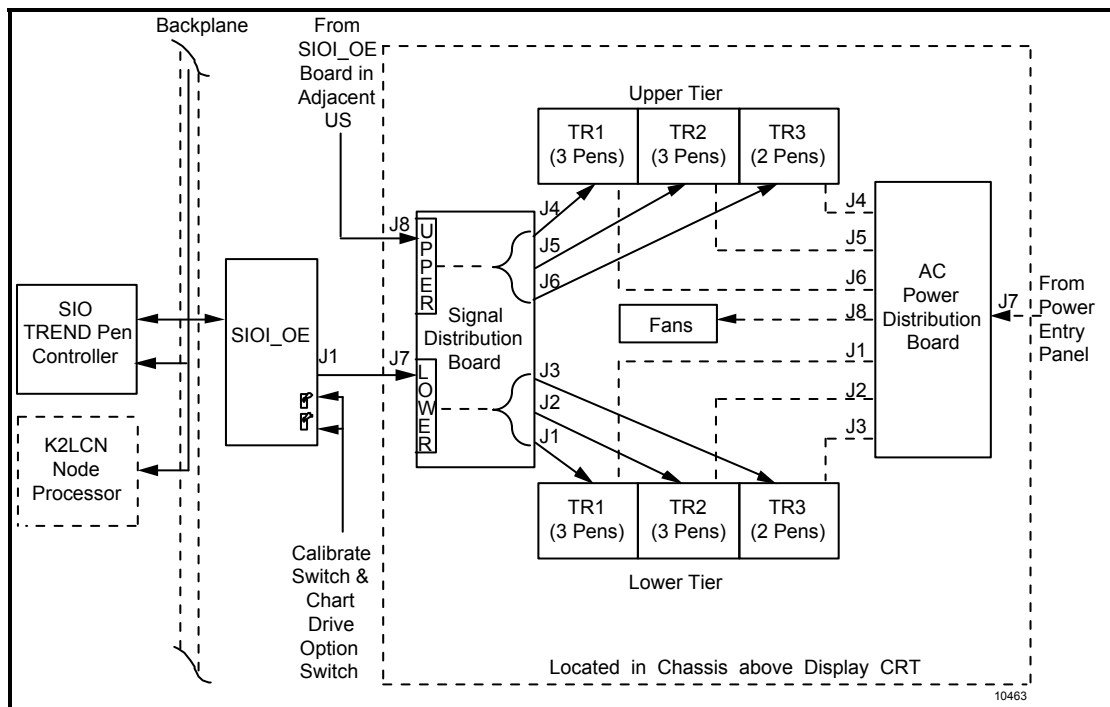


Figure 21-3 — Trend Pen Recorder Cabling

21.6 Recorder Removal

NOTE

The Trend Pen recorder is removed as a unit. There are two screwjacks: one on top and one on the bottom. See Figure 21-4.

Use the procedure in Table 21-2 to remove a Trend Pen recorder.

Table 21-2 — Trend Pen Recorder Removal Procedure

Step	Action
1	Remove the rear cover of the Trend Pen recorder pod.
2	Disconnect the recorder's ac power cord from the ac power distribution panel. See Figure 21-2.
3	Disconnect the ac power cord from the back of the recorder.
4	Tag the input signal wires that are connected to the recorder. Disconnect the signal wires from the input terminals
5	Using a blade screwdriver with a long shaft, loosen the top screwjack 1 inch, then push it forward and slide the rear of the screwjack to the side to disengage it from its retaining slot on the recorder. Remove the screwjack.
6	Loosen the bottom screwjack 1 inch, then push it forward and slide the rear of the screwjack to the side to disengage it from its retaining slot. Remove the screwjack.
7	Slide the recorder out through the front of the pod.

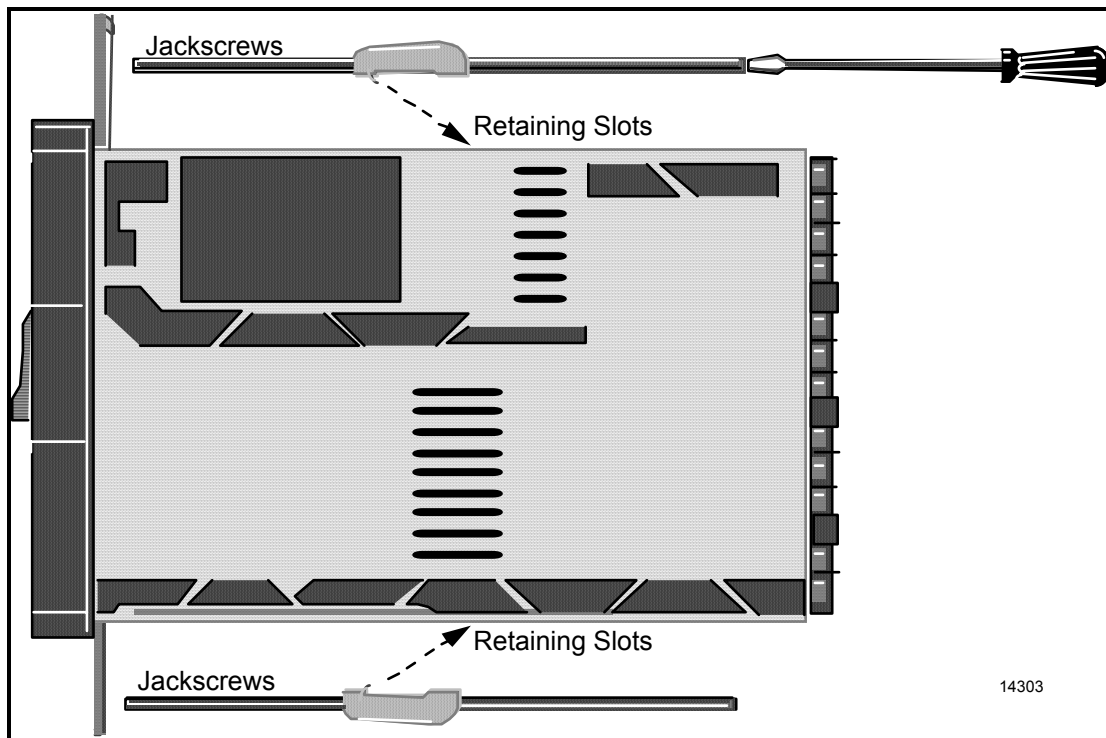


Figure 21-4 — Mounting the Trend Pen Recorder

21.7 Recorder Installation

NOTE

The Trend Pen recorder is replaced as a unit. There are two screwjacks; one on top and one on the bottom. Refer to Figure 21-4.

Use the procedure in Table 21-3 to install a Trend Pen recorder.

Table 21-3 — Trend Pen Recorder Installation Procedure

Step	Action
1	Remove the rear cover of the Trend Pen recorder pod.
2	Being careful not to damage the input boards at the rear of the recorder chassis, slide the recorder into the cabinet until it makes contact with the front plate.
3	Insert the bottom screwjack into its retaining slot on the recorder. Using a blade screwdriver with a long shaft, tighten the jackscrew until the recorder is supported horizontally. Refer to Figure 21-4.
4	Insert the top screwjack into its retaining slot on the recorder. Tighten the jackscrew until the recorder is securely mounted.
5	Connect the signal wires to the recorder's input terminals.
6	Verify that the ac power cord is disconnected from the ac distribution panel. Refer to Figure 21-2.
7	Connect the ac wiring to the ac input terminals that are located at the upper right hand corner of the recorder.
8	Connect the ac power cord to the ac power distribution panel.

21.8 Recorder Calibration Check

The Trend Pen recorder is supplied with resident factory calibration data stored in nonvolatile memory for each available input actuation. This calibration data is not affected by hardware changes, and therefore, recalibration of the recorder is not necessary.

A scale check is built into the SIOI_E board. The right-hand switch on the SIOI_E board has three switch positions:

- NORMAL—normal operation
- LOW—positions pen at 2% of scale, overrides input data
- HIGH—positions pen at 98% of scale, overrides input data

System software must be active with no print inhibits active. The calibration check can be run whenever these conditions are met because the calibrate switch overrides the input.

21.9 Configuration

The Trend Pen recorder takes full advantage of digital technology to provide maximum configuration flexibility. All software configuration is affected by the keys that are on the front face of the recorder. Configuration involves the selection of the required parameter from the recorder's menu, or the entry of a numerical value or alphanumeric sequence.

The configuration procedure is fully explained in Sections 6 and 7 of the *DPR100 Product* manual, US11 6137, which is included with the recorder. Section 9 of the same manual details how to load the configuration with the optional PC Loader software application.

21.10 Battery Replacement

The Lithium battery has a long life. It is located under a metal plate on a circuit board behind the paper cassette .

Use the procedure in Table 21-4 to replace the battery.

Table 21-4 — Battery Replacement Procedure

Step	Action
1	Disconnect the recorder's ac power cord from the ac power distribution panel.
2	Connect an ESD wriststrap to the bare metal on the Universal Station furniture.
3	Open the recorder's front cover and remove the recording paper.
4	Remove the two screws shown in Figure 21-5.
5	Lift the metal plate to expose the circuit board.
6	Replace the battery in the battery holder.
7	Replace the cover plate and secure the plate with the two screws that were previously removed.
8	Replace the recording paper.
9	Reconnect the ac power cord to the ac distribution panel.
10	Reload the recorder's configuration.

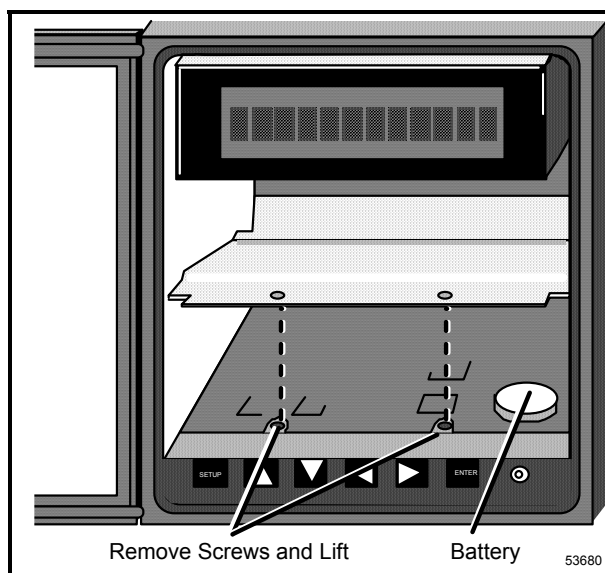


Figure 21-5 — Battery Replacement

21.11 REPLACEMENT PARTS

NOTE

A spare DPR100 Trend Pen recorder is capable of both 120 Vac and 240 Vac operation.

Table 21-5 contains a list of replacement parts for the DPR100 Trend Pen recorder.

Table 21-5 — DPR100 Trend Pen Recorder Replacement Parts

Part Number	Description
*51196674-100	DPR100 Trend Pen Recorder
51196674-101	Ink Cartridge Multipoint (6 color)
51196674-102	Chart Roll Paper (100 divisions)
51196674-103	Fan Fold Paper (100 divisions)
51196674-104	Fuse (85 to 264 Vac Power Supply (Europe) 5 x 20)
51196674-105	Fuse (85 to 264 Vac Power Supply (US) 6.3 x 32)
51196674-106	Front Label (2 Channel)
51196674-107	Front Label (3 Channel)
51196674-108	Chart Cassette — Roll
51196674-109	Chart Cassette — Fan-fold Multipoint
51196674-110	PC Loader Interface
51196674-111	PC Loader Application Software
51196674-112	Panel Mounting Kit
51196674-113	Lithium Battery, 3.0 V (Renata CR2430)
51308069-200	Power Cable (120 Vac)
51308070-200	Power Cable (240 Vac)
*51400855-100	SIO (Serial Interface Output) Board
*51304814-100	SIOI_E (SIO I/O Board)—non-CE Compliant
*51304814-200	SIOI_E (SIO I/O Board)—CE Compliant

Table 21-5 — DPR100 Trend Pen Recorder Replacement Parts

Part Number	Description
*51308075-100	Trend Signal Distribution Board
*51308012-100	Upper Power Distribution Board (120 Vac)
*51308014-100	Upper Power Distribution Board (240 Vac)
51308076-200	Trend Recorder Interface Cable (same furniture bay)
51308076-400	Trend Recorder Interface Cable (adjacent furniture bay)
51308077-300	Trend Recorder Interface Jumper Cable
51308078-300	Trend Power Jumper Cable 120 Vac
51308079-300	Trend Power Jumper Cable 240 Vac
51308087-200	AC Fan Power Cable

* ORU = Optimum Replaceable Unit

22 CE COMPLIANT REMOTE USER LCN ADAPTER (RULA)

22.1 Overview

Access to the LCN from a remote Universal Station is possible with the use of RULA Installation Kit, MP-RLKT11. The kit contains two replacement cables, an additional cable, and a cable mounting bracket.

The required DaynaPORT SCSI/LINK-3 adapter must be provided by the user.

The cabling of a RULA in a Universal Station is illustrated in Figure 22-1.

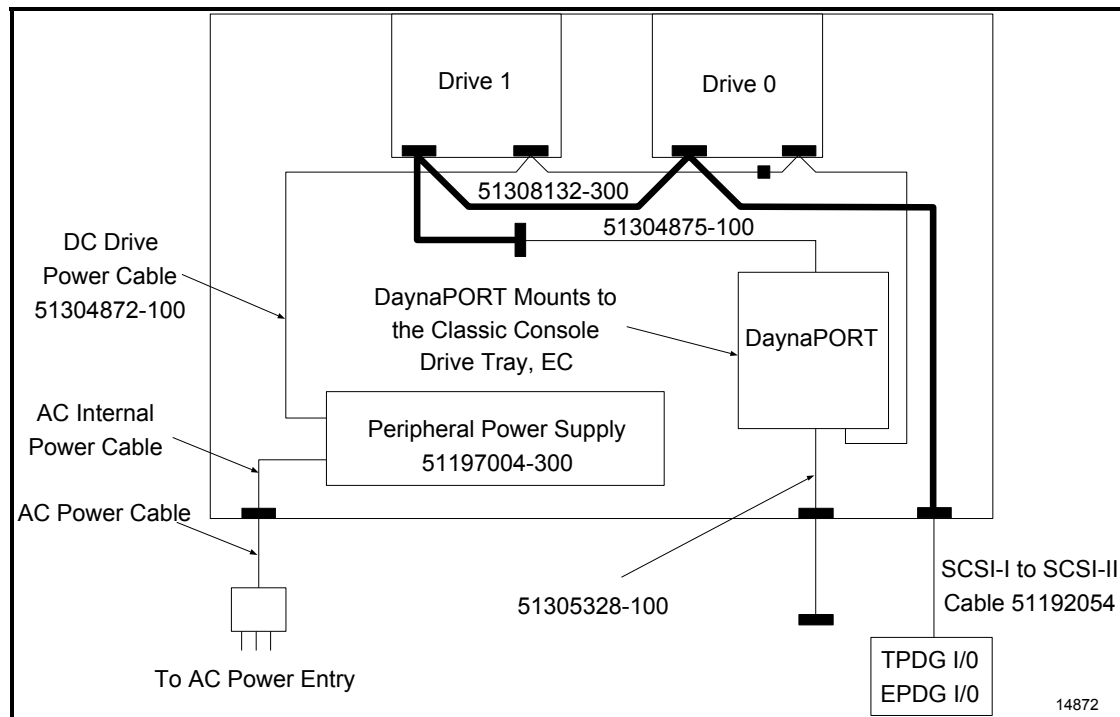


Figure 22-1 — Universal Station RULA Cabling

22.2 Dynaport SCSI/Link-3 Adapter Description

Figure 22-2 illustrates a DaynaPORT SCSI/Link-3 adapter.

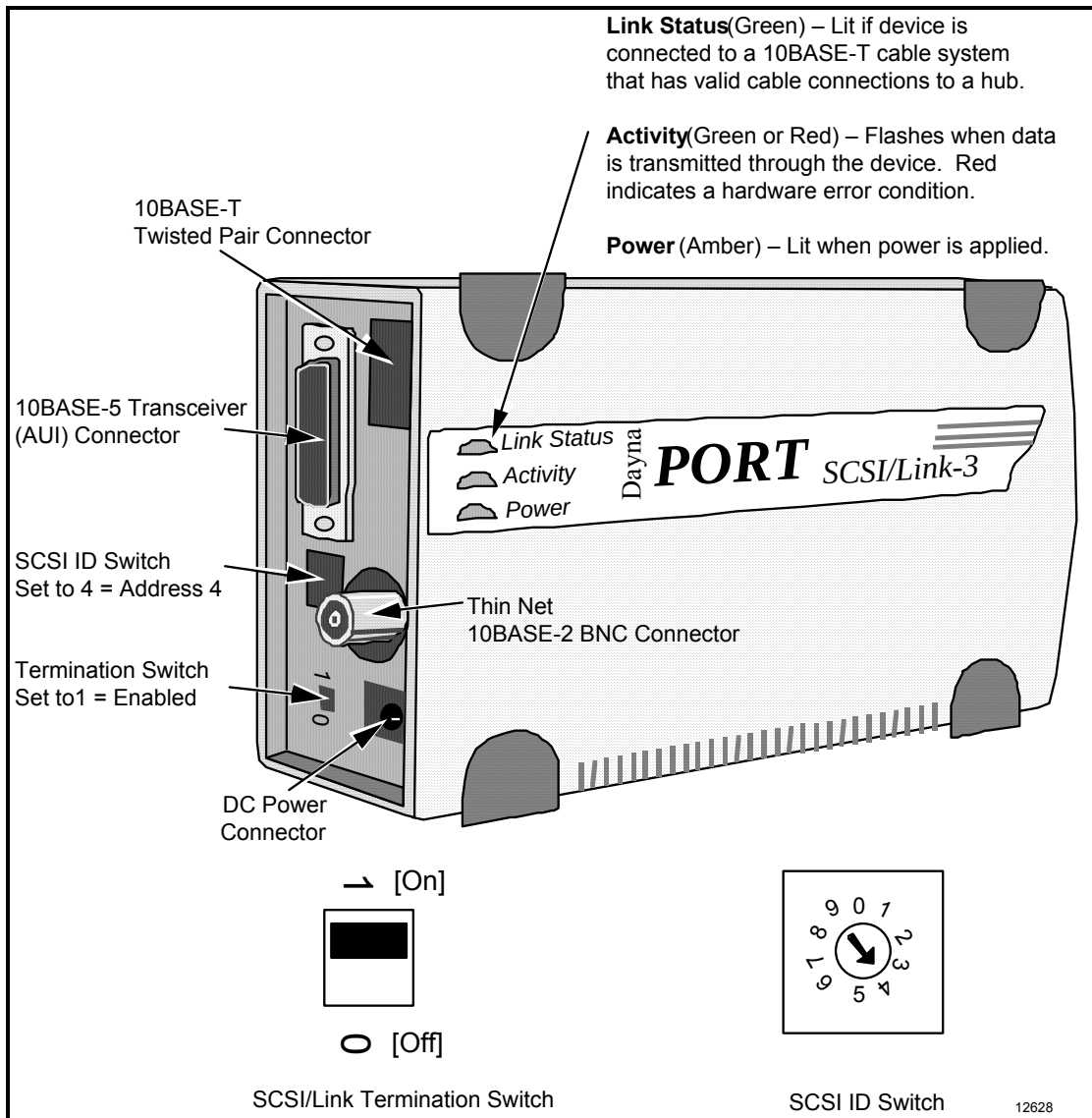


Figure 22-2 — DynaPort SCSI/Link-3 Adapter

22.3 Rula Cabling

Figure 22-3 illustrates the RULA cabling in a cartridge enclosure.

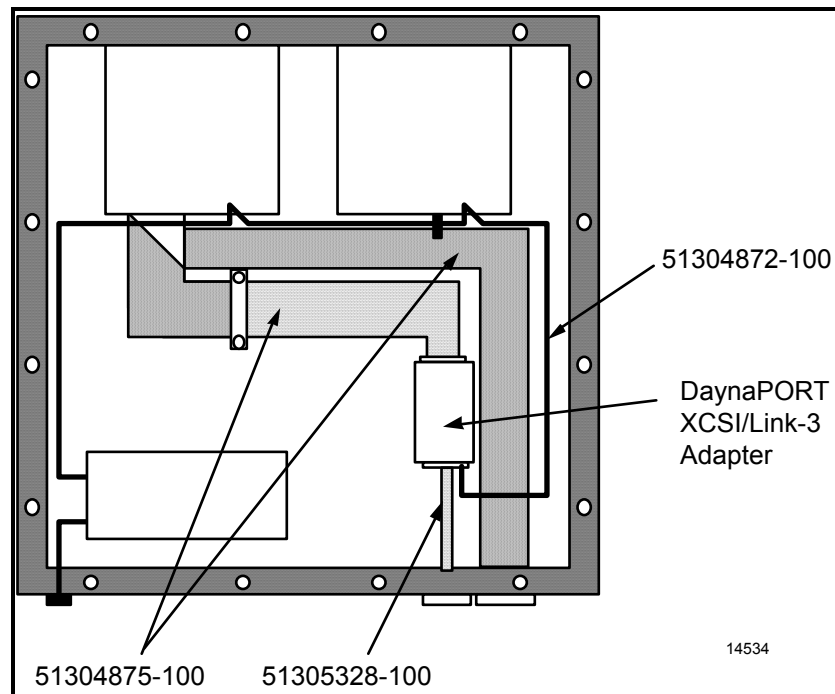


Figure 22-3 — Cartridge Enclosure with RULA Installed

22.4 Special Cartridge Drive Pinning for RULA

Figure 22-4 illustrates the cartridge drive pinning that is required when RULA is installed in the US.

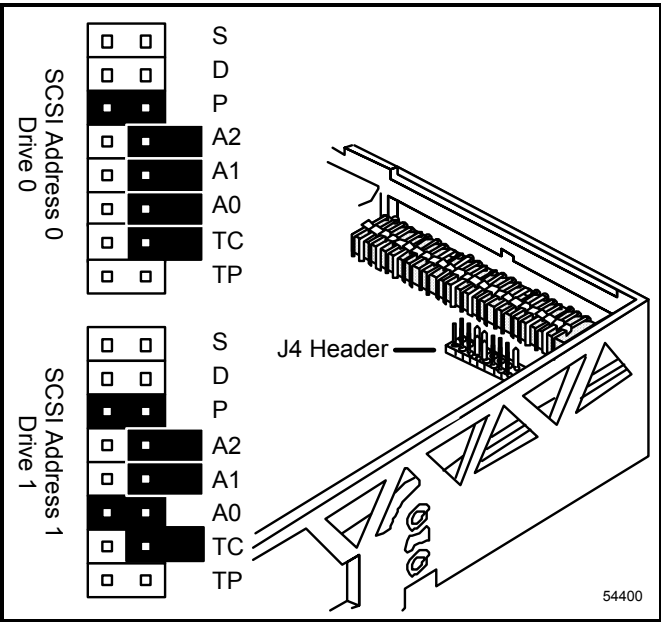


Figure 22--4 — Cartridge Drive Pinning for RULA Installation

22.5 Replacement Parts

Table 22-1 contains a list of RULA replacement parts.

Table 22-1 — RULA Replacement Parts

Part Number	Description
N/A	DaynaPORT SCSI/Link-3 Adapter — order from DaynaPORT
51304875-100	RULA SCSI Adapter Cable
51305328-100	RULA Transceiver Cable
51304872-100	RULA DC Power Cable

23 SONY OPEN-FRAME CRT-51197149-100

23.1 Overview

23.1.1 Scope

This 19-inch flat screen high-resolution color display monitor with digital multiscan capability is an Optimum Replaceable Unit (ORU). The removal/replacement procedure and screen adjustments are given in this section.

23.1.2 Illustration



Figure 23-1 — Sony 19-inch Open-Frame CRT—51197149-100

23.2 The Monitor Assembly

23.2.1 Description

The monitor assembly is composed of the following:

- Monitor is a Flat Square Tube (FST) enclosed in an EMI shield.
- Monitor is virtually a flat screen [radius of curvature – 50,000 mm (164 ft)] that minimizes image distortion and reflective glare.
- Digital panel control and a full suite of on-screen display (OSD) programming/calibration menus.
- The lower tier monitor slides out of the Classic furniture on slide rails.
- The upper tier monitor is bolted to a CRT mounting bracket.
- A Control Panel connector is bracket-mounted on the side of the monitor mounting chassis for the cable plug from the Control Panel assembly on the bezel.

In general, the monitor:

- Is secured to the monitor mounting chassis and onto the slide tray with screws and PEM nuts.
- Can be replaced by removing the monitor mounting chassis from the slide tray. The replacement monitor is secured to the slide tray.
- Can be removed from the station without the disassembly of the touchscreen assembly or monitor control assembly.

23.2.2 Specifications

Table 23-1 Monitor Specifications

Type	High Resolution
Resolution	Up to 1800 x 1440 @ 75 Hz
Frequency	30 to 110 KHz
Interface Video Sync Video Impedance	HD15, 0.7 Vp-p, positive Separate HD/VD, TTL Polarity Free 75 ohm
AC Input Voltage Frequency Power	90 to 132 Vac 198 to 264 Vac 47 to 63 Hz ≤ 140 W
Weight	25.47 Kgs (56.15 lbs.)

23.3 Cleaning the Monitor CRT

23.3.1 Monitor screen

An anti-reflective coating is provided on the CRT screen to reduce eyestrain. Avoid touching the CRT screen because soil and fingerprints reduce legibility and the effectiveness of the anti-reflective treatment. To remove soil or fingerprints from the screen, do the following:

- Use a cleaner (such as window cleaning solution) to clean the screen face. Be careful not to scratch or mar the anti-reflective treatment. Before using an untested cleaner, spot test an area of the screen to determine if the results are satisfactory.
- Remove the cleaner with a damp cloth and wipe the surface with a soft, dry cloth.

23.4 Replacing the Classic Console Monitor

23.4.1 Procedure

Table 23-2 Procedure for Replacing the Lower and Upper Monitor

Step	Action
1	Shut down the Windows NT operating system. Go to another station and do a second shutdown on this node number to prevent the LCN from isolating nodes.
2	Turn off the ac breaker, located in the front of the station, under the keyboard.
3	Remove the screws retaining the rear cover on the monitor and slide the rear cover off of the station.
4	Disconnect the ac power cable from the back of the monitor. See Figure 23-2.
5	Disconnect the video interface cable from the monitor.
6	Disconnect the Control Panel assembly cable from the monitor.
7	Lower Monitor: • First, remove the two hex head screws securing the slide trays restraint bar to the console frame. • Release the two detent buttons located on each side of the rear of the monitor-mounting tray. • Remove the six hex-head screws from the bottom of the slide tray. NOTE: Once these 6 hex-head screws are removed, the monitor may slide off the slide tray. • Carefully slide the monitor out of the cabinet, ensuring that the cable in the cabinet does not get caught or pinched. Upper Monitor: This monitor is mounted directly to the support tray. Go to the next step.

Table 23-2 Procedure for Replacing the Lower and Upper Monitor

Step	Action
8	UPPER MONITOR ONLY: - Loosen the four vertical oriented hex-head screws on one side of the monitor. - Remove the three horizontally oriented hex-head screws from each side of the monitor-mounting bracket. - Lift the monitor from the mounting tray. - Place the new monitor on the support tray between the two mounting brackets. - Replace the three horizontally mounted hex-head screws on each side of the monitor. - Tighten the four vertically oriented hex-head screws that were previously loosened
9	LOWER MONITOR ONLY: - Place the new monitor assembly on the slide tray. - Replace the six screws and washers in the slide tray/monitor. - Install the six screws through the bottom of the slide tray and into the PEM nuts mounted on the monitor assembly. - Align the monitor with the bezel before tightening the screws.
10	Reconnect the monitor Control Panel cable to the connector on the left side of the monitor. Reconnect the video interface cable and ac power cord.
11	Press the button on each side of the slide mechanism and push the monitor forward until it contacts the bezel and the side mechanism clicks into the detent.
12	Install the monitor support channel assembly, 51201879-100, by inserting a screw and grounding washer through a wide flat washer and spacer, and into the slide rail assembly.
13	Reinstall the lower pod rear cover and retaining screws.

23.4.2 Illustration

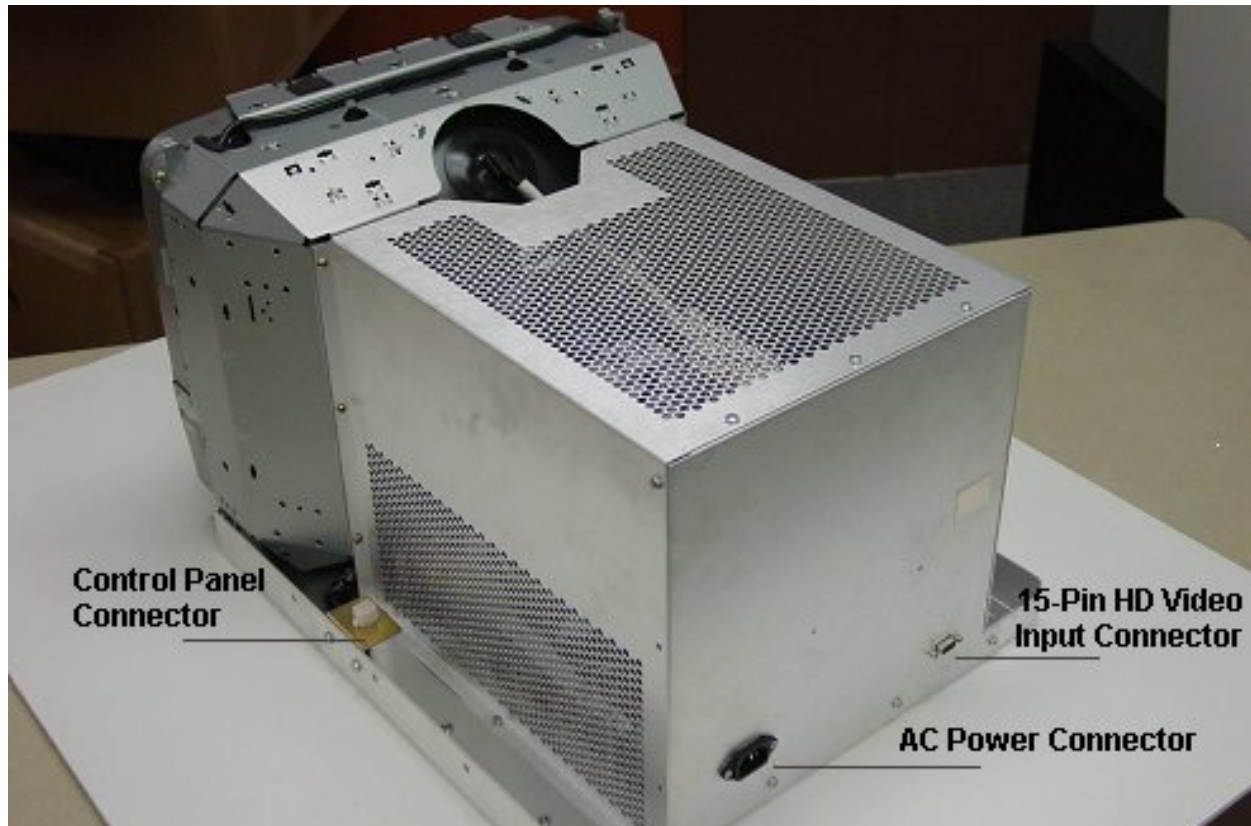


Figure 23-2 Monitor Cabling

23.5 Replacing the Monitor Control Panel Assembly

23.5.1 Preliminary

To replace the monitor control panel assembly, the rear cover and the monitor must be removed first. Refer to Table 23-3.

23.5.2 Procedure

NOTE

If the monitor contains a touchscreen, the bezel/control panel assembly must be removed first before the Control Panel can be accessed and removed.

Table 23-3 Replacing the Monitor Control Assembly

Step	Action
1	Remove the monitor, as detailed in Table 23-2.
2	Remove the screws holding the monitor control assembly into the touchscreen bezel. See Figure 23-3.
3	Remove the monitor Control Panel assembly from the bezel.
4	Install the new monitor control panel assembly on the bezel. See Figure 23-3. Orient assembly so switches appear at front of bezel as shown in Figure 23-4.
5	Install the four screws to hold the assembly to the bezel.
6	Replace the monitor, as detailed in Table 23-2.

23.5.3 Illustration

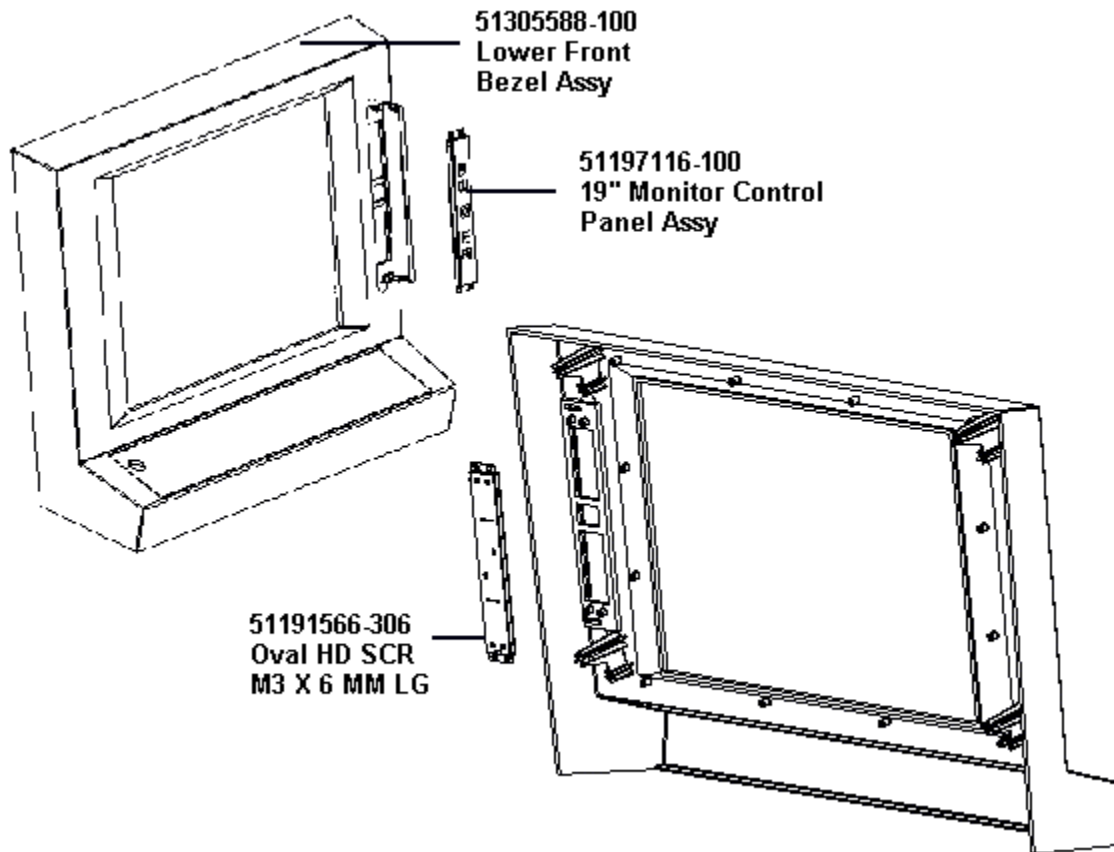


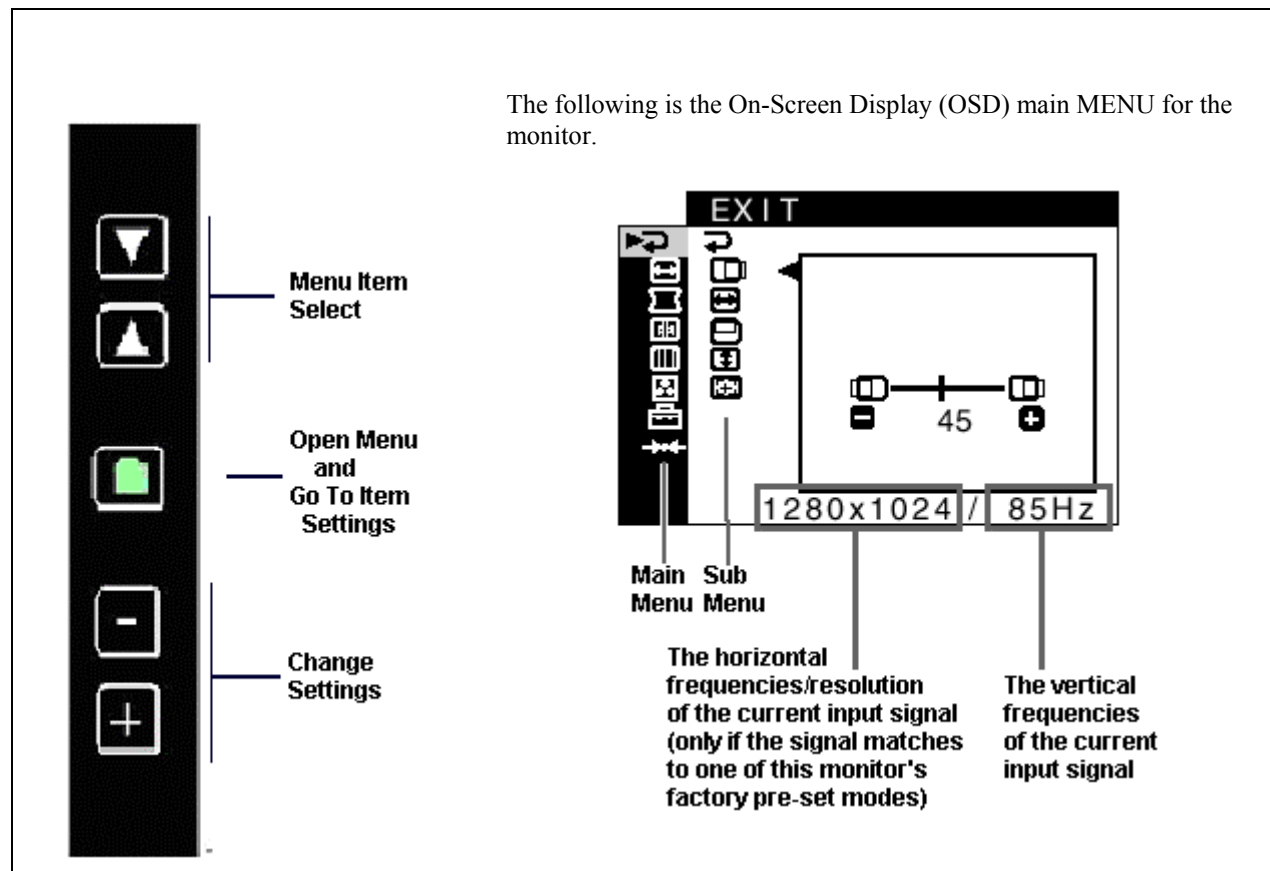
Figure 23-3 Control Panel Assembly for 19-Inch Monitor

23.6 Monitor Adjustments

23.6.1 Introduction

The monitor screen adjustments are all controlled by an on-screen menu. Access to these on-screen menus are made using the membrane switches on the monitor Control Panel as shown in Figure 23-4.

23.6.2 Illustration



23.6.3 Procedure

Table 23-4 Using the Monitor Control Panel Buttons

Step	Action
1	Displaying the main MENU: Press the  button to display the main MENU on your screen.
2	Selecting the SUB MENU containing the item you want to adjust: Press the  button to highlight the desired sub menu. Press the  button to select the menu item.
3	Selecting the adjustment item from the SUB MENU: Press the  buttons to select the desired adjustment item. Press the  buttons to make the adjustment.
4	Closing the MENU: Press the  button once to return to the main MENU, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.

23.6.4 Other adjustments

Other adjustments are available from the MAIN and SUB menus.

- Changing the OSD menu position (self explanatory)
- Locking functions
 - o If the LOCK is ON (active), selecting any other function will cause a  (key mark) to appear on the screen.
 - o To lock or unlock the functions: select LOCK and press either one of the  buttons again.

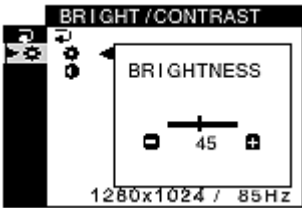
The SCREEN menu contains:

- Degaussing the monitor (see Table 23-6).
- Adjusting the moiré cancellation (see Table 23-7).

23.6.5 Adjusting the brightness and contrast

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu.

Table 23-5 Adjusting the Brightness and Contrast

Step	Action
1	Press the control button  (☀) or the  (☾) to display the BRIGHT/CONTRAST menu.
	The BRIGHTNESS/CONTRAST menu appears on the screen. 
2	Press the  /  to select the brightness (☀) or contrast (☾). Then move the control button  /  to make adjustments.

23.7 Degaussing the Monitor

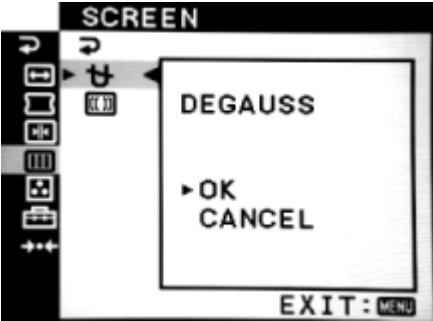
23.7.1 Scope

The monitor is automatically demagnetized (degaussed) each time the monitor is powered on. The degaussing eliminates color impurities and other distortions by neutralizing the effects of magnetic fields in the surrounding environment.

Monitors can pick up additional magnetic flux when left on for long periods. Whenever color purity is in question, degauss the CRT tube by performing the steps in Table 23-6. If a second degauss cycle is needed, allow a minimum interval of 20 minutes between attempts to degauss the monitor.

23.7.2 Procedure

Table 23-6 Degaussing the Monitor

Step	Action
1	Press the  button. The main MENU appears on the screen.
2	Press the / to highlight the SCREEN menu entry and press the  button again. The SCREEN menu appears on the screen. 
3	Press  to select the DEGAUSS menu entry and then press the  button again to degauss. The screen is degaussed for about 2 seconds. If a second degauss cycle is needed, allow a 20-minute interval (minimum) for the best results.

23.8 Adjusting The Moire (Picture Quality)

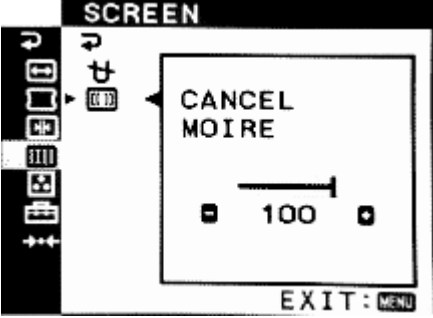
23.8.1 Scope

Moiré is a type of natural interference that produces soft, wavy lines on your screen. It may appear due to interference between the pattern of the picture on the screen and the phosphor pitch pattern of the monitor. If elliptical or wavy patterns appear on the screen, cancel the moire (CANCEL MOIRE).

The SCREEN settings allow you to adjust the quality of the picture by controlling the moiré. The CANCEL MOIRE and MOIRE ADJUST settings are stored in memory for the current input signal. All other settings are stored in memory for all input signals.

23.8.2 Procedure

Table 23-7 Moire Adjustments

Step	Action
1	Press the  button. The main MENU appears on the screen.
2	Press the / to highlight the SCREEN menu entry and press the  button again.
3	Press / buttons to select Moiré option and press  again. 
4	Press the  button to turn the moiré cancellation ON or to increase the amount of moiré cancellation, and then press the  button. Press the  button to reduce the moiré cancellation or to turn the moiré cancellation OFF (reduced to 0), and then press the  button. NOTE: Use of the moiré adjustment may make the picture fuzzy.

23.9 Spare Parts List

Table 23.2 Monitor Spare Parts

Part Number	Description
*51197149-100	Replacement Monitor Unit

* = Optimum Replaceable Unit (ORU)

23. SONY OPEN-FRAME CRT-51197149-100

23.9. Spare Parts List

24 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.1 18.1-inch Open Frame LCD Overview

24.1.1 Description

The 18.1-inch (viewable) LCD monitor features a TFT active matrix LCD technology packaged in sheet metal with drilled tabs for mounting to the Honeywell Classic Console bezel. The sheet metal also contains self-clinching nuts on the tabs to allow mounting to the Honeywell Classic Console CRT slide tray using brackets. An anti-glare and shatterproof glass lens is attached to the front of the display.

There is no on/off switch or “power on” indicator as the LCD is designed to be “always on,” and it is powered from an AC to DC converter that is permanently mounted on the backside of the display. The AC connection to the converter is an IEC 301 type three-prong connector.

The LCD operates from VGA, SVGA, and XGA video sources as well as a psuedo VGA source that is specific to Honeywell equipment. The video interface is a 15-pin High density D-connector

The display has an on-screen display menu accessed through control panel that uses a 9-pin D-sub connector.

24.1.2 Specifications

Table 24-1 Specifications for 18.1inch Open Frame LCD (51199193-100)

Type	Specifications
Viewable image size	Approximately 359.04 x 287.232 mm (w/h) (14.14 x 11.31 inches)
Resolution	<u>Recommended:</u> Horizontal: 1280 dots Vertical: 1024 lines
Refresh Rate	60-75 Hz (60 Hz recommended)
Input signal levels Video signal Sync signal	Analog RGB: 0.700 Vp-p (positive), 75 Ω Separate H/V, or composite sync: TTL 2Ω, Polarity Free Sync on green: 0.3 Vp-p (negative)
Standard image area	Approximately 388 x 291 mm (15.375 inches x 11.50 inches)
AC input voltage/current	100 to 240 V, 50-60 Hz, 2.0-1.0A
Power consumption	Approximately 60 W
Operating temperature	10 °C to 40°C
Dimensions	Height: 323.85 mm (12.75 inches)
	Width: 491.49 mm (19.35 inches)
	Depth: 130.18 mm (5.125 inches)
Weight	7.26 kg (16.0 lbs)

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.2. Installing the 18.1-inch LCD

24.1.3 18.1-inch LCD part numbers

The following table lists the optimal replacement parts for the 18.1-inch LCD.

Description	Part Number
18.1-inch LCD	51199193-100
Upper bezel assembly without touchscreen	51403685-100
Upper bezel assembly with touchscreen	51403685-200
Lower bezel assembly without touchscreen	51403683-100
Lower bezel assembly with touchscreen	51403683-200

24.2 Installing the 18.1-inch LCD

24.2.1 Description

This section provides procedures for installing a new 18.1-inch LCD.

24.2.2 Tasks for installing the LCD

No	Task Description
1	Remove the rear cover if you have not done so already. See the procedure in "Remove the rear cover" in Section 24.3.
2	Install the FPD assembly.
3	Connect the cables.
4	Replace all covers.
5	Power up the station.
6	Make initial monitor adjustments.

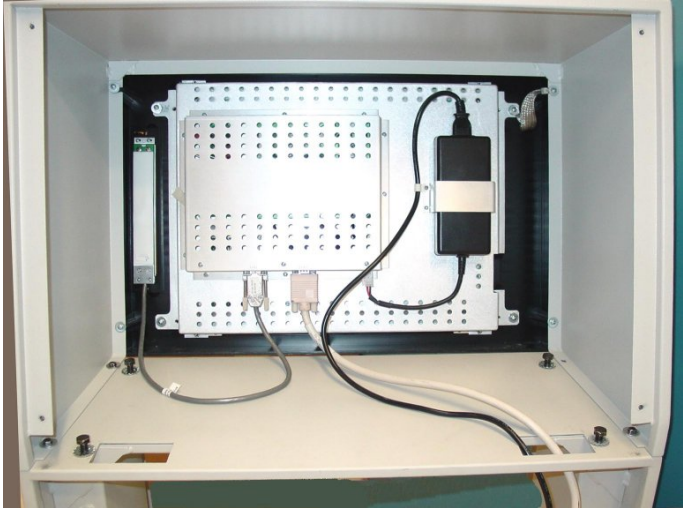
24.2.3 Install the LCD

Step	Action
1	Attach the LCD to the bezel assembly with the grounding strap in the upper right corner of the LCD as viewed from the back of the monitor. 
	ATTENTION Attach all grounding straps using the washer orientation depicted in the photograph. 

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.2. Installing the 18.1-inch LCD

Step	Action
2	Set the bezel/monitor assembly on the console, aligning the edge of the bezel with the edge of the console. 
3	Secure the bezel/monitor assembly to the monitor enclosure mounting brackets with the four Phillips head screws and serrated washers. 

Step	Action
4	Attach the grounding strap securely to the monitor enclosure mounting bracket using proper washer position.  A close-up photograph showing a metal mounting bracket with a grid of holes. A grounding strap is being attached to the bracket using a washer and nut. The strap is connected to a black component, likely the LCD monitor.
5	Reconnect the power and video cables to the LCD. If there is a touchscreen, reconnect the touchscreen.  A photograph showing the LCD monitor installed in a rack. The monitor is mounted on a metal bracket. Several cables (power, video, and touchscreen) are connected to the back of the monitor. The monitor is facing away from the camera, showing its internal components and connections.

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.2. Installing the 18.1-inch LCD

24.2.4 Replace the rear cover

Use the following procedure to replace the cover on the rear of the console.

Step	Action
<i>Replacing the upper tier cover</i>	
1	Set the upper tier cover in place and attach it to the console using four screws.  Secure with 2 screws on each side

Step	Action
Replacing the lower tier cover	
1	Set the cover box into position on the console and secure with the two hex head screws. 
2	Place rear console door back on its hinges and gently close door. 

**REFERENCE - INTERNAL**

Power up the station and make initial monitor adjustments in the following sections:

- For a Universal Station, see Section 24.5, "Recommended Monitor Settings for a Universal Station"
- For initial configuration, see Section 24.6 Servicing and Configuring the 18.1-inch LCD

24.3 Replacing the 18.1-inch LCD

24.3.1 Description

This section provides procedures for replacing a new 18.1-inch LCD with the same model. If you are replacing another monitor type with the 18.1-inch LCD, you will need to obtain the appropriate upgrade kit.

24.3.2 Tasks for replacing the LCD

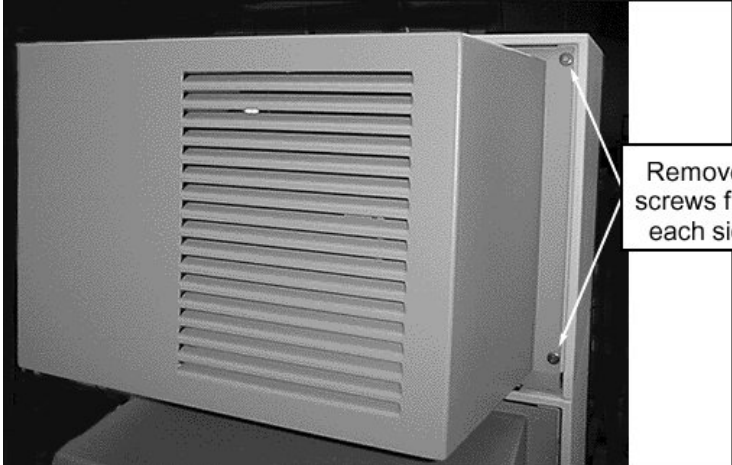
No	Task Description
1	Shutdown the station.
2	Remove the rear cover.
3	Replace the LCD or bezel assembly.
<i>See Section 24.2 for the following procedures</i>	
4	Install the LCD.
5	Replace the rear cover.
6	Make initial monitor adjustments.

24.3.3 Shutdown the station

Step	Action
1	Shut down all the existing nodes in the station as follows. • Universal Station – Do a double shutdown of the node. • Universal Station^x – Shut down the coprocessor and then do a double shutdown of the node. • GUS – Do a SHUTDOWN sequence on the PC software.
2	Place the ac power switch in the OFF position.

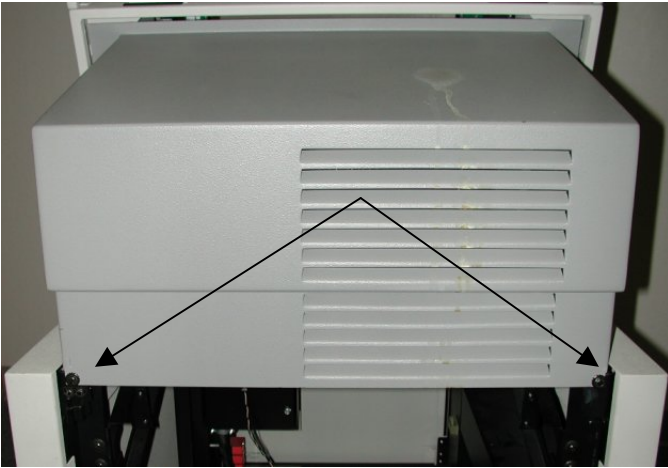
24.3.4 Remove the rear cover

Use the following procedure to remove the cover from the back of the console.

<i>Removing the upper tier cover</i>	
Step	Action
1	Remove the four screws on the rear of the console that secure the upper tier cover, and remove the cover.  Remove 2 screws from each side

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.3. Replacing the 18.1-inch LCD

<i>Removing the lower tier cover</i>	
Step	Action
1	Open console door slightly and lift the door off hinges to remove. 
2	Remove the two hex head screws securing the lower tier cover and lift the cover from the console. 

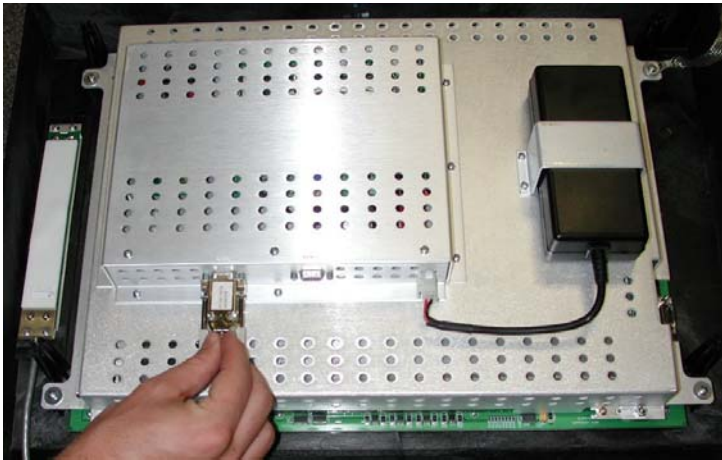
24.3.5 Replace the LCD or bezel assembly

Use the following procedure to replace the LCD or bezel assembly on either the upper or the lower section of the console. The exact location and shape of various parts are slightly different for the upper and lower assembly, but the steps for replacement are the same.

Step	Action
1	Disconnect the power and video cables from the rear of the monitor. Disconnect the touchscreen if there is one present.
2	Detach the monitor assembly from the monitor enclosure mounting brackets by removing the four Phillips-head screws and serrated washers. 
3	Remove the bezel/monitor assembly from the console. Do this slowly to prevent damage to any wires that may still be connected to the rear of the bezel. 

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.3. Replacing the 18.1-inch LCD

Step	Action
4	Disconnect the Control Panel cable from the rear of the monitor assembly.  A close-up photograph showing a hand pulling a small, gold-colored connector from a port on the rear of a silver metal monitor bezel. A black cable is attached to the connector. The bezel has a grid of mounting holes and a green PCB is visible at the bottom.
5	Remove the LCD from the front bezel assembly by removing the four Phillips head self-tapping screws.  A photograph showing a hand using a screwdriver to remove a screw from the bottom edge of the silver metal bezel. The bezel is resting on a green PCB. A black cable is connected to the top of the bezel.
6	Mount the replacement LCD or bezel assembly to the existing part and attach using the four Phillips head self-tapping screws.  A photograph showing a hand holding a replacement silver metal bezel assembly over a green PCB. The bezel has a black cable connected to its top. The hand is positioning the bezel to be mounted onto the PCB.
7	Follow the procedures in Section 24.2 to complete the installation.

24.4 Using the Control Panel and On-screen Display

24.4.1 Overview

This section contains generic procedures for using the control panel and on-screen display to adjust the monitor. See also Section 24.5, “Recommended Monitor Settings for a Universal Station” for a list of adjustments for the monitor after it is installed.

24.4.2 Using the control panel

Use the control panel and the on-screen display (OSD) to make necessary adjustments to the flat panel display.

Icon	Name	Use to . . .
	Menu/select	• Open the on-screen display, or • Select an option
	Right/Increase	• Move cursor to the right, or • Increase a value
	Left/Decrease	• Move cursor to the left, or • Decrease a value

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.4. Using the Control Panel and On-screen Display

24.4.3 General procedures for using the on-screen display menus

This procedure lists the general steps for using the on-screen display and control panel to choose menus and make selections.

Step	Action
1	Press the menu/select  button to open the on-screen display. 
2	Press the right/increase  button or the left/decrease  button to navigate to a submenu. 
3	Press the menu/select  button to display the submenu. 
4	Press the right/increase  button or the left/decrease  button to navigate to a selection. 
5	Press the menu/select  button to select the option.
6	Press the right/increase  button or the left/decrease  button to change the value.
7	Press the menu/select  button to select the value.
8	Select the exit  icon to close the sub-menu.

24.4.4 On-screen display adjustment options

The following table lists the types of adjustments you can make using the on-screen display. See subsequent sections for step-by-step procedures.

Icon	Description	Use to . . .
	Input interface setting	Select the major signal input.
	Brightness/ Contrast/ Black Level	Adjust the brightness, contrast or black level.
	Color	Adjust the color temperature, skin tone, hue or color saturation.
	Image Setting	Adjust the image width, signal phase, horizontal position and vertical position of the screen.
	Miscellaneous tools	Adjust the amount of time the on-screen menu displays or the position of the on-screen menu. Zoom in on the on-screen display. Reset all user settings back to factory defaults.
	Exit	Exit the menu or sub-menu once you make a selection. Values are automatically saved after pressing the Exit icon.

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.4. Using the Control Panel and On-screen Display

24.4.5 Input interface setting

Step	Action						
1	Press the menu/select  button to open the on-screen display. 						
2	Press the right/increase  button to select the input interface icon. 						
3	Press the menu/select  button to display the submenu. 						
4	Select one of the following menu options:						
	<table><tr><th>Icon</th><th>Selection</th></tr><tr><td></td><td>Analog input</td></tr><tr><td></td><td>Not Applicable</td></tr></table>	Icon	Selection		Analog input		Not Applicable
	Icon	Selection					
	Analog input						
	Not Applicable						
5	Press the right/increase  button or the left/decrease  button to change the value.						
6	Press the menu/select  button to select the value.						
7	Select the exit  icon to close the sub-menu.						

24.4.6 Brightness/contrast/black level adjustment

Step	Action								
1	Press the menu/select  button to open the on-screen display. 								
2	Press the right/increase  button to select the brightness/contrast icon. 								
3	Press the menu/select  button to display the submenu. 								
4	Select one of the following menu options:								
	<table><tr><th>Icon</th><th>Selection</th></tr><tr><td></td><td>Brightness</td></tr><tr><td></td><td>Contrast</td></tr><tr><td></td><td>Black level adjustment</td></tr></table>	Icon	Selection		Brightness		Contrast		Black level adjustment
	Icon	Selection							
		Brightness							
		Contrast							
	Black level adjustment								
5	Press the right/increase  button or the left/decrease  button to change the value.								
6	Press the menu/select  button to select the value.								
7	Select the exit  icon to close the sub-menu.								

24.4.7 Color adjustment

Step	Action	
1	Press the menu/select  button to open the on-screen display. 	
2	Press the right/increase  button to select the color adjustment icon. 	
3	Press the menu/select  button to display the color adjustment submenu. 	
4	Select one of the following menu options:	
5	Icon	Selection
		Auto color setup
		Standard color setup
		RGB color temperature setting & color adjustment
		RGB color skin tone adjustment
		RGB color hue adjustment
		RGB color saturation adjustment
6	If the selection has a submenu, press the right/increase  button or the left/decrease  button to navigate to a submenu and press the menu/select  button to select it.	
7	If necessary, press the right/increase  button or the left/decrease  button to change the value.	
8	Press the menu/select  button to select the value.	
9	Select the exit  icon to close the sub-menu.	

24.4.8 Image setting

Step	Action	
1	Press the menu/select  button to open the on-screen display. 	
2	Press the right/increase  button to select the image setting  icon.	
3	Press the menu/select  button to display the image setting submenu. 	
4	Select one of the following menu options:	
5	Icon	Selection
		Auto configure: automatically configure an optimal display setting
		Image width: adjust the width of the image across the screen
		Phase adjustment: adjust the signal phase in the analog to digital converter
		Horizontal position: adjust the horizontal position of the image on the screen
		Vertical position: adjust the vertical position of the image on the screen
6	Press the right/increase  button or the left/decrease  button to change the value.	
7	Press the menu/select  button to select the value.	
8	Select the exit  icon to close the sub-menu.	

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.4. Using the Control Panel and On-screen Display

24.4.9 Miscellaneous adjustments

Step	Action	
1	Press the menu/select  button to open the on-screen display. 	
2	Press the right/increase  button to select the tools  icon.	
3	Press the menu/select  button to display the miscellaneous tools submenu. 	
4	Select one of the following menu options:	
5	Icon	Selection
		<i>Display adjustments</i>
		 OSD timeout: decrease or increase the amount of time that elapses before the menu disappears.
		 OSD horizontal position
		 OSD vertical position
		 OSD Screen Zoom Setting - Press S1 to enable OSD Zoom feature.
		Factory reset: Reset all user settings and then confirm user settings.
6	Press the right/increase  button or the left/decrease  button to change the value.	
7	Press the menu/select  button to select the value.	
8	Select the exit  icon to close the sub-menu.	

24.5 Recommended Monitor Settings for a Universal Station

24.5.1 Overview

The following table lists the recommended monitor adjustments for the 18.1-inch LCD on a Universal Station, and indicates if it is preferable to use the automatic or manual adjustment for the setting.

Monitor Setting	Preferred Adjustment Type
Color adjustment	Auto adjustment
Image width	Manual adjustment
Image phase setting	Manual adjustment
Image horizontal position	Manual adjustment
Image vertical position	Manual adjustment

ATTENTION

The auto adjustment for image and phase settings will not work properly on the Universal Station. Use the manual adjustment procedures to change the image settings.

24.5.2 Auto adjust monitor color setting

Use the control panel and on-screen display to automatically adjust the monitor color setting.

Step	Action
1	Press the menu/select  button to open the on-screen display. 
2	Press the right/increase  button to select the color adjustment icon. 
3	Press the menu/select  button to display the color adjustment submenu. 
4	Press the right/increase  button to select the auto color setup icon. 
5	Press the menu/select  button to select the option.

24.5.3 Manually adjust image width

Use the control panel and on-screen display to manually adjust the image width setting.

Step	Action
1	Press the menu/select  button to open the on-screen display. 
2	Press the right/increase  button to select the image setting  icon.
3	Press the menu/select  button to display the image setting submenu. 
4	Press the right/increase  button to select the image width  icon and press the menu/select  button to select it.
5	Press the right/increase  button or the left/decrease  button to adjust the image width to 73. If there are still any blurry areas, adjust the value slightly to create a clear image.
6	Press the menu/select  button to accept the setting.

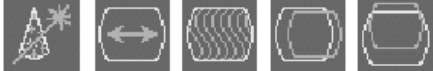
24.5.4 Manually adjust phase setting

Use the control panel and on-screen display to manually adjust the phase setting.

Step	Action
1	Press the menu/select  button to open the on-screen display. 
2	Press the right/increase  button to select the image setting  icon.
3	Press the menu/select  button to display the image setting submenu. 
4	Press the right/increase  button to select the phase adjustment  icon and press the menu/select  button to select it.
5	Press the right/increase  button or the left/decrease  button to adjust the phase to 63.
6	Press the menu/select  button to accept the setting.

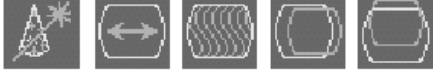
24.5.5 Manually adjust image horizontal position

Use the control panel and on-screen display to manually adjust the image horizontal position.

Step	Action
1	Press the menu/select  button to open the on-screen display. 
2	Press the right/increase  button to select the image setting  icon.
3	Press the menu/select  button to display the image setting submenu. 
4	Press the right/increase  button to select the horizontal position  icon and press the menu/select  button to select it.
5	Press the right/increase  button or the left/decrease  button to adjust the horizontal position to 231.
6	Press the menu/select  button to accept the setting.

24.5.6 Manually adjust image vertical position

Use the control panel and on-screen display to manually adjust the image vertical position.

Step	Action
1	Press the menu/select  button to open the on-screen display. 
2	Press the right/increase  button to select the image setting  icon.
3	Press the menu/select  button to display the image setting submenu. 
4	Press the right/increase  button to select the vertical position  icon and press the menu/select  button to select it.
5	Press the right/increase  button or the left/decrease  button to adjust the vertical position to 47.
6	Press the menu/select  button to accept the setting.

24.6 Servicing and Configuring the 18.1-inch LCD

24.6.1 Recommendations for initial monitor settings

Following are the recommended settings for the 18.1-inch LCD.

- **Native resolution:** 1280X1024
- **Color:** High color (16 bit)
- **Refresh:** 60 HZ

ATTENTION

When using an 18.1-inch LCD in conjunction with a CRT monitor, the LCD should be set to the maximum resolution of the CRT.

24.6.2 Configure monitor settings

Step	Action
1	Select Start > Settings > Control Panel .
2	Double-click the Display icon and select the Settings tab.
3	Under the Colors area, select High Color (16 bit).
4	Click Apply .
5	Under the Screen area, drag the slider to the appropriate resolution.
6	Click Apply .
7	From the Display Properties window, click the Advanced button and select the Monitor tab.
8	Select 60 Hertz from the Refresh Frequency drop down menu.
9	Click OK to apply changes.

24.6.3 Monitor procedures for dual consoles

If you have a dual console, you will also need to perform the following procedures.

- Correct color mismatch
- Configure adjoining monitor displays

24.6.4 Correct color mismatch

Step	Action
NOTE Perform these steps if there appears to be a color mismatch on adjacent console monitors.	
1	Push the PROCEED button to activate Auto Adjust.
2	Highlight AUTO by using the arrow control buttons.
3	Press PROCEED again to execute the Auto Adjust.

24.6.5 Configure adjoining monitor displays

Step	Action
1	Select Start > Settings > Control Panel .
2	Double-click the Display icon and select the Settings tab.
3	Identify your primary monitor.
4	Drag the monitor icons to align them in a dual configuration as they are physically aligned on the console.
5	Place the cursor on each monitor to view the monitor coordinates.
6	Verify that the monitors are aligned horizontally and vertically with one another.

24. 18.1 INCH LCD MONITOR INSTALLATION, OPERATION AND SERVICE

24.6. Servicing and Configuring the 18.1-inch LCD

25 CLASSIC CONSOLE TOUCHSCREEN SERVICE

25.1 Overview

25.1.1 Scope

The touchscreen assembly consists of a printed wiring assembly (PWA) and infrared (IR) windows. It is assembled to the inside of the plastic front bezel. The touchscreen provides cursor position information signals based on the position of an operator's finger or other stylus being placed in proximity to the monitor screen surface.

25.1.2 Description

The touchscreen uses a matrix of optoelectronic devices operating in the infrared spectrum and controller electronics to provide cursor position information to the computer. The touchscreen assembly is mounted inside the plastic front bezel assembly for mounting to the screen of a 19-inch flat screen monitor. An interconnecting cable is provided for connection to the computer.

25.1.3 Illustration

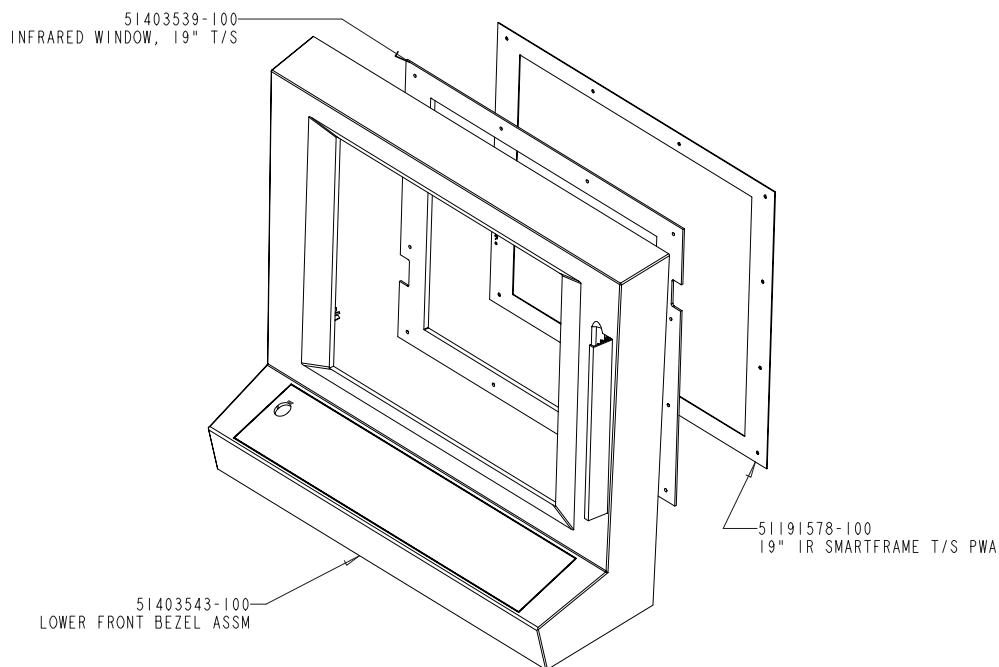


Figure 24-1 Exploded View of 19-inch Touchscreen Assembly

25.2 Cleaning the Touchscreen Window

25.2.1 Procedure

Follow these steps to clean the touchscreen window.

Table 24-1 Cleaning the Touchscreen Window

Step	Action
1	Gently wipe the four inside edges of the touchscreen window with a clean, slightly damp, soft cloth.
2	Ensure that the surface of the touchscreen window is dry by wiping all four sides with a clean, soft, dry cloth.
3	If the touchscreen still does not function correctly, disassemble the touchscreen assembly and clean the individual LEDs and sensors on the touchscreen frame before replacing the touchscreen assembly.

25.3 Touchscreen Alignment Verification

25.3.1 Procedure

Follow these steps to clean the touchscreen window.

Table 24-2 Touchscreen Alignment Verification

Step	Action
1	After Autoboot completes, select CT PANEL from the Honeywell GUS (Common) window. This opens the Carroll Touch control panel.
2	Choose Calibration from the drop-down menu.
3	A large “+” mark appears in the upper-left corner of the screen. Follow the instructions that are displayed in the center of the screen to complete the process.

25.4 Removing the Touchscreen Assembly

25.4.1 Scope

The touchscreen assembly is attached to the inside of the monitor bezel. The monitor must be removed from the station before the bezel assembly can be removed. See Section 3 for instructions for access and retraction of the monitor.

25.4.2 Removal procedure

To remove a touchscreen assembly from the monitor, follow the procedure below and reference the illustration that follows.

Table 24-3 Replacing the Touchscreen

Step	Action
1	Place the breaker switch inside the station in the OFF position.
2	Disconnect all cables from the monitor and remove the monitor assembly from the station. If you do not remove it now, you will have to later. After the monitor is removed, slide the monitor rails out of the way.
3	Disconnect the interface cable from the touchscreen.
4	Disconnect the power cable from the touchscreen.
5	Disconnect the two ground wires.
6	After removing the four screws that fasten the four corners of the touchscreen to the bezel, remove the touchscreen.
7	Slide the touchscreen out of the bezel.

25.5 Replacing the Touchscreen Assembly

25.5.1 Replacement procedure

The touchscreen frame can be installed by reversing the previous procedure.

25.6 Touchscreen Cabling

25.6.1 Scope

The connection point for the PC-end of the cabling between the touchscreen and the IKBI2 board in the Global User Station electronic module is documented in Figure 2-2.

25.7 Touchscreen Calibration

25.7.1 Scope

After a touchscreen has been replaced, it should be calibrated to establish correct registration with the monitor.

25.7.2 Procedure

Follow this procedure to calibrate the touchscreen.

Table 24-4 Calibrating the Touchscreen

Step	Action
1	Double-click the Carroll Touch Control Panel icon added to the Honeywell GUS program group. The calibration program is turned on. ATTENTION If there is a problem communicating with the touchscreen, an error message will be displayed. If this happens, reboot the system to enable the driver and start the calibration again.
2	Select the CT Panel item on the menu bar and then the Calibration... item. The calibration screen will appear, containing the following instructions: "If the touch system does not respond to your touch after a few seconds, exit the calibration screen by pressing ESC, then select the Touch Event Settings menu item."
3	If the check box in the Enable Mouse/Touch section reads "No Touch," select the Communication Settings ... menu item and make sure that the communication settings match those of your touch system.
4	Touch the center of the upper-left calibration target using only one finger, inserting it straight into the touchscreen. Note: Even if beeps do <u>not</u> occur in the steps below, continue holding finger in place until next target appears. Hold your finger in place until you hear a beep or target changes, then remove your finger.
5	Follow the above instructions and after the beep or target changes, the following new instructions appear: "Touch the center of the lower-right calibration target using only one finger, inserting it straight into the touchscreen. Hold your finger in place until you hear a beep, then remove your finger."
6	After you hear the second beep or target changes, keep the finger in place until the screen returns to the "Carroll Touch Control Panel" screen clearing the second target.
7	Close the window to exit the Carroll Touch Control Panel program.
8	The touchscreen is now calibrated.

25.8 Spare Parts List

Table 24.5 Touchscreen Spare Parts

Part Number	Description
*51191578-100	19" IR SmartFrame T/S PWA
*51403539-100	Infrared Window, 19" T/S
*51403543-100	Lower Front Bezel Assm

* = Optimum Replaceable Unit (ORU)

25. CLASSIC CONSOLE TOUCHSCREEN SERVICE

25.8. Spare Parts List

26 VUTRONIC TREND PEN RECORDER MAINTENANCE Addendum AD

AD.1 RECORDER SERVICE DATA

The *Recorder Service Data* (RO-02-00) manual should be referenced when replacing chart paper, pen capsules and other maintenance items in the Vutronics Trend Pen recorder. The manual also contains lubrication, cleaning, calibration, troubleshooting, and parts replacement instructions. The manual can be ordered from the Honeywell Technical Resources Distribution Center, (602) 313-3176.

AD.2 RECORDER PARTS LIST

The *Recorder Parts List* (RO-04-01) manual contains part numbers and detailed parts-placement drawings that are useful in identifying and ordering replacement parts. The manual can be ordered from the Honeywell Technical Resources Distribution Center, (602) 313-3176.

26. VUTRONIC TREND PEN RECORDER MAINTENANCE

Addendum AD

25.8. Spare Parts List
